6. If the error signal is not the difference between input and output, by what general name can we describe the error signal?
7. Name two advantages of having a computer in the loop.
8. Name the three major design criteria for control systems.
9. Name the two parts of a system's response.
10. Physically, what happens to a system that is unstable?
11. Instability is attributable to what part of the total response?
12. Describe a typical control system analysis task.
13. Describe a typical control system design task.
14. Adjustments of the forward path gain can cause changes in the transient response. True or false?
15. Name three approaches to the mathematical modeling of control systems.
16. Briefly describe each of your answers to Question 15.

Problems

1. A variable resistor, called a *potentiometer*, is shown in Figure P1.1. The resistance is varied by moving a wiper arm along a fixed resistance. The resistance from A to C is fixed, but the resistance from B to C varies with the position of the wiper arm. If it takes 10 turns to move the wiper arm from A to C, draw a block diagram of the potentiometer showing the input variable, the output variable, and (inside the block) the gain, which is a constant and is the amount by which the input is multiplied to obtain the output. [Section 1.4: Introduction to a Case Study]

WileyPLUS [5] WPCS Control Solutions

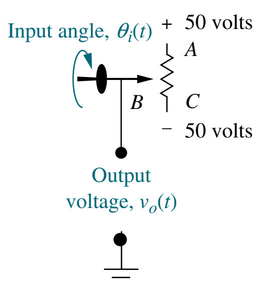


FIGURE P1.1 Potentiometer

2. A temperature control system operates by sensing the difference between the thermostat setting and the actual temperature and then opening a fuel valve an amount proportional to this difference. Draw a functional closed-loop block diagram similar to Figure 1.9(d) identifying the input and output transducers, the controller, and the plant. Further, identify the input and output signals of all subsystems previously described. [Section 1.4: Introduction to a Case Study]

3. An aircraft's attitude varies in roll, pitch, and yaw as defined in Figure P1.2. Draw a functional block diagram for a closed-loop system that stabilizes the roll as follows: The system measures the actual roll angle with a gyro and compares the actual roll angle with the desired roll angle. The ailerons respond to the roll-angle error by undergoing an angular deflection. The aircraft responds to this angular deflection, producing a roll angle rate. Identify the input and output transducers, the controller, and the plant. Further, identify the nature of each signal. [Section 1.4: Introduction to a Case Study]

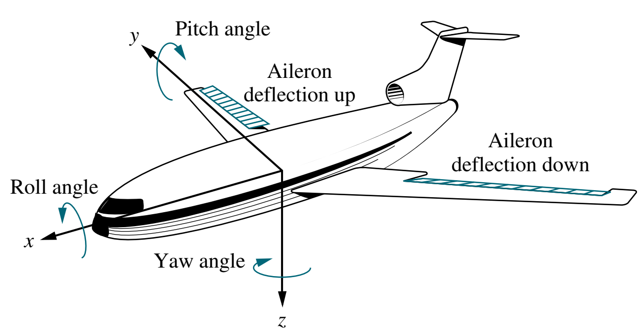


FIGURE P1.2 Aircraft attitude defined

[5] The WileyPLUS icon identifies interactive worked examples and problems. These problems, developed by JustAsk, are worked in detail and offer explanations of every facet of the solution. The identified examples and problems can be accessed at www.wiley.com/college/nise.

4. Many processes operate on rolled material that moves from a supply reel to a take-up reel. Typically, these systems, called *winders*, control the material so that it travels at a constant velocity. Besides velocity, complex winders also control tension, compensate for roll inertia while accelerating or decelerating, and regulate acceleration due to sudden changes. A winder is shown in Figure P1.3. The force transducer measures tension; the winder pulls against the nip rolls, which provide an opposing force; and the bridle provides slip. In order to compensate for changes in speed, the material is looped around a *dancer*. The loop prevents rapid changes from causing excessive slack or damaging the material. If the dancer position is sensed by a potentiometer or other device, speed variations due to buildup on the take-up reel or other causes can be controlled by comparing the potentiometer voltage to the commanded speed. The system then corrects the speed and resets the dancer to the desired position (*Ayers, 1988*). Draw a functional block diagram for the speed control system, showing each component and signal. [Section 1.4: Introduction to a Case Study]

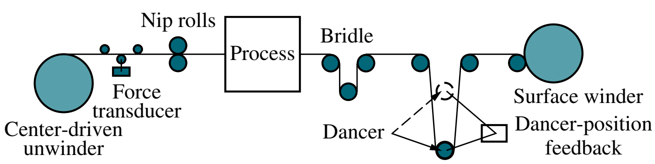


FIGURE P1.3 Winder

5. In a nuclear power generating plant, heat from a reactor is used to generate steam for turbines. The rate of the fission reaction determines the amount of heat generated, and this rate is controlled by rods inserted into the radioactive core. The rods regulate the flow of neutrons. If the rods are lowered into the core, the rate of fission will diminish; if the rods are raised, the fission rate will increase. By automatically controlling the position of the rods, the amount of heat generated by the reactor can be regulated. Draw a functional block diagram for the nuclear reactor control system shown in Figure P1.4. Show all blocks and signals. [Section 1.4: Introduction to a Case Study]

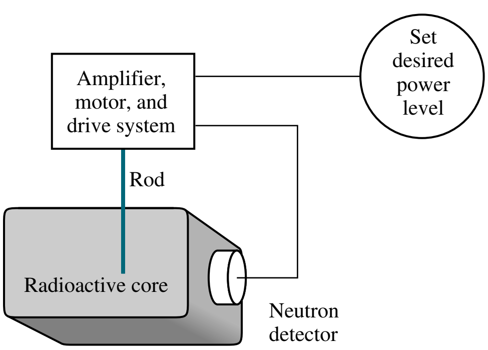


FIGURE P1.4 Control of a nuclear reactor

6. A university wants to establish a control system model that represents the student population as an output, with the desired student population as an input. The administration determines the rate of admissions by comparing the current and desired student populations. The admissions office then uses this rate to admit students. Draw a functional block diagram showing the administration and the admissions office as blocks of the system. Also show the following signals: the desired student population, the actual student population, the desired student rate as determined by the administration, the actual student rate as generated by the admissions office, the dropout rate, and the net rate of influx. [Section 1.4: Introduction to a Case Study]

7. We can build a control system that will automatically adjust a motorcycle's radio volume as the noise generated by the motorcycle changes. The noise generated by the motorcycle increases with speed. As the noise increases, the system increases the volume of the radio. Assume that the amount of noise can be represented by a voltage generated by the speedometer cable, and the volume of the radio is controlled by a dc voltage (*Hogan, 1988*). If the dc voltage represents the desired volume disturbed by the motorcycle noise, draw the functional block diagram of the automatic volume control system, showing the input transducer, the volume control circuit, and the speed transducer as blocks. Also show the following signals: the desired volume as an input, the actual volume as an output, and voltages representing speed, desired volume, and actual volume. [Section 1.4: Introduction to a Case Study]

WileyPLUS **WPCS** Control Solutions

8. Your bathtub at home is a control system that keeps the water level constant. A constant flow from the tap yields a constant water level, because the flow rate through the drain increases as the water level increases, and decreases as the water level decreases. After equilibrium has been reached, the level can be controlled by controlling the input flow rate. A low input flow rate yields a lower level, while a higher input flow rate yields a higher level. [Section 1.4: Introduction to a Case Study]

a. Sketch a control system that uses this principle to precisely control the fluid level in a tank. Show the intake and drain valves, the tank, any sensors and transducers, and the interconnection of all components.

b. Draw a functional block diagram of the system, identifying the input and output signals of each block.

9. A dynamometer is a device used to measure torque and speed and to vary the load on rotating devices. The dynamometer operates as follows to control the amount of torque: A hydraulic actuator attached to the axle presses a tire against a rotating flywheel. The greater the displacement of the actuator, the more force that is applied to the rotating flywheel. A strain gage load cell senses the force. The displacement of the actuator is controlled by an electrically operated valve whose displacement regulates fluid flowing into the actuator (*D'Souza, 1988*). Draw a functional block diagram of a closed-loop system that uses the described dynamometer to regulate the force against the tire during testing. Show all signals and systems. Include amplifiers that power the valve, the valve, the actuator and load, and the tire. [Section 1.4: Introduction to a Case Study]

10. During a medical operation an anesthesiologist controls the depth of unconsciousness by controlling the concentration of isoflurane in a vaporized mixture with oxygen and nitrous oxide. The depth of anesthesia is measured by the patient's blood pressure. The anesthesiologist also regulates ventilation, fluid balance, and the administration of other drugs. In order to free the anesthesiologist to devote more time to the latter tasks, and in the interest of the patient's safety, we wish to automate the depth of anesthesia by automating the control of isoflurane concentration. Draw a functional block diagram of the system showing pertinent signals and subsystems (*Meier, 1992*). [Section 1.4: Introduction to a Case Study]

WileyPLUS WPCS Control Solutions

11. The vertical position, $x(t)$, of the grinding wheel shown in Figure P1.5 is controlled by a closed-loop system. The input to the system is the desired depth of grind, and the output is the actual depth of grind. The difference between the desired depth and the actual depth drives the motor, resulting in a force applied to the work. This force results in a feed velocity for the grinding wheel (*Jenkins, 1997*). Draw a closed-loop functional block diagram for the grinding process, showing the input, output, force, and grinder feed rate. [Section 1.4: Introduction to a Case Study]

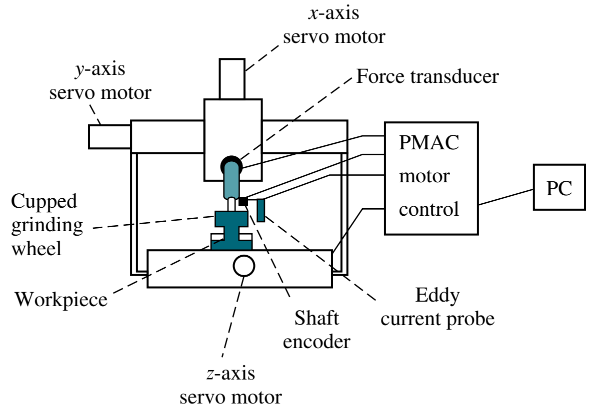


FIGURE P1.5 Grinder system (Reprinted with permission of ASME.)

12. A high-speed proportional solenoid valve is shown in Figure P1.6. A voltage proportional to the desired position of the spool is applied to the coil. The resulting magnetic field produced by the current in the coil causes the armature to move. A push pin connected to the armature moves the spool. A linear voltage differential transformer (LVDT) that outputs a voltage proportional to displacement senses the spool's position. This voltage can be used in a feedback path to implement closed-loop operation (*Vaughan, 1996*). Draw a functional block diagram of the valve, showing input and output positions, coil voltage, coil current, and spool force. [Section 1.4: Introduction to a Case Study]

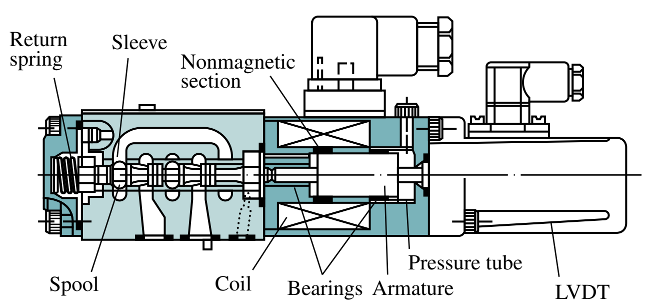


FIGURE P1.6 High-speed proportional solenoid valve (Reprinted with permission of ASME.)

13. The human eye has a biological control system that varies the pupil diameter to maintain constant light intensity to the retina. As the light intensity increases, the optical nerve sends a signal to the brain, which commands internal eye muscles to decrease the pupil's eye diameter. When the light intensity decreases, the pupil diameter increases.

a. Draw a functional block diagram of the light-pupil system indicating the input, output, and intermediate signals; the sensor; the controller; and the actuator. [Section 1.4: Introduction to a Case Study]

b. Under normal conditions the incident light will be larger than the pupil, as shown in Figure P1.7(*a*). If the incident light is smaller than the diameter of the pupil as shown in Figure P1.7(*b*), the feedback path is broken (*Bechhoefer*, *2005*). Modify your block diagram from Part **a.** to show where the loop is broken. What will happen if the narrow beam of light varies in intensity, say in a sinusoidal fashion?

c. It has been found (*Bechhoefer*, *2005*) that it takes the pupil about 300 milliseconds to react to a change in the incident light. If light shines off center to the retina as shown in Figure P1.7(*c*), describe the response of the pupil with delay present and then without delay present.

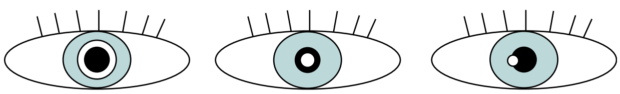


FIGURE P1.7 Pupil is shown black; light beam is shown white. **a.** Light beam diameter is larger than pupil. **b.** Light beam diameter is smaller than pupil. **c.** Narrow light beam is illuminated at pupil's edge.

14. A Segway®[6] Personal Transporter (PT) (Figure P1.8) is a two-wheeled vehicle in which the human operator stands vertically on a platform. As the driver leans left, right, forward, or backward, a set of sensitive gyroscopic sensors sense the desired input. These signals are fed to a computer that amplifies them and commands motors to propel the vehicle in the desired direction. One very important feature of the PT is its safety: The system will maintain its vertical position within a specified angle despite road disturbances, such as uphills and downhills or even if the operator over-leans in any direction. Draw a functional block diagram of the PT system that keeps the system in a vertical position. Indicate the input and output signals, intermediate signals, and main subsystems. (http://segway.com)

FIGURE P1.8 The Segway Personal Transporter (PT)

15. In humans, hormone levels, alertness, and core body temperature are synchronized through a 24-hour circadian cycle. Daytime alertness is at its best when sleep/wake cycles are in synch with the circadian cycle. Thus alertness can be easily affected with a distributed work schedule, such as the one to which astronauts are subjected. It has been shown that the human circadian cycle can be delayed or advanced through light stimulus. To ensure optimal alertness, a system is designed to track astronauts' circadian cycles and increase the quality of sleep during missions. Core body temperature can be used as an indicator of the circadian cycle. A computer model with optimum circadian body temperature variations can be compared to an astronaut's body temperatures. Whenever a difference is detected, the astronaut is subjected to a light stimulus to advance or delay the astronaut's circadian cycle (*Mott*, *2003*). Draw a functional block diagram of the system. Indicate the input and output signals, intermediate signals, and main subsystems.

WileyPLUS WPCS Control Solutions

16. Tactile feedback is an important component in the learning of motor skills such as dancing, sports, and physical rehabilitation. A suit with white dots recognized by a vision system to determine arm joint positions with millimetric precision was developed. This suit is worn by both teacher and

[6] Segway is a registered trademark of Segway, Inc. in the United States and/or other countries.

student to provide position information. (*Lieberman, 2007*). If there is a difference between the teacher's positions and that of the student, vibrational feedback is provided to the student through 8 strategically placed vibrotactile actuators in the wrist and arm, which take advantage of a sensory effect known as *cutaneous rabbit* that tricks the subject to feel uniformly spaced stimuli in places where the actuators are not present. These stimuli help the student adjust to correct the motion. In summary, the system consists of an instructor and a student having their movements followed by the vision system. Their movements are fed into a computer that finds the differences between their joint positions and provides proportional vibrational strength feedback to the student. Draw a block diagram describing the system design.

17. Given the electric network shown in Figure P1.9. [Review] WileyPLUS WPCS Control Solutions

a. Write the differential equation for the network if $v(t) = u(t)$, a unit step.

b. Solve the differential equation for the current, $i(t)$, if there is no initial energy in the network.

c. Make a plot of your solution if $R/L = 1$.

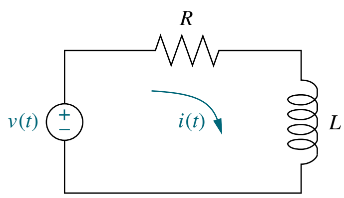


FIGURE P1.9 *RL* network

18. Repeat Problem 17 using the network shown in Figure P1.10. Assume $R = 2\ \Omega$, $L = 1$ H, and $1/LC = 25$. [Review]

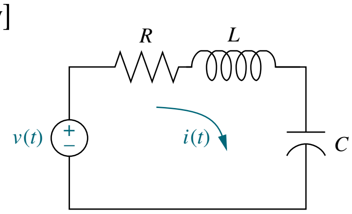


FIGURE P1.10 *RLC* network

19. Solve the following differential equations using classical methods. Assume zero initial conditions. [Review] WileyPLUS WPCS Control Solutions

a. $\dfrac{dx}{dt} + 7x = 5\cos 2t$

b. $\dfrac{d^2x}{dt^2} + 6\dfrac{dx}{dt} + 8x = 5\sin 3t$

c. $\dfrac{d^2x}{dt^2} + 8\dfrac{dx}{dt} + 25x = 10u(t)$

20. Solve the following differential equations using classical methods and the given initial conditions: [Review] WileyPLUS WPCS Control Solutions

a. $\dfrac{d^2x}{dt^2} + 2\dfrac{dx}{dt} + 2x = \sin 2t$

$x(0) = 2;\ \dfrac{dx}{dt}(0) = -3$

b. $\dfrac{d^2x}{dt^2} + 2\dfrac{dx}{dt} + x = 5e^{-2t} + t$

$x(0) = 2;\ \dfrac{dx}{dt}(0) = 1$

c. $\dfrac{d^2x}{dt^2} + 4x = t^2$

$x(0) = 1;\ \dfrac{dx}{dt}(0) = 2$

PROGRESSIVE ANALYSIS AND DESIGN PROBLEMS

21. High-speed rail pantograph. Some high-speed rail systems are powered by electricity supplied to a pantograph on the train's roof from a catenary overhead, as shown in Figure P1.11. The force applied by the pantograph to the catenary is regulated to avoid loss of contact due to excessive transient motion. A proposed method to regulate the force uses a closed-loop feedback system, whereby a force, F_{up}, is applied to the bottom of the pantograph, resulting in an output

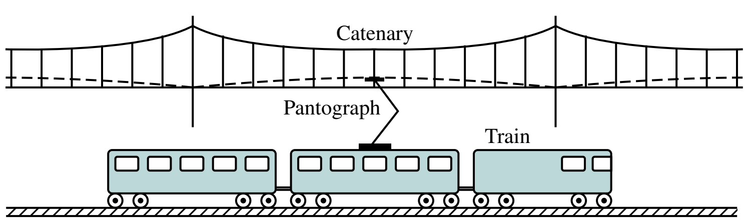


FIGURE P1.11 High-speed rail system showing pantograph and catenary (Reprinted with permission of ASME.)

force applied to the catenary at the top. The contact between the head of the pantograph and the catenary is represented by a spring. The output force is proportional to the displacement of this spring, which is the difference between the catenary and pantograph head vertical positions (*O'Connor, 1997*). Draw a functional block diagram showing the following signals: the desired output force as the input; the force, F_{up}, applied to the bottom of the pantograph; the difference in displacement between the catenary and pantograph head; and the output contact force. Also, show blocks representing the input transducer, controller, actuator generating F_{up}, pantograph dynamics, spring described above, and output sensor. All forces and displacements are measured from equilibrium.

22. **Contol of HIV/AIDS.** As of 2005, the number of people living worldwide with Human Immunodeficiency Virus/Acquired Immune Deficiency Syndrome (HIV/AIDS) was estimated at 40 million, with 5 million new infections per year and 3 million deaths due to the disease (*UNAIDS, 2005*). Currently there is no known cure for the disease, and HIV cannot be completely eliminated in an infected individual. Drug combinations can be used to maintain the virus numbers at low levels, which helps prevent AIDS from developing. A common treatment for HIV is the administration of two types of drugs: reverse transcriptase inhibitors (RTIs) and protease inhibitors (PIs). The amount in which each of these drugs is administered is varied according to the amount of HIV viruses in the body (*Craig, 2004*). Draw a block diagram of a feedback system designed to control the amount of HIV viruses in an infected person. The plant input variables are the amount of RTIs and PIs dispensed. Show blocks representing the controller, the system under control, and the transducers. Label the corresponding variables at the input and output of every block.

23. **Hybrid vehicle.** The use of hybrid cars is becoming increasingly popular. A hybrid electric vehicle (HEV) combines electric machine(s) with an internal combustion engine (ICE), making it possible (along with other fuel consumption–reducing measures, such as stopping the ICE at traffic lights) to use smaller and more efficient gasoline engines. Thus, the efficiency advantages of the electric drivetrain are obtained, while the energy needed to power the electric motor is stored in the onboard fuel tank and not in a large and heavy battery pack.

There are various ways to arrange the flow of power in hybrid car. In a serial HEV (Figure P1.12), the ICE is not connected to the drive shaft. It drives only the generator, which charges the batteries and/or supplies power to the electric motor(s) through an inverter or a converter.

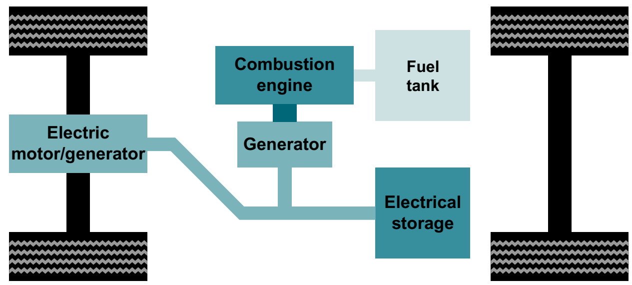


FIGURE P1.12 Serial hybrid-electric vehicle

The HEVs sold today are primarily of the parallel or split-power variety. If the combustion engine can turn the drive wheels as well as the generator, then the vehicle is referred to as a *parallel* hybrid, because both an electric motor and the ICE can drive the vehicle. A parallel hybrid car (Figure P1.13) includes a relatively small battery pack (electrical storage) to put out extra power to the electric motor when fast acceleration is needed. See (*Bosch 5th ed., 2000*), (*Bosch 7th ed., 2007*), (*Edelson, 2008*), (*Anderson, 2009*) for more detailed information about HEV.

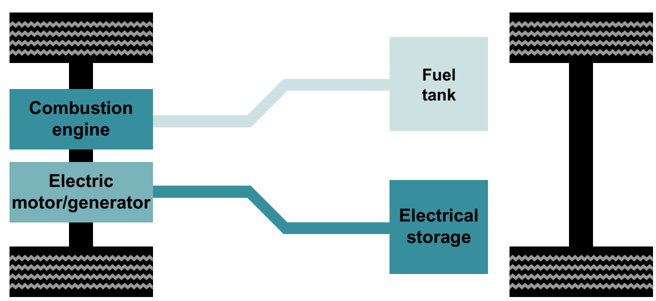


FIGURE P1.13 Parallel hybrid drive

As shown in Figure P1.14, split-power hybrid cars utilize a combination of series and parallel drives (*Bosch, 5th ed., 2007*). These cars use a planetary gear (3) as a split-power transmission to allow some of the ICE power to be applied mechanically to the drive. The other part is converted into electrical energy through the alternator (7) and the inverter (5) to feed the electric motor (downstream of the transmission) and/or to charge the high-voltage battery (6). Depending upon driving conditions, the ICE, the electric motor, or both propel the vehicle.

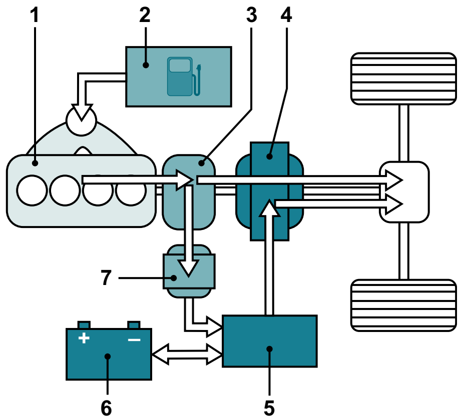


FIGURE P1.14 Split-power hybrid electric vehicle

Draw a functional block diagram for the cruise (speed) control system of:

a. A serial hybrid vehicle, showing its major components, including the speed sensor, electronic control unit (ECU), inverter, electric motor, and vehicle dynamics; as well as all signals, including the desired vehicle speed, actual speed, control command (ECU output), controlled voltage (inverter output), the motive force (provided by the electric motor), and running resistive force[7];

b. A parallel hybrid vehicle, showing its major components, which should include also a block that represents the accelerator, engine, and motor, as well as the signals (including accelerator displacement and combined engine/motor motive force);

c. A split-power HEV, showing its major components and signals, including, in addition to those listed in Parts **a** and **b**, a block representing the planetary gear and its control, which, depending upon driving conditions, would allow the ICE, the electric motor, or both to propel the vehicle, that is, to provide the necessary total motive force.

[7] These include the aerodynamic drag, rolling resistance, and climbing resistance. The aerodynamic drag is a function of car speed, whereas the other two are proportional to car weight.

Cyber Exploration Laboratory

Experiment 1.1

Objective To verify the behavior of closed-loop systems as described in the Chapter 1 Case Study.

Minimum Required Software Packages LabVIEW and the LabVIEW Control Design and Simulation Module. *Note*: While no knowledge of LabVIEW is required for this experiment, see Appendix D to learn more about LabVIEW, which will be pursued in more detail in later chapters.

Prelab

1. From the discussion in the Case Study, describe the effect of the gain of a closed-loop system upon transient response.
2. From the discussion in the Case Study about steady-state error, sketch a graph of a step input superimposed with a step response output and show the steady-state error. Assume any transient response. Repeat for a ramp input and ramp output response. Describe the effect of gain upon the steady-state error.

Lab

1. Launch LabVIEW and open **Find Examples . . .**

ONE

Introduction

ANSWERS TO REVIEW QUESTIONS

1. Guided missiles, automatic gain control in radio receivers, satellite tracking antenna

2. Yes - power gain, remote control, parameter conversion; No - Expense, complexity

3. Motor, low pass filter, inertia supported between two bearings

4. Closed-loop systems compensate for disturbances by measuring the response, comparing it to the input response (the desired output), and then correcting the output response.

5. Under the condition that the feedback element is other than unity

6. Actuating signal

7. Multiple subsystems can time share the controller. Any adjustments to the controller can be implemented with simply software changes.

8. Stability, transient response, and steady-state error

9. Steady-state, transient

10. It follows a growing transient response until the steady-state response is no longer visible. The system will either destroy itself, reach an equilibrium state because of saturation in driving amplifiers, or hit limit stops.

11. Natural response

12. Determine the transient response performance of the system.

13. Determine system parameters to meet the transient response specifications for the system.

14. True

15. Transfer function, state-space, differential equations

16. <u>Transfer function</u> - the Laplace transform of the differential equation

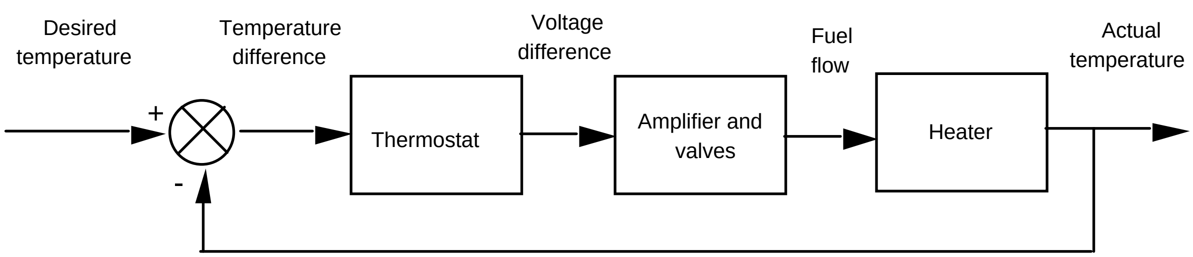
<u>State-space</u> - representation of an nth order differential equation as n simultaneous first-order differential equations

<u>Differential equation</u> - Modeling a system with its differential equation

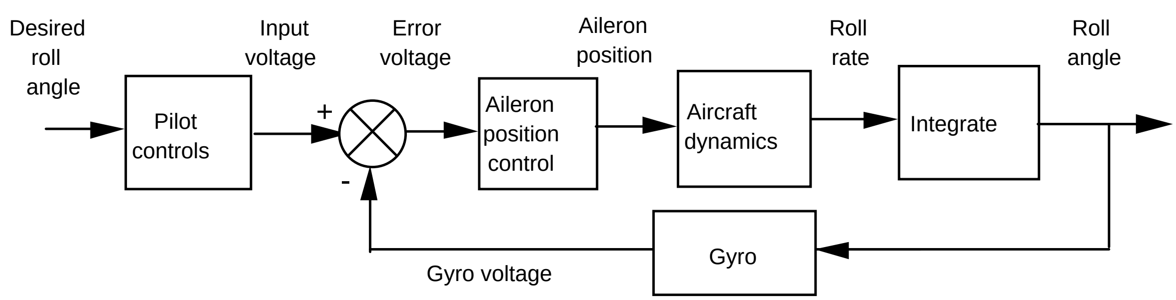
SOLUTIONS TO PROBLEMS

1. Five turns yields 50 v. Therefore K = $\frac{50\text{ volts}}{5 \text{ x } 2\pi \text{ rad}}$ **= 1.59**

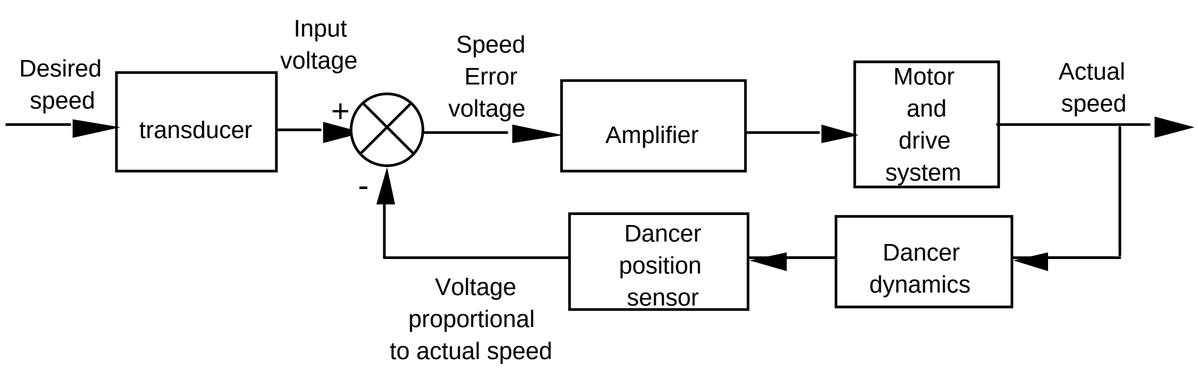
2.


Desired
temperature
Temperature
difference
Voltage
difference
Fuel
flow
Actual
temperature
+
-
Thermostat
Amplifier and
valves
Heater

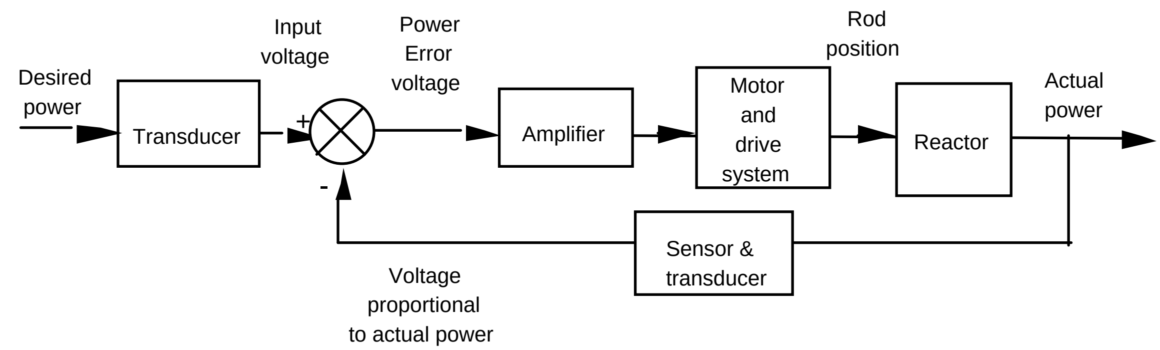
3.


Desired
roll
angle
Input
voltage
Error
voltage
Aileron
position
Roll
rate
Roll
angle
Pilot
controls
+
-
Aileron
position
control
Aircraft
dynamics
Integrate
Gyro
Gyro voltage

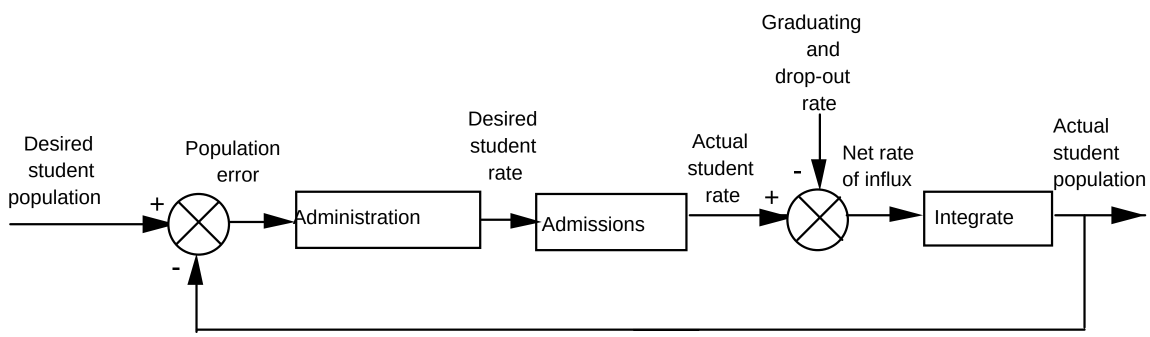
4.


Input
voltage
Desired
speed
Speed
Error
voltage
Actual
speed
transducer
+
-
Amplifier
Motor
and
drive
system
Dancer
position
sensor
Dancer
dynamics
Voltage
proportional
to actual speed

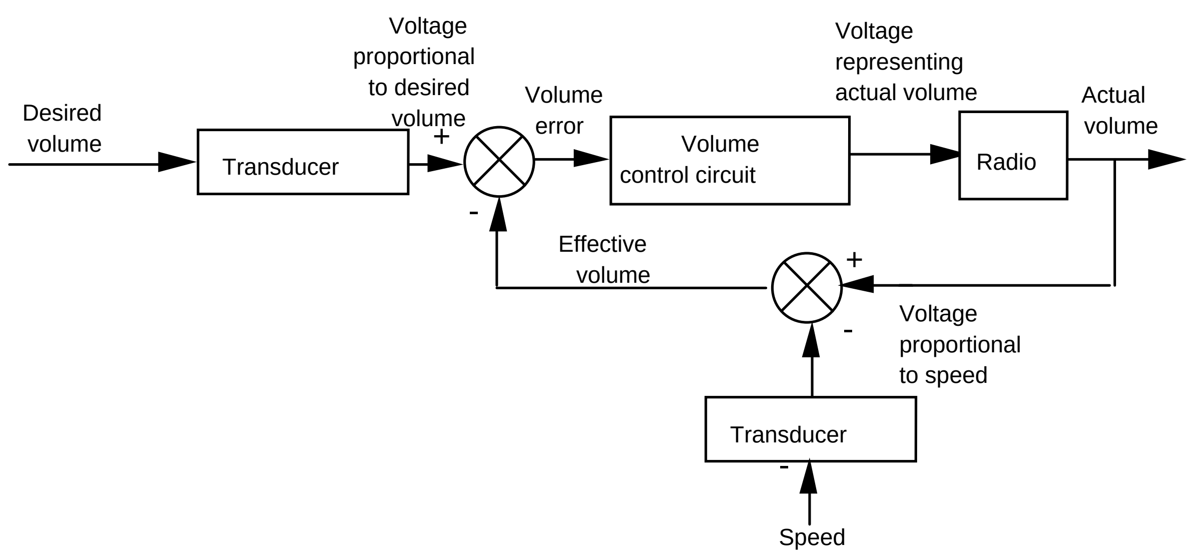
5.


Desired
power
Transducer
Input
voltage
+
-
Power
Error
voltage
Amplifier
Motor
and
drive
system
Rod
position
Reactor
Actual
power
Sensor &
transducer
Voltage
proportional
to actual power

6.

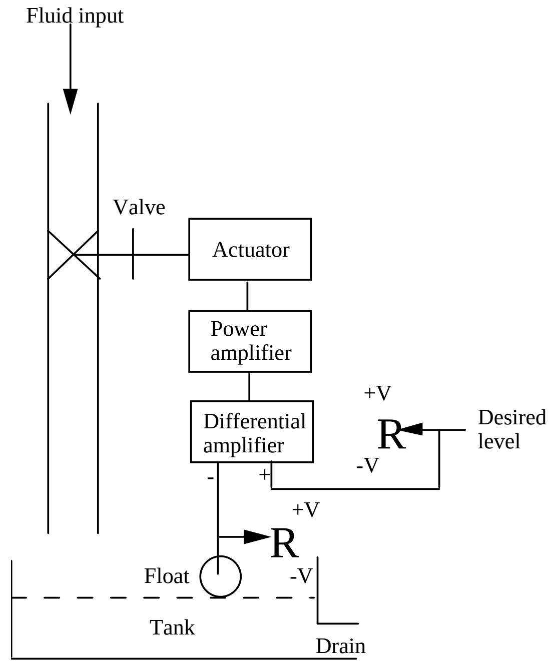

Desired
student
population
+
-
Population
error
Administration
Desired
student
rate
Admissions
Actual
student
rate
+
Graduating
and
drop-out
rate
-
Net rate
of influx
Integrate
Actual
student
population

7.


Desired
volume
Transducer
Voltage
proportional
to desired
volume
+
-
Volume
error
Volume
control circuit
Voltage
representing
actual volume
Radio
Actual
volume
Effective
volume
+
-
Voltage
proportional
to speed
Transducer
-
Speed

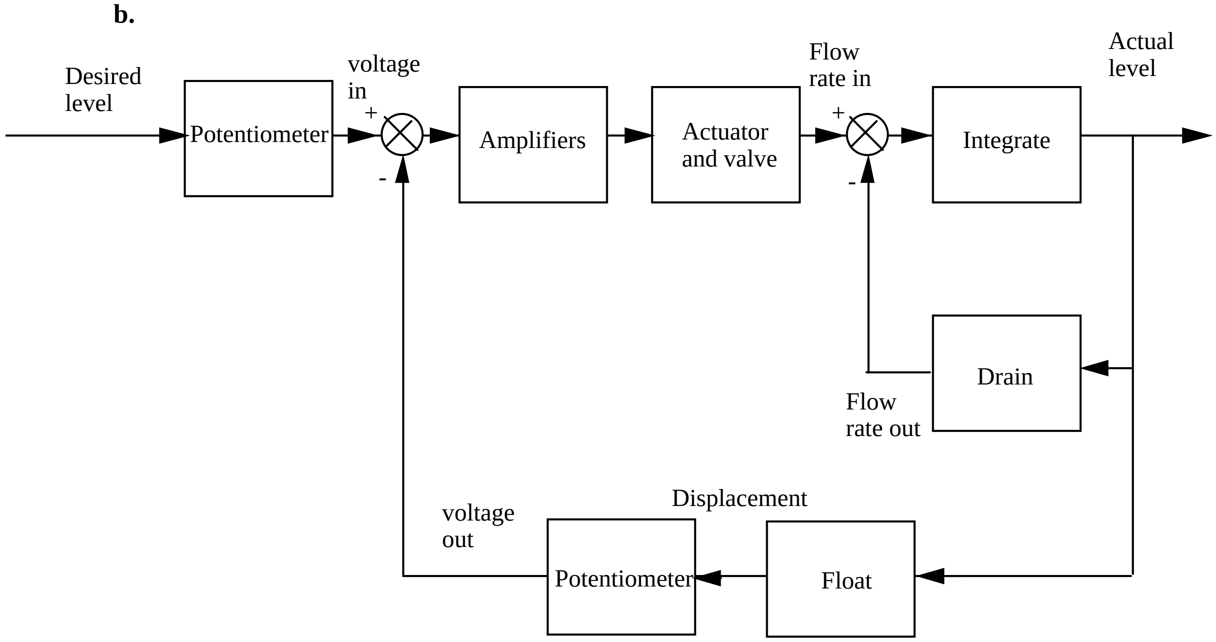
8.

a.



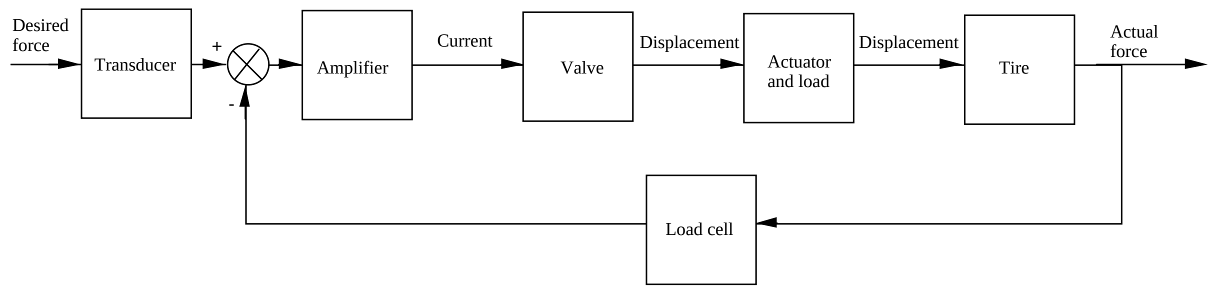
Fluid input
Valve
Actuator
Power amplifier
Differential amplifier
+V
R
-V
Desired level
-
+
+V
R
-V
Float
Tank
Drain

b.

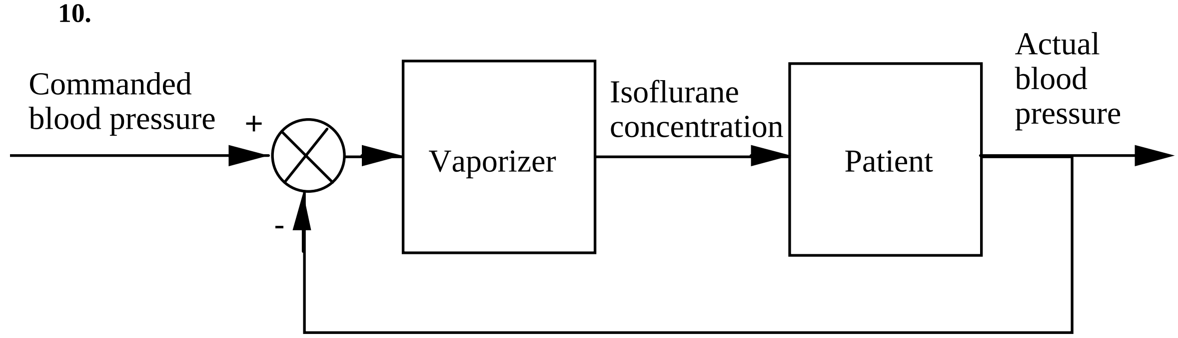


Desired level
Potentiometer
voltage in
+
-
Amplifiers
Actuator and valve
Flow rate in
+
-
Integrate
Actual level
Drain
Flow rate out
Displacement
voltage out
Potentiometer
Float

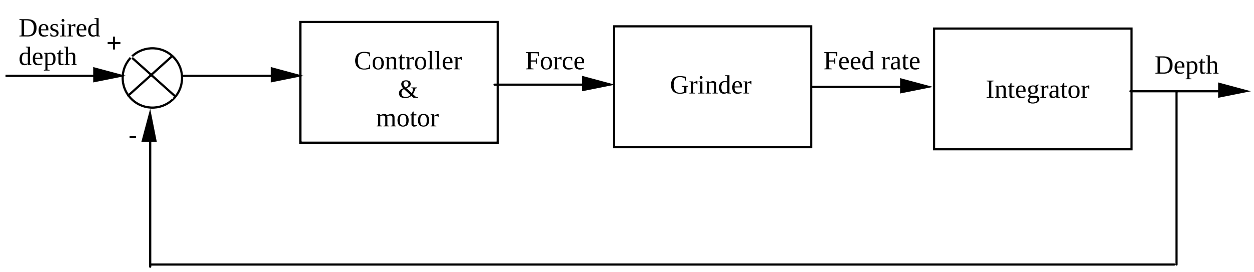
9.


Desired force
Transducer
+
-
Amplifier
Current
Valve
Displacement
Actuator and load
Displacement
Tire
Actual force
Load cell

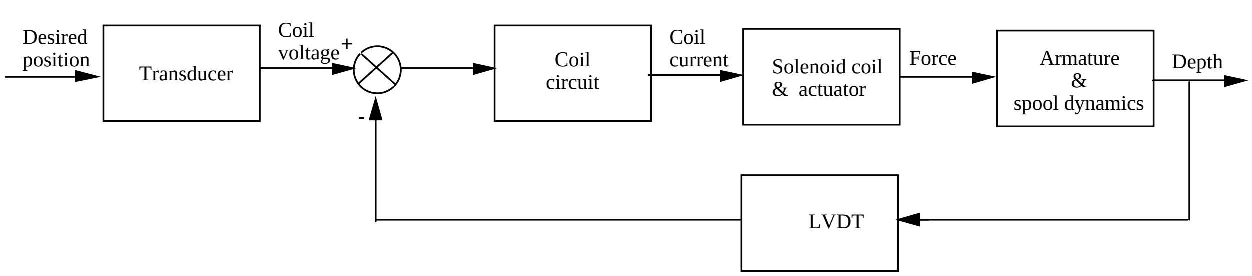
10.


Commanded blood pressure
+
-
Vaporizer
Isoflurane concentration
Patient
Actual blood pressure

11.

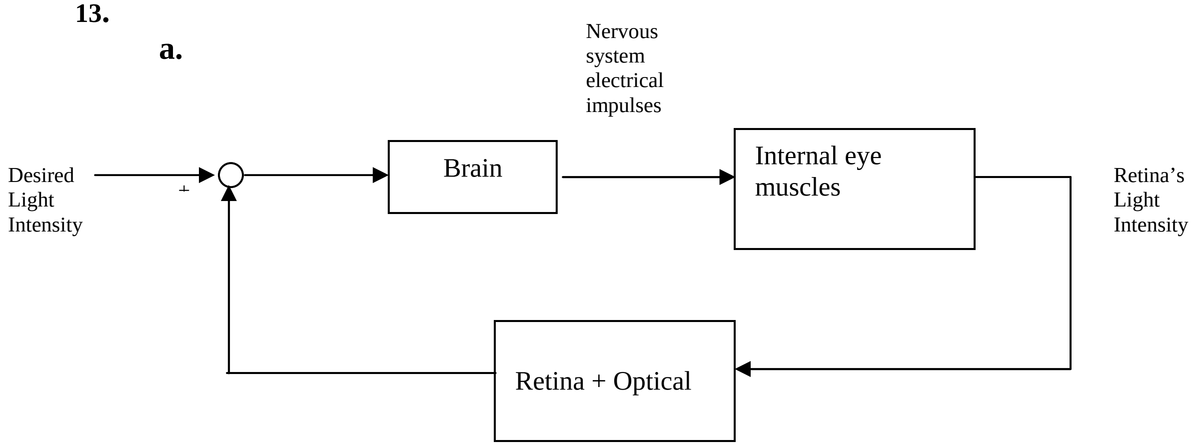

Desired depth
+
-
Controller & motor
Force
Grinder
Feed rate
Integrator
Depth

12.

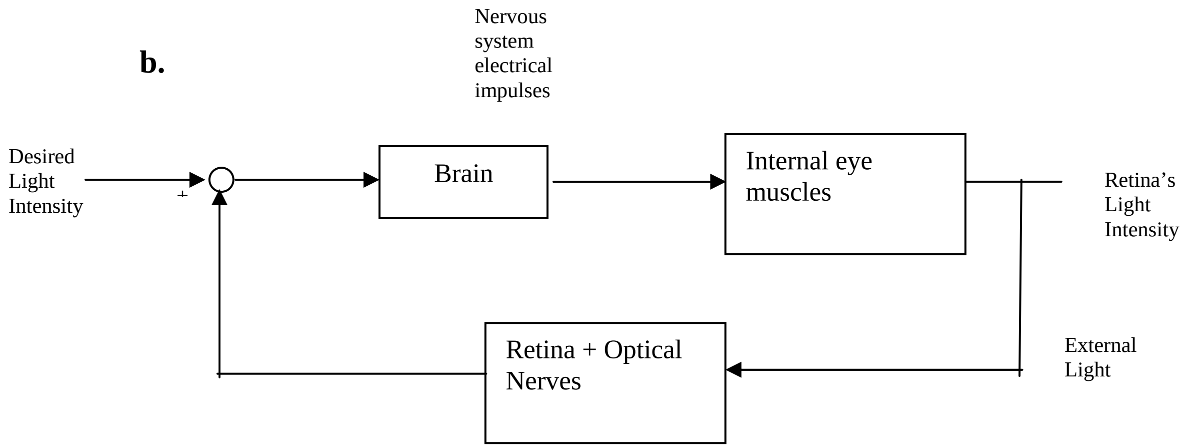

Desired position
Transducer
Coil voltage
+
-
Coil circuit
Coil current
Solenoid coil & actuator
Force
Armature & spool dynamics
Depth
LVDT

13.

a.



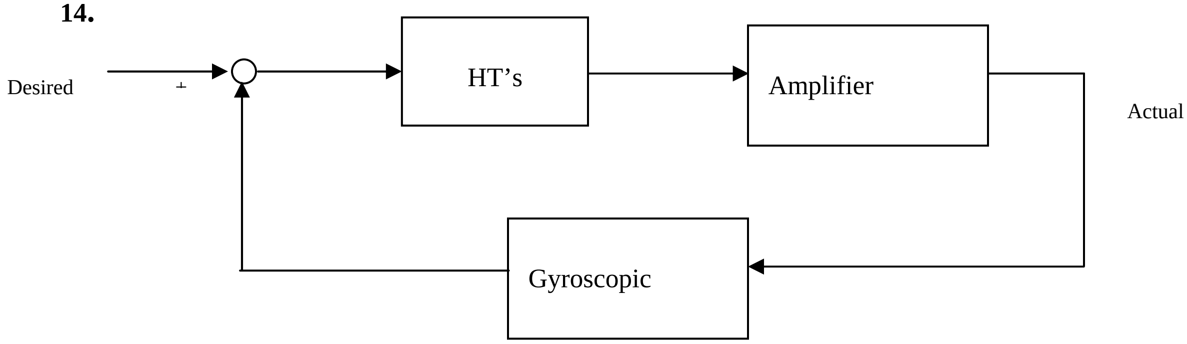
b.



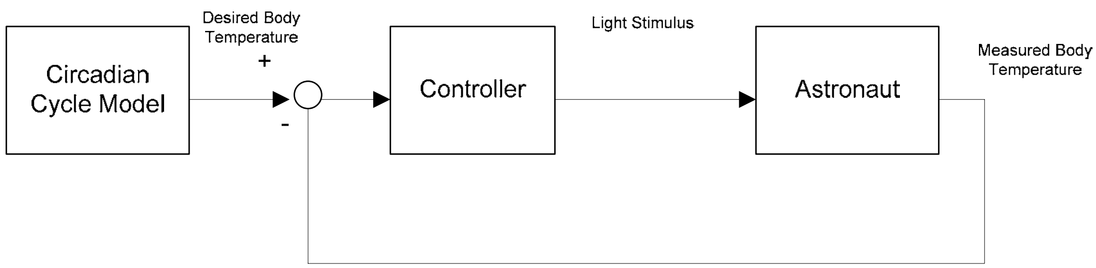
If the narrow light beam is modulated sinusoidally the pupil's diameter will also vary sinusoidally (with a delay see part c) in problem)

c. If the pupil responded with no time delay the pupil would contract only to the point where a small amount of light goes in. Then the pupil would stop contracting and would remain with a fixed diameter.

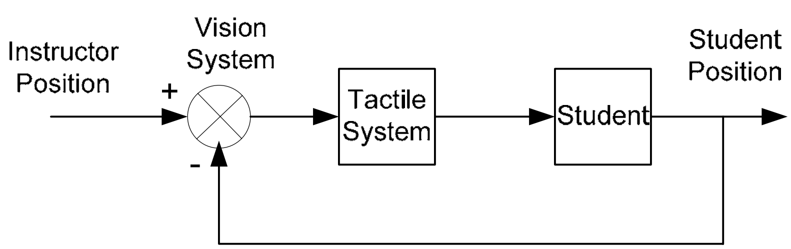
14.



15.



16.



17.

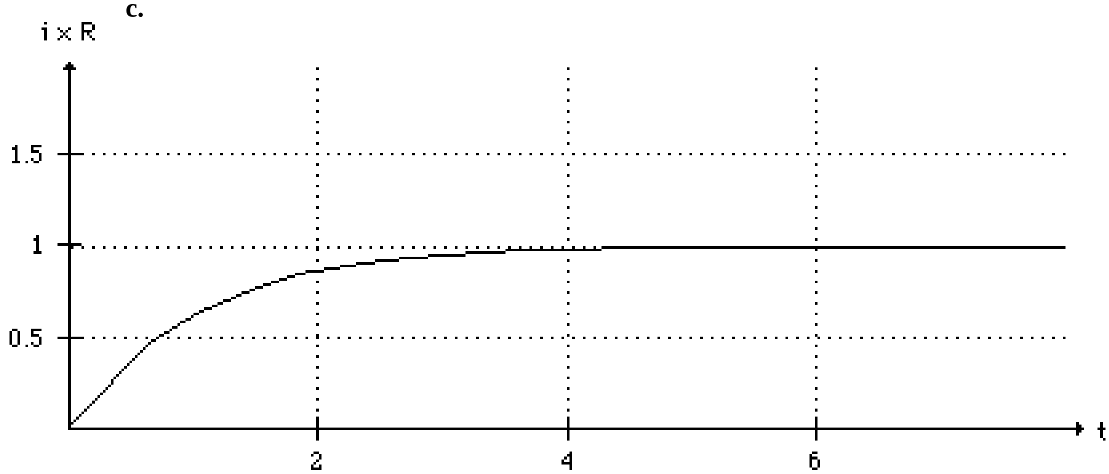
a. $L\dfrac{di}{dt} + Ri = u(t)$

b. Assume a steady-state solution $i_{SS} = B$. Substituting this into the differential equation yields RB =

1,

from which $B = \dfrac{1}{R}$. The characteristic equation is LM + R = 0, from which $M = -\dfrac{R}{L}$. Thus, the total solution is $i(t) = Ae^{-(R/L)t} + \frac{1}{R}$. Solving for the arbitrary constants, $i(0) = A + \frac{1}{R} = 0$. Thus, $A = -\frac{1}{R}$. The final solution is $i(t) = \frac{1}{R} - \frac{1}{R} e^{-(R/L)t} = \frac{1}{R}(1 - e^{-(R/L)t})$.

c.



18.

a. Writing the loop equation, $Ri + L\frac{di}{dt} + \frac{1}{C}\int idt + v_C(0) = v(t)$

b. Differentiating and substituting values, $\frac{d^2i}{dt^2} + 2\frac{di}{dt} + 25i = 0$

Writing the characteristic equation and factoring,

$$M^2 + 2M + 25 = (M + 1 + \sqrt{24}i)(M + 1 - \sqrt{24}i).$$

The general form of the solution and its derivative is

$$i = Ae^{-t}\cos(\sqrt{24}t) + Be^{-t}\sin(\sqrt{24}t)$$

$$\frac{di}{dt} = (-A + \sqrt{24}B)e^{-t}\cos(\sqrt{24}t) - (\sqrt{24}A + B)e^{-t}\sin(\sqrt{24}t)$$

Using $i(0) = 0; \frac{di}{dt}(0) = \frac{v_L(0)}{L} = \frac{1}{L} = 1$

$i(0) = A = 0$

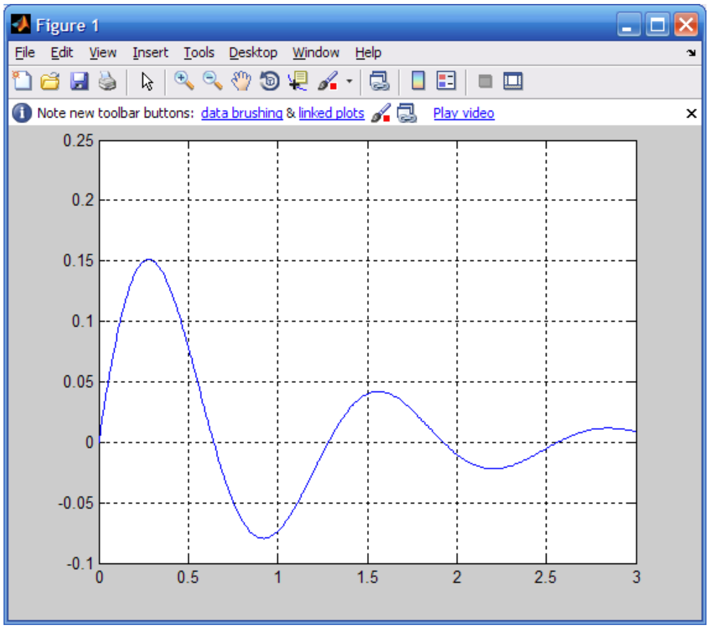
$$\frac{di}{dt}(0) = -A + \sqrt{24}B = 1$$

Thus, $A = 0$ and $B = \frac{1}{\sqrt{24}}$.

The solution is

$$i = \frac{1}{\sqrt{24}}e^{-t}\sin(\sqrt{24}t)$$

c.



19.

a. Assume a particular solution of

Substitute into the differential equation and obtain

$$(7C + 2D)\cos(2t) + (-2C + 7D)\sin(2t) = 5\cos(2t)$$

Equating like coefficients,

$$7C + 2D = 5$$

$$-2C + 7D = 0$$

From which, C = $\frac{35}{53}$ and D = $\frac{10}{53}$.

The characteristic polynomial is

$$M + 7 = 0$$

Thus, the total solution is

$$x(t) = A\,e^{-7t} + \left(\frac{35}{53}\cos[2t] + \frac{10}{53}\sin[2t]\right)$$

Solving for the arbitrary constants, x(0) = A +$\frac{35}{53}$ = 0. Therefore, A = -$\frac{35}{53}$. The final solution is

$$x(t) = \left(-\frac{35}{53}\right)e^{-7t} + \left(\frac{35}{53}\cos[2t] + \frac{10}{53}\sin[2t]\right)$$

b. Assume a particular solution of

$$x_p = A\sin 3t + B\cos 3t$$

Substitute into the differential equation and obtain

$$(18A - B)\cos(3t) - (A + 18B)\sin(3t) = 5\sin(3t)$$

Therefore, 18A – B = 0 and –(A + 18B) = 5. Solving for A and B we obtain

$$x_p = (-1/65)\sin 3t + (-18/65)\cos 3t$$

The characteristic polynomial is

$$M^2 + 6\,M + 8 = (M + 4)\,(M + 2)$$

Thus, the total solution is

$$x = C\,e^{-4\,t} + D\,e^{-2\,t} + \left(-\frac{18}{65}\cos(3\,t) - \frac{1}{65}\sin(3\,t)\right)$$

Solving for the arbitrary constants, $x(0) = C + D - \frac{18}{65} = 0$.

Also, the derivative of the solution is

$$\frac{dx}{dt} = -\frac{3}{65}\cos(3\,t) + \frac{54}{65}\sin(3\,t) - 4\,C\,e^{-4\,t} - 2\,D\,e^{-2\,t}$$

Solving for the arbitrary constants, $\dot{x}(0)\ -\frac{3}{65} - 4C - 2D = 0$, or $C = -\frac{3}{10}$ and $D = \frac{15}{26}$.

The final solution is

$$x = -\frac{18}{65}\cos(3\,t) - \frac{1}{65}\sin(3\,t) - \frac{3}{10}\,e^{-4\,t} + \frac{15}{26}\,e^{-2\,t}$$

c. Assume a particular solution of

$$x_p = A$$

Substitute into the differential equation and obtain 25A = 10, or A = 2/5.

The characteristic polynomial is

$$M^2 + 8\,M + 25 = (M + 4 + 3\,i)\,(M + 4 - 3\,i)$$

Thus, the total solution is

$$x = \frac{2}{5} + e^{-4\,t}\,(B\,\sin(3\,t) + C\,\cos(3\,t))$$

Solving for the arbitrary constants, x(0) = C + 2/5 = 0. Therefore, C = -2/5. Also, the derivative of the solution is

$$\frac{dx}{dt} = ((3B - 4C)\cos(3t) - (4B + 3C)\sin(3t))\,e^{-4t}$$

Solving for the arbitrary constants, $\dot{x}(0) = 3B - 4C = 0$. Therefore, B = -8/15. The final solution is

$$x(t) = \frac{2}{5} - e^{-4t}\left(\frac{8}{15}\sin(3t) + \frac{2}{5}\cos(3t)\right)$$

20.

a. Assume a particular solution of

$$x_p(t) = C\cos(2t) + D\sin(2t)$$

Substitute into the differential equation and obtain

$$-2(C - 2D)\cos(2t) - 4\left(C + \frac{1}{2}D\right)\sin(2t) = \sin(2t)$$

Equating like coefficients,

$$-2(C - 2D) = 0$$
$$-4\left(C + \frac{1}{2}D\right) = 1$$

From which, C = $-\frac{1}{5}$ and D = $-\frac{1}{10}$.

The characteristic polynomial is

$$M^2 + 2M + 2 = (M + 1 + i)(M + 1 - i)$$

Thus, the total solution is

$$x = -\frac{1}{5}\cos(2t) - \frac{1}{10}\sin(2t) + e^{-t}(A\cos[t] + B\sin[t])$$

Solving for the arbitrary constants, x(0) = A $-\frac{1}{5}$ = 2. Therefore, A = $\frac{11}{5}$. Also, the derivative of the solution is

$$\frac{dx}{dt} = -\frac{1}{5}\cos(2t) + \frac{2}{5}\sin(2t) + (-A + B)e^{-t}\cos(t) - (A + B)e^{-t}\sin(t)$$

Solving for the arbitrary constants, $\dot{x}(0) = -A + B - 0.2 = -3$. Therefore, B = $-\frac{3}{5}$. The final solution is

$$x(t) = -\frac{1}{5}\cos(2t) - \frac{1}{10}\sin(2t) + e^{-t}\left(\frac{11}{5}\cos(t) - \frac{3}{5}\sin(t)\right)$$

b. Assume a particular solution of

$$x_p = Ce^{-2t} + Dt + E$$

Substitute into the differential equation and obtain

$$Ce^{-2t} + Dt + 2D + E = 5e^{-2t} + t$$

Equating like coefficients, C = 5, D = 1, and 2D + E = 0.

From which, C = 5, D = 1, and E = - 2.

The characteristic polynomial is

$$M^2 + 2M + 1 = (M + 1)^2$$

Thus, the total solution is

$$x(t) = Ae^{-t} + Be^{-t}t + 5e^{-2t} + t - 2$$

Solving for the arbitrary constants, x(0) = A + 5 - 2 = 2 Therefore, A = -1. Also, the derivative of the solution is

$$\frac{dx}{dt} = (-A + B)e^{-t} - Bte^{-t} - 10e^{-2t} + 1$$

Solving for the arbitrary constants, $\dot{x}(0)$ = B - 8 = 1. Therefore, B = 9. The final solution is

$$x(t) = -e^{-t} + 9te^{-t} + 5e^{-2t} + t - 2$$

c. Assume a particular solution of

$$x_p = Ct^2 + Dt + E$$

Substitute into the differential equation and obtain

$$4Ct^2 + 4Dt + 2C + 4E = t^2$$

Equating like coefficients, C = $\frac{1}{4}$, D = 0, and 2C + 4E = 0.

From which, C = $\frac{1}{4}$, D = 0, and E = - $\frac{1}{8}$.

The characteristic polynomial is

$$M^2 + 4 = (M + 2i)(M - 2i)$$

Thus, the total solution is

$$x(t) = A\cos(2t) + B\sin(2t) + \frac{1}{4}t^2 - \frac{1}{8}$$

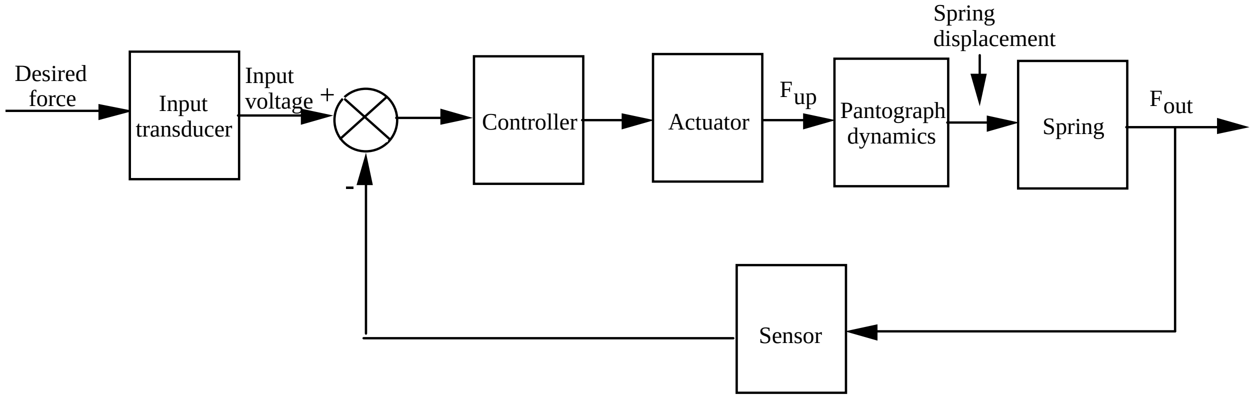
Solving for the arbitrary constants, x(0) = A - $\frac{1}{8}$ = 1 Therefore, A = $\frac{9}{8}$. Also, the derivative of the solution is

$$\frac{dx}{dt} = 2B\cos(2t) - 2A\sin(2t) + \frac{1}{2}t$$

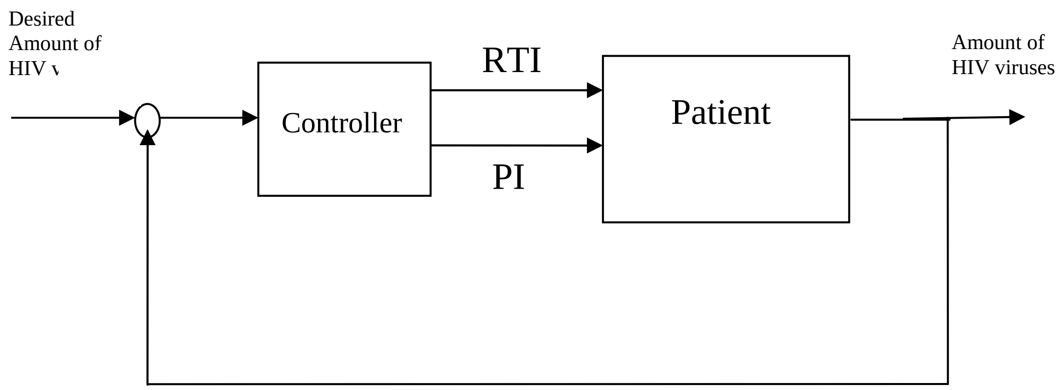
Solving for the arbitrary constants, $\dot{x}(0)$ = 2B = 2. Therefore, B = 1. The final solution is

$$x(t) = \frac{9}{8}\cos(2t) + \sin(2t) + \frac{1}{4}t^2 - \frac{1}{8}$$

21.

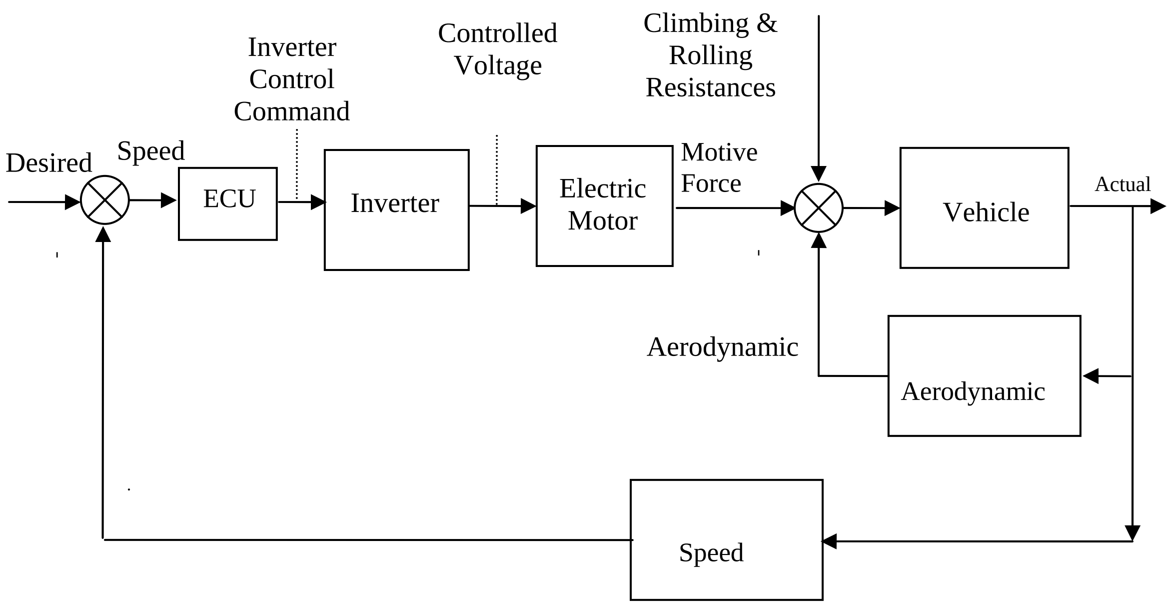


22.



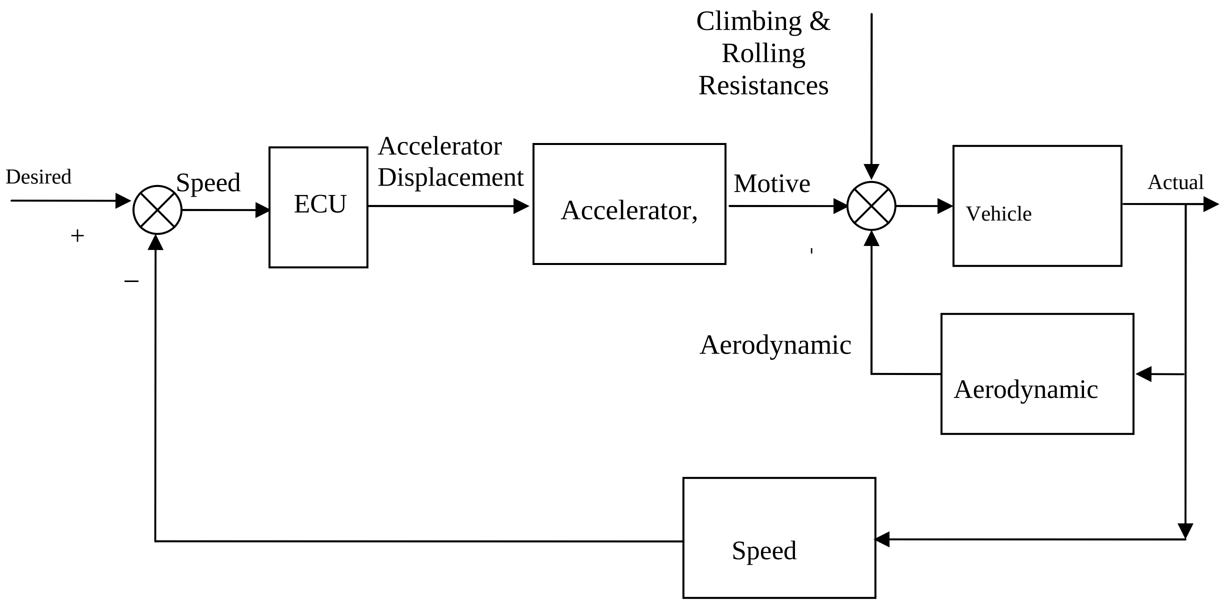
23.

a.

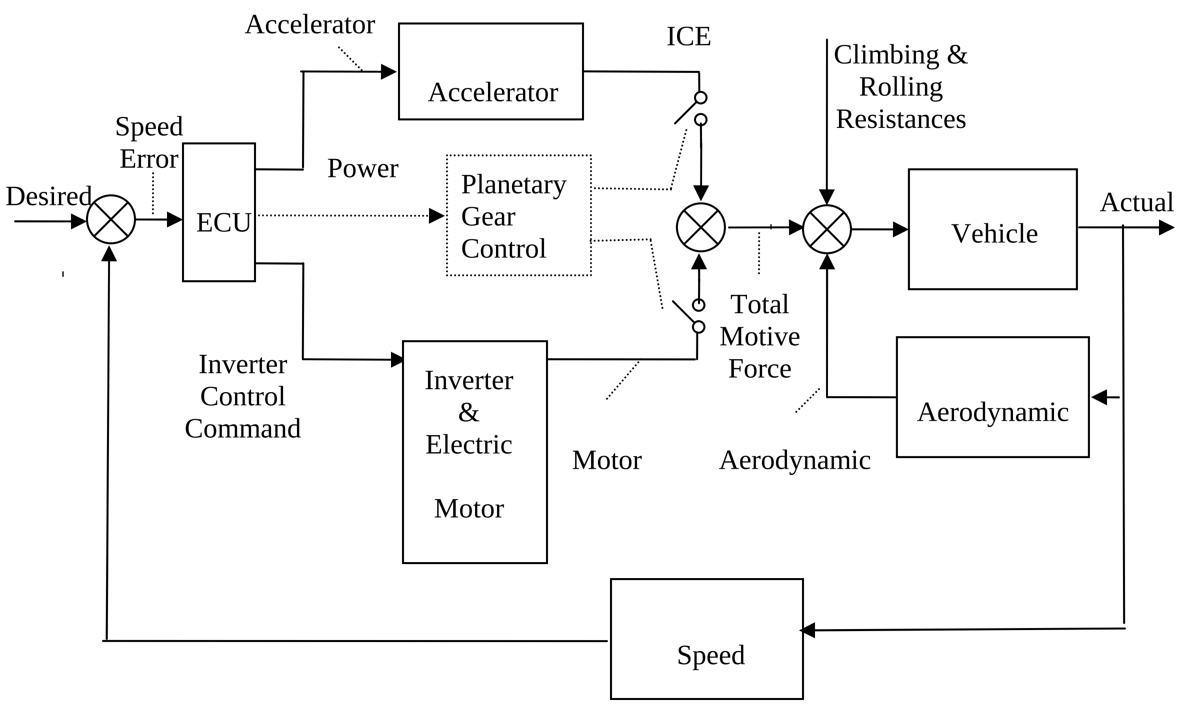


Inverter Control Command
Controlled Voltage
Climbing & Rolling Resistances
Desired
Speed
ECU
Inverter
Electric Motor
Motive Force
Vehicle
Actual
Aerodynamic
Aerodynamic
Speed

b.


Climbing &
Rolling
Resistances
Desired
Speed
+
–
ECU
Accelerator
Displacement
Accelerator,
Motive
Vehicle
Actual
Aerodynamic
Aerodynamic
Speed

c.


Accelerator
Accelerator
ICE
Climbing & Rolling Resistances
Speed Error
Power
Planetary Gear Control
Desired
ECU
Vehicle
Actual
Total Motive Force
Inverter Control Command
Inverter & Electric Motor
Aerodynamic
Motor
Aerodynamic
Speed

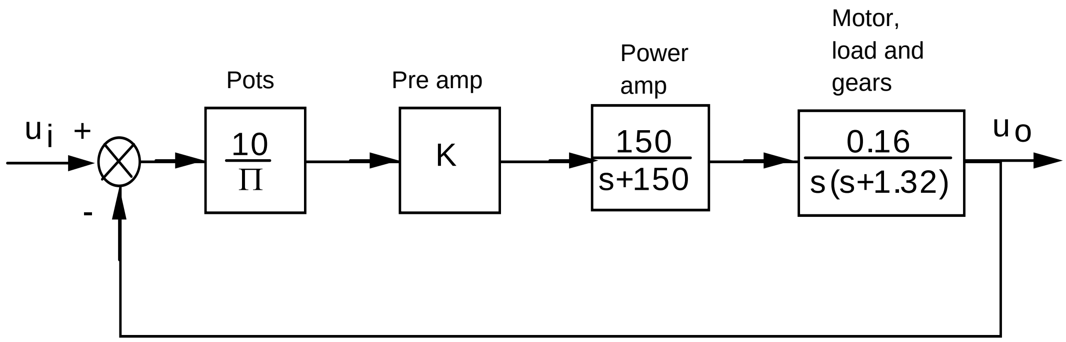
F I V E

Reduction of Multiple Subsystems

SOLUTIONS TO CASE STUDIES CHALLENGES

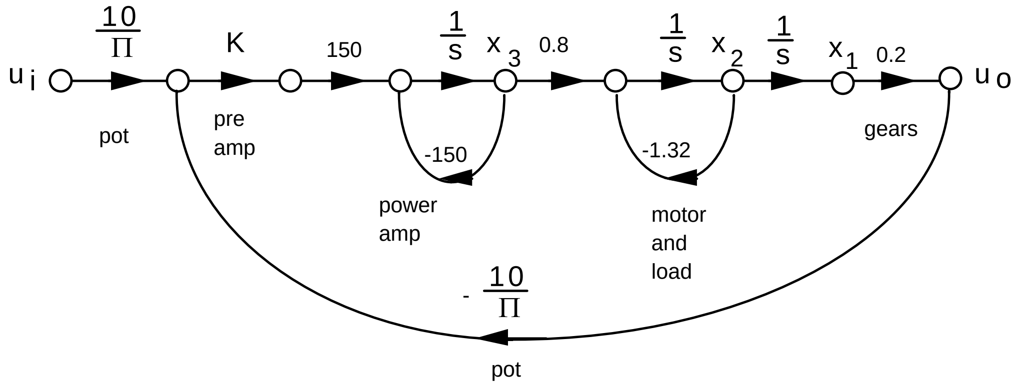
Antenna Control: Designing a Closed-Loop Response

a. Drawing the block diagram of the system:



Thus, $T(s) = \dfrac{76.39K}{s^3+151.32s^2+198s+76.39K}$

b. Drawing the signal flow-diagram for each subsystem and then interconnecting them yields:



$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -1.32x_2 + 0.8x_3$$

$$\dot{x}_3 = -150x_3 + 150K\left(\frac{10}{\pi}(q_i - 0.2x_1)\right) = -95.49Kx_1 - 150x_3 + 477.46K\theta_i$$

$$\theta_o = 0.2x_1$$

In vector-matrix notation,

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1.32 & 0.8 \\ -95.49K & 0 & -150 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 477.46K \end{bmatrix}\theta_i$$

$$\theta_o = \begin{bmatrix} 0.2 & 0 & 0 \end{bmatrix}\mathbf{x}$$

c. $T_1 = \left(\frac{10}{\pi}\right)(K)(150)\left(\frac{1}{s}\right)(0.8)\left(\frac{1}{s}\right)\left(\frac{1}{s}\right)(0.2) = \frac{76.39}{s^3}$

$$G_{L1} = \frac{-150}{s};\ G_{L2} = \frac{-1.32}{s};\ G_{L3} = (K)(150)\left(\frac{1}{s}\right)(0.8)\left(\frac{1}{s}\right)\left(\frac{1}{s}\right)(0.2)\left(\frac{-10}{\pi}\right) = \frac{-76.39K}{s^3}$$

Nontouching loops:

$$G_{L1}G_{L2} = \frac{198}{s^2}$$

$$\Delta = 1 - [G_{L1} + G_{L2} + G_{L3}] + [G_{L1}G_{L2}] = 1 + \frac{150}{s} + \frac{1.32}{s} + \frac{76.39K}{s^3} + \frac{198}{s^2}$$

$$\Delta_1 = 1$$

$$T(s) = \frac{T_1\Delta_1}{\Delta} = \frac{76.39K}{s^3+151.32s^2+198s+76.39K}$$

d. The equivalent forward path transfer function is $G(s) = \dfrac{\frac{10}{\pi}0.16K}{s(s+1.32)}$.

Therefore,

$$T(s) = \frac{2.55}{s^2+1.32s+2.55}$$

The poles are located at $-0.66 \pm j1.454$. $\omega_n = \sqrt{2.55} = 1.597$ rad/s; $2\zeta\omega_n = 1.32$, therefore, $\zeta = 0.413$.

$\%OS = e^{-\zeta\pi/\sqrt{1-\zeta^2}} x100 = 24\%$; $T_s = \frac{4}{\zeta\omega_n} = \frac{4}{0.66} = 6.06$ seconds; $T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{1.454} =$ 2.16 seconds; Using Figure 4.16, the normalized rise time is 1.486. Dividing by the natural frequency, $T_r = \frac{1.486}{\sqrt{2.55}} = 0.93$ seconds.

e.

$$C(s) = \frac{2.55}{s\left(s^2 + 1.32s + 2.55\right)}$$

$$C(s) = \frac{1}{s} - \frac{1}{25}\,\frac{25s + 33}{s^2 + 1.32s + 2.55}$$

$$C(s) = \frac{1}{s} - \frac{1}{25}\,\frac{25(s + 0.66) + 11.347\sqrt{2.1144}}{(s + 0.66)^2 + \sqrt{2.1144}^2}$$

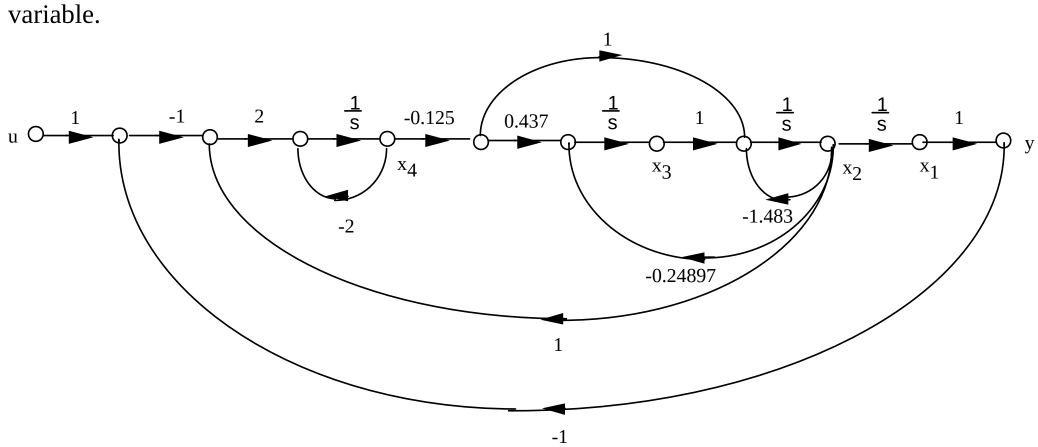
$$c(t) = 1 - e^{-0.66t}\left(\cos[1.454t] + 0.454\sin[1.454t]\right)$$

f. Since $G(s) = \frac{0.51K}{s(s+1.32)}$, $T(s) = \frac{0.51K}{s^2+1.32s+0.51K}$. Also, $\zeta = \frac{-\ln\left(\frac{\%OS}{100}\right)}{\sqrt{\pi^2 + \ln^2\left(\frac{\%OS}{100}\right)}} = 0.517$ for 15% overshoot; $\omega_n = \sqrt{0.51K}$; and $2\zeta\omega_n = 1.32$. Therefore, $\omega_n = \frac{1.32}{2\zeta} = \frac{1.32}{2(0.5147)} = 1.277 = \sqrt{0.51K}$.

Solving for K, K=3.2.

UFSS Vehicle: Pitch-Angle Control Representation

a. Use the observer canonical form for the vehicle dynamics so that the output yaw rate is a state variable.



b. Using the signal flow graph to write the state equations:

$$\begin{aligned}
\dot{x}_1 &= x_2 \\
\dot{x}_2 &= -1.483x_2 + x_3 - 0.125x_4 \\
\dot{x}_3 &= -0.24897x_2 - (0.125*0.437)x_4 \\
\dot{x}_4 &= 2x_1 + 2x_2 - 2x_4 - 2u
\end{aligned}$$

In vector-matrix form:

$$\begin{aligned}
\dot{\mathbf{x}} &= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -1.483 & 0 & -0.125 \\ 0 & -0.24897 & 0 & -0.054625 \\ 2 & 2 & 0 & -2 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ -2 \end{bmatrix} \\
y &= [1 \quad 0 \quad 0 \quad 0]\mathbf{x}
\end{aligned}$$

c.

Program:

```
numg1=-0.25*[1 0.437];
deng1=poly([-2 -1.29 -0.193 0]);
'G(s)'
G=tf(numg1,deng1)
numh1=[-1 0];
denh1=[0 1];
'H(s)'
H=tf(numh1,denh1)
'Ge(s)'
Ge=feedback(G,H)
'T(s)'
T=feedback(-1*Ge,1)
[numt,dent]=tfdata(T,'V');
[Acc,Bcc,Ccc,Dcc]=tf2ss(numt,dent)
```

Computer response:

```
ans =

G(s)

Transfer function:
           -0.25 s - 0.1093
--------------------------------------
s^4 + 3.483 s^3 + 3.215 s^2 + 0.4979 s

ans =

H(s)

Transfer function:
-s

ans =

Ge(s)
```

```
Transfer function:
           -0.25 s - 0.1093
-------------------------------------
s^4 + 3.483 s^3 + 3.465 s^2 + 0.6072 s

ans =

T(s)

Transfer function:
                0.25 s + 0.1093
----------------------------------------------
s^4 + 3.483 s^3 + 3.465 s^2 + 0.8572 s + 0.1093

Acc =

   -3.4830   -3.4650   -0.8572   -0.1093
    1.0000         0         0         0
         0    1.0000         0         0
         0         0    1.0000         0

Bcc =

     1
     0
     0
     0

Ccc =

         0         0    0.2500    0.1093

Dcc =

     0
```

ANSWERS TO REVIEW QUESTIONS

1. Signals, systems, summing junctions, pickoff points

2. Cascade, parallel, feedback

3. Product of individual transfer functions, sum of individual transfer functions, forward gain divided by one plus the product of the forward gain times the feedback gain

4. Equivalent forms for moving blocks across summing junctions and pickoff points

5. As K is varied from 0 to ∞, the system goes from overdamped to critically damped to underdamped. When the system is underdamped, the settling time remains constant.

6. Since the real part remains constant and the imaginary part increases, the radial distance from the origin is increasing. Thus the angle θ is increasing. Since $\zeta = \cos\theta$ the damping ratio is decreasing.

7. Nodes (signals), branches (systems)

8. Signals flowing into a node are added together. Signals flowing out of a node are the sum of signals flowing into a node.

9. One

10. Phase-variable form, cascaded form, parallel form, Jordan canonical form, observer canonical form

11. The Jordan canonical form and the parallel form result from a partial fraction expansion.

12. Parallel form

13. The system poles, or eigenvalues

14. The system poles including all repetitions of the repeated roots

15. Solution of the state variables are achieved through decoupled equations. i.e. the equations are solvable individually and not simultaneously.

16. State variables can be identified with physical parameters; ease of solution of some representations

17. Systems with zeros

18. State-vector transformations are the transformation of the state vector from one basis system to another. i.e. the same vector represented in another basis.

19. A vector which under a matrix transformation is collinear with the original. In other words, the length of the vector has changed, but not its angle.

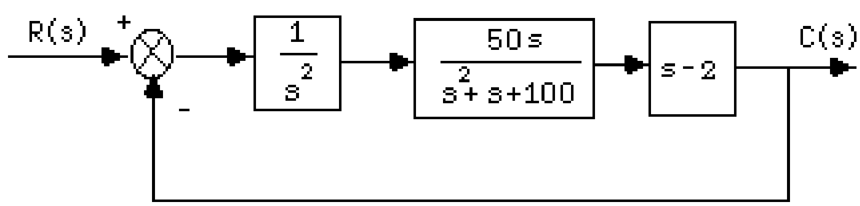
20. An eigenvalue is that multiple of the original vector that is the transformed vector.

21. Resulting system matrix is diagonal.

SOLUTIONS TO PROBLEMS

1.

a. Combine the inner feedback and the parallel pair.



Multiply the blocks in the forward path and apply the feedback formula to get,

$$T(s) = \frac{50(s-2)}{s^3+s^2+150s-100} \quad .$$

b.

Program:

```
'G1(s)'
G1=tf(1,[1 0 0])
'G2(s)'
G2=tf(50,[1 1])
'G3(s)'
G3=tf(2,[1 0])
'G4(s)'
G4=tf([1 0],1)
'G5(s)'
G5=2
'Ge1(s)=G2(s)/(1+G2(s)G3(s))'
Ge1=G2/(1+G2*G3)
'Ge2(s)=G4(s)-G5(s)'
```

```
Ge2=G4-G5
'Ge3(s)=G1(s)Ge1(s)Ge2(s)'
Ge3=G1*Ge1*Ge2
'T(s)=Ge3(s)/(1+Ge3(s))'
T=feedback(Ge3,1);
T=minreal(T)
```

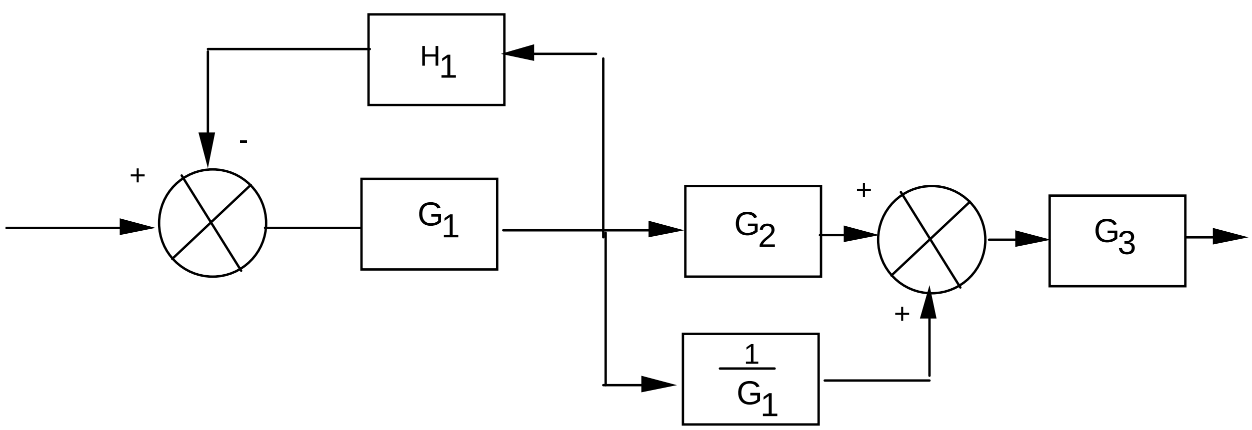
Computer response:

```
ans =

G1(s)

Transfer function:
 1
---
s^2

ans =

G2(s)

Transfer function:
 50
-----
s + 1

ans =

G3(s)

Transfer function:
2
-
s

ans =

G4(s)

Transfer function:
s

ans =

G5(s)

G5 =

     2

ans =

Ge1(s)=G2(s)/(1+G2(s)G3(s))


Transfer function:
      50 s^2 + 50 s
-------------------------
s^3 + 2 s^2 + 101 s + 100

ans =

Ge2(s)=G4(s)-G5(s)
```

```
Transfer function:
s - 2

ans =

Ge3(s)=G1(s)Ge1(s)Ge2(s)

Transfer function:
    50 s^3 - 50 s^2 - 100 s
-------------------------------
s^5 + 2 s^4 + 101 s^3 + 100 s^2

ans =

T(s)=Ge3(s)/(1+Ge3(s))

Transfer function:
      50 s - 100
-----------------------
s^3 + s^2 + 150 s - 100
```

2.

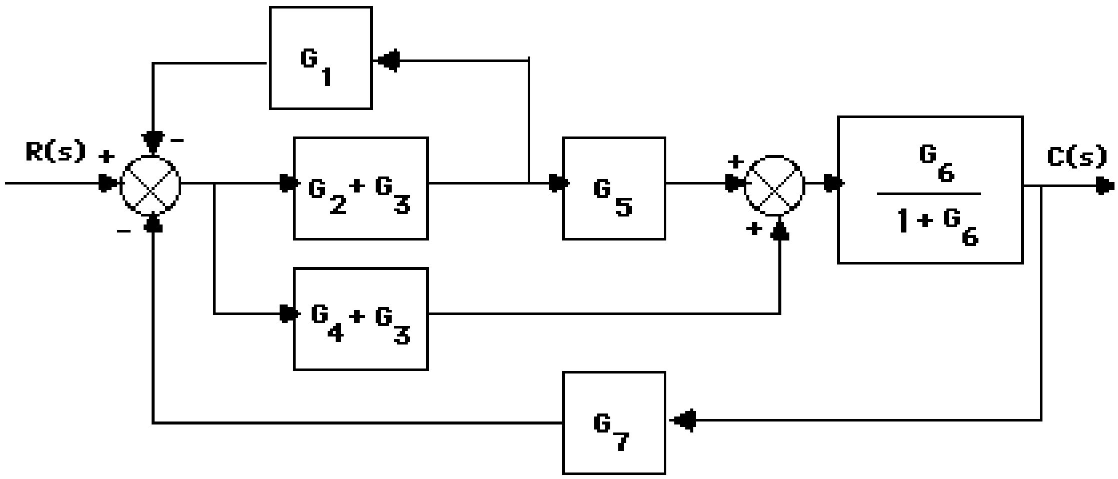
Push $G_1(s)$ to the left past the pickoff point.



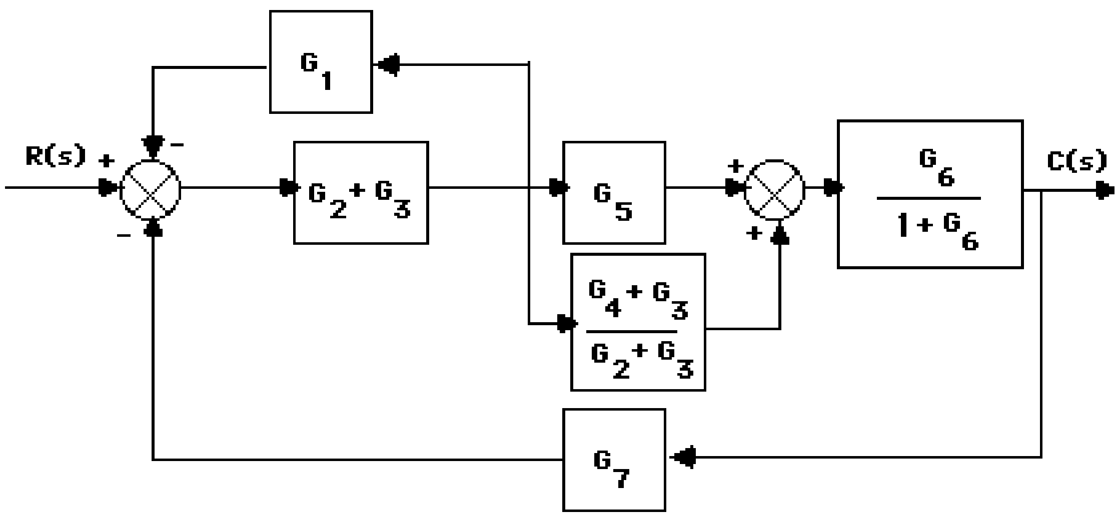
Thus, $T(s)=\left(\dfrac{G_1}{1+G_1H_1}\right)\left(G_2+\dfrac{1}{G_1}\right)G_3=\dfrac{(G_1G_2+1)G_3}{(1+G_1H_1)}$

3.

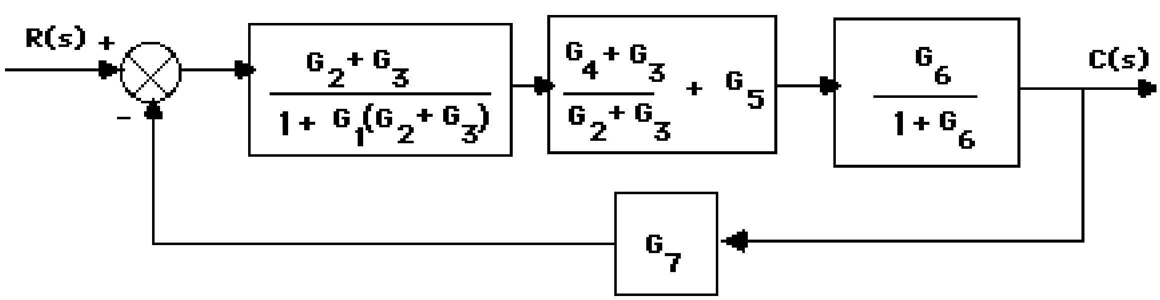
a. Split G_3 and combine with G_2 and G_4. Also use feedback formula on G_6 loop.



Push $G_2 + G_3$ to the left past the pickoff point.



Using the feedback formula and combining parallel blocks,

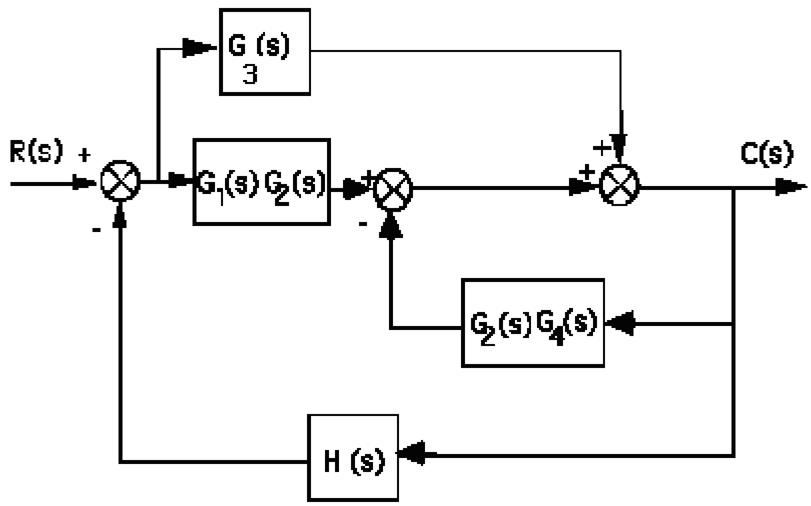


Multiplying the blocks of the forward path and applying the feedback formula,

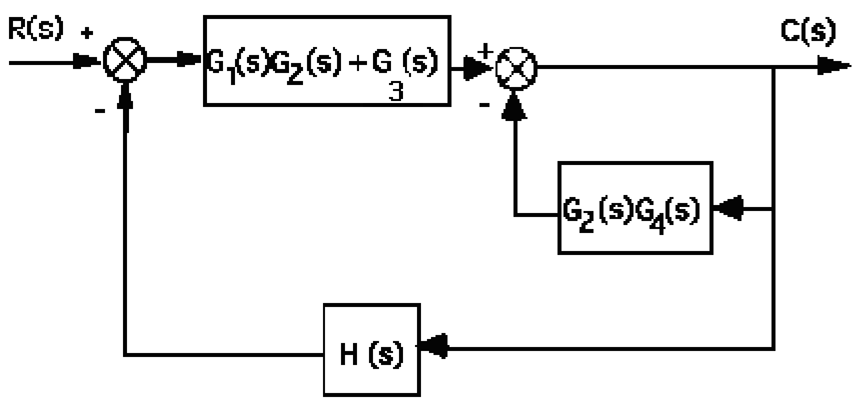
$$T(s) = \frac{G_6 G_4 + G_6 G_3 + G_6 G_5 G_3 + G_6 G_5 G_2}{1 + G_6 + G_3 G_1 + G_2 G_1 + G_7 G_6 G_4 + G_7 G_6 G_3 + G_7 G_6 G_5 G_3 + G_7 G_6 G_5 G_2 + G_6 G_3 G_1 + G_6 G_2 G_1}$$

4.

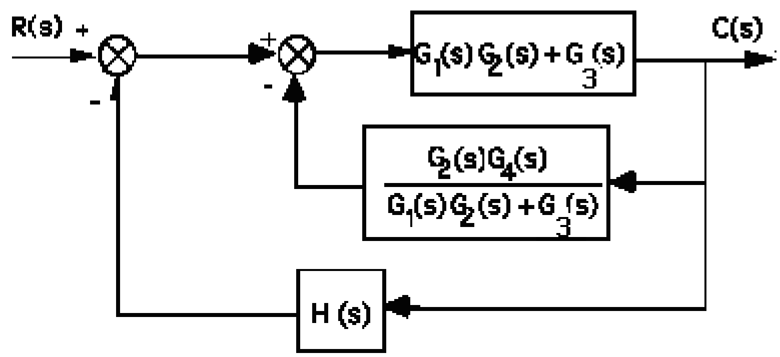
Push $G_2(s)$ to the left past the summing junction.



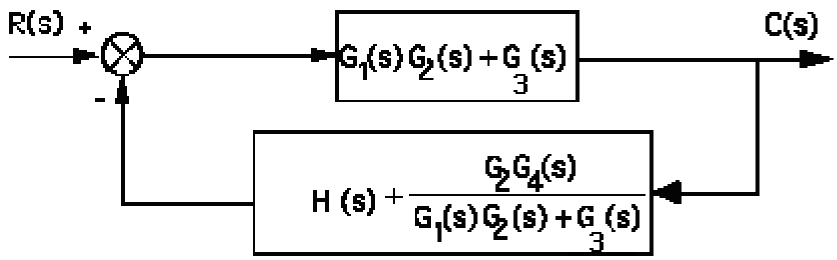
Collapse the summing junctions and add the parallel transfer functions.



Push $G_1(s)G_2(s) + G_5(s)$ to the right past the summing junction.



Collapse summing junctions and add feedback paths.

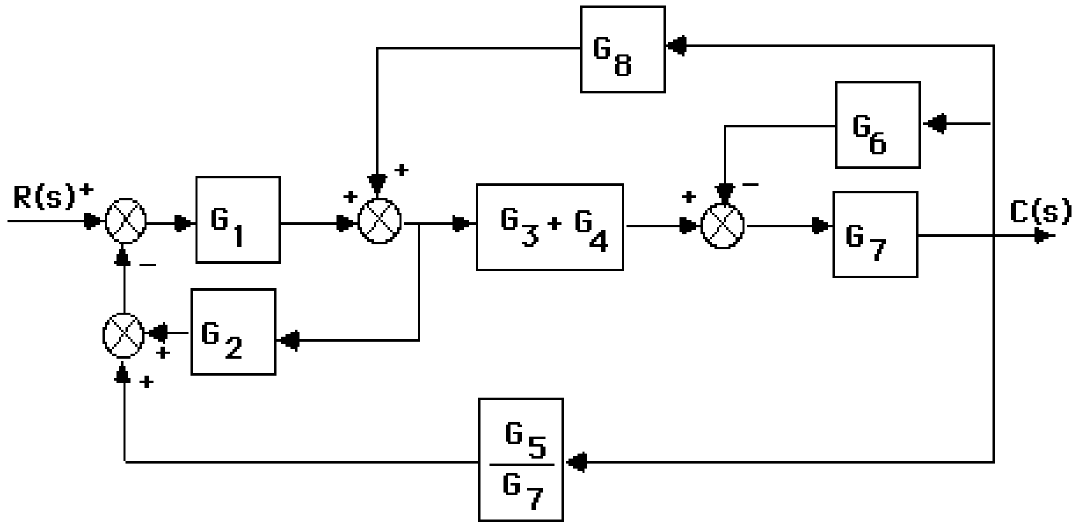


Applying the feedback formula,

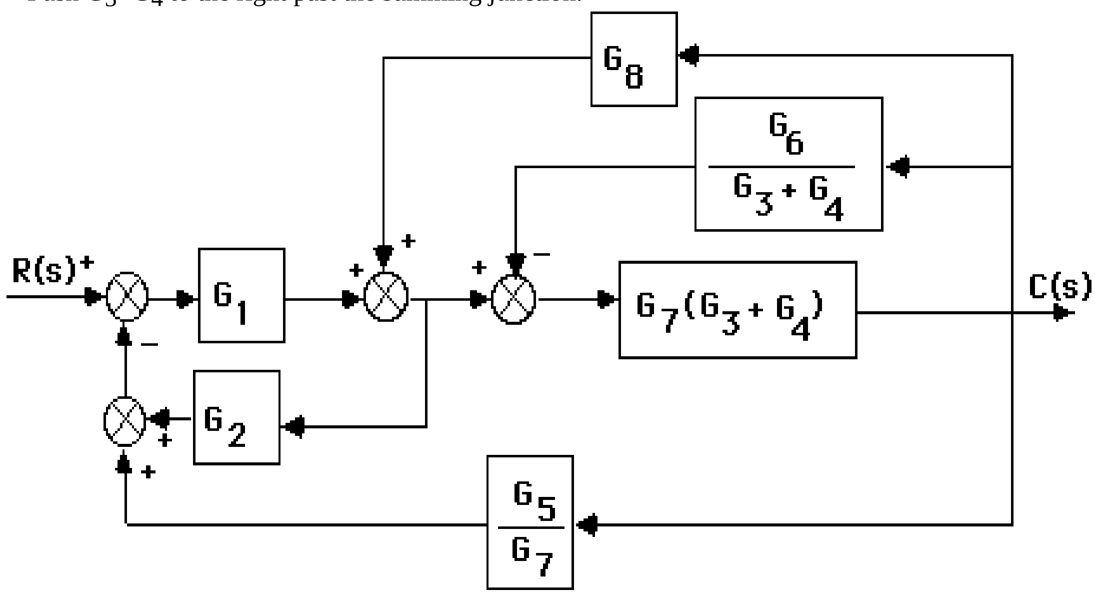
$$T(s) = \frac{G_3(s)+G_1(s)G_2(s)}{1+[G_3(s)+G_1(s)G_2(s)]\left[H+\dfrac{G_2(s)G_4(s)}{G_3(s)+G_1(s)G_2(s)}\right]}$$
$$= \frac{G_3(s)+G_1(s)G_2(s)}{1+H[G_3(s)+G_1(s)G_2(s)]+G_2(s)G_4(s)}$$

5.

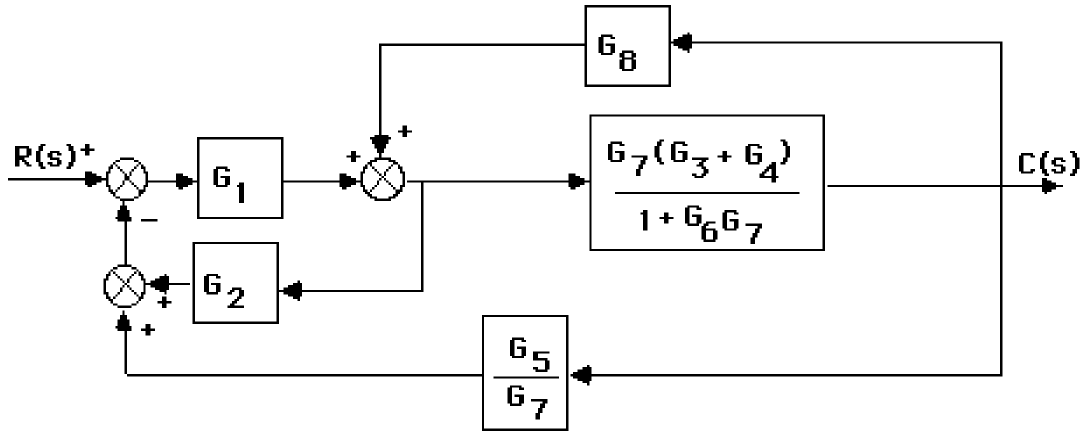
a. Push G_7 to the left past the pickoff point. Add the parallel blocks, G_3+G_4.



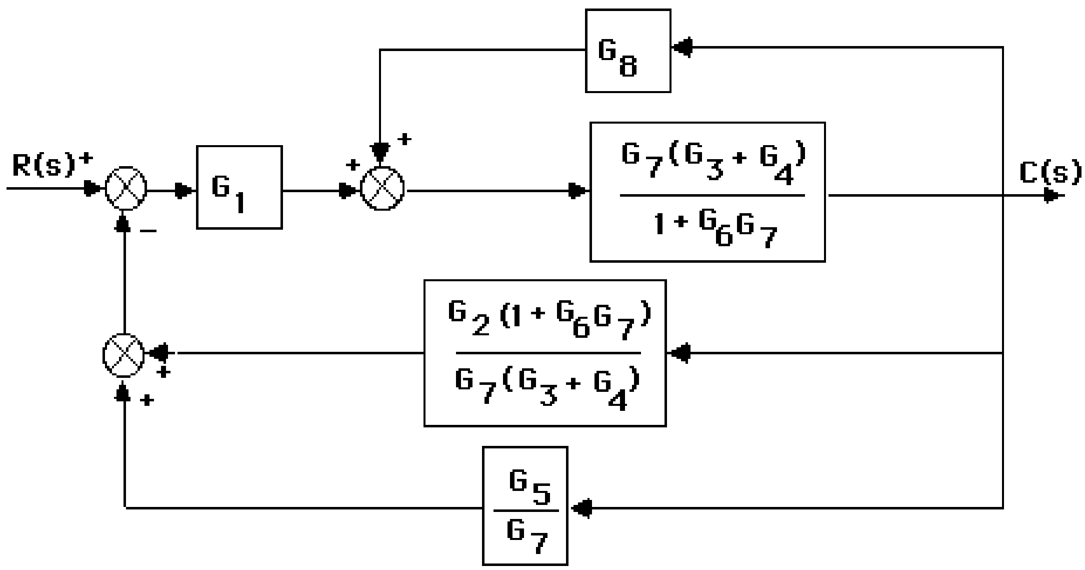
Push G_3+G_4 to the right past the summing junction.



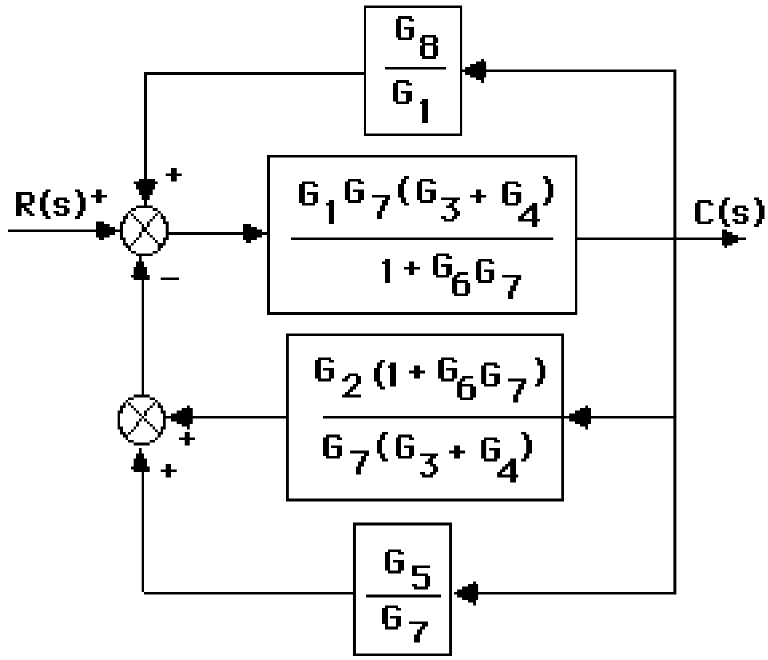
Collapse the minor loop feedback.



Push $\frac{G_7(G_3+G_4)}{1+G_6G_7}$ to the left past the pickoff point.



Push G_1 to the right past the summing junction.



Add the parallel feedback paths to get the single negative feedback,

$$H(s) = \frac{G_5}{G_7} + \frac{G_2(1+G_6G_7)}{G_7(G_3+G_4)} - \frac{G_8}{G_1} \text{. Thus,}$$

$$T(s) = \frac{G}{1+GH} = \frac{G_7 G_1 (G_4+G_3)}{([G_7 G_6 + 1] G_2 G_1 + [G_4 + G_3][G_5 G_1 - G_8 G_7]) + (G_7 G_6 + 1)}$$

b.

Program:

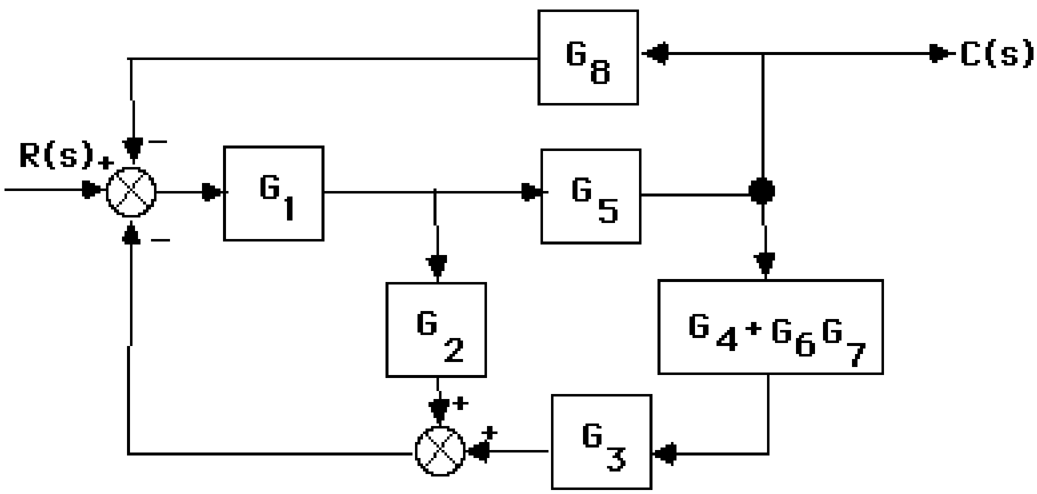
```
G1=tf([0 1],[1 7]);                   %G1=1/s+7 input transducer
G2=tf([0 0 1],[1 2 3]);               %G2=1/s^2+2s+3
G3=tf([0 1],[1 4]);                   %G3=1/s+4
G4=tf([0 1],[1 0]);                   %G4=1/s
G5=tf([0 5],[1 7]);                   %G5=5/s+7
G6=tf([0 0 1],[1 5 10]);              %G6=1/s^2+5s+10
G7=tf([0 3],[1 2]);                   %G7=3/s+2
G8=tf([0 1],[1 6]);                   %G8=1/s+6
G9=tf([1],[1]);                       %Add G9=1 transducer at the input
T1=append(G1,G2,G3,G4,G5,G6,G7,G8,G9);
Q=[1 -2 -5 9
2 1 8 0
3 1 8 0
4 1 8 0
5 3 4 -6
6 7 0 0
7 3 4 -6
8 7 0 0];
inputs=9;
outputs=7;
Ts=connect(T1,Q,inputs,outputs);
T=tf(Ts)
```

Computer response:

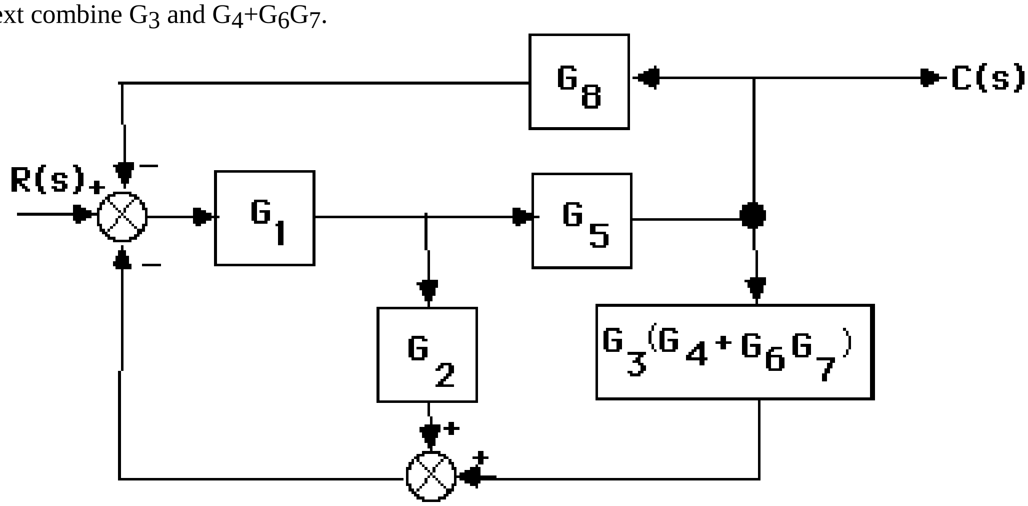
```
Transfer function:

6 s^7 + 132 s^6 + 1176 s^5 + 5640 s^4 + 1.624e004 s^3

                + 2.857e004 s^2 + 2.988e004 s + 1.512e004

-----------------------------------------------------------

s^10 + 33 s^9 + 466 s^8 + 3720 s^7 + 1.867e004 s^6

        + 6.182e004 s^5 + 1.369e005 s^4 + 1.981e005 s^3

                + 1.729e005 s^2 + 6.737e004 s - 1.044e004
```

6.

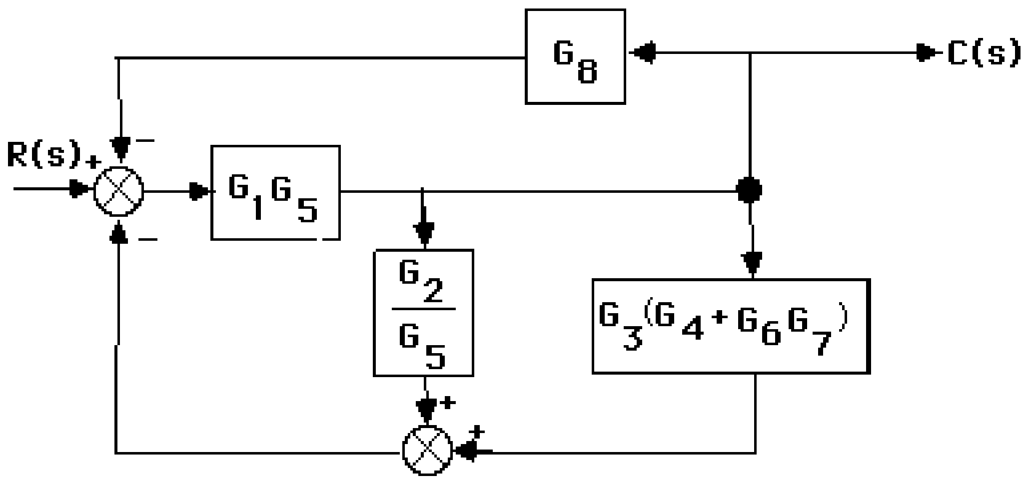
Combine G_6 and G_7 yielding G_6G_7. Add G_4 and obtain the following diagram:



Next combine G_3 and $G_4+G_6G_7$.



Push G_5 to the left past the pickoff point.



Notice that the feedback is in parallel form. Thus the equivalent feedback, $H_{eq}(s) = \frac{G_2}{G_5} + G_3(G_4+G_6G_7) + G_8$. Since the forward path transfer function is $G(s) = G_{eq}(s) = G_1G_5$, the closed-loop transfer function is

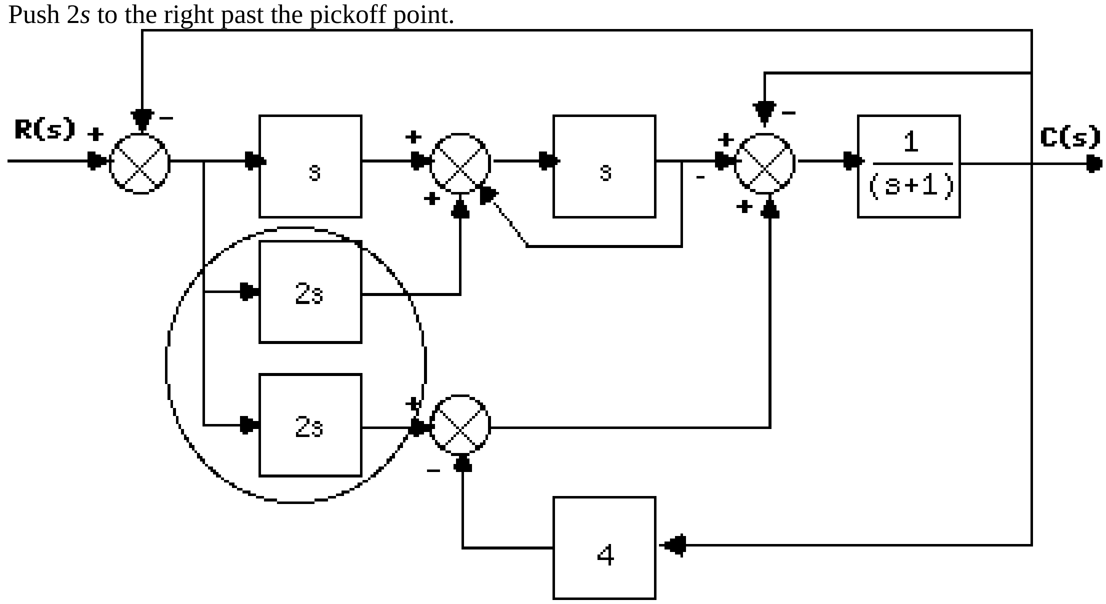
$$T(s) = \frac{G_{eq}(s)}{1+G_{eq}(s)H_{eq}(s)}\ .$$

Hence,

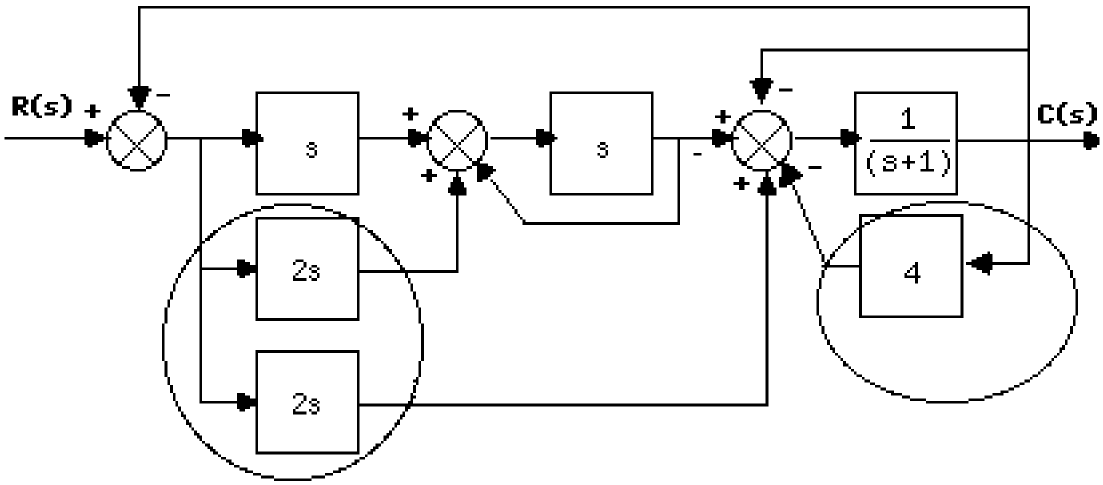
$$T(s) = \frac{G_5 G_1}{1+G_1\left(G_8 G_5 + G_7 G_6 G_5 G_3 + G_5 G_4 G_3 + G_2\right)}$$

7.

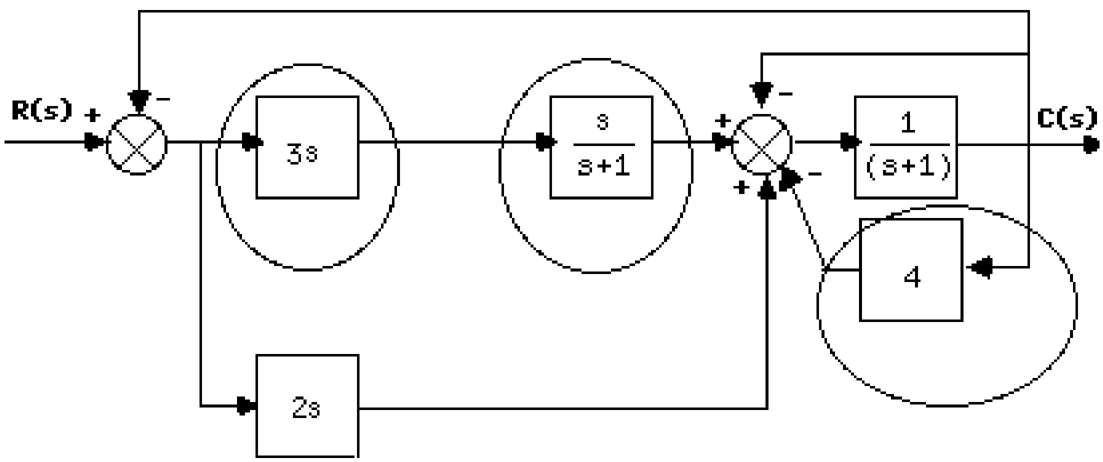
Push 2*s* to the right past the pickoff point.



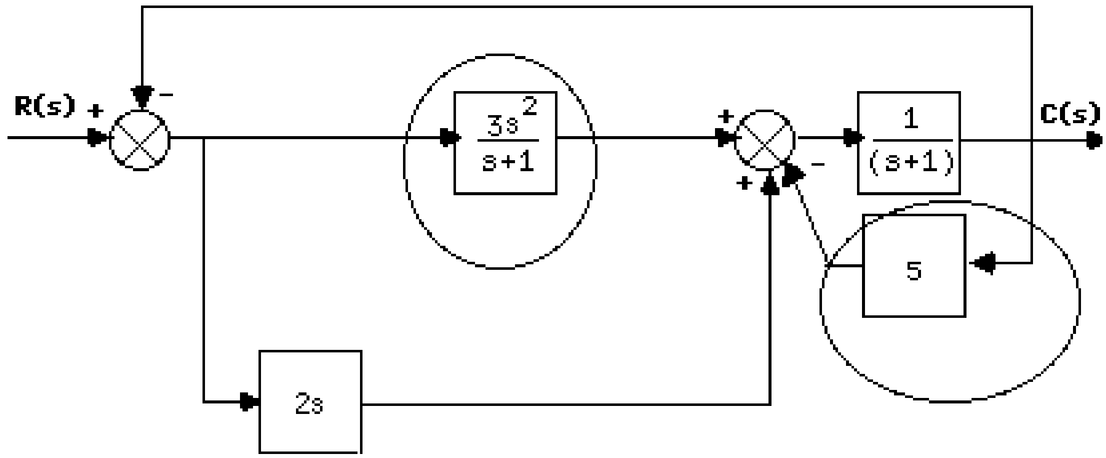
Combine summing junctions.



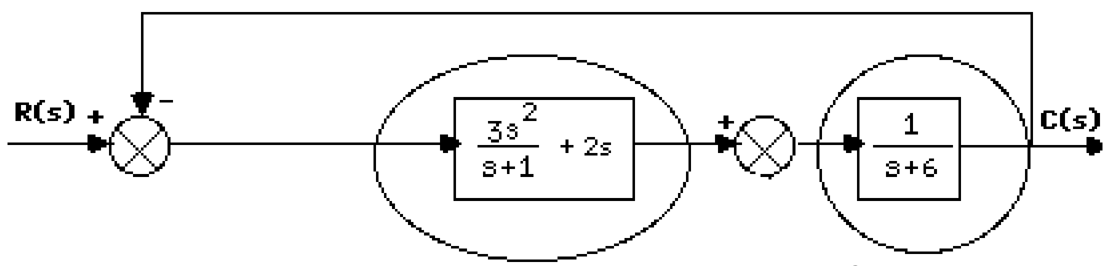
Combine parallel 2s and s. Apply feedback formula to unity feedback with G(s) = s.



Combine cascade pair and add feedback around 1/(s+1).



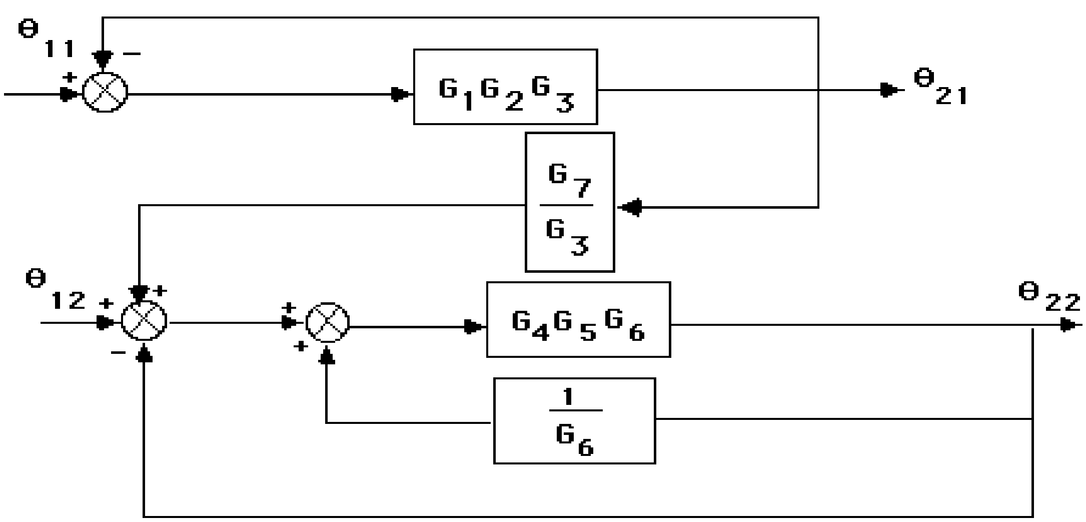
Combine parallel pair and feedback in forward path.



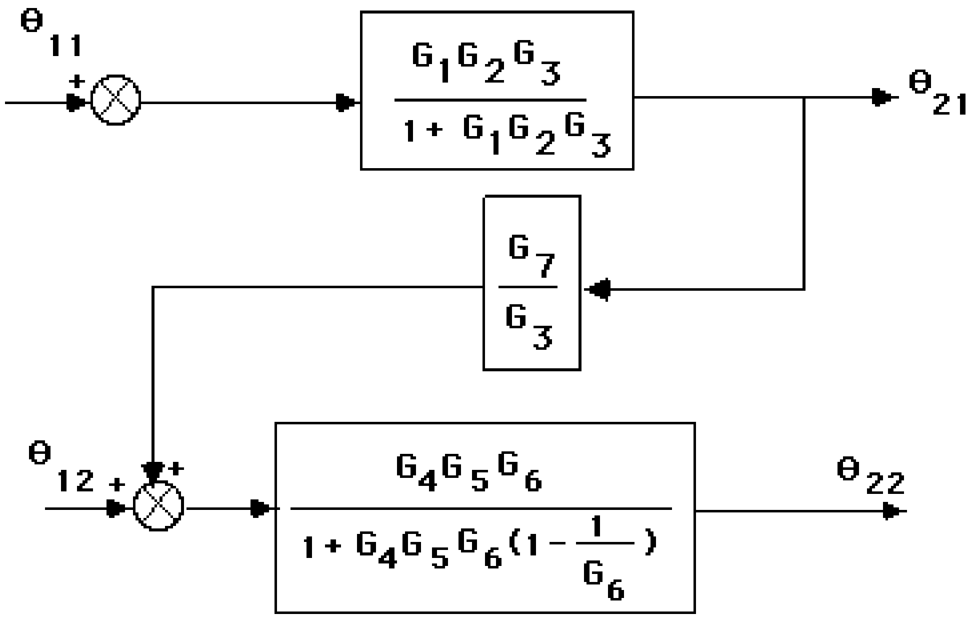
Combine cascade pair and apply final feedback formula yielding $T(s) = \dfrac{5s^2 + 2s}{6s^2 + 9s + 6}$.

8.

Push G_3 to the left past the pickoff point. Push G_6 to the left past the pickoff point.



Hence,

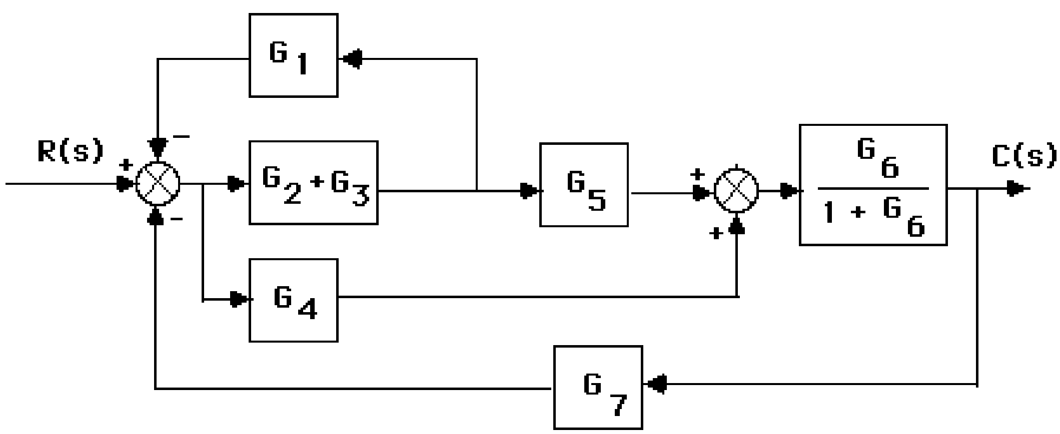


Thus the transfer function is the product of the functions, or

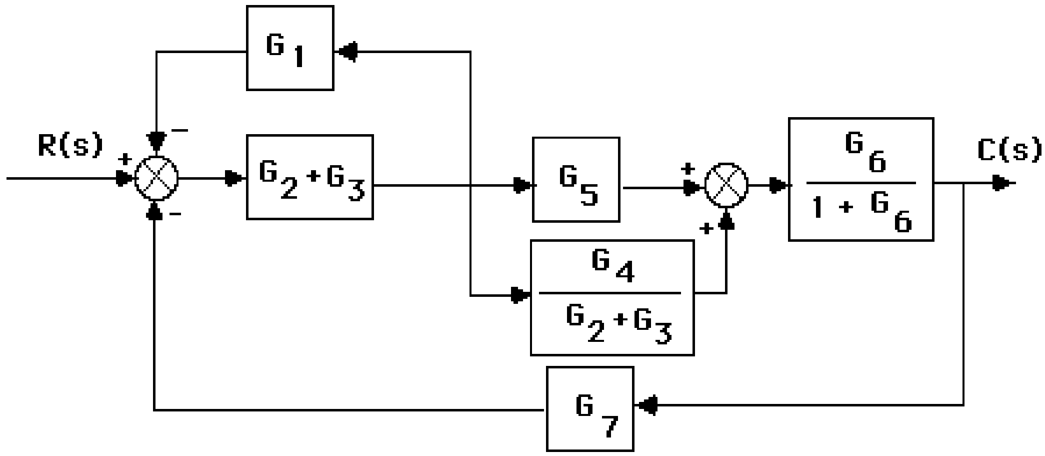
$$\frac{\theta_{22}(s)}{\theta_{11}(s)} = \frac{G_1G_2G_4G_5G_6G_7}{1 - G_4G_5 + G_4G_5G_6 + G_1G_2G_3 - G_1G_2G_3G_4G_5 + G_1G_2G_3G_4G_5G_6}$$

9.

Combine the feedback with G_6 and combine the parallel G_2 and G_3.



Move G_2+G_3 to the left past the pickoff point.



Combine feedback and parallel pair in the forward path yielding an equivalent forward-path transfer function of

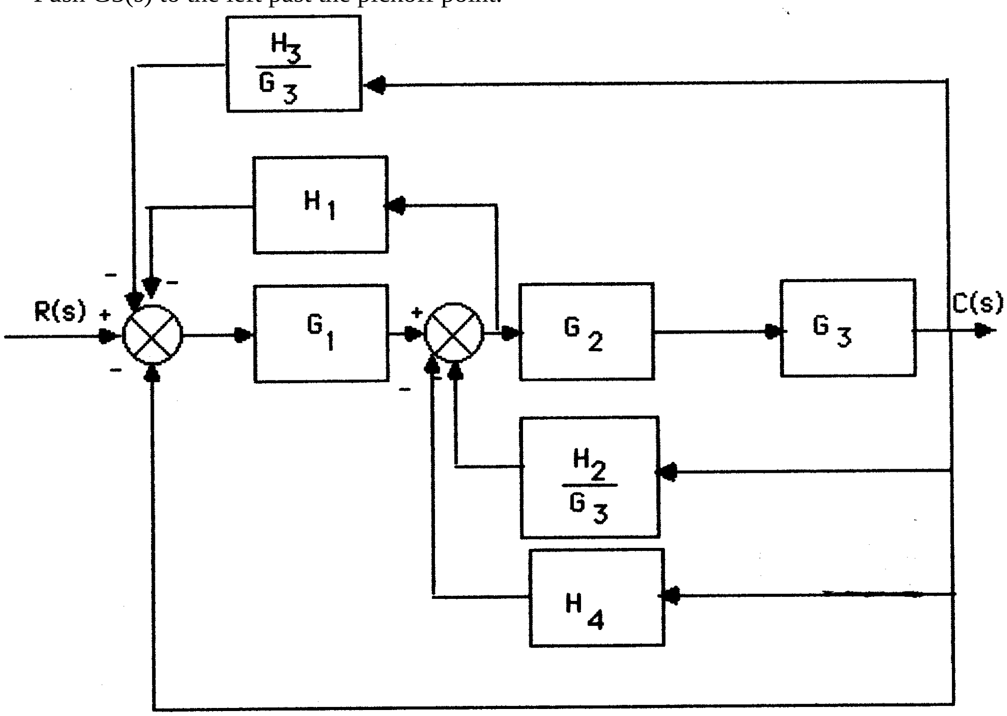
$$G_e(s) = \left(\frac{G_2+G_3}{1+G_1(G_2+G_3)}\right)\left(G_5+\frac{G_4}{G_2+G_3}\right)\left(\frac{G_6}{1+G_6}\right)$$

But, $T(s) = \dfrac{G_e(s)}{1+G_e(s)G_7(s)}$. Thus,

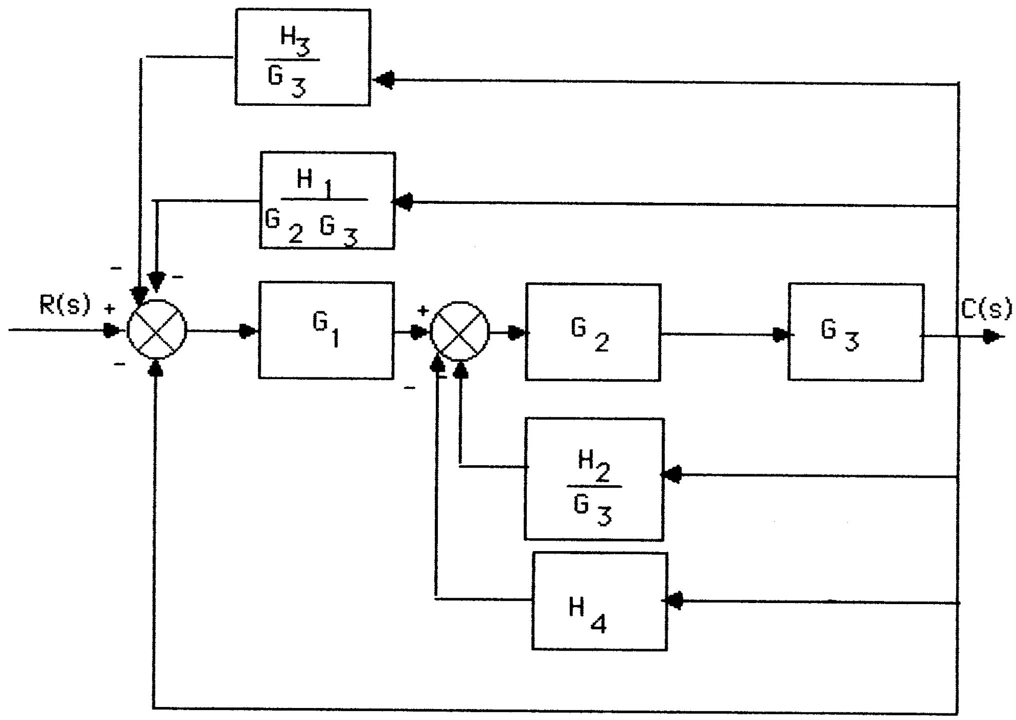
$$T(s) = \frac{G_6\,(G_4+G_5\,G_3+G_5\,G_2)}{G_6\,(G_7\,G_4+G_7\,G_5\,G_3+G_7\,G_5\,G_2+G_3\,G_1+G_2\,G_1+1)+G_1\,(G_3+G_2)+1}$$

10.

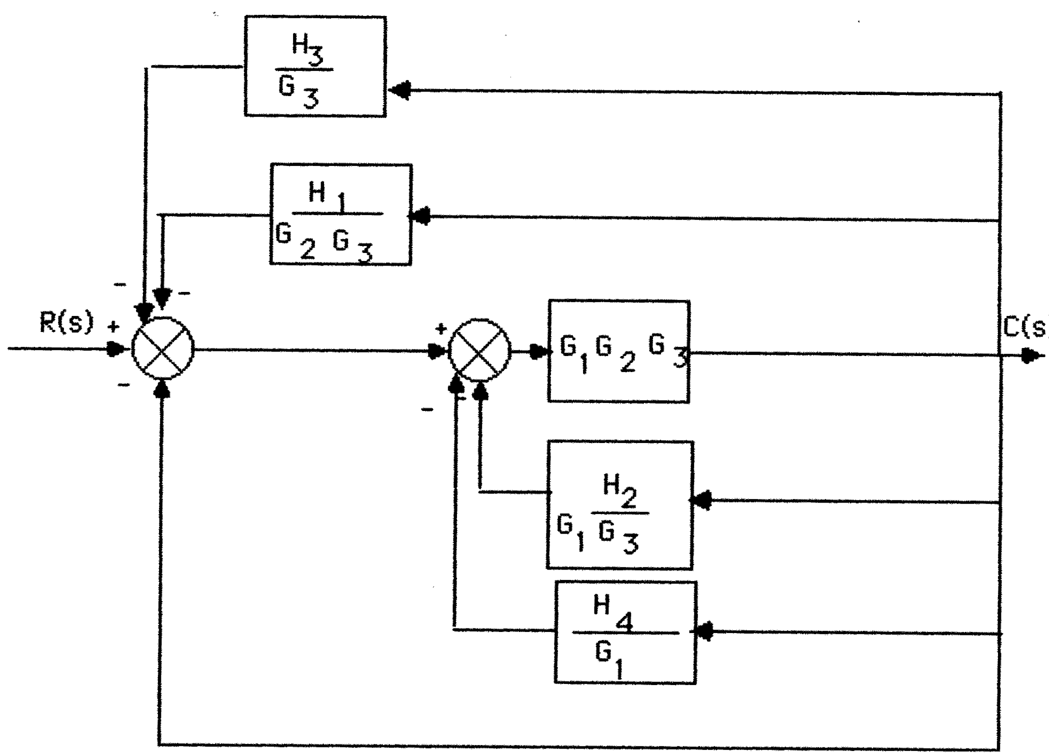
Push G3(s) to the left past the pickoff point.



Push G2(s)G3(s) to the left past the pickoff point.



Push G1(s) to the right past the summing junction.



Collapsing the summing junctions and adding the feedback transfer functions,

$$T(s) = \frac{G_1(s)G_2(s)G_3(s)}{1 + G_1(s)G_2(s)G_3(s)H_{eq}(s)}$$

where

$$H_{eq}(s) = \frac{H_3(s)}{G_3(s)} + \frac{H_1(s)}{G_2(s)G_3(s)} + \frac{H_2(s)}{G_1(s)G_3(s)} + \frac{H_4(s)}{G_1(s)} + 1$$

11.

$T(s) = \dfrac{225}{s^2 + 15s + 225}$. Therefore, $2\zeta\omega_n = 15$, and $\omega_n = 15$. Hence, $\zeta = 0.5$.

$\%OS = e^{-\zeta\pi/\sqrt{1-\zeta^2}} x100 = 16.3\%$; $T_s = \dfrac{4}{\zeta\omega_n} = 0.533$; $T_p = \dfrac{\pi}{\omega_n\sqrt{1-\zeta^2}} = 0.242$.

12.

$$C(s) = \frac{5}{s(s^2+2s+5)} = \frac{A}{s} + \frac{Bs+C}{s^2+2s+5}$$

$$A = 1$$

$$5 = s^2 + 2s + 5 + Bs^2 + Cs$$

$$\therefore B = -1,\ C = -2$$

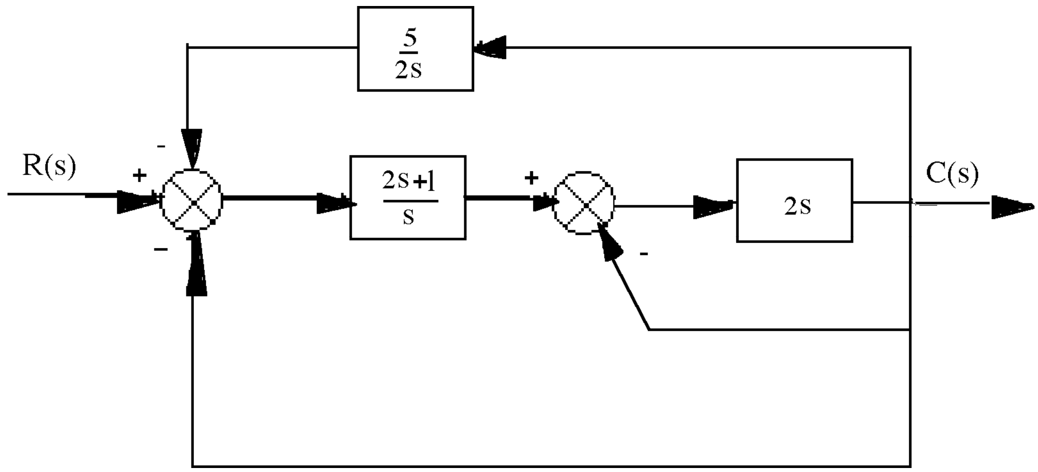
$$C(s) = \frac{1}{s} - \frac{s+2}{s^2+2s+5} = \frac{1}{s} - \frac{s+2}{(s+1)^2+4}$$

$$= \frac{1}{s} - \frac{(s+1)+1}{(s+1)^2+4} = \frac{1}{s} - \frac{(s+1)+\frac{1}{2}2}{(s+1)^2+4}$$

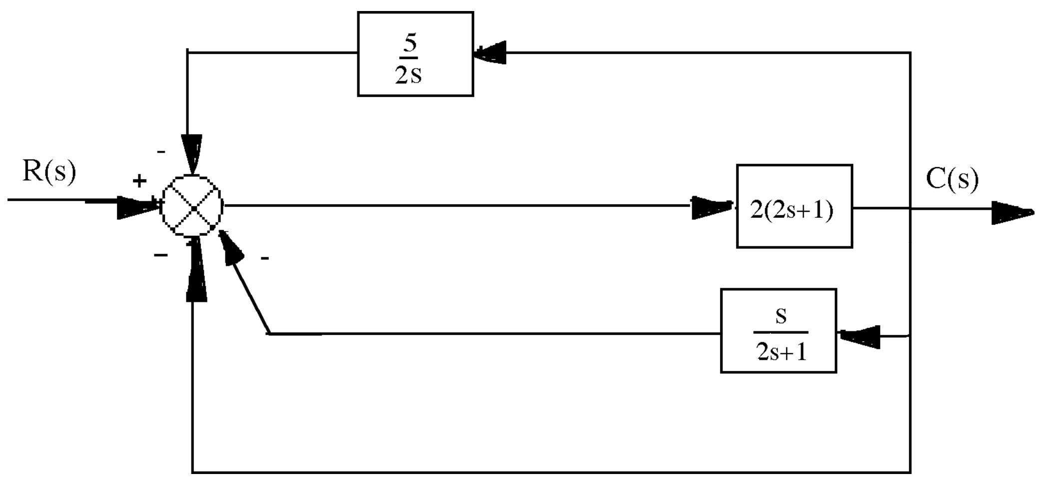
$$c(t) = 1 - e^{-t}(\cos 2t + \frac{1}{2}\sin 2t)$$

13.

Push 2s to the left past the pickoff point and combine the parallel combination of 2 and 1/s.



Push (2s+1)/s to the right past the summing junction and combine summing junctions.



Hence, $T(s)=\dfrac{2(2s+1)}{1+2(2s+1)H_{eq}(s)}$, where $H_{eq}(s)=1+\dfrac{s}{2s+1}+\dfrac{5}{2s}$.

14.

Since $G(s)=\dfrac{K}{s(s+30)}$, $T(s)=\dfrac{G(s)}{1+G(s)}=\dfrac{K}{s^2+30s+K}$. Therefore, $2\zeta\omega_n = 30$. Thus, $\zeta = 15/\omega_n = 0.5912$ (i.e. 10% overshoot). Hence, $\omega_n = 25.37 = \sqrt{K}$. Therefore K = 643.6.

15.

$$T(s)=\frac{K}{s^2+\alpha s+K};\ \zeta=\frac{-\ln(\frac{\%OS}{100})}{\sqrt{\pi^2+\ln^2(\frac{\%OS}{100})}}=0.358;T_s=\frac{4}{\zeta\omega_n}=0.15$$

. Therefore, ω_n = 74.49. K = ω_n^2 = 5549. $\alpha = 2\zeta\omega_n$ = 53.33.

16.

The equivalent forward-path transfer function is $G(s)=\dfrac{10K_1}{s[s+(10K_2+2)]}$. Hence,

$T(s)=\dfrac{G(s)}{1+G(s)}=\dfrac{10K_1}{s^2+(10K_2+2)s+10K_1}$. Since

$T_s=\dfrac{4}{\text{Re}}=3.2,\ \therefore \text{Re}=1.25;\ \text{and } T_p=\dfrac{\pi}{\text{Im}}=1.5,\ \therefore \text{Im}=2.09$. The poles are thus at $-1.25+j2.09$. Hence, $\omega_n=\sqrt{1.25^2+2.09^2}=\sqrt{10K_1}$. Thus, K_1 = 0.593. Also, $(10K_2 + 2)/2$ = Re = 1.25. Hence, K_2 = 0.05.

17.

a. For the inner loop, $G_e(s)=\dfrac{20}{s(s+12)}$, and $H_e(s)$ = 0.2s. Therefore, $T_e(s)=\dfrac{G_e(s)}{1+G_e(s)H_e(s)}$ = $\dfrac{20}{s(s+16)}$. Combining with the equivalent transfer function of the parallel pair, $G_p(s)$ = 20, the system is reduced to an equivalent unity feedback system with $G(s) = G_p(s)\,T_e(s)=\dfrac{400}{s(s+16)}$. Hence, $T(s)=\dfrac{G(s)}{1+G(s)}=\dfrac{400}{s^2+16s+400}$.

b. ω_n^2 = 400; $2\zeta\omega_n$ = 16. Therefore, ω_n = 20, and ζ = 0.4. $\%OS=e^{-\zeta\pi/\sqrt{1-\zeta^2}}x100=25.38$; $T_s=\dfrac{4}{\zeta\omega_n}$ =0.5; $T_p=\dfrac{\pi}{\omega_n\sqrt{1-\zeta^2}}$ =0.171. From Figure 4.16, ω_nT_r = 1.463. Hence, T_r = 0.0732.

$\omega_d=\text{Im}=\omega_n\sqrt{1-\zeta^2}$ = 18.33.

18.

$T(s)=\dfrac{38343}{s^2+200s+38343}$; from which, $2\zeta\omega_n$ = 200 and $\omega_n=\sqrt{38343}$ = 195.81. Hence,

$\zeta=0.511$. $\%OS=e^{-\zeta\pi/\sqrt{1-\zeta^2}}x100=15.44\%$ **;** $T_p=\dfrac{\pi}{\omega_n\sqrt{1-\zeta^2}}=0.0187$ s**.**

Also, $T_s=\dfrac{4}{\zeta\omega_n}=0.04$ s**.**

19.

For the generator, $E_g(s) = K_fI_f(s)$. But, $I_f(s)=\dfrac{E_i(s)}{R_f+L_fs}$. Therefore, $\dfrac{E_g(s)}{E_i(s)}=\dfrac{2}{s+1}$. For the motor, consider R_a = 2 Ω, the sum of both resistors. Also, $J_e=J_a+J_L(\frac{1}{2})^2=0.75+4x\frac{1}{4}$ = 1.75; $D_e=D_L(\frac{1}{2})^2$ = 1. Therefore,

$$\frac{\theta_m(s)}{E_g(s)} = \frac{\dfrac{K_t}{R_a J_e}}{s(s+\dfrac{1}{J_e}(D_e+\dfrac{K_t K_b}{R_a}))} = \frac{0.286}{s(s+0.857)}.$$

But, $\frac{\theta_o(s)}{\theta_m(s)} = \frac{1}{2}$. Thus, $\frac{\theta_o(s)}{E_g(s)} = \frac{0.143}{s(s+0.857)}$. Finally,

$$\frac{\theta_o(s)}{E_i(s)} = \frac{E_g(s)}{E_i(s)}\frac{\theta_o(s)}{E_g(s)} = \frac{0.286}{s(s+1)(s+0.857)}.$$

20.

For the mechanical system, $J\left(\frac{N_2}{N_1}\right)^2 s^2\theta_2(s) = T\left(\frac{N_2}{N_1}\right)$. For the potentiometer, $E_i(s) = 10\frac{\theta_2(s)}{2\pi}$, or

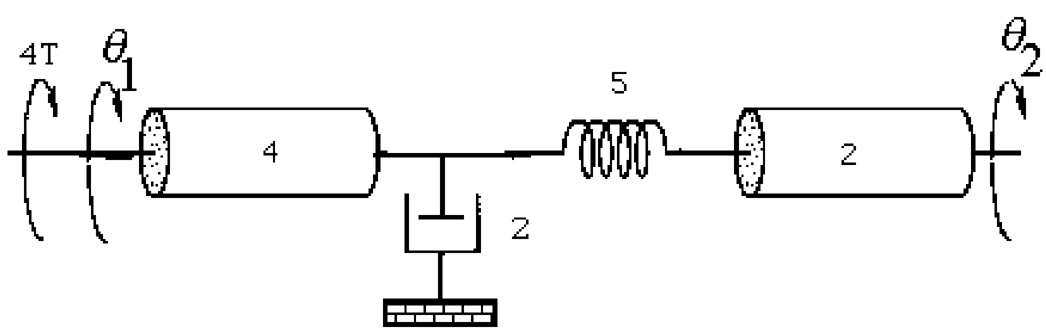
$\theta_2(s) = \frac{\pi}{5}E_i(s)$. For the network, $E_o(s) = E_i(s)\frac{R}{R+\frac{1}{Cs}} = E_i(s)\frac{s}{s+\frac{1}{RC}}$, or $E_i(s) = E_o(s)\frac{s+\frac{1}{RC}}{s}$.

Therefore, $\theta_2(s) = \frac{\pi}{5}E_o(s)\frac{s+\frac{1}{RC}}{s}$. Substitute into mechanical equation and obtain,

$$\frac{E_o(s)}{T(s)} = \frac{\dfrac{5N_1}{J\pi N_2}}{s\left(s+\dfrac{1}{RC}\right)}.$$

21.

The equivalent mechanical system is found by reflecting mechanical impedances to the spring.



Writing the equations of motion:

$$(4s^2+2s+5)\theta_1(s) - 5\theta_2(s) = 4T(s)$$
$$-5\theta_1(s) + (2s^2+5)\theta_2(s) = 0$$

Solving for $\theta_2(s)$,

$$\theta_2(s) = \frac{\begin{vmatrix} (4s^2+2s+5) & 4T(s) \\ -5 & 0 \end{vmatrix}}{\begin{vmatrix} (4s^2+2s+5) & -5 \\ -5 & (2s^2+5) \end{vmatrix}} = \frac{20T(s)}{8s^4+4s^3+30s^2+10s}$$

The angular rotation of the pot is 0.2 that of θ_2, or

$$\frac{\theta_p(s)}{T(s)} = \frac{2}{s(4s^3+2s^2+15s+5)}$$

For the pot:

$$\frac{E_p(s)}{\theta_p(s)} = \frac{50}{5(2\pi)} = \frac{5}{\pi}$$

For the electrical network: Using voltage division,

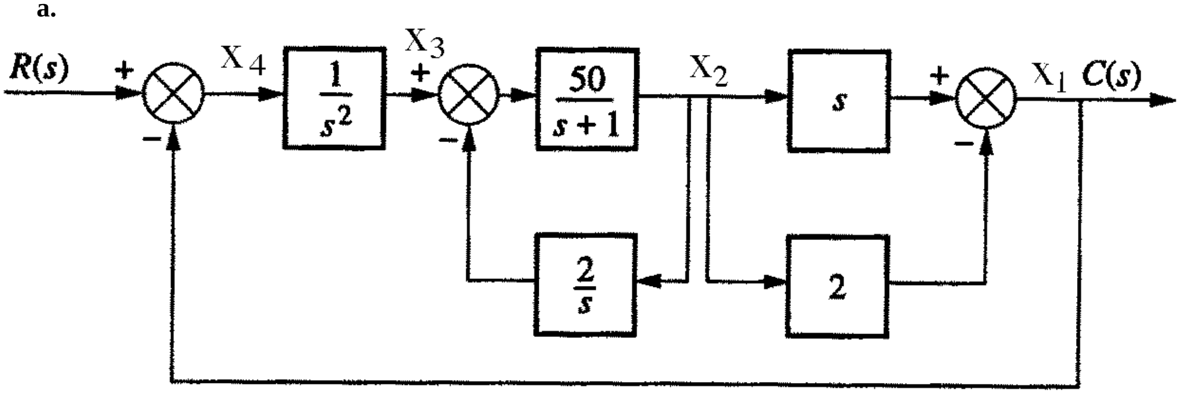
$$\frac{E_o(s)}{E_p(s)} = \frac{200,000}{\frac{1}{10^{-5}s}+200,000} = \frac{s}{s+\frac{1}{2}}$$

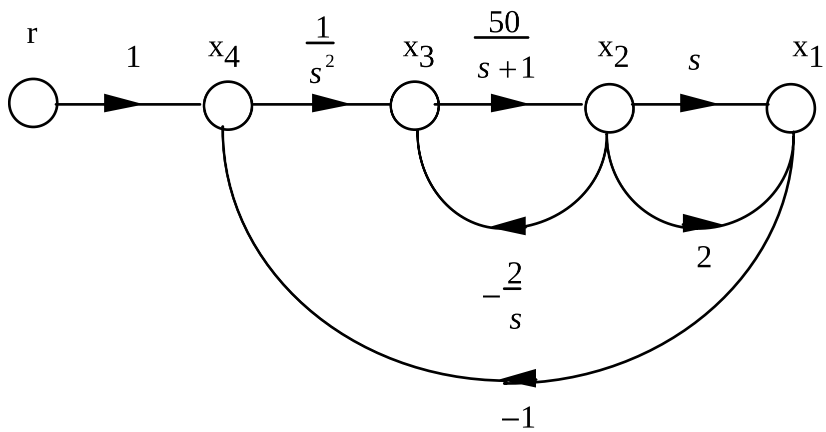
Substituting the previously obtained values,

$$\frac{E_o(s)}{T(s)} = \left(\frac{\theta_p(s)}{T(s)}\right)\left(\frac{E_p(s)}{\theta_p(s)}\right)\left(\frac{E_o(s)}{E_p(s)}\right) = \frac{\frac{10}{\pi}s}{s\left(s+\frac{1}{2}\right)(4s^3+2s^2+15s+5)}$$

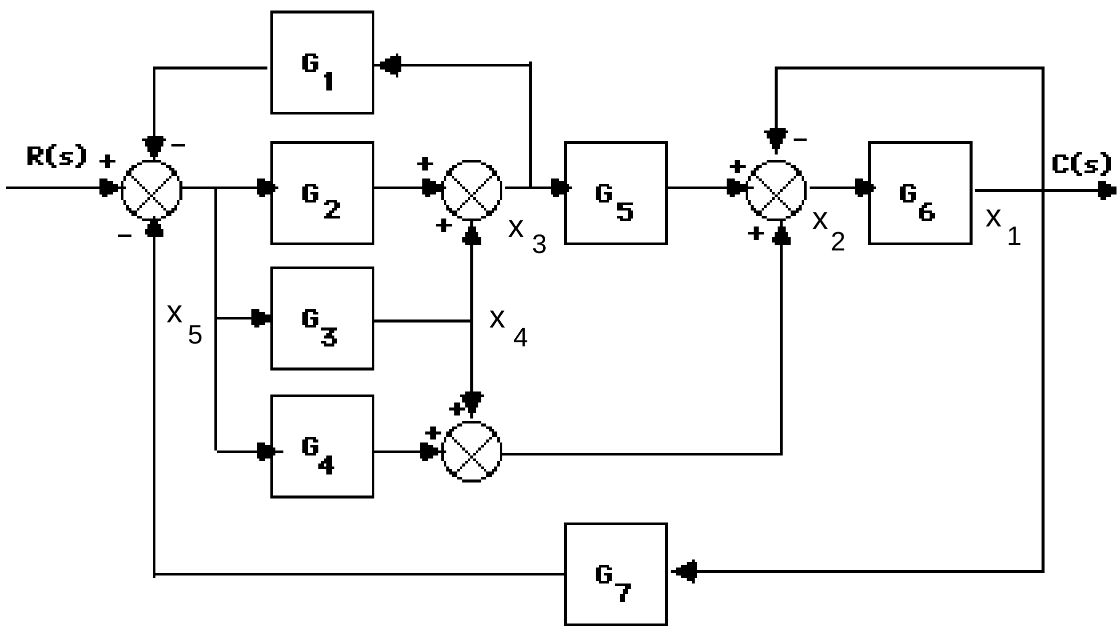
22.

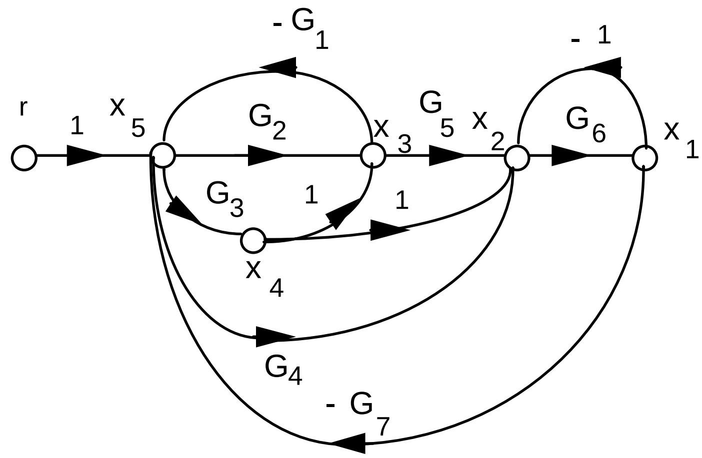
a.



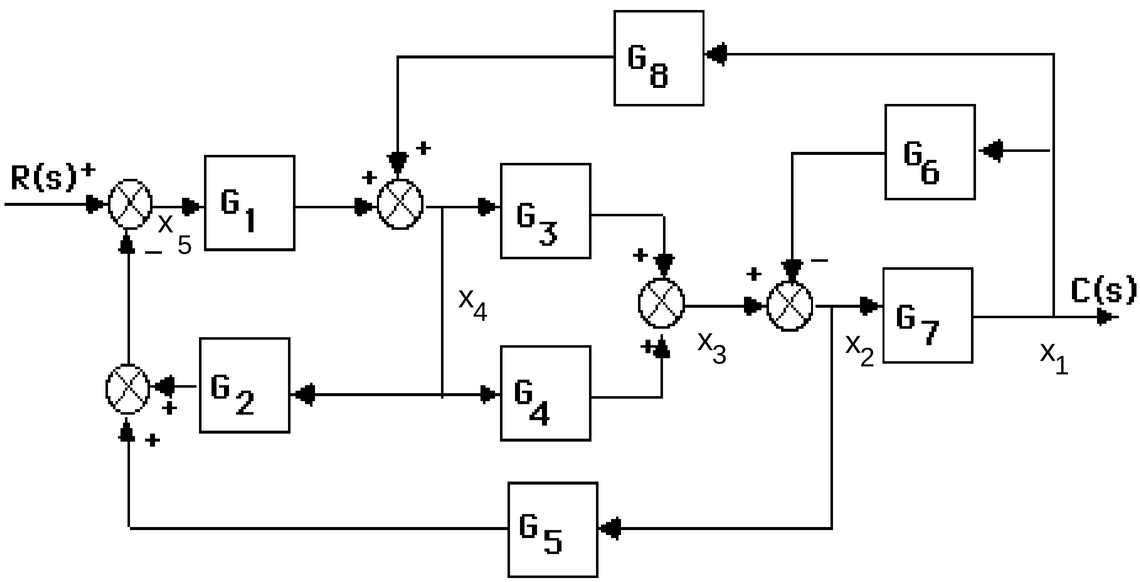

r
1
x4
1
s^2
x3
50
$s+1$
x2
s
x1
$-\frac{2}{s}$
2
−1

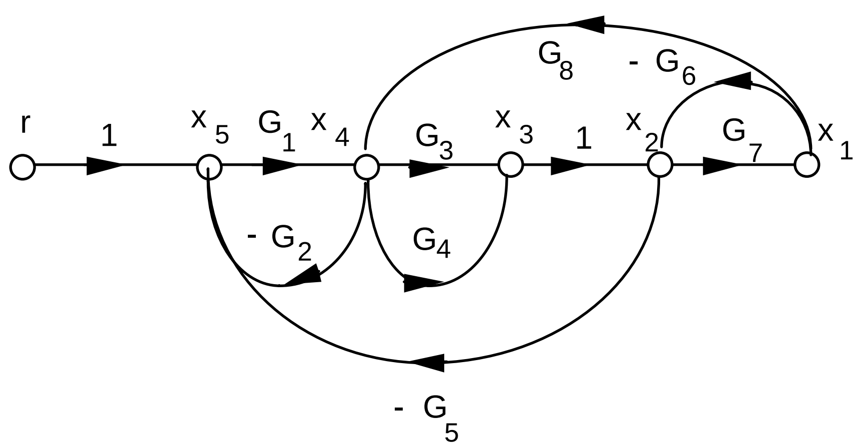
b.


G1
R(s)
G2
G5
G6
C(s)
X 3
X 2
X 1
G3
X 5
X 4
G4
G7


- G1
- 1
r
1
x 5
G2
x 3
G 5
x 2
G6
x 1
G3
1
1
x 4
G4
- G 7

c.

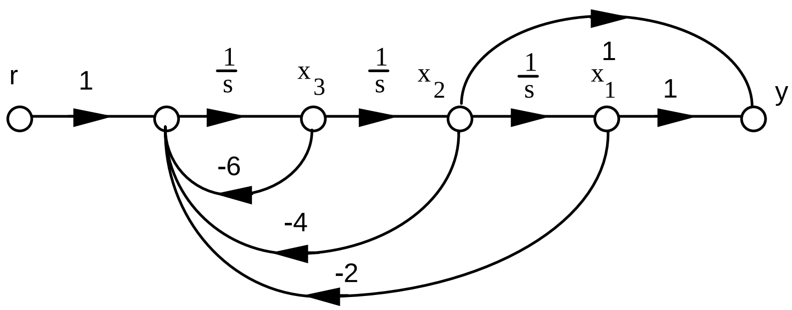

G8
R(s)+
G1
x 5
–
+
+
G3
+
+
–
C(s)
G6
x4
G7
x3
x2
x1
G2
G4
+
+
+
+
G5


r
1
x 5
G 1
x 4
G3
x 3
1
x 2
G 7
x 1
G8
- G6
- G2
G4
- G 5

23.

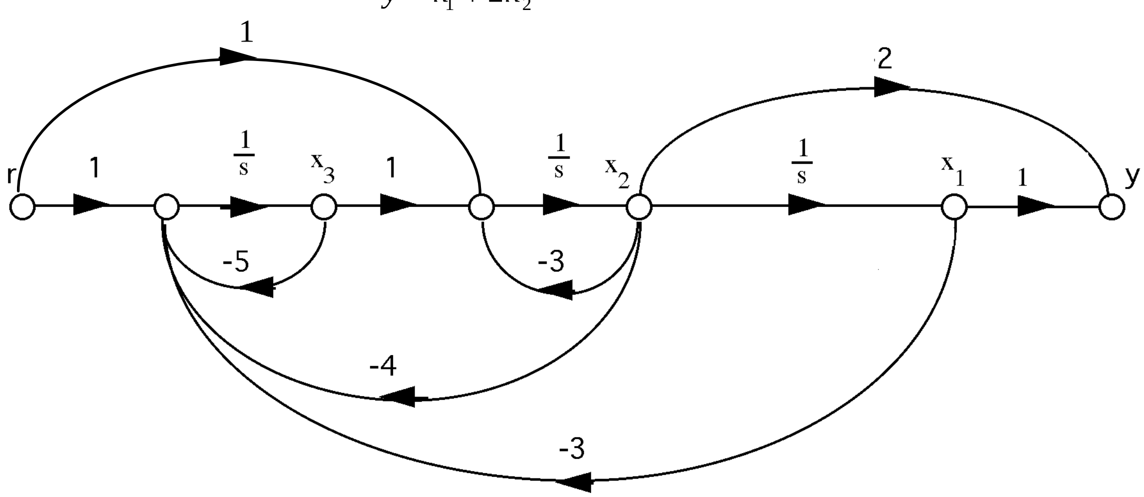
a.

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$\dot{x}_3 = -2x_1 - 4x_2 - 6x_3 + r$$
$$y = x_1 + x_2$$



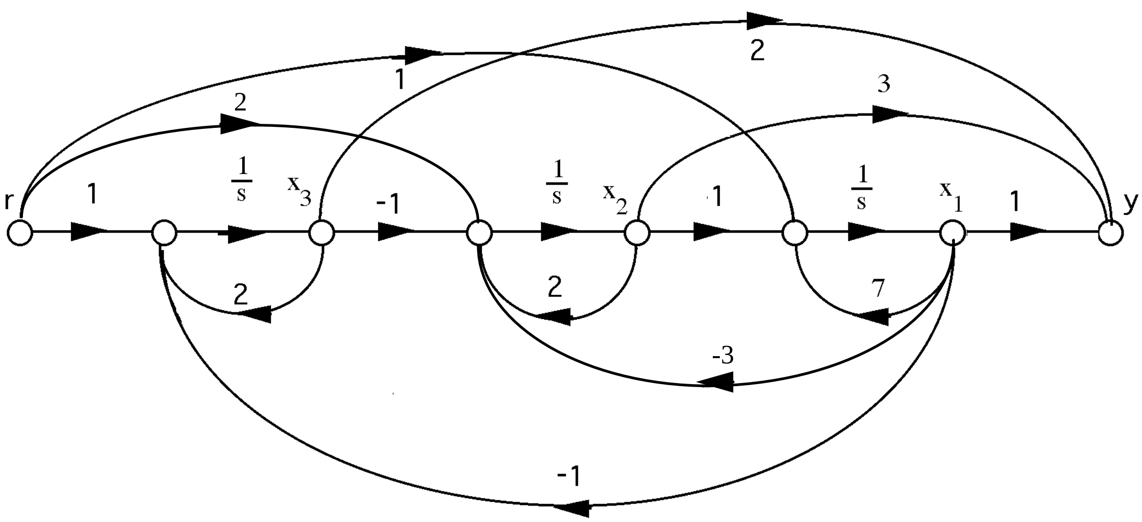
b.

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -3x_2 + x_3 + r$$
$$\dot{x}_3 = -3x_1 - 4x_2 - 5x_3 + r$$
$$y = x_1 + 2x_2$$



c.

$$\dot{x}_1 = 7x_1 + x_2 + r$$
$$\dot{x}_2 = -3x_1 + 2x_2 - x_3 + 2r$$
$$\dot{x}_3 = -x_1 + 2x_3 + r$$
$$y = x_1 + 3x_2 + 2x_3$$



24.

a. Since G(s) = $\dfrac{10}{s^3 + 24s^2 + 191s + 504} = \dfrac{C(s)}{R(s)}$,

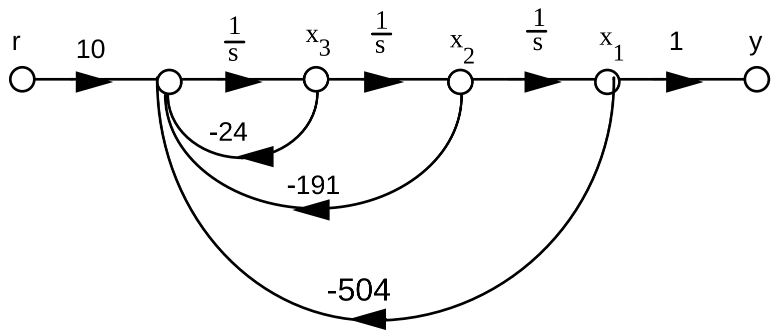
$$\dddot{c} + 24\ddot{c} + 191\dot{c} + 504c = 10r$$

Let,

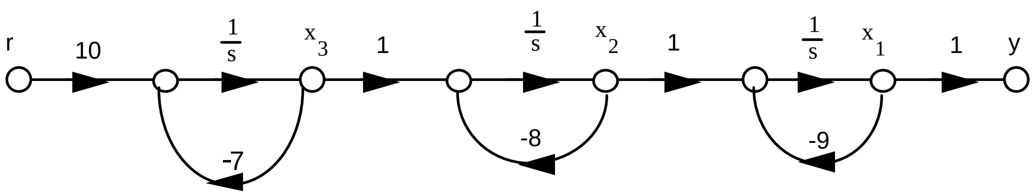
$$c = x_1$$
$$\dot{c} = x_2$$
$$\ddot{c} = x_3$$

Therefore,

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$\dot{x}_3 = -504x_1 - 191x_2 - 24x_3 + 10r$$
$$y = x_1$$



b. G(s) = $\left(\frac{10}{s+7}\right)\left(\frac{1}{s+8}\right)\left(\frac{1}{s+9}\right)$



Therefore,

$$\dot{x}_1 = -9x_1 + x_2$$
$$\dot{x}_2 = -8x_2 + x_3$$
$$\dot{x}_3 = -7x_3 + 10r$$
$$y = x_1$$

25.

a. Since G(s) = $\frac{20}{s^4 + 15s^3 + 66s^2 + 80s} = \frac{C(s)}{R(s)}$,

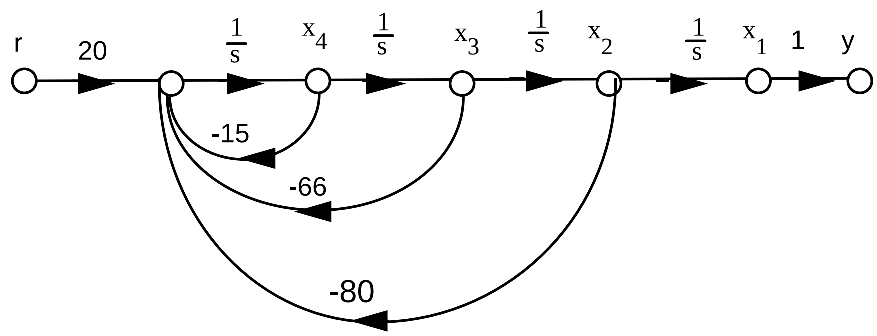
$$\ddddot{c} + 15\dddot{c} + 66\ddot{c} + 80\dot{c} = 20r$$

Let,

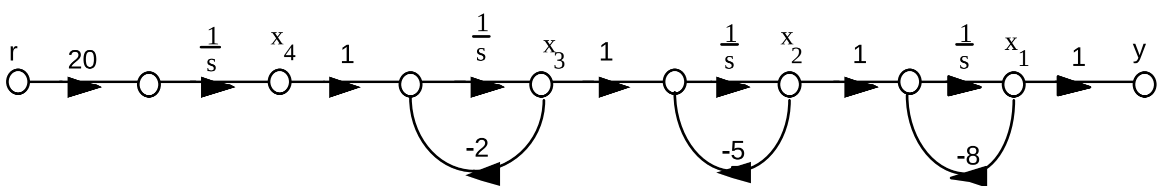
$$c = x_1$$
$$\dot{c} = x_2$$
$$\ddot{c} = x_3$$
$$\dddot{c} = x_4$$

Therefore,

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$\dot{x}_3 = x_4$$
$$\dot{x}_4 = -80x_2 - 66x_3 - 15x_4 + 20r$$
$$y = x_1$$



b. $G(s) = \left(\frac{20}{s}\right)\left(\frac{1}{s+2}\right)\left(\frac{1}{s+5}\right)\left(\frac{1}{s+8}\right)$. Hence,



From which,

$$\dot{x}_1 = -8x_1 + x_2$$
$$\dot{x}_2 = -5x_2 + x_3$$
$$\dot{x}_3 = -2x_3 + x_4$$
$$\dot{x}_4 = 20r$$
$$y = x_1$$

26.

$\Delta = 1 + [G_2G_3G_4 + G_3G_4 + G_4 + 1] + [G_3G_4 + G_4]$; $T_1 = G_1G_2G_3G_4$; $\Delta_1 = 1$. Therefore,

$$T(s) = \frac{T_1\Delta_1}{\Delta} = \frac{G_1G_2G_3G_4}{2 + G_2G_3G_4 + 2G_3G_4 + 2G_4}$$

27.

Closed-loop gains: $G_2G_4G_6G_7H_3$; $G_2G_5G_6G_7H_3$; $G_3G_4G_6G_7H_3$; $G_3G_5G_6G_7H_3$; G_6H_1; G_7H_2

Forward-path gains: $T_1 = G_1G_2G_4G_6G_7$; $T_2 = G_1G_2G_5G_6G_7$; $T_3 = G_1G_3G_4G_6G_7$; $T_4 =$

$G_1G_3G_5G_6G_7$

Nontouching loops 2 at a time: $G_6H_1G_7H_2$

$\Delta = 1 - [H_3G_6G_7(G_2G_4 + G_2G_5 + G_3G_4 + G_3G_5) + G_6H_1 + G_7H_2] + [G_6H_1G_7H_2]$

$\Delta_1 = \Delta_2 = \Delta_3 = \Delta_4 = 1$

$$T(s) = \frac{T_1\Delta_1 + T_2\Delta_2 + T_3\Delta_3 + T_4\Delta_4}{\Delta}$$

$$= \frac{G_1G_2G_4G_6G_7 + G_1G_2G_5G_6G_7 + G_1G_3G_4G_6G_7 + G_1G_3G_5G_6G_7}{1 - H_3G_6G_7(G_2G_4 + G_2G_5 + G_3G_4 + G_3G_5) - G_6H_1 - G_7H_2 + G_6H_1G_7H_2}$$

28.

Closed-loop gains: $-s^2; -\frac{1}{s}\ ;\ -\frac{1}{s}\ ; -s^2$

Forward-path gains: $T_1 = s$; $T_2 = \frac{1}{s^2}$

Nontouching loops: None

$\Delta = 1 - (-s^2 - \frac{1}{s} - \frac{1}{s} - s^2)$

$\Delta_1 = \Delta_2 = 1$

$$G(s) = \frac{T_1\Delta_1 + T_2\Delta_2}{\Delta} = \frac{s + \frac{1}{s^2}}{1 + (s^2 + \frac{1}{s} + \frac{1}{s} + s^2)} = \frac{s^3+1}{2s^4+s^2+2s}$$

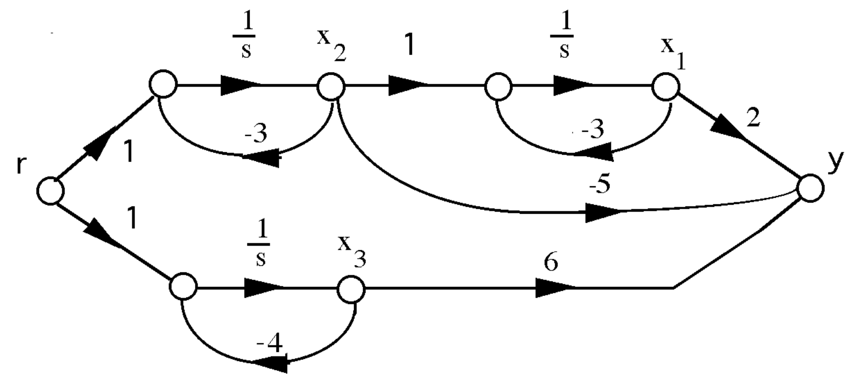
29.

$$T(s) = \frac{G_1\left(\frac{G_2G_3G_4G_5}{(1-G_2H_1)(1-G_4H_2)}\right)}{1 - \frac{G_2G_3G_4G_5G_6G_7G_8}{(1-G_2H_1)(1-G_4H_2)(1-G_7H_4)}} =$$

$$\frac{G_1G_2G_3G_4G_5(1-G_7H_4)}{1-G_2H_1-G_4H_2+G_2G_4H_1H_2-G_7H_4+G_2G_7H_1H_4+G_4G_7H_2H_4-G_2G_4G_7H_1H_2H_4-G_2G_3G_4G_5G_6G_7G_8}$$

30.

a. $G(s) = \dfrac{(s+1)(s+2)}{(s+3)^2(s+4)} = \dfrac{2}{(s+3)^2} - \dfrac{5}{s+3} + \dfrac{6}{s+4}$



Writing the state and output equations,

$$\dot{x}_1 = -3x_1 + x_2$$

$$\dot{x}_2 = -3x_2 + r$$

$$\dot{x}_3 = -4x_3 + r$$

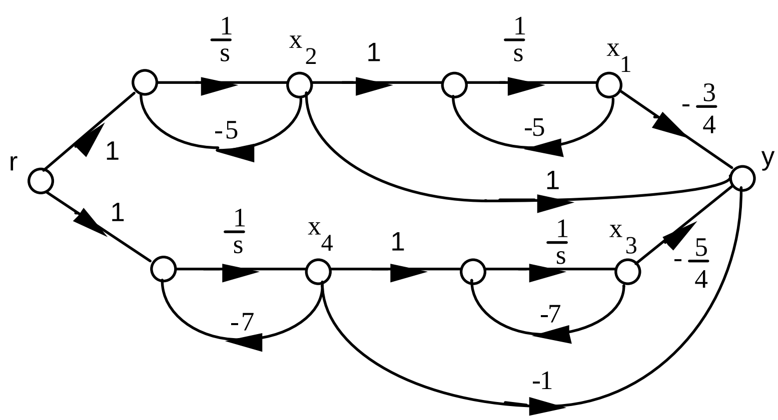
$$y = 2x_1 - 5x_2 + 6x_3$$

In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} -3 & 1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -4 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} r$$

$$y = \begin{bmatrix} 2 & -5 & 6 \end{bmatrix}$$

b. G(s) = $G(s) = \dfrac{(s+2)}{(s+5)^2(s+7)^2} = -\dfrac{3/4}{(s+5)^2} + \dfrac{1}{s+5} - \dfrac{5/4}{(s+7)^2} - \dfrac{1}{s+7}$



Writing the state and output equations,

$$\dot{x}_1 = -5x_1 + x_2$$

$$\dot{x}_2 = -5x_2 + r$$

$$\dot{x}_3 = -7x_3 + x_4$$

$$\dot{x}_4 = -7x_4 + r$$

$$y = -\frac{3}{4}x_1 + x_2 - \frac{5}{4}x_3 - x_4$$

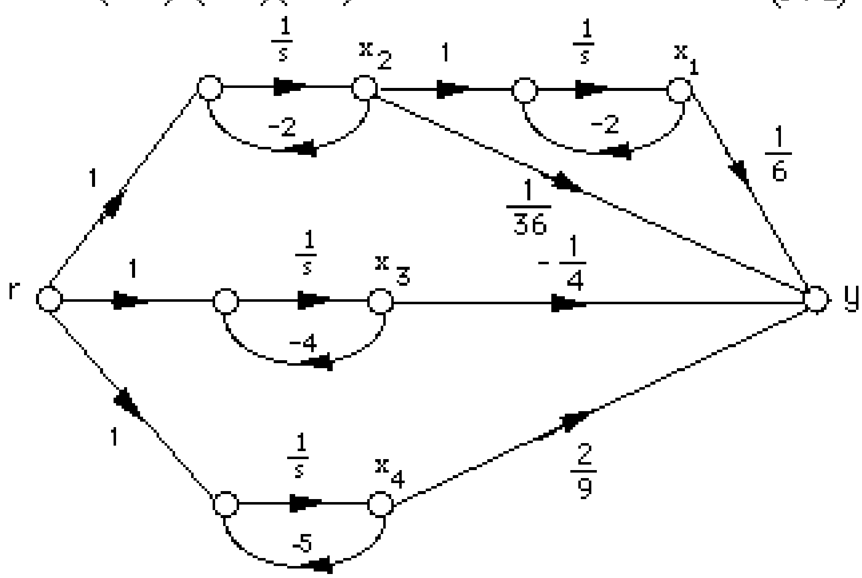
In vector matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} -5 & 1 & 0 & 0 \\ 0 & -5 & 0 & 0 \\ 0 & 0 & -7 & 1 \\ 0 & 0 & 0 & -7 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} r$$

$$y = \begin{bmatrix} -\frac{3}{4} & 1 & -\frac{5}{4} & -1 \end{bmatrix}\mathbf{x}$$

c.

$$G(s) = \frac{s+3}{(s+2)^2(s+4)(s+5)} = \frac{2}{9}\frac{1}{s+5} - \frac{1}{4}\frac{1}{s+4} + \frac{1}{36}\frac{1}{s+2} + \frac{1}{6}\frac{1}{(s+2)^2}$$



Writing the state and output equations,

$$\dot{x}_1 = -2x_1 + x_2$$

$$\dot{x}_2 = -2x_2 + r$$

$$\dot{x}_3 = -4x_3 + r$$

$$\dot{x}_4 = -5x_4 + r$$

$$y = \frac{1}{6}x_1 + \frac{1}{36}x_2 - \frac{1}{4}x_3 + \frac{2}{9}x_4$$

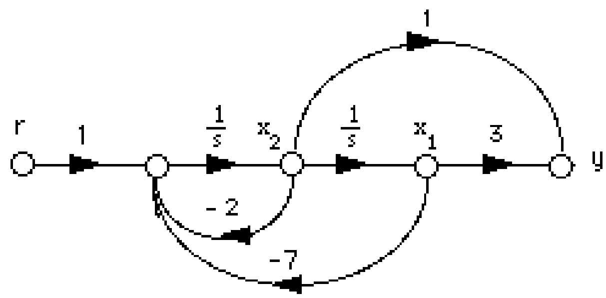
In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} -2 & 1 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -4 & 0 \\ 0 & 0 & 0 & -5 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} r$$

$$y = \begin{bmatrix} \frac{1}{6} & \frac{1}{36} & -\frac{1}{4} & \frac{2}{9} \end{bmatrix} \mathbf{x}$$

31.

a.



Writing the state equations,

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -7x_1 - 2x_2 + r$$

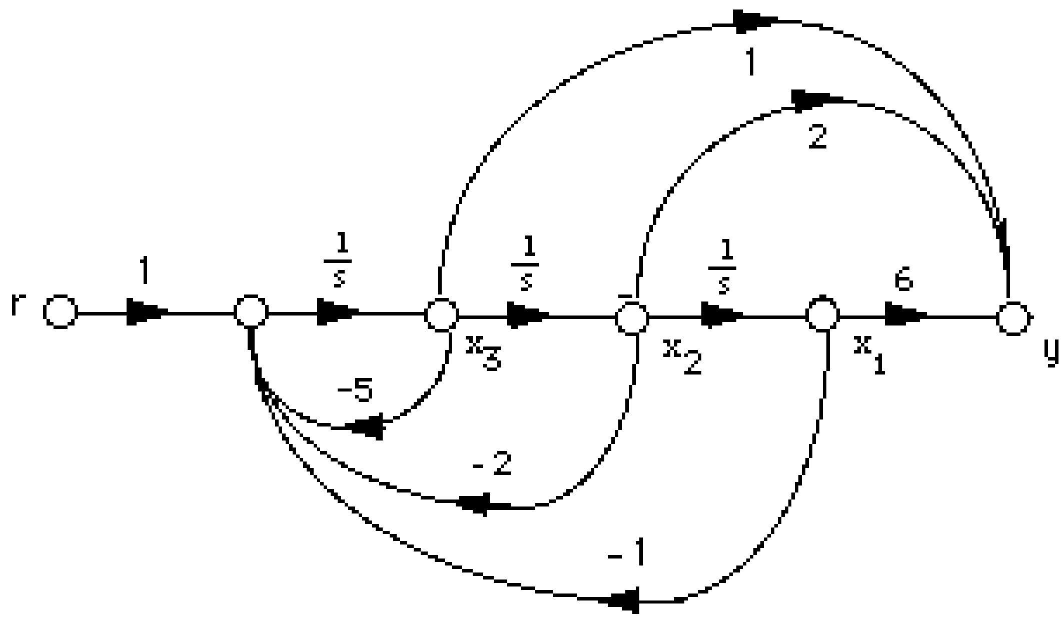
$$y = 3x_1 + x_2$$

In vector matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -7 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r$$

$$y = \begin{bmatrix} 3 & 1 \end{bmatrix} \mathbf{x}$$

b.



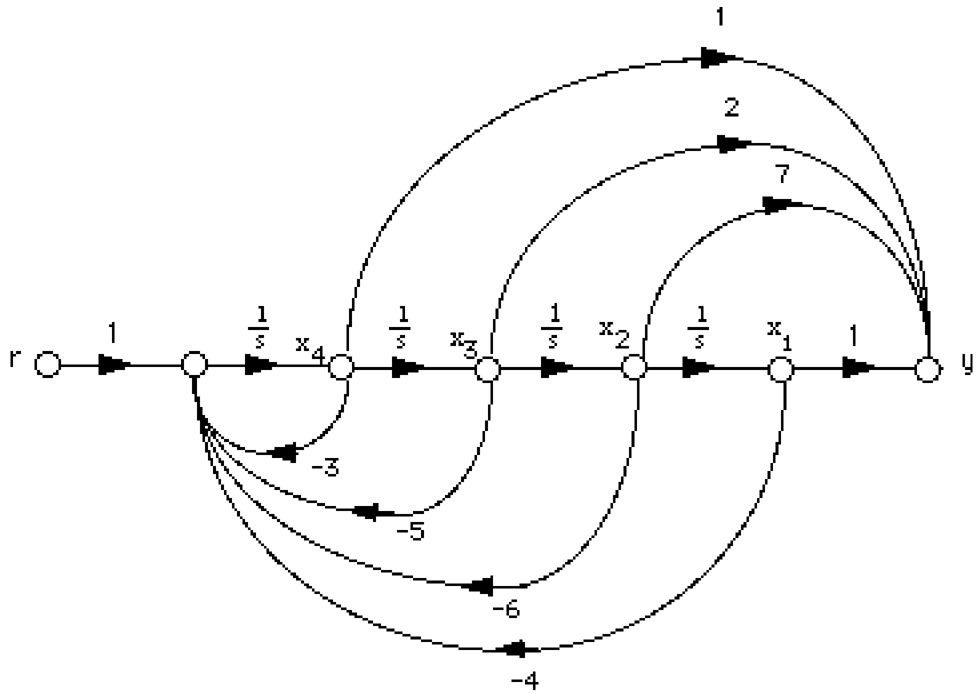
Writing the state equations,

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$\dot{x}_3 = -x_1 - 2x_2 - 5x_3 + r$$
$$y = 6x_1 + 2x_2 + x_3$$

In vector matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -5 \end{bmatrix} \mathbf{X} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r$$
$$y = \begin{bmatrix} 6 & 2 & 1 \end{bmatrix} \mathbf{x}$$

c.



$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = x_4$$

$$\dot{x}_4 = -4x_1 - 6x_2 - 5x_3 - 3x_4 + r$$

$$y = x_1 + 7x_2 + 2x_3 + x_4$$

In vector matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -4 & -6 & -5 & -3 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 7 & 2 & 1 \end{bmatrix}\mathbf{x}$$

32.

a. Controller canonical form:

From the phase-variable form in Problem 5.31(a), reverse the order of the state variables and obtain,

$$\dot{x}_2 = x_1$$

$$\dot{x}_1 = -7x_2 - 2x_1 + r$$

$$y = 3x_2 + x_1$$

Putting the equations in order,

$$\dot{x}_1 = -2x_1 - 7x_2 + r$$

$$\dot{x}_2 = x_1$$

$$y = x_1 + 3x_2$$

In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} -2 & -7 \\ 1 & 0 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 3 \end{bmatrix}\mathbf{x}$$

Observer canonical form:

$G(s) = \dfrac{s+3}{s^2+2s+7}$. Divide each term by $\dfrac{1}{s^2}$ and get

$$G(s) = \frac{\dfrac{1}{s} + \dfrac{3}{s^2}}{1 + \dfrac{2}{s} + \dfrac{7}{s^2}} = \frac{C(s)}{R(s)}$$

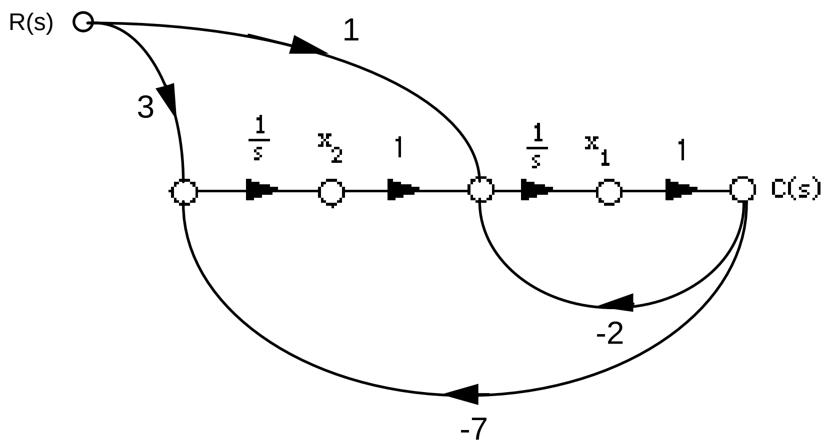
Cross multiplying,

$$\left(\frac{1}{s} + \frac{3}{s^2}\right) R(s) = \left(1 + \frac{2}{s} + \frac{7}{s^2}\right) C(s)$$

Thus,

$$\frac{1}{s}(R(s) - 2C(s)) + \frac{1}{s^2}(3R(s) - 7C(s)) = C(s)$$

Drawing the signal-flow graph,



Writing the state and output equations,

$$\dot{x}_1 = -2x_1 + x_2 + r$$

$$\dot{x}_2 = -7x_1 + 3r$$

$$y = x_1$$

In vector matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} -2 & 1 \\ -7 & 0 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 1 \\ 3 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix}\mathbf{x}$$

b. Controller canonical form:

From the phase-variable form in Problem 5.31(b), reverse the order of the state variables and obtain,

$$\dot{x}_3 = x_2$$

$$\dot{x}_2 = x_1$$

$$\dot{x}_1 = -x_3 - 2x_2 - 5x_1$$

$$y = 6x_3 + 2x_2 + x_1$$

Putting the equations in order,

$$\dot{x}_1 = -5x_1 - 2x_2 - x_3$$

$$\dot{x}_2 = x_1$$

$$\dot{x}_3 = x_2$$

$$y = x_1 + 2x_2 + 6x_3$$

In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} -5 & -2 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}r$$

$$y = \begin{bmatrix} 1 & 2 & 6 \end{bmatrix}\mathbf{x}$$

Observer canonical form:

$G(s) = \dfrac{s^2+2s+6}{s^3+5s^2+2s+1}$. Divide each term by $\dfrac{1}{s^3}$ and get

$$G(s) = \frac{\frac{1}{s}+\frac{2}{s^2}+\frac{6}{s^3}}{1+\frac{5}{s}+\frac{2}{s^2}+\frac{1}{s^3}} = \frac{C(s)}{R(s)}$$

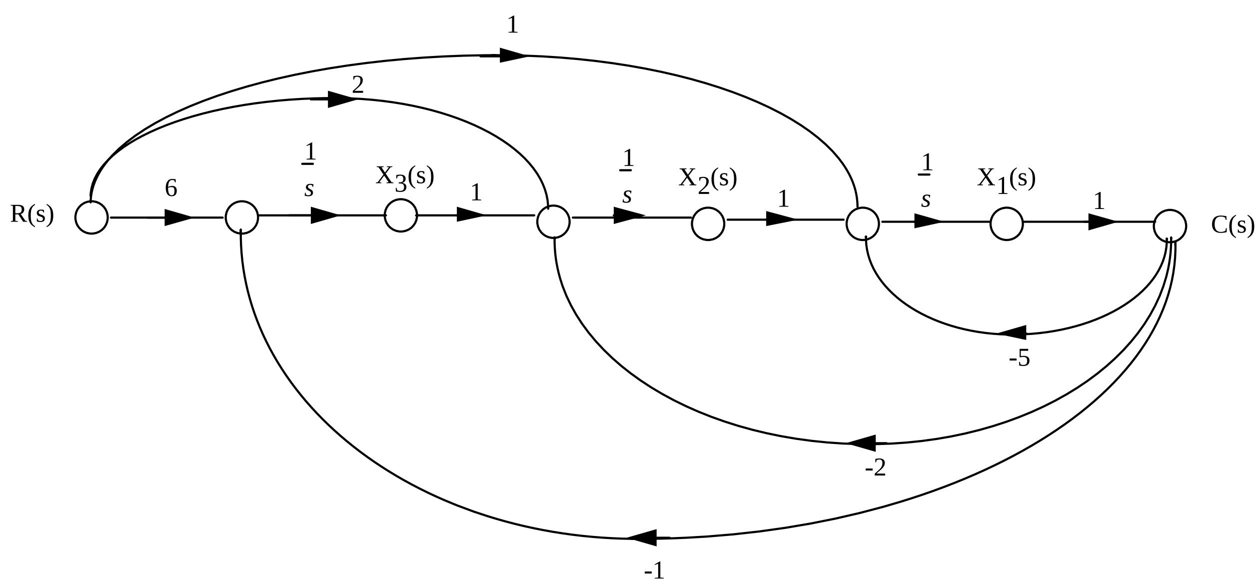
Cross-multiplying,

$$\left(\frac{1}{s}+\frac{2}{s^2}+\frac{6}{s^3}\right)R(s) = \left(1+\frac{5}{s}+\frac{2}{s^2}+\frac{1}{s^3}\right)C(s)$$

Thus,

$$\frac{1}{s}(R(s)-5c(s)) + \frac{1}{s^2}(2R(s)-2C(s)) + \frac{1}{s^3}(6R(s)-C(s)) = C(s)$$

Drawing the signal-flow graph,



Writing the state and output equations,

$$\dot{x}_1 = -5x_1 + x_2 + r$$

$$\dot{x}_2 = -2x_1 + x_3 + 2r$$

$$\dot{x}_3 = -x_1 + 6r$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\mathbf{x}$$

In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} -5 & 1 & 0 \\ -2 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\mathbf{x}$$

c. Controller canonical form:

From the phase-variable form in Problem 5.31(c), reverse the order of the state variables and obtain,

$$\dot{x}_4 = x_3$$

$$\dot{x}_3 = x_2$$

$$\dot{x}_2 = x_1$$

$$\dot{x}_1 = -4x_4 - 6x_3 - 5x_2 - 3x_1 + r$$

$$y = x_4 + 7x_3 + 2x_2 + x_1$$

Putting the equations in order,

$$\dot{x}_1 = -3x_1 - 5x_2 - 6x_3 - 4x_4 + r$$

$$\dot{x}_2 = x_1$$

$$\dot{x}_3 = x_2$$

$$\dot{x}_4 = x_3$$

$$y = x_1 + 2x_2 + 7x_3 + x_4$$

In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} -3 & -5 & -6 & -4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}\mathbf{X} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 2 & 7 & 1 \end{bmatrix}\mathbf{x}$$

Observer canonical form:

$G(s) = \dfrac{s^3+2s^2+7s+1}{s^4+3s^3+5s^2+6s+4}$. Divide each term by $\dfrac{1}{s^2}$ and get

$$G(s) = \frac{\frac{1}{s} + \frac{2}{s^2} + \frac{7}{s^3} + \frac{1}{s^4}}{1 + \frac{3}{s} + \frac{5}{s^2} + \frac{6}{s^3} + \frac{4}{s^4}} = \frac{C(s)}{R(s)}$$

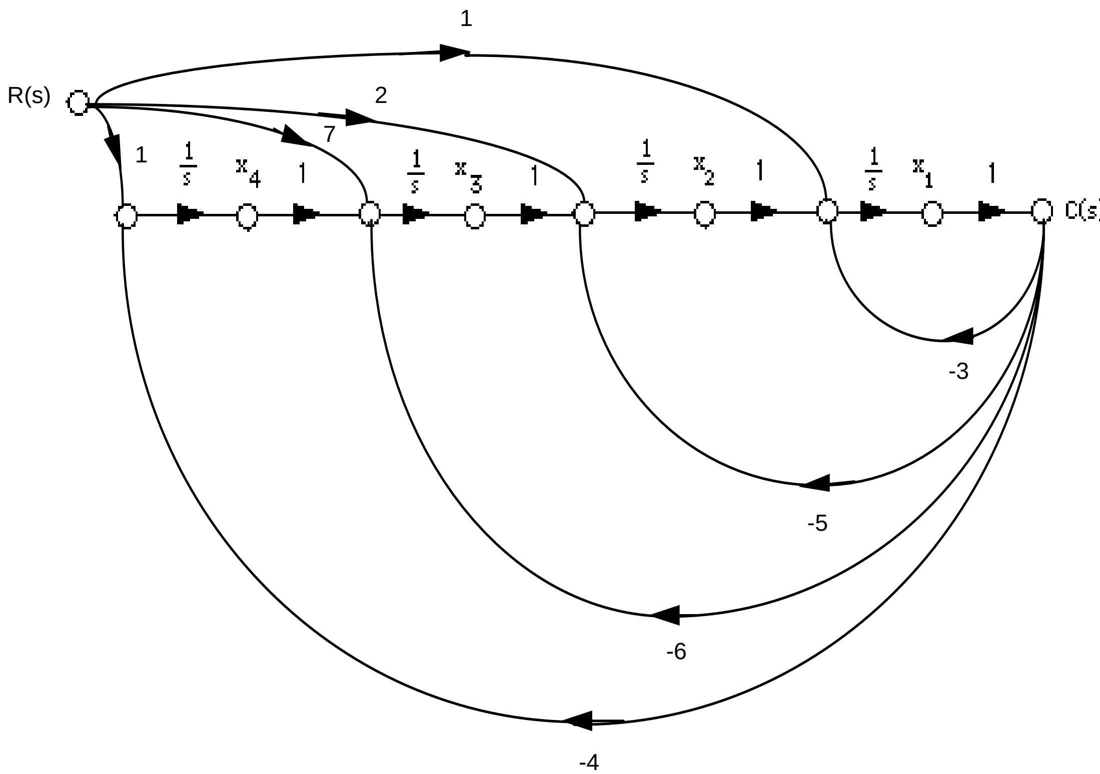
Cross multiplying,

$$\left(\frac{1}{s}+\frac{2}{s^2}+\frac{7}{s^3}+\frac{1}{s^4}\right)R(s)=\left(1+\frac{3}{s}+\frac{5}{s^2}+\frac{6}{s^3}+\frac{4}{s^4}\right)C(s)$$

Thus,

$$\frac{1}{s}(R(s)-3C(s))+\frac{1}{s^2}(2R(s)-5C(s))+\frac{1}{s^3}(7R(s)-6C(s))+\frac{1}{s^4}(R(s)-4C(s))=C(s)$$

Drawing the signal-flow graph,



Writing the state and output equations,

$$\dot{x}_1 = -3x_1 + x_2 + r$$
$$\dot{x}_2 = -5x_1 + x_3 + 2r$$
$$\dot{x}_3 = -6x_1 + x_4 + 7r$$
$$\dot{x}_4 = -4x_1 + r$$
$$y = x_1$$

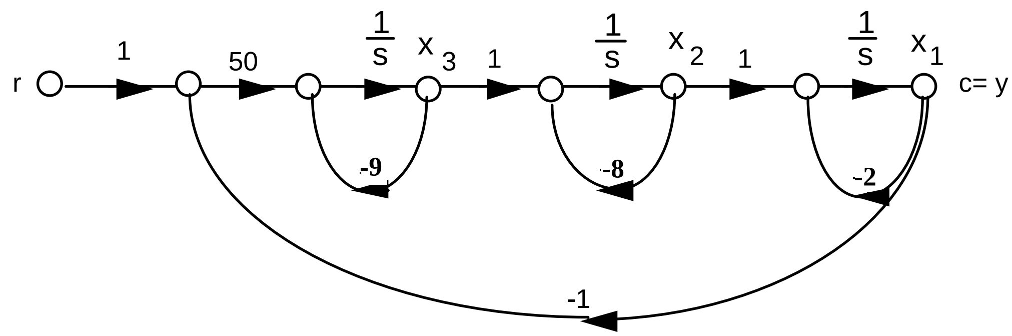
In vector matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} -3 & 1 & 0 & 0 \\ -5 & 0 & 1 & 0 \\ -6 & 0 & 0 & 1 \\ -4 & 0 & 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 2 \\ 7 \\ 1 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \mathbf{x}$$

33.

a.



Writing the state equations,

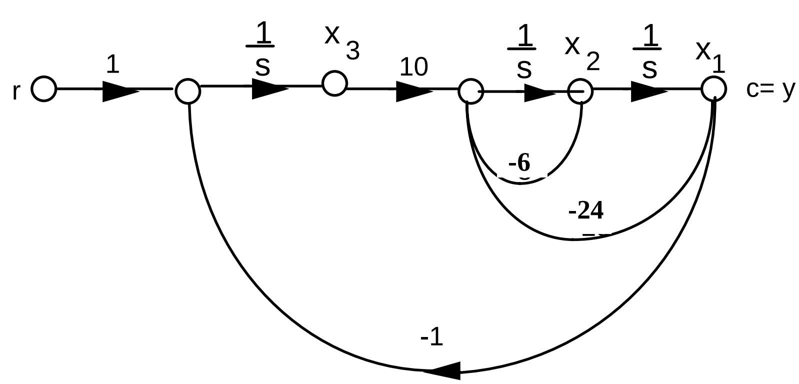
$$\dot{x}_1 = -2x_1 + x_2$$
$$\dot{x}_2 = -8x_2 + x_3$$
$$\dot{x}_3 = -50x_1 - 9x_3 + 50r$$
$$y = x_1$$

In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -8 & 1 \\ -50 & 0 & -9 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 50 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x}$$

b.



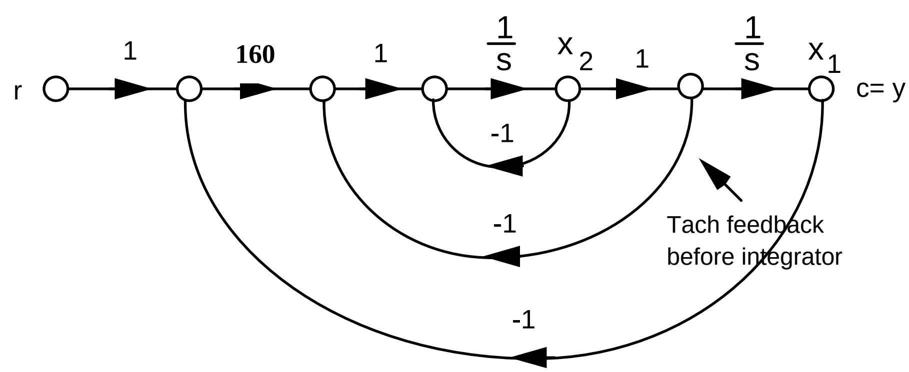
Writing the state equations,

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -24x_1 - 6x_2 + 10x_3$$
$$\dot{x}_3 = -x_1 + r$$
$$y = x_1$$

In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ -24 & -6 & 10 \\ -1 & 0 & 0 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r$$
$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\mathbf{x}$$

c.



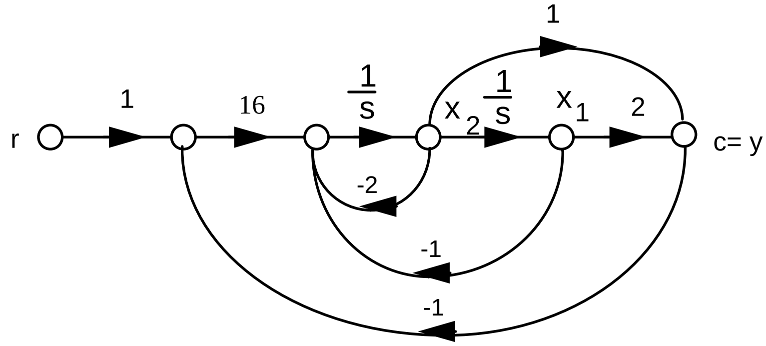
$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -x_2 - x_2 + 160(r - x_1) = -160x_1 - 2x_2 + 160r$$
$$y = x_1$$

In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -160 & -2 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ -160 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}$$

d. Since $\frac{1}{(s+1)^2} = \frac{1}{s^2+2s+1}$, we draw the signal-flow as follows:



Writing the state equations,

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1 - 2x_2 + 16(r-c) = -x_1 - 2x_2 + 16(r - (2x_1+x_2)) = -33x_1 - 18x_2 + 16r$$

$$y = 2x_1 + x_2$$

In vector-matrix form,

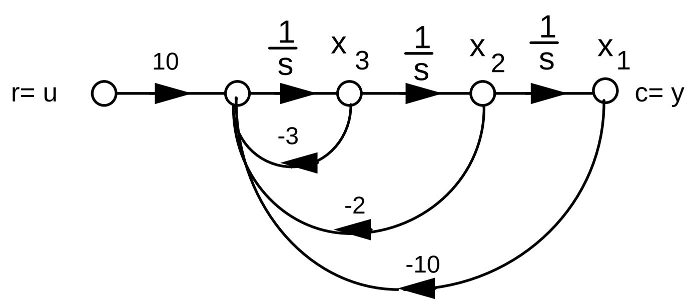
$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -33 & -18 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 16 \end{bmatrix} r$$

$$\dot{y} = [2 \quad 1]\boldsymbol{x}$$

34.

a. Phase-variable form:

$$T(s) = \frac{10}{s^3+3s^2+2s+10}$$



Writing the state equations,

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -10x_1 - 2x_2 - 3x_3 + 10u$$

$$y = x_1$$

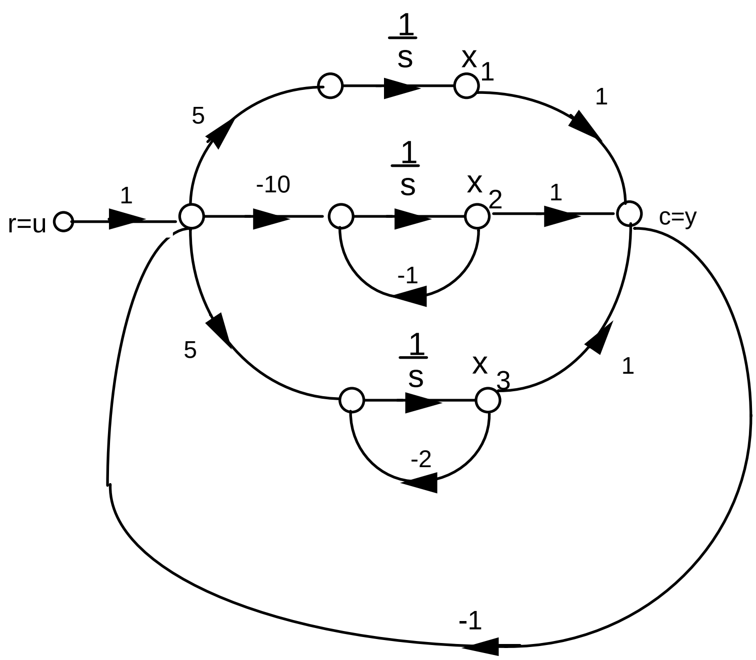
In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -10 & -2 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x}$$

b. Parallel form:

$$G(s) = \frac{5}{s} + \frac{-10}{s+1} + \frac{5}{s+2}$$



Writing the state equations,

$$\dot{x}_1 = 5(u - x_1 - x_2 - x_3) = -5x_1 -5x_2 -5x_3 +5u$$

$$\dot{x}_2 = -10(u - x_1 - x_2 - x_3) - x_2 = 10x_1 + 9x_2 + 10x_3 - 10u$$

$$\dot{x}_3 = 5(u - x_1 - x_2 - x_3) - 2x_3 = -5x_1 -5x_2 -7x_3 +5u$$

$$y = x_1 + x_2 + x_3$$

In vector-matrix form,

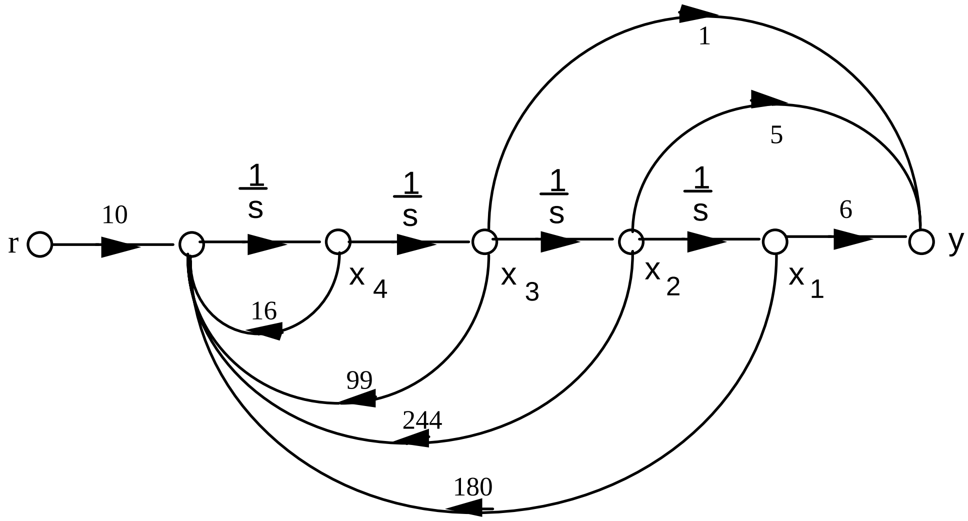
$$\dot{\mathbf{x}} = \begin{bmatrix} -5 & -5 & -5 \\ 10 & 9 & 10 \\ -5 & -5 & -7 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 5 \\ -10 \\ 5 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \mathbf{x}$$

35.

a. $T(s) = \dfrac{10(s^2 + 5s + 6)}{s^4 + 16s^3 + 99s^2 + 244s + 180}$

Drawing the signal-flow diagram,



Writing the state and output equations,

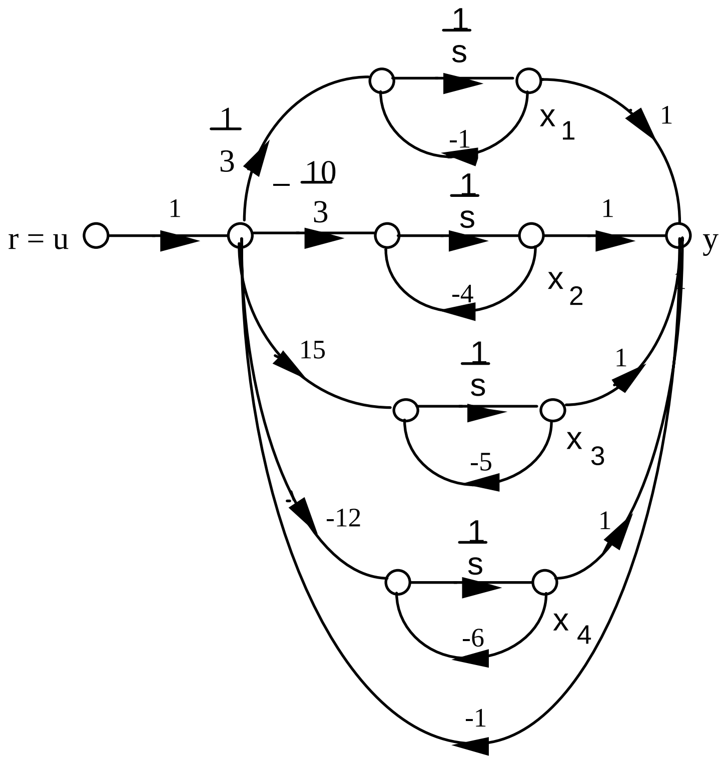
$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = x_3$$
$$\dot{x}_3 = x_4$$
$$\dot{x}_1 = -180x_1 - 244x_2 - 99x_3 - 16x_4 + 10r$$
$$y = 6x_1 + 5x_2 + x_3$$

In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -180 & -244 & -99 & -16 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 10 \end{bmatrix} r$$
$$y = \begin{bmatrix} 6 & 5 & 1 & 0 \end{bmatrix}\mathbf{x}$$

b. $G(s) = \dfrac{10(s+2)(s+3)}{(s+1)(s+4)(s+5)(s+6)} = \dfrac{1/3}{s+1} - \dfrac{10/3}{s+4} + \dfrac{15}{s+5} - \dfrac{12}{s+6}$

Drawing the signal-flow diagram and including the unity-feedback path,



Writing the state and output equations,

$$\dot{x}_1 = \frac{1}{3}(u - x_1 - x_2 - x_3 - x_4) - x_1$$

$$\dot{x}_2 = \frac{-10}{3}(u - x_1 - x_2 - x_3 - x_4) - 4x_2$$

$$\dot{x}_3 = 15(u - x_1 - x_2 - x_3 - x_4) - 5x_3$$

$$\dot{x}_4 = -12(u - x_1 - x_2 - x_3 - x_4) - 12x_4$$

$$y = x_1 + x_2 + x_3 + x_4$$

In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} -\frac{4}{3} & -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \\ \frac{10}{3} & -\frac{2}{3} & \frac{10}{3} & \frac{10}{3} \\ -15 & -15 & -20 & -15 \\ 12 & 12 & 12 & 0 \end{bmatrix}\mathbf{x} + \begin{bmatrix} \frac{1}{3} \\ -\frac{10}{3} \\ 15 \\ -12 \end{bmatrix}u$$

$$y = \begin{bmatrix}1 & 1 & 1 & 1\end{bmatrix}\mathbf{x}$$

36.

Program:

```
'(a)'
'G(s)'
G=zpk([-2 -3],[-1 -4 -5 -6],10)
```

```
'T(s)'
T=feedback(G,1,-1)
[numt,dent]=tfdata(T,'v');
'Find controller canonical form'
[Acc,Bcc,Ccc,Dcc]=tf2ss(numt,dent)
A1=flipud(Acc);
'Transform to phase-variable form'
Apv=fliplr(A1)
Bpv=flipud(Bcc)
Cpv=fliplr(Ccc)
'(b)'
'G(s)'
G=zpk([-2 -3],[-1 -4 -5 -6],10)
'T(s)'
T=feedback(G,1,-1)
[numt,dent]=tfdata(T,'v');
'Find controller canonical form'
[Acc,Bcc,Ccc,Dcc]=tf2ss(numt,dent)
'Transform to modal form'
[A,B,C,D]=canon(Acc,Bcc,Ccc,Dcc,'modal')
```

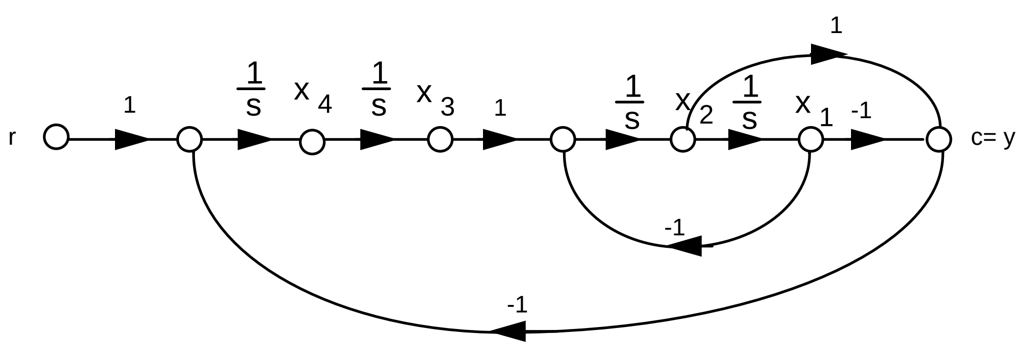
Computer response:

```
ans =

(a)

ans =

G(s)

Zero/pole/gain:
    10 (s+2) (s+3)
-----------------------
(s+1) (s+4) (s+5) (s+6)

ans =

T(s)

Zero/pole/gain:

              10 (s+2) (s+3)
------------------------------------------
(s+1.264) (s+3.412) (s^2 + 11.32s + 41.73)

ans =

Find controller canonical form

Acc =

  -16.0000  -99.0000 -244.0000 -180.0000
    1.0000         0         0         0
         0    1.0000         0         0
         0         0    1.0000         0
Bcc =

     1
     0
     0
     0
```

```
Ccc =

         0   10.0000   50.0000   60.0000

Dcc =

     0

ans =

Transform to phase-variable form

Apv =

         0    1.0000         0         0
         0         0    1.0000         0
         0         0         0    1.0000
 -180.0000 -244.0000  -99.0000  -16.0000

Bpv =

     0
     0
     0
     1

Cpv =

   60.0000   50.0000   10.0000         0

ans =

(b)

ans =

G(s)

Zero/pole/gain:
    10 (s+2) (s+3)
-----------------------
(s+1) (s+4) (s+5) (s+6)

ans =

T(s)

Zero/pole/gain:
              10 (s+2) (s+3)
------------------------------------------
(s+1.264) (s+3.412) (s^2 + 11.32s + 41.73)

ans =

Find controller canonical form

Acc =

  -16.0000  -99.0000 -244.0000 -180.0000
    1.0000         0         0         0
         0    1.0000         0         0
         0         0    1.0000         0
```

```
Bcc =

     1
     0
     0
     0

Ccc =

         0   10.0000   50.0000   60.0000

Dcc =

     0

ans =

Transform to modal form

A =

   -5.6618    3.1109         0         0
   -3.1109   -5.6618         0         0
         0         0   -3.4124         0
         0         0         0   -1.2639

B =

   -4.1108
    1.0468
    1.3125
    0.0487
C =

    0.1827    0.6973   -0.1401    4.2067

D =

     0
```

37.



Writing the state equations,

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1 + x_3$$

$$\dot{x}_3 = x_4$$

$$\dot{x}_4 = x_1 - x_2 + r$$

$$y = -x_1 + x_2$$

In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & -1 & 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \mathbf{r}$$

$$y = c = [-1 \quad 1 \quad 0 \quad 0]\,\mathbf{x}$$

38.

a.

$$\ddot{\theta}_1 + 5\dot{\theta}_1 + 6\theta_1 - 3\dot{\theta}_2 - 4\theta_2 = 0$$

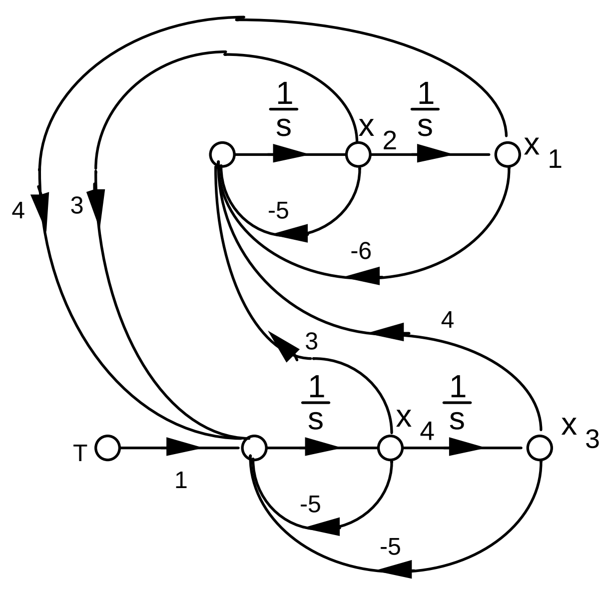
$$-3\dot{\theta}_1 - 4\theta_1 + \ddot{\theta}_2 + 5\dot{\theta}_2 + 5\theta_2 = T$$

or

$$\ddot{\theta}_1 = -5\dot{\theta}_1 - 6\theta_1 + 3\dot{\theta}_2 + 4\theta_2$$

$$\ddot{\theta}_2 = 3\dot{\theta}_1 + 4\theta_1 - 5\dot{\theta}_2 - 5\theta_2 + T$$

Letting, $\theta_1 = x_1$; $\dot{\theta}_1 = x_2$; $\theta_2 = x_3$; $\dot{\theta}_2 = x_4$,



where $x = \theta$.

b. Using the signal-flow diagram,

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -6x_1 - 5x_2 + 4x_3 + 3x_4$$

$$\dot{x}_3 = x_4$$

$$\dot{x}_4 = 4x_1 + 3x_2 - 5x_3 - 5x_4 + T$$

$$y = x_3$$

In vector-matrix form,

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -6 & -5 & 4 & 3 \\ 0 & 0 & 0 & 1 \\ 4 & 3 & -5 & -5 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} T$$

$$y = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \mathbf{x}$$

39.

Program:

```
numg=7;
deng=poly([0 -9 -12]);
G=tf(numg,deng);
T=feedback(G,1)
[numt,dent]=tfdata(T,'v')
[A,B,C,D]=tf2ss(numt,dent); %Obtain controller canonical form
'(a)'                        %Display label
A=flipud(A);                 %Convert to phase-variable form
A=fliplr(A)                  %Convert to phase-variable form
B=flipud(B)                  %Convert to phase-variable form
C=fliplr(C)                  %Convert to phase-variable form
'(b)'                        %Display label
[a,b,c,d]=canon(A,B,C,D)     %Convert to parallel form
```

Computer response:

```
Transfer function:
           7
------------------------
s^3 + 21 s^2 + 108 s + 7

numt =

     0     0     0     7

dent =

     1    21   108     7
```

```
ans =

(a)

A =

     0     1     0
     0     0     1
    -7  -108   -21

B =

     0
     0
     1

C =

     7     0     0

ans =

(b)


a =

   -0.0657         0         0
         0  -12.1807         0
         0         0   -8.7537


b =

   -0.0095
   -3.5857
    2.5906


c =

   -6.9849   -0.0470   -0.0908


d =

     0
```

40.

$$\dot{\mathbf{x}}_1 = \mathbf{A_1 x_1} + \mathbf{B_1} r \tag{1}$$

$$y_1 = \mathbf{C_1 x_1} \tag{2}$$

$$\dot{\mathbf{x}}_2 = \mathbf{A_2 x_2} + \mathbf{B_2} y_1 \tag{3}$$

$$y_2 = \mathbf{C_2 x_2} \tag{4}$$

Substituting Eq. (2) into Eq. (3),

$$\dot{\mathbf{x}}_1 = \mathbf{A_1 x_1} + \mathbf{B_1} r$$

$$\dot{\mathbf{x}}_2 = \mathbf{B_2C_1x_1} + \mathbf{A_2x_2}$$

$$y_2 = \mathbf{C_2x_2}$$

In vector-matrix notation,

$$\begin{bmatrix} \dot{\mathbf{x}}_1 \\ \dot{\mathbf{x}}_2 \end{bmatrix} = \left[\begin{array}{c:c} \mathbf{A_1} & \mathbf{O} \\ \mathbf{B_2 C_1} & \mathbf{A_2} \end{array}\right] \begin{bmatrix} \mathbf{x_1} \\ \mathbf{x_2} \end{bmatrix} + \begin{bmatrix} \mathbf{B_1} \\ \mathbf{O} \end{bmatrix} r$$

$$\mathbf{y}_2 = \left[\begin{array}{c:c} \mathbf{O} & \mathbf{C}_2 \end{array}\right] \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix}$$

41.

$$\dot{\mathbf{x}}_1 = \mathbf{A_1x_1} + \mathbf{B_1}r \tag{1}$$

$$y_1 = \mathbf{C_1x_1} \tag{2}$$

$$\dot{\mathbf{x}}_2 = \mathbf{A_2x_2} + \mathbf{B_2}r \tag{3}$$

$$y_2 = \mathbf{C_2x_2} \tag{4}$$

In vector-matrix form,

$$\begin{bmatrix} \dot{\mathbf{x}}_1 \\ \dot{\mathbf{x}}_2 \end{bmatrix} = \left[\begin{array}{c:c} \mathbf{A_1} & \mathbf{O} \\ \mathbf{O} & \mathbf{A_2} \end{array}\right] \begin{bmatrix} \mathbf{x_1} \\ \mathbf{x_2} \end{bmatrix} + \begin{bmatrix} \mathbf{B_1} \\ \mathbf{B_2} \end{bmatrix} r$$

$$\mathbf{y} = \mathbf{y}_1 + \mathbf{y}_2 = \left[\begin{array}{c:c} \mathbf{C}_1 & \mathbf{C}_2 \end{array}\right] \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix}$$

42.

$$\dot{\mathbf{x}}_1 = \mathbf{A_1x_1} + \mathbf{B_1}e \tag{1}$$

$$y = \mathbf{C_1x_1} \tag{2}$$

$$\dot{\mathbf{x}}_2 = \mathbf{A_2x_2} + \mathbf{B_2}y \tag{3}$$

$$p = \mathbf{C_2x_2} \tag{4}$$

Substituting e = r - p into Eq. (1) and substituting Eq. (2) into (3), we obtain,

$$\dot{\mathbf{x}}_1 = \mathbf{A_1x_1} + \mathbf{B_1}(r - p) \tag{5}$$

$$y = \mathbf{C_1x_1} \tag{6}$$

$$\dot{\mathbf{x}}_2 = \mathbf{A_2x_2} + \mathbf{B_2C_1x_1} \tag{7}$$

$$p = \mathbf{C_2x_2} \tag{8}$$

Substituting Eq. (8) into Eq. (5),

$$\dot{\mathbf{x}}_1 = \mathbf{A_1x_1} - \mathbf{B_1C_2x_2} + \mathbf{B_1r}$$

$$\dot{\mathbf{x}}_2 = \mathbf{B}_2\mathbf{C}_1\mathbf{x}_1 + \mathbf{A}_2\mathbf{x}_2$$

$$\mathbf{y} = \mathbf{C}_1\mathbf{x}_1$$

In vector-matrix form,

$$\begin{bmatrix} \dot{\mathbf{x}}_1 \\ \dot{\mathbf{x}}_2 \end{bmatrix} = \left[\begin{array}{c:c} \mathbf{A}_1 & -\mathbf{B}_1\mathbf{C}_2 \\ \hdashline \mathbf{B}_2\mathbf{C}_1 & \mathbf{A}_2 \end{array}\right] \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} + \begin{bmatrix} \mathbf{B}_1 \\ \mathbf{0} \end{bmatrix} \mathbf{r}$$

$$y = \left[\begin{array}{c:c} \mathbf{C}_1 & \mathbf{0} \end{array}\right] \begin{bmatrix} \dot{\mathbf{x}}_1 \\ \dot{\mathbf{x}}_2 \end{bmatrix}$$

43.

$$\dot{\mathbf{z}} = \mathbf{P}^{-1}\mathbf{APz} + \mathbf{P}^{-1}\mathbf{B}u$$

$$y = \mathbf{CPz}$$

$$\mathbf{P}^{-1} = \begin{bmatrix} -4 & 9 & -3 \\ 0 & -4 & 7 \\ -1 & -4 & -9 \end{bmatrix}; \ \therefore\ \mathbf{P} = \begin{bmatrix} -0.2085 & -0.3029 & -0.1661 \\ 0.0228 & -0.1075 & -0.0912 \\ 0.0130 & 0.0814 & -0.0521 \end{bmatrix}$$

$$\mathbf{P}^{-1}\mathbf{AP} = \begin{bmatrix} 18.5961 & 25.4756 & 5.6156 \\ -12.9023 & -28.8893 & -8.3909 \\ -0.5733 & 11.4169 & 5.2932 \end{bmatrix}; \ \mathbf{P}^{-1}\mathbf{B} = \begin{bmatrix} -58 \\ 63 \\ -12 \end{bmatrix}; \ \mathbf{CP} = \begin{bmatrix} 1.5668 & 3.0423 & 2.7329 \end{bmatrix}$$

44.

$$\dot{\mathbf{z}} = \mathbf{P}^{-1}\mathbf{APz} + \mathbf{P}^{-1}\mathbf{B}u$$

$$y = \mathbf{CPz}$$

$$\mathbf{P}^{-1} = \begin{bmatrix} 5 & -4 & 9 \\ 6 & -7 & 6 \\ 6 & -5 & -3 \end{bmatrix}; \ \mathrm{P} = \begin{bmatrix} 0.3469 & -0.3878 & 0.2653 \\ 0.3673 & -0.4694 & 0.1633 \\ 0.0816 & 0.0068 & -0.0748= \end{bmatrix}$$

$$\mathbf{P}^{-1}\mathbf{AP} = \begin{bmatrix} -28.2857 & 40.8095 & -40.9048 \\ -18.3061 & 28.2245 & -37.4694 \\ 5.3878 & -6.5510 & -5.9388 \end{bmatrix}; \ \mathbf{P}^{-1}\mathbf{B} = \begin{bmatrix} 61 \\ 82 \\ 74 \end{bmatrix}; \ \mathbf{CP} = \begin{bmatrix} -2.0816 & 2.6599 & -1.2585 \end{bmatrix}$$

45.

Eigenvalues are -1, -2, and -3 since,

$$|\lambda \mathbf{I} - \mathrm{A}| = (\lambda + 3)(\lambda + 2)(\lambda + 1)$$

Solving for the eigenvectors, $\mathbf{Ax} = \lambda\mathbf{x}$

or,

$$4x_3 - 5x_2 - (\lambda + 5)x_1 = 0$$
$$-2x_3 - x_2\lambda + 2x_1 = 0$$
$$-2x_2 - (\lambda + 1)x_3 = 0$$

For $\lambda = -1$, $x_2 = 0$, $x_1 = x_3$. For $\lambda = -2$, $x_1 = x_2 = \frac{x_3}{2}$. For $\lambda = -3$, $x_1 = -\frac{x_2}{2}$, $x_2 = x_3$. Thus,

$\dot{\mathbf{z}} = \mathbf{P}^{-1}\mathbf{APz} + \mathbf{P}^{-1}\mathbf{B}u$; $y = \mathbf{CPz}$, where

$$P^{-1}AP = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{pmatrix} ; \quad P^{-1}B = \begin{pmatrix} -12 \\ 8 \\ -3 \end{pmatrix} ; \quad CP = (1, 4, 7)$$

46.

Eigenvalues are 1, -2, and 3 since,

$$|\lambda\mathbf{I} - \mathbf{A}| = (\lambda - 3)(\lambda + 2)(\lambda - 1)$$

Solving for the eigenvectors, $\mathbf{Ax} = \lambda\mathbf{x}$

or,

$$(-\lambda - 10)x_1 + 7x_3 - 3x_2 = 0$$
$$\frac{73}{4}x_1 + \left(-\lambda + \frac{25}{4}\right)x_2 - \frac{47}{4}x_3 = 0$$
$$-\frac{29}{4}x_1 - \frac{9}{4}x_2 + \left(-\lambda + \frac{23}{4}\right)x_3 = 0$$

For $\lambda = 1$, $x_1 = x_2 = \frac{x_3}{2}$. For $\lambda = -2$, $x_1 = 2x_3$, $x_2 = -3x_3$. For $\lambda = 3$, $x_1 = x_3$, $x_2 = -2x_3$. Thus,

$\dot{\mathbf{z}} = \mathbf{P}^{-1}\mathbf{APz} + \mathbf{P}^{-1}\mathbf{B}u$; $y = \mathbf{CPz}$, where

$$P^{-1}AP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{pmatrix} ; \quad P^{-1}B = \begin{pmatrix} \frac{3}{2} \\ \frac{1}{2} \\ -\frac{3}{2} \end{pmatrix} ; \quad CP = (7, 12, 9)$$

47.

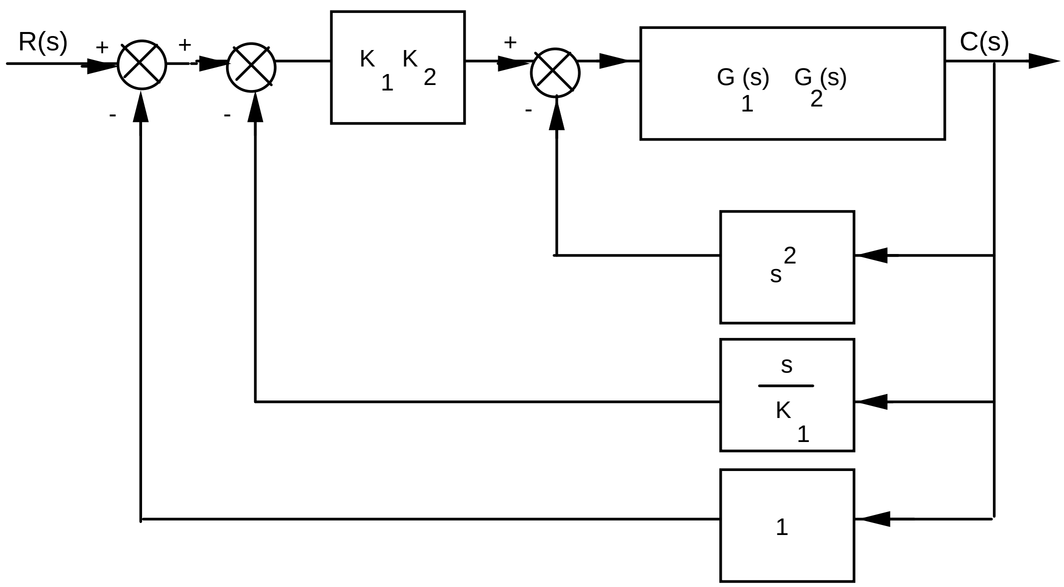
Program:
```
A=[-10 -3 7;18.25 6.25 -11.75;-7.25 -2.25 5.75];
B=[1;3;2];
C=[1 -2 4];
[P,d]=eig(A);
Ad=inv(P)*A*P
Bd=inv(P)*B
Cd=C*P
```

Computer response:

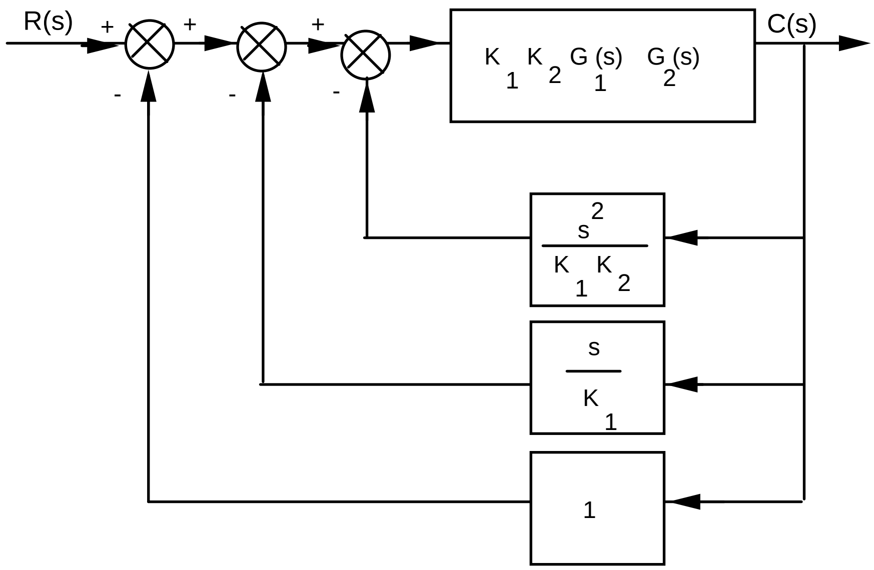
```
Ad =

   -2.0000    0.0000    0.0000
   -0.0000    3.0000   -0.0000
    0.0000    0.0000    1.0000


Bd =

    1.8708
   -3.6742
    3.6742


Cd =

    3.2071    3.6742    2.8577
```

48.

a. Combine $G_1(s)$ and $G_2(s)$. Then push K_1 to the right past the summing junction:

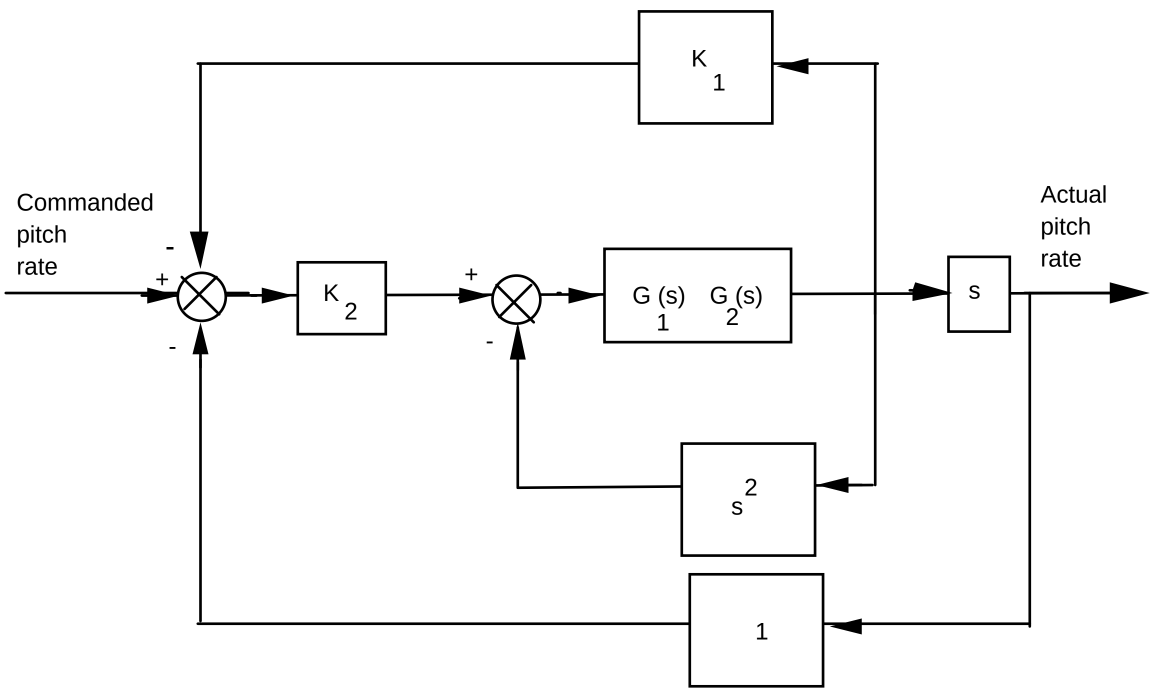


Push K_1K_2 to the right past the summing junction:

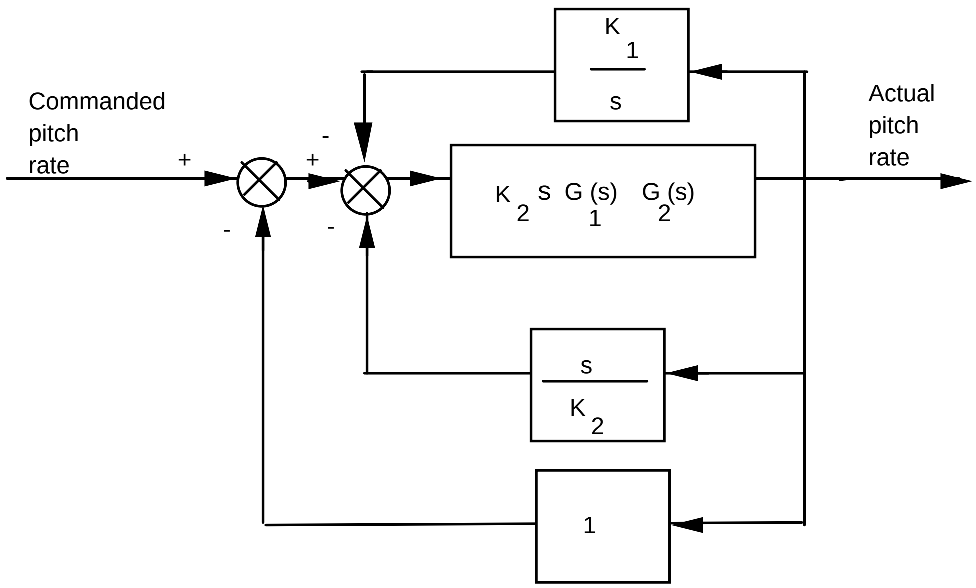


Hence, $T(s) = \dfrac{K_1K_2G_1(s)G_2(s)}{1 + K_1K_2G_1(s)G_2(s)\left(1 + \dfrac{s}{K_1} + \dfrac{s^2}{K_1K_2}\right)}$

b. Rearranging the block diagram to show commanded pitch rate as the input and actual pitch rate as the output:



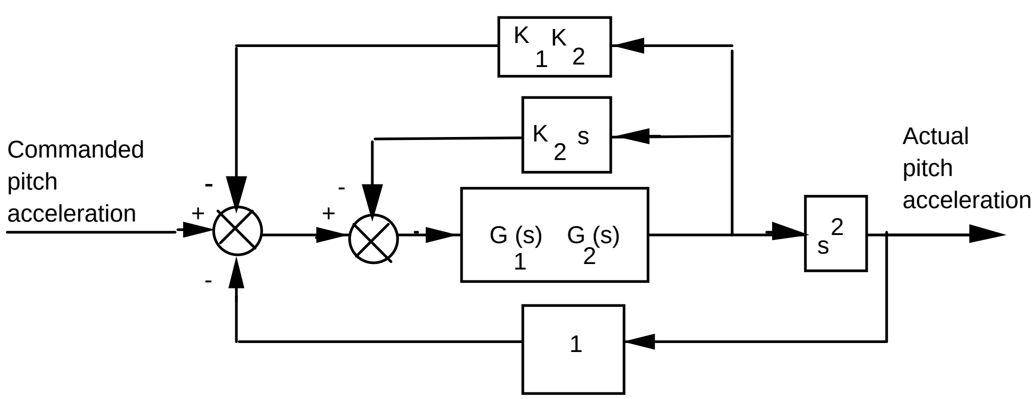
Pushing K_2 to the right past the summing junction; and pushing s to the left past the pick-off point yields,



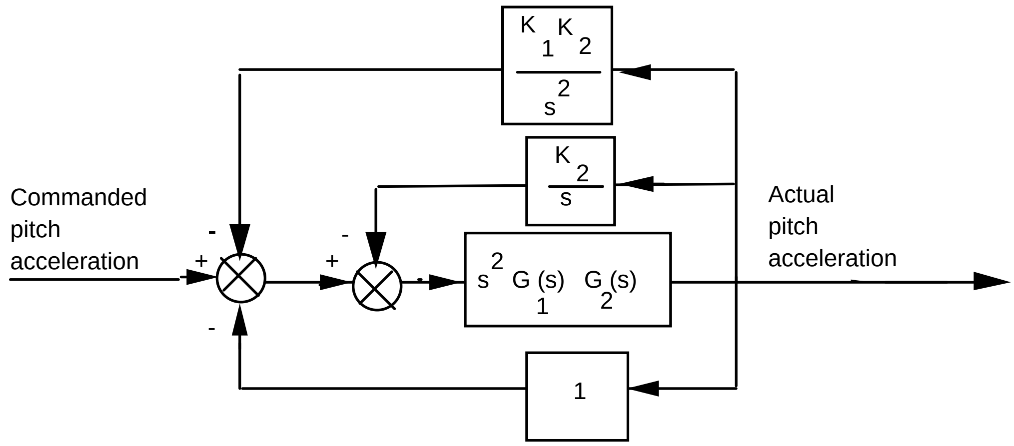
Finding the closed-loop transfer function:

$$T(s) = \frac{K_2 s G_1(s) G_2(s)}{1 + K_2 s G_1(s) G_2(s)\left(1 + \frac{s}{K_2} + \frac{K_1}{s}\right)} = \frac{K_2 s G_1(s) G_2(s)}{1 + G_1(s) G_2(s)(s^2 + K_2 s + K_1 K_2)}$$

c. Rearranging the block diagram to show commanded pitch acceleration as the input and actual pitch acceleration as the output:



Pushing s^2 to the left past the pick-off point yields,

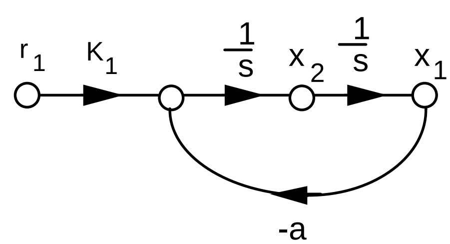


Finding the closed-loop transfer function:

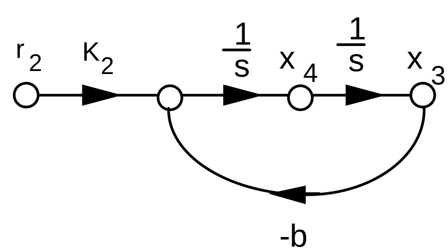
$$T(s) = \frac{s^2 G_1(s) G_2(s)}{1 + s^2 G_1(s) G_2(s)\left(1 + \frac{K_1 K_2}{s^2} + \frac{K_2}{s}\right)} = \frac{s^2 G_1(s) G_2(s)}{1 + G_1(s) G_2(s)(s^2 + K_2 s + K_1 K_2)}$$

49.

Establish a sinusoidal model for the carrier: $T(s) = \frac{K_1}{s^2+a^2}$



Establish a sinusoidal model for the message: $T(s) = \frac{K_2}{s^2+b^2}$



Writing the state equations,

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -a^2x_1 + K_1r$$

$$\dot{x}_3 = x_4$$

$$\dot{x}_4 = -b^2x_3 + K_2r$$

$$y = x_1x_3$$

50.

The equivalent forward transfer function is $G(s) = \frac{K_1K_2}{s(s+a_1)}$. The equivalent feedback transfer function is

$H(s) = K_3 + \frac{K_4s}{s+a_2}$. Hence, the closed-loop transfer function is

$$T(s) = \frac{G(s)}{1 + G(s)H(s)} = \frac{K_1K_2(s+a_2)}{s^3 + (a_1+a_2)s^2 + (a_1a_2+K_1K_2K_3+K_1K_2K_4)s + K_1K_2K_3a_2}$$

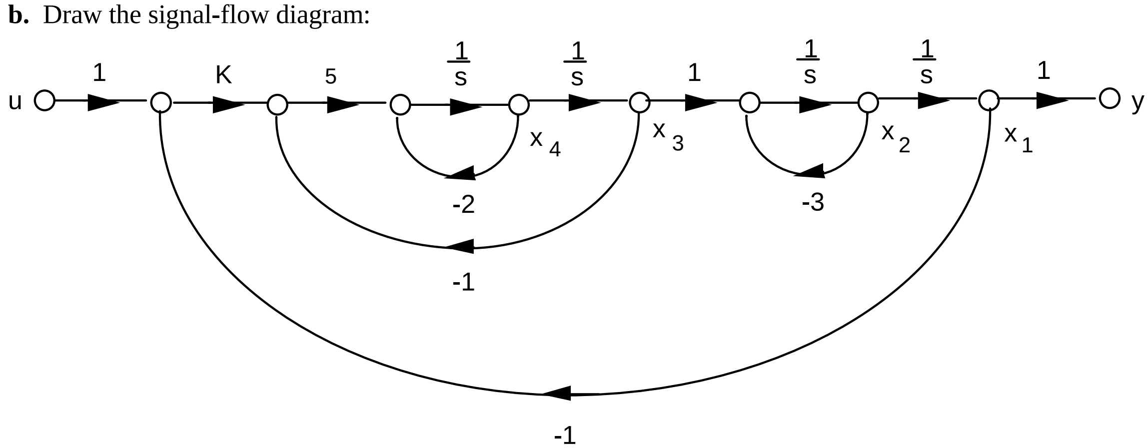
51.

a. The equivalent forward transfer function is

$$G_e(s) = K\frac{\frac{5}{s(s+2)}}{1+\frac{5}{s(s+2)}}\frac{1}{s(s+3)} = \frac{5K}{s(s+3)(s^2+2s+5)}$$

$$T(s) = \frac{G_e}{1+G_e} = \frac{5K}{s^4+5s^3+11s^2+15s+5K}$$

b. Draw the signal-flow diagram:



Writing the state and output equations from the signal-flow diagram:

$$\dot{x}_1 = x_2$$
$$\dot{x}_2 = -3x_2 + x_3$$
$$\dot{x}_3 = x_4$$
$$\dot{x}_4 = -5Kx_1 - 5x_3 - 2x_4 + 5Ku$$
$$y = x_1$$

In vector-matrix form:

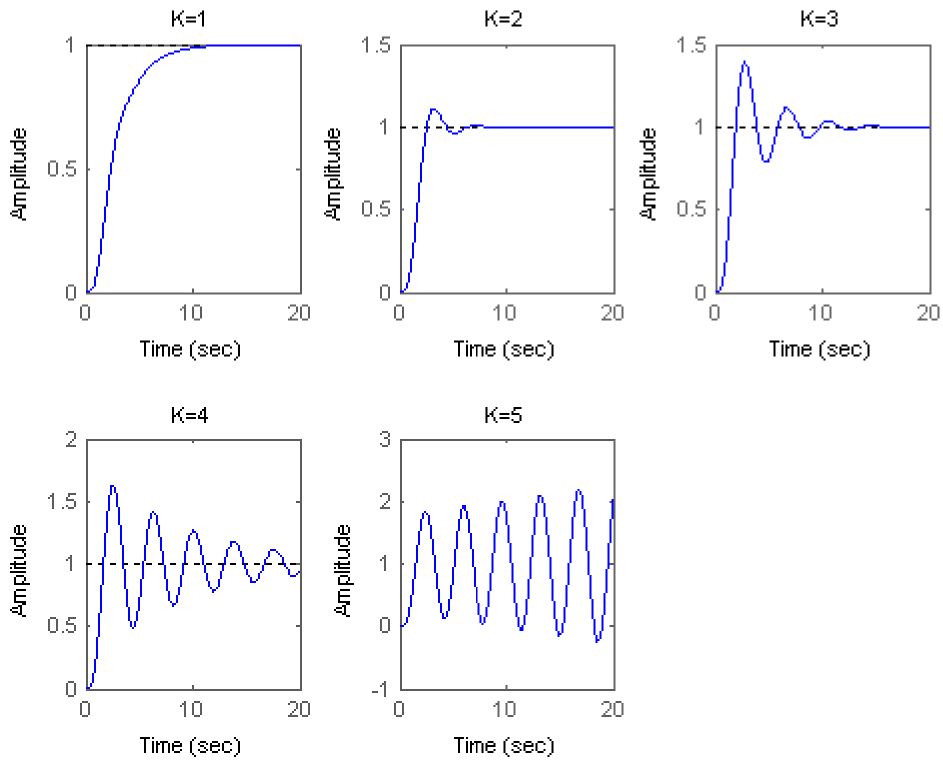
$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -3 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -5K & 0 & -5 & -2 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 5\mathrm{K} \end{bmatrix}u$$
$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}\mathbf{x}$$

c.

Program:

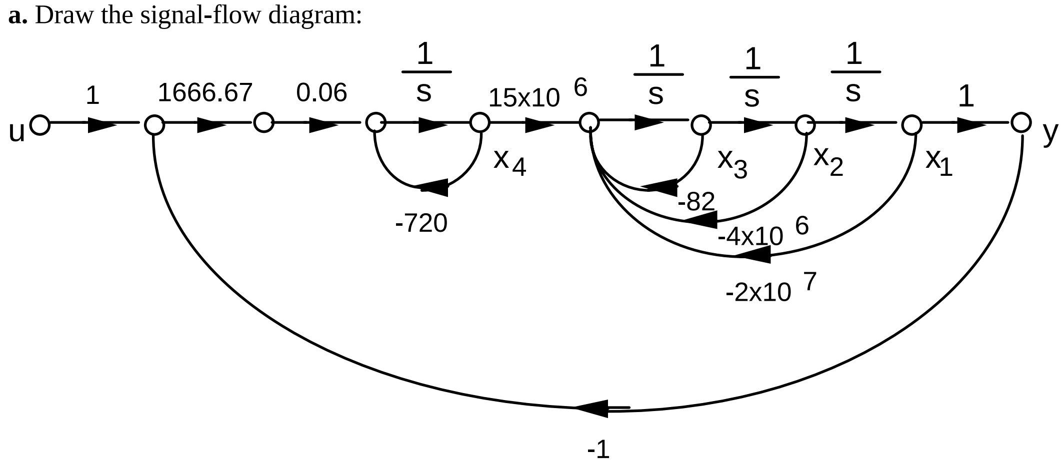
```
for K=1:1:5
numt=5*K;
dent=[1 5 11 15 5*K];
T=tf(numt,dent);
hold on;
subplot(2,3,K);
step(T,0:0.01:20)
title(['K=',int2str(K)])
end
```

Computer response:



52.

a. Draw the signal-flow diagram:



Write state and output equations from the signal-flow diagram:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -2*10^7 x_1 - 4*10^6 x_2 - 82x_3 + 15*10^6 x_4$$

$$\dot{x}_4 = -100x_1 - 720x_4 + 100u$$

$$y = x_1$$

In vector-matrix form:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -2*10^7 & -4*10^6 & -82 & 15*10^6 \\ -100 & 0 & 0 & -720 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 100 \end{bmatrix}u$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}\mathbf{x}$$

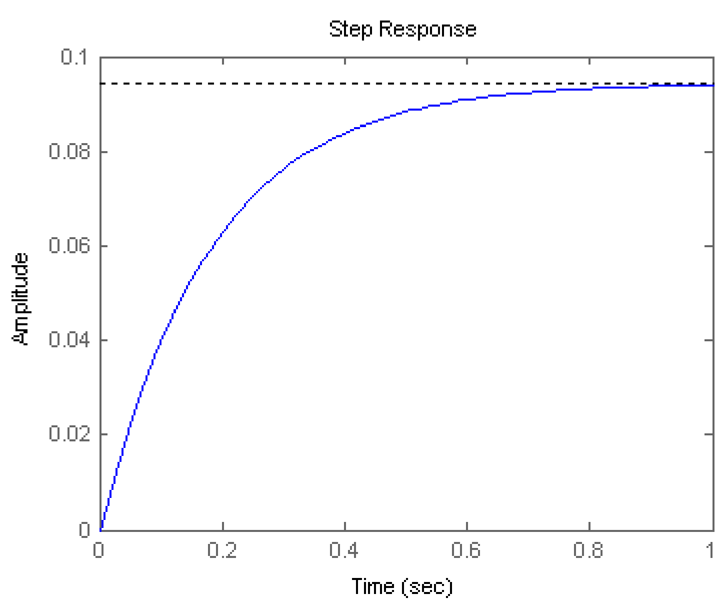
b.

Program:

```
numg=1666.67*0.06*15e6;
deng=conv([1 720],[1 82 4e6 2e7]);
'G(s)'
G=tf(numg,deng)
'T(s)'
T=feedback(G,1)
step(T)
```

Computer response:

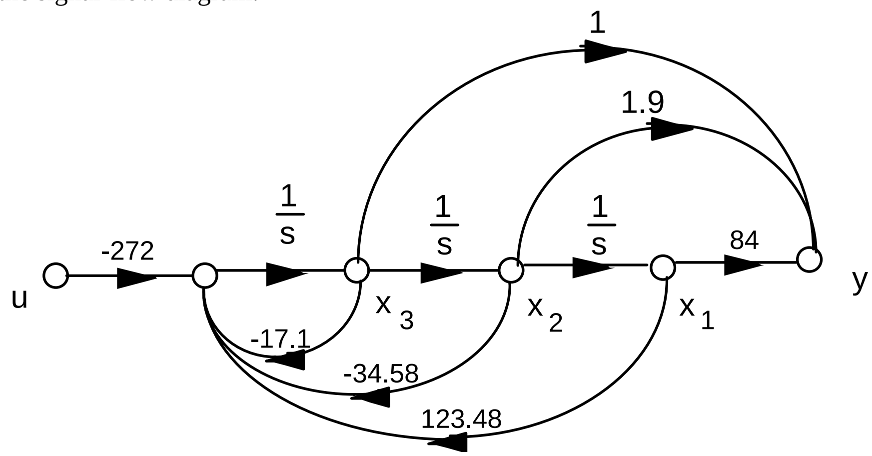
```
ans =

G(s)

Transfer function:
                       1.5e009
----------------------------------------------------
s^4 + 802 s^3 + 4.059e006 s^2 + 2.9e009 s + 1.44e010

ans =

T(s)

Transfer function:
                       1.5e009
----------------------------------------------------
s^4 + 802 s^3 + 4.059e006 s^2 + 2.9e009 s + 1.59e010
```



53.

a. Phase-variable from: $G(s) = \dfrac{-272(s^2+1.9s+84)}{s^3+17.1s^2+34.58s-123.48}$

Drawing the signal-flow diagram:



Writing the state and output equations:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = 123.48x_1 - 34.58x_2 - 17.1x_3 - 272u$$
$$y = 84x_1 + 1.9x_2 + x_3$$

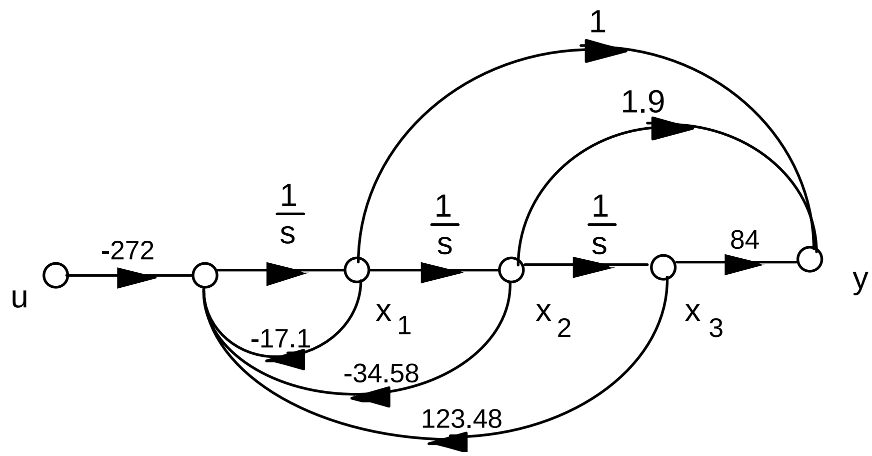
In vector-matrix form:

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 123.48 & -34.58 & -17.1 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ -272 \end{bmatrix}u$$

$$y = \begin{bmatrix} 84 & 1.9 & 1 \end{bmatrix}\mathbf{x}$$

b. Controller canonical form: $G(s) = \dfrac{-272(s^2+1.9s+84)}{s^3+17.1s^2+34.58s-123.48}$

Drawing the signal-flow diagram:



Writing the state and output equations:

$$\dot{x}_1 = -17.1x_1 - 34.58x_2 + 123.48x_3 - 272u$$

$$\dot{x}_2 = x_1$$

$$\dot{x}_3 = x_2$$

$$y = x_1 + 1.9x_2 + 84x_3$$

In vector-matrix form:

$$\dot{\mathbf{x}} = \begin{bmatrix} -17.1 & -34.58 & 123.48 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}\mathbf{x} + \begin{bmatrix} -272 \\ 0 \\ 0 \end{bmatrix}u$$

$$y = \begin{bmatrix} 1 & 1.9 & 84 \end{bmatrix}\mathbf{x}$$

c. Observer canonical form: Divide by highest power of s and obtain

$$G(s) = \frac{\frac{-272}{s} - \frac{516.8}{s^2} - \frac{22848}{s^3}}{1 + \frac{17.1}{s} + \frac{34.58}{s^2} - \frac{123.48}{s^3}}$$

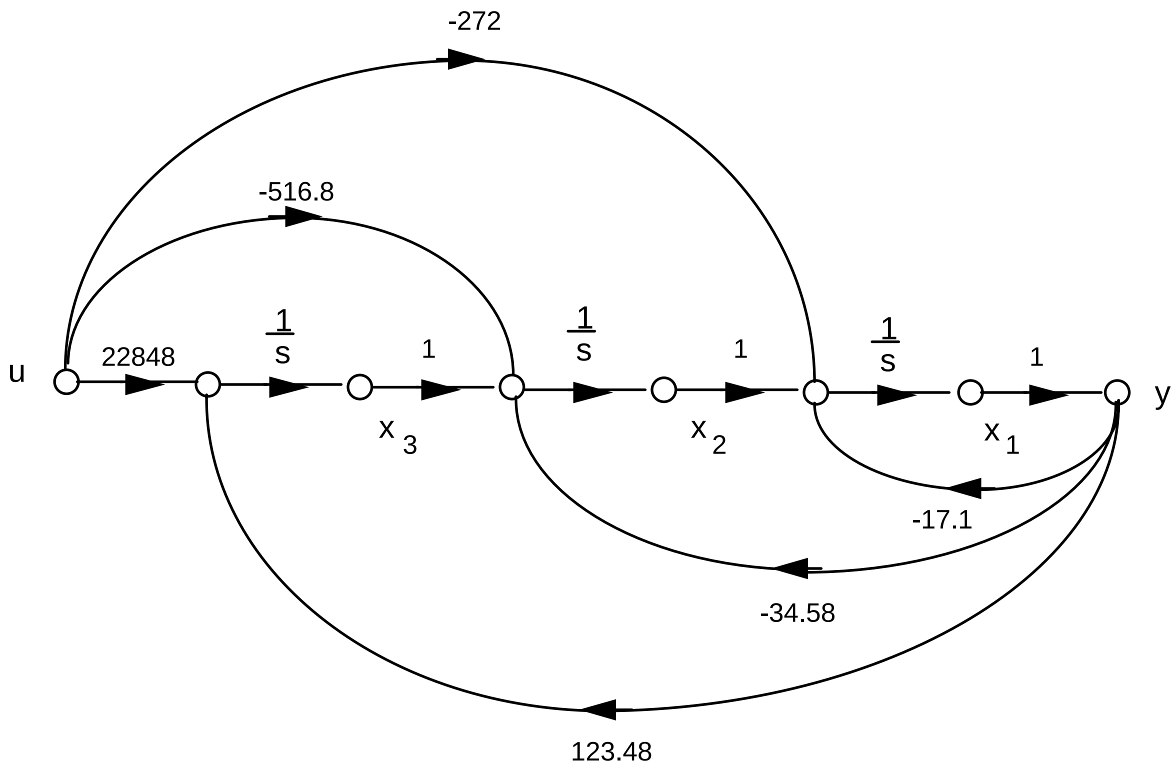
Cross multiplying,

$$[\frac{-272}{s} - \frac{516.8}{s^2} - \frac{22848}{s^3}]R(s) = [1 + \frac{17.1}{s} + \frac{34.58}{s^2} - \frac{123.48}{s^3}]C(s)$$

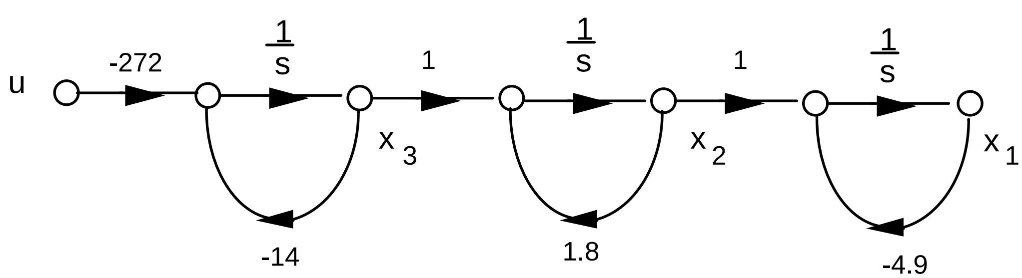
Rearranging,

$$C(s) = \frac{1}{s}[-272R(s) - 17.1C(s)] + \frac{1}{s^2}[-516.8R(s) - 34.58C(s)] + \frac{1}{s^3}[-22848R(s) + 123.48C(s)]$$

Drawing the signal-flow diagram, where r = u and y = c:



d. Draw signal-flow ignoring the polynomial in the numerator:



Write the state equations:

$$\dot{x}_1 = -4.9x_1 + x_2$$

$$\dot{x}_2 = 1.8x_2 + x_3$$

$$\dot{x}_3 = -14x_3 - 272u$$

The output equation is

$$y = \ddot{x}_1 + 1.9\dot{x}_1 + 84x_1 \tag{1}$$

But,

$$\dot{x}_1 = -4.9x_1 + x_2 \tag{2}$$

and

$$\ddot{x}_1 = -4.9\dot{x}_1 + \dot{x}_2 = -4.9(-4.9x_1 + x_2) + 1.8x_2 + x_3 \tag{3}$$

Substituting Eqs. (2) and (3) into (1) yields,

$$y = 98.7x_1 - 1.2x_2 + x_3$$

In vector-matrix form:

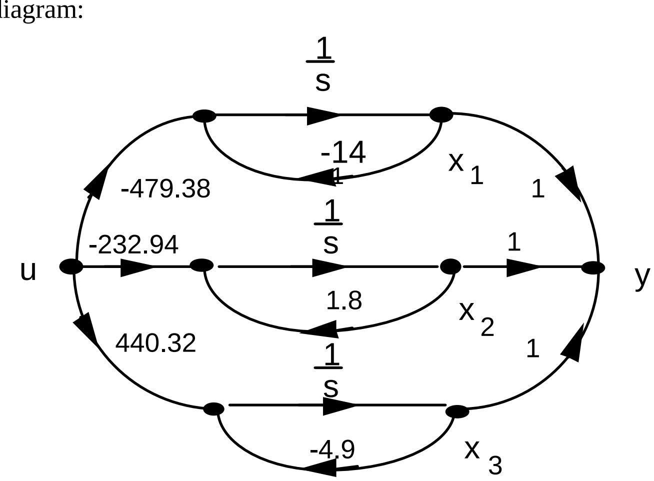
$$\dot{\mathbf{x}} = \begin{bmatrix} -4.9 & 1 & 0 \\ 0 & 1.8 & 1 \\ 0 & 0 & -14 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ -272 \end{bmatrix}u$$

$$y = \begin{bmatrix} 98.7 & -1.2 & 1 \end{bmatrix}\mathbf{x}$$

e. Expand as partial fractions:

$$G(s) = -479.38\frac{1}{s+14} - 232.94\frac{1}{s-1.8} + 440.32\frac{1}{s+4.9}$$

Draw signal-flow diagram:



Write state and output equations:

$$\dot{x}_1 = -14x_1 + -479.38u$$

$$\dot{x}_2 = 1.8x_2 - 232.94u$$

$$\dot{x}_3 = -4.9x_3 + 440.32u$$

$$y = x_1 + x_2 + x_3$$

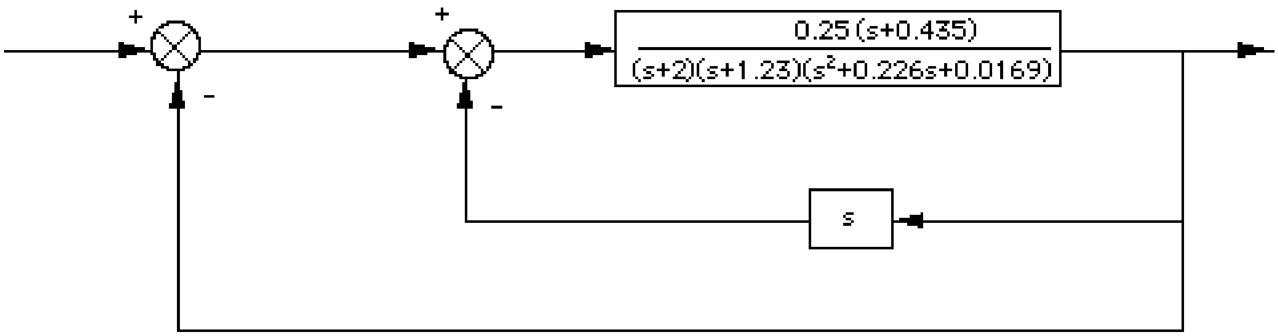
In vector-matrix form:

$$\dot{\mathbf{x}} = \begin{bmatrix} -14 & 0 & 0 \\ 0 & 1.8 & 0 \\ 0 & 0 & -4.9 \end{bmatrix}\mathbf{x} + \begin{bmatrix} -479.38 \\ -232.94 \\ 440.32 \end{bmatrix}u$$

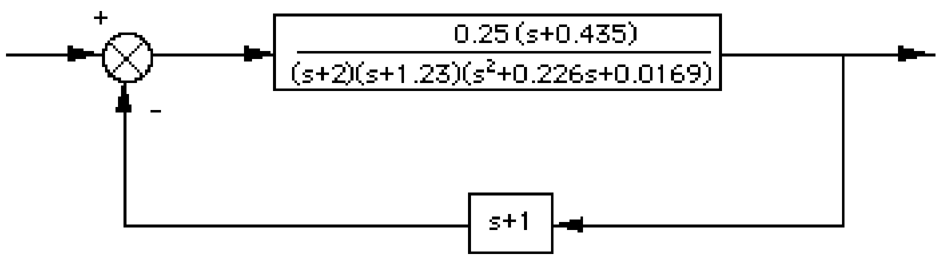
$$y = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}\mathbf{x}$$

54.

Push Pitch Gain to the right past the pickoff point.



Collapse the summing junctions and add the feedback transfer functions.



Apply the feedback formula and obtain,

$$T(s) = \frac{G(s)}{1+G(s)H(s)} = \frac{0.25(s+0.435)}{s^4+3.4586s^3+3.4569s^2+0.9693s+0.15032}$$

55.

Program:

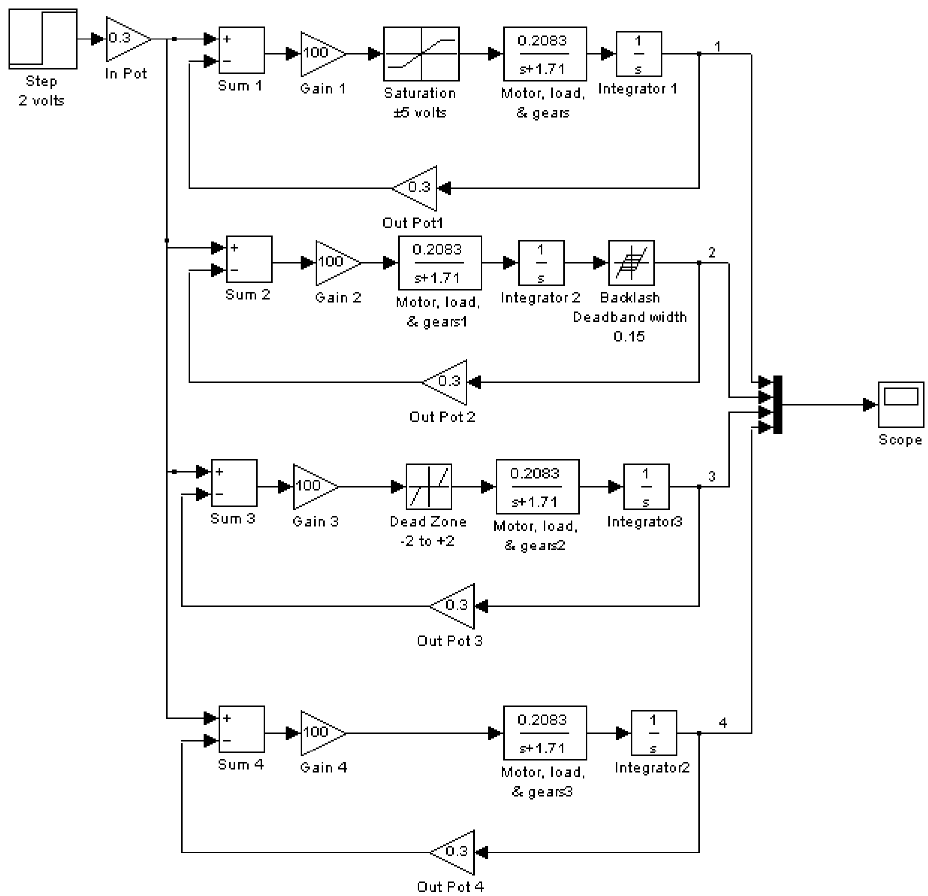
```
numg1=-0.125*[1 0.435]
deng1=conv([1 1.23],[1 0.226 0.0169])
'G1'
G1=tf(numg1,deng1)
'G2'
G2=tf(2,[1 2])
G3=-1
'H1'
H1=tf([-1 0],1)
'Inner Loop'
Ge=feedback(G1*G2,H1)
'Closed-Loop'
T=feedback(G3*Ge,1)
```

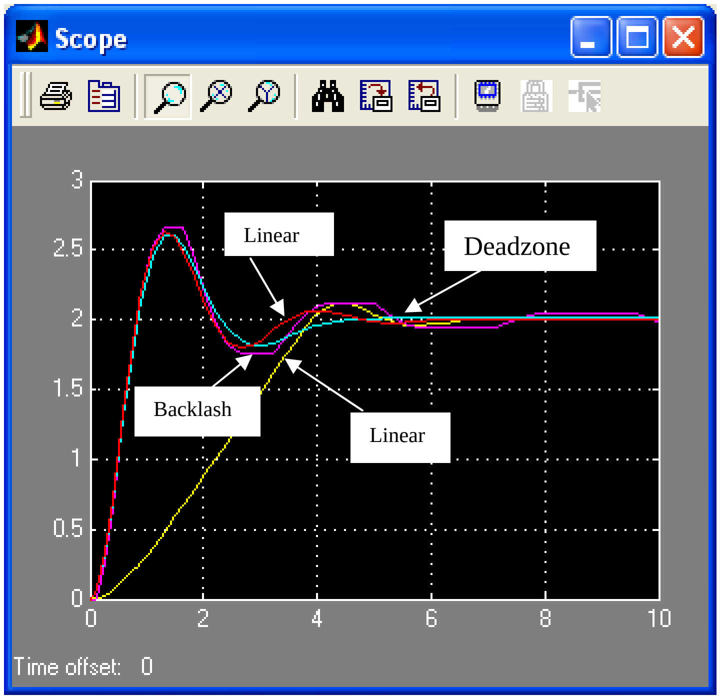
Computer response:

```
numg1 =

   -0.1250   -0.0544
```

```
deng1 =

    1.0000    1.4560    0.2949    0.0208


ans =

G1


Transfer function:
         -0.125 s - 0.05438
------------------------------------
s^3 + 1.456 s^2 + 0.2949 s + 0.02079


ans =

G2


Transfer function:
  2
-----
s + 2


G3 =

    -1


ans =

H1


Transfer function:
-s


ans =

Inner Loop


Transfer function:
                -0.25 s - 0.1088
------------------------------------------------
s^4 + 3.456 s^3 + 3.457 s^2 + 0.7193 s + 0.04157


ans =

Closed-Loop


Transfer function:
                0.25 s + 0.1088
-----------------------------------------------
s^4 + 3.456 s^3 + 3.457 s^2 + 0.9693 s + 0.1503
```

56.


Step
2 volts
0.3
In Pot
Sum 1
100
Gain 1
Saturation
±5 volts
0.2083
s+1.71
Motor, load, & gears
1
s
Integrator 1
1
0.3
Out Pot1
Sum 2
100
Gain 2
0.2083
s+1.71
Motor, load, & gears1
1
s
Integrator 2
Backlash
Deadband width 0.15
2
0.3
Out Pot 2
Scope
Sum 3
100
Gain 3
Dead Zone
-2 to +2
0.2083
s+1.71
Motor, load, & gears2
1
s
Integrator3
3
0.3
Out Pot 3
Sum 4
100
Gain 4
0.2083
s+1.71
Motor, load, & gears3
1
s
Integrator2
4
0.3
Out Pot 4



57.

a. Since $V_L(s) = V_g(s) - V_R(s)$, the summing junction has $V_g(s)$ as the positive input and $V_R(s)$ as the negative input, and $V_L(s)$ as the error. Since $I(s) = V_L(s)\,(1/(Ls))$, $G(s) = 1/(Ls)$. Also, since $V_R(s) = I(s)R$, the feedback is $H(s) = R$. Summarizing, the circuit can be modeled as a negative feedback system, where $G(s) = 1/(Ls)$, $H(s) = R$, input $= V_g(s)$, output $= I(s)$, and error $= V_L(s)$, where the negative input to the summing junction is $V_R(s)$.

b. $T(s) = \dfrac{I(s)}{V_g(s)} = \dfrac{G(s)}{1+G(s)H(s)} = \dfrac{\frac{1}{Ls}}{1+\frac{1}{Ls}R} = \dfrac{1}{Ls+R}$. Hence, $I(s) = V_g(s)\dfrac{1}{Ls+R}$.

c. Using circuit analysis, $I(s) = \dfrac{V_g(s)}{Ls+R}$.

58.

a.

$T_1 = \dfrac{\alpha\beta}{s^2}$, $L_1 = -\dfrac{\alpha}{s}$, $L_2 = -\dfrac{\beta}{s}$, No non-touching loops.

$\Delta = 1 + \dfrac{\alpha}{s} + \dfrac{\beta}{s}$, $\Delta_1 = 1$

$$\frac{H_m}{H_r}(s) = \frac{T_1\Delta_1}{\Delta} = \frac{\frac{\alpha\beta}{s^2}}{1+\frac{\alpha}{s}+\frac{\beta}{s}} = \frac{\alpha\beta}{s[s+(\alpha+\beta)]}$$

b.

$T_1 = \frac{\alpha}{s}$, $L_1 = -\frac{\alpha}{s}$, $L_2 = -\frac{\beta}{s}$, No non-touching loops.

$\Delta = 1+\frac{\alpha}{s}+\frac{\beta}{s}$, $\Delta_1 = 1$

$$\frac{H_f}{H_r}(s) = \frac{T_1\Delta_1}{\Delta} = \frac{\frac{\alpha}{s}}{1+\frac{\alpha}{s}+\frac{\beta}{s}} = \frac{\alpha}{s+(\alpha+\beta)}$$

c.

$T_1 = \alpha$, $L_1 = -\frac{\alpha}{s}$, $L_2 = -\frac{\beta}{s}$, No non-touching loops.

$\Delta = 1+\frac{\alpha}{s}+\frac{\beta}{s}$, $\Delta_1 = 1$

$$\frac{Q_i}{H_r}(s) = \frac{T_1\Delta_1}{\Delta} = \frac{\alpha}{1+\frac{\alpha}{s}+\frac{\beta}{s}} = \frac{\alpha s}{s+(\alpha+\beta)}$$

d.

$T_1 = \frac{\alpha\beta}{s}$, $L_1 = -\frac{\alpha}{s}$, $L_2 = -\frac{\beta}{s}$, No non-touching loops.

$\Delta = 1+\frac{\alpha}{s}+\frac{\beta}{s}$, $\Delta_1 = 1$

$$\frac{Q_o}{H_r}(s) = \frac{T_1\Delta_1}{\Delta} = \frac{\frac{\alpha\beta}{s}}{1+\frac{\alpha}{s}+\frac{\beta}{s}} = \frac{\alpha\beta}{s+(\alpha+\beta)}$$

e.

Let $H_r(s) = \frac{K}{s}$. From part d) $q_o(\infty) = \lim_{s\to 0} sQ_o(s) = \lim_{s\to 0} s\frac{\alpha\beta}{s+(\alpha+\beta)}H_r(s) = \frac{\alpha\beta}{\alpha+\beta}K =$ constant

$$h_m(t) = \mathcal{L}^{-1}\{H_m(s)\} = \mathcal{L}^{-1}\left\{\frac{\alpha\beta}{s[s+(\alpha+\beta)]}H_r(s)\right\} = \mathcal{L}^{-1}\left\{\frac{\alpha\beta K}{s^2[s+(\alpha+\beta)]}\right\}$$

From (a)

$$= \mathcal{L}^{-1}\left\{\frac{K_1}{s^2}+\frac{K_2}{s}+\frac{K_3}{s+(\alpha+\beta)}\right\} = K_1 t + K_2 + K_3 e^{-(\alpha+\beta)t}$$

So $\lim_{t\to\infty} h_m(t) = K_1 t + K_2$. It is clear that $h_m(t)$ increases at constant speed.

59.

a.

We start by finding $\frac{U}{U_1}$ and $\frac{U}{U_2}$ using Mason's Rule

$\frac{U}{U_1}$

$T_1 = G_c G_p$; Loops: $L_1 = L_2 = A^2 G_l G_v$, no non-touching loops. $\Delta = 1 - 2A^2 G_l G_v$, $\Delta_1 = 1$ Follows

$$\frac{U}{U_1} = \frac{G_C G_p}{1 - 2A^2 G_l G_v}$$

$\frac{U}{U_2}$

$T_1 = -G_c G_p$; Loops: $L_1 = L_2 = A^2 G_l G_v$, no non-touching loops. $\Delta = 1 - 2A^2 G_l G_v$, $\Delta_1 = 1$ Follows

$$\frac{U}{U_2} = \frac{-G_C G_p}{1 - 2A^2 G_l G_v}$$

By superposition $U = \frac{G_C G_p}{1 - 2A^2 G_l G_v}(U_1 - U_2)$

b.

Now we find $\frac{U}{F_{ext}}$

Two forward paths: $T_1 = T_2 = A^2 G_l G_v$ Loops: $L_1 = L_2 = A^2 G_l G_v$, no non-touching loops.

$\Delta = 1 - 2A^2 G_l G_v$. $\Delta_1 = \Delta_2 = 1$

$$\frac{U}{F_{ext}} = \frac{2A^2 G_l G_v}{1 - 2A^2 G_l G_v}$$

c. From (a) to get $u = 0$ it is required that $u_1 = u_2$.

60.

a. The first equation follows from the schematic. The second equation is obtained by applying the voltage divider rule at the op-amp's inverting terminal, noting that since the op-amp considered is ideal, there is no current demand there.

b. $T_1 = A$; $L = -A\frac{R_i}{R_i + R_f}$; $\Delta = 1 + A\frac{R_i}{R_i + R_f}$; $\Delta_1 = 1$

$$\frac{v_o}{v_i}=\frac{T_1\Delta_1}{\Delta}=\frac{A}{1+A\dfrac{R_i}{R_i+R_f}}$$

c. $$\frac{v_o}{v_i}=\lim_{A\to\infty}\frac{A}{1+A\dfrac{R_i}{R_i+R_f}}=\frac{1}{\dfrac{R_i}{R_i+R_f}}=1+\frac{R_f}{R_i}$$

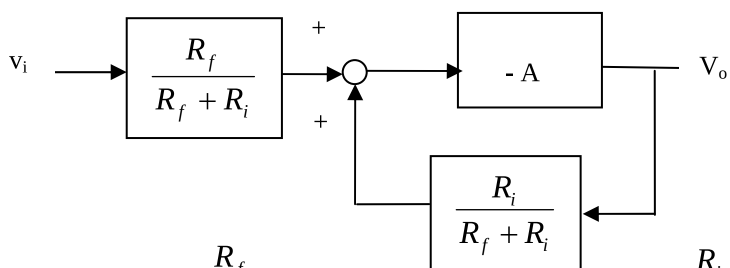
61.

a. Adding currents at the op-amp's inverting terminal, under ideal condition we get

$\dfrac{v_i-v_1}{R_i}=\dfrac{v_1-v_o}{R_f}$ which after some algebraic manipulations gives $v_1=\dfrac{R_f}{R_f+R_i}v_i+\dfrac{R_i}{R_f+R_i}v_o$

Also from the circuits diagram $v_o=-Av_1$

b. These equations can be represented by the following block diagram



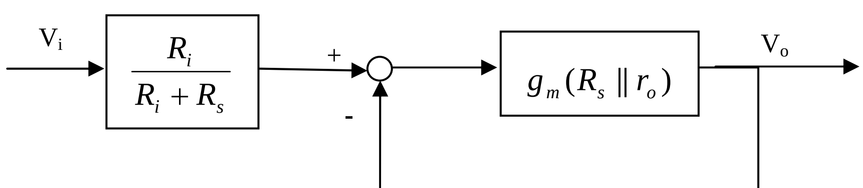
We have that $T_1=-A\dfrac{R_f}{R_f+R_i}$; $L=-A\dfrac{R_i}{R_i+R_f}$; $\Delta=1+A\dfrac{R_i}{R_i+R_f}$; $\Delta_1=1$

$$\frac{v_o}{v_i}=\frac{T_1\Delta_1}{\Delta}=\frac{-A\dfrac{R_f}{R_i+R_f}}{1+A\dfrac{R_i}{R_i+R_f}}$$

c. $$\frac{v_o}{v_i}=\lim_{A\to\infty}\frac{-A\dfrac{R_f}{R_i+R_f}}{1+A\dfrac{R_i}{R_i+R_f}}=\frac{-\dfrac{R_f}{R_i+R_f}}{\dfrac{R_i}{R_i+R_f}}=\frac{R_f}{R_i}$$

62.

a. The three equations follow by direct observation from the small signal circuit.
b. The block diagram is given by



c. From the block diagram we get

$$\frac{v_o}{v_i} = \frac{g_m(R_s \| r_o)}{1+g_m(R_s \| r_o)}\frac{R_i}{R_i+R_s}$$

63.

a. Using Mason's rule

$T_1 = \frac{K_t}{s^2+\omega_0^2}\frac{1}{M_{US}} = \frac{\omega_0^2}{s^2+\omega_0^2}$; Loops $L_1 = +\frac{\omega_0^2}{s^2+\omega_0^2}\frac{\varepsilon}{s+\varepsilon}m_r$ and $L_2 = -\frac{\varepsilon m_r}{s+\varepsilon}$, no non-touching loops. $\Delta_1 = 1$

$$\frac{X_3}{R} = \frac{T_1\Delta_1}{\Delta} = \frac{\frac{\omega_0^2}{s^2+\omega_0^2}}{1+m_r\frac{\varepsilon}{s+\varepsilon} - \frac{\omega_0^2}{s^2+\omega_0^2}\frac{\varepsilon}{s+\varepsilon}m_r} = \frac{\frac{\omega_0^2}{s^2+\omega_0^2}}{1+\frac{\varepsilon m_r s^2}{(s+\varepsilon)(s^2+\omega_0^2)}}$$

a. From part (a)

$$\frac{X_1}{R} = \frac{X_3}{R}\frac{\varepsilon}{s+\varepsilon} = \frac{\frac{\omega_0^2}{s^2+\omega_0^2}\frac{\varepsilon}{s+\varepsilon}}{1+\frac{\varepsilon m_r s^2}{(s+\varepsilon)(s^2+\omega_0^2)}}$$

64.

a.

```
>> A=[-100.2 -20.7 -30.7 200.3; 40 -20.22 49.95 526.1;...
0 10.22 -59.95 -526.1; 0 0 0 0];
>> B=[208; -208; -108.8; -1];
>> C = [0 1570 1570 59400];
>> D = -6240;
>> [n,d]=ss2tf(A,B,C,D)
```

```
n =

  1.0e+009 *

  Columns 1 through 3

  -0.00000624000000  -0.00168228480000  -0.14206098728000

  Columns 4 through 5

  -3.91955218234127  -9.08349454230472


d =

  1.0e+005 *

  Columns 1 through 3

   0.00001000000000   0.00180370000000   0.09562734000000

  Columns 4 through 5

   1.32499100000000                  0
```

```
>> roots(n)

ans =


  1.0e+002 *


  -1.34317654991673
  -0.78476212102923
  -0.54257777928519
  -0.02545278053809


>> roots(d)


ans =


                 0
 -92.38329312886714
 -66.38046756013043
 -21.60623931100260
```

Note that $s\frac{Y}{U}(0) = 68555.14$, follows that

$$\frac{Y}{U}(s) = -\frac{6348.17(s+2.5)(s+54.3)(s+78.5)(s+134.3)}{s(s+21.6)(s+66.4)(s+92.4)}$$

b.

```
>> [r,p,k]=residue(n,d)
```

r =

1.0e+005 *

-0.73309459854184

-0.51344619392820

-3.63566779304453

-0.68555141448543

p =

-92.38329312886714

-66.38046756013043

-21.60623931100260

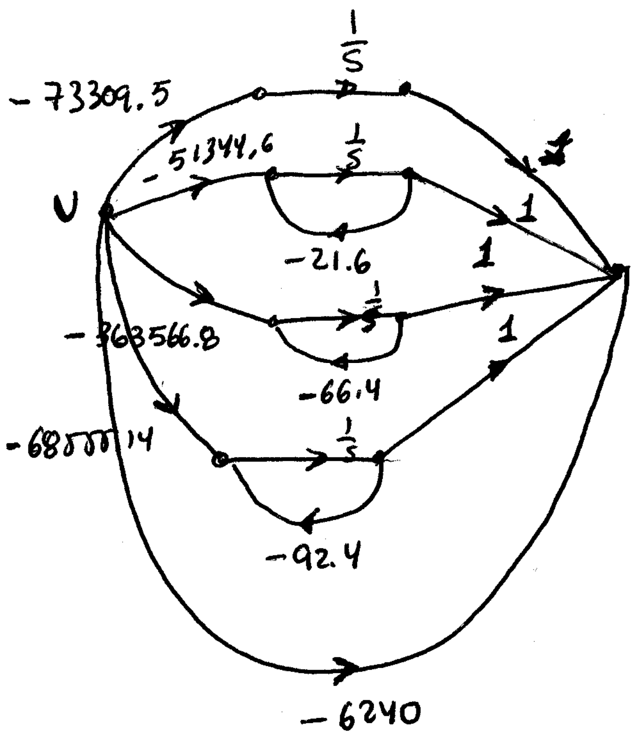
0

k =

-6240

or $\frac{Y}{U}(s) = -6240 - \frac{73309.46}{s} - \frac{51344.6}{s+21.6} - \frac{363566.8}{s+66.4} - \frac{68555.14}{s+92.4}$

c.



d. The corresponding state space representation is:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\\ \dot{x}_4\end{bmatrix}=\begin{bmatrix}0 & 0 & 0 & 0\\ 0 & -21.6 & 0 & 0\\ 0 & 0 & -66.4 & 0\\ 0 & 0 & 0 & -9.4\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\\ x_4\end{bmatrix}+\begin{bmatrix}-73309.5\\ -51344.6\\ -363566.8\\ -68555.14\end{bmatrix}u(t)$$

$$y=\begin{bmatrix}1 & 1 & 1 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\\ x_4\end{bmatrix}-6240u(t)$$

65.

a.

```
>> A = [0 1 0; 0 -68.3 -7.2; 0 3.2 -0.7]
```

```
A =

     0    1.0000         0
     0  -68.3000   -7.2000
     0    3.2000   -0.7000

>> [V,D]=eig(A)

V =

  1.0000    0.0147   -0.1016
     0   -0.9988    0.1059
     0    0.0475   -0.9892

D =

     0         0         0
     0  -67.9574         0
     0         0   -1.0426
```

Matrix V is the sought similarity transformation.

b.

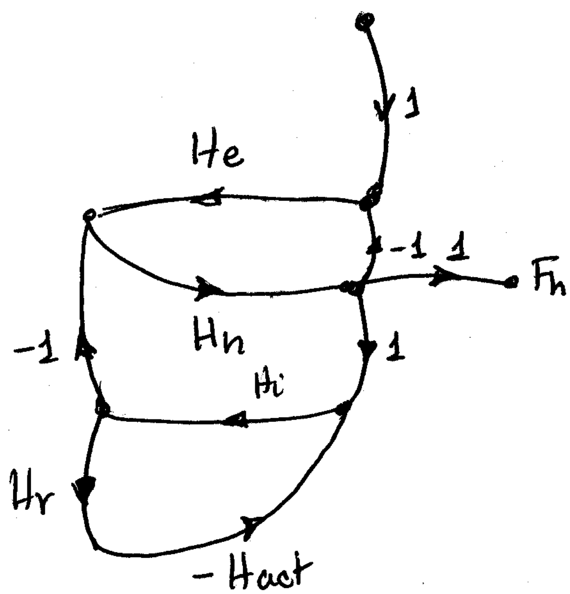
```
>> Ad = inv(V)*A*V

Ad =

         0   -0.0000   -0.0000
         0  -67.9574    0.0000
         0   -0.0000   -1.0426

>> B = [0;425.4;0]

B =

         0
  425.4000
         0

>> Bd = inv(V)*B

Bd =

    4.2030
 -428.1077
  -20.5661
```

The diagonalized system is:

$$\begin{bmatrix}\dot z_1\\ \dot z_2\\ \dot z_3\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\\ 0 & -67.96 & 0\\ 0 & 0 & -1.0426\end{bmatrix}\begin{bmatrix}z_1\\ z_2\\ z_3\end{bmatrix}+\begin{bmatrix}4.2\\ -428.11\\ -20.57\end{bmatrix}e_m$$

66.

a.



b. There is only one forward path $T_1 = H_e H_h$

There are three loops: $L_1 = -H_e H_h$; $L_2 = -H_i H_h$ and $L_3 = -H_i H_r H_{act}$

L_1 and L_3 are non-touching loops so

$\Delta = 1 - L_1 - L_2 - L_3 + L_1 L_3 = 1 + H_e H_h + H_i H_h + H_i H_r H_{act} + H_e H_h H_i H_r H_{act}$

When T_1 is eliminated only L_3 is left so $\Delta_1 = 1 - L_3 = 1 + H_r H_i H_{act}$

Finally $\dfrac{F_h}{D}(s) = \dfrac{T_1\Delta_1}{\Delta} = \dfrac{H_e H_h\left(1 + H_r H_i H_{act}\right)}{1 + H_e H_h + H_i H_h + H_i H_r H_{act} + H_e H_h H_i H_r H_{act}}$

67.

Enter Numerators and Denominator of System's Transfer Function--recall reverse order!

Numerator

0	2	7	1

Denominator

0	24	26	9	1

Transfer Function

$$\frac{s^2 + 7s + 2}{s^3 + 9s^2 + 26s + 24}$$

CONTROLLER and OBSERVER CANONICAL FORM SYSTEM PAGES 249-250

CONTROLLER CANONICAL FORM

$$\mathbf{dx/dt} = \begin{bmatrix} -9 & -26 & -24 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \mathbf{x(t)} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \mathbf{u(t)}$$

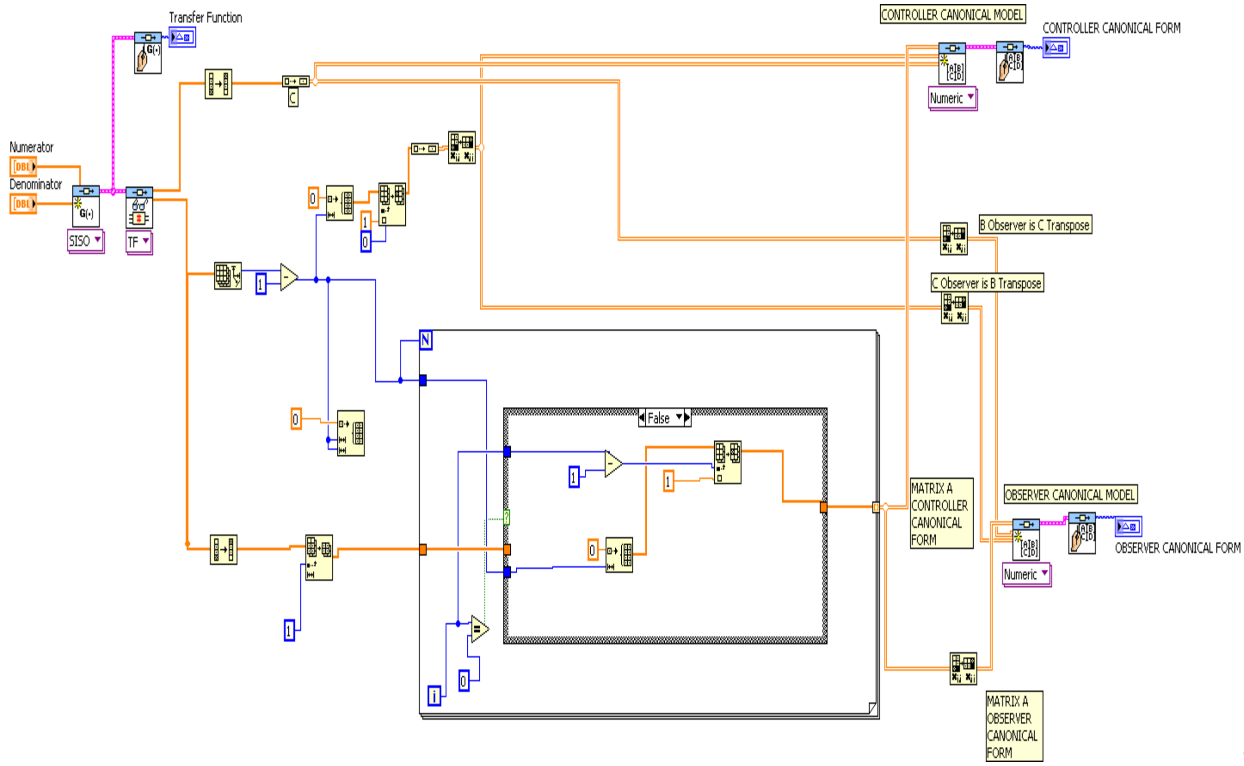
$$\mathbf{y(t)} = \begin{bmatrix} 1 & 7 & 2 \end{bmatrix} \mathbf{x(t)} + \begin{bmatrix} 0 \end{bmatrix} \mathbf{u(t)}$$

OBSERVER CANONICAL FORM

$$\mathbf{dx/dt} = \begin{bmatrix} -9 & 1 & 0 \\ -26 & 0 & 1 \\ -24 & 0 & 0 \end{bmatrix} \mathbf{x(t)} + \begin{bmatrix} 1 \\ 7 \\ 2 \end{bmatrix} \mathbf{u(t)}$$

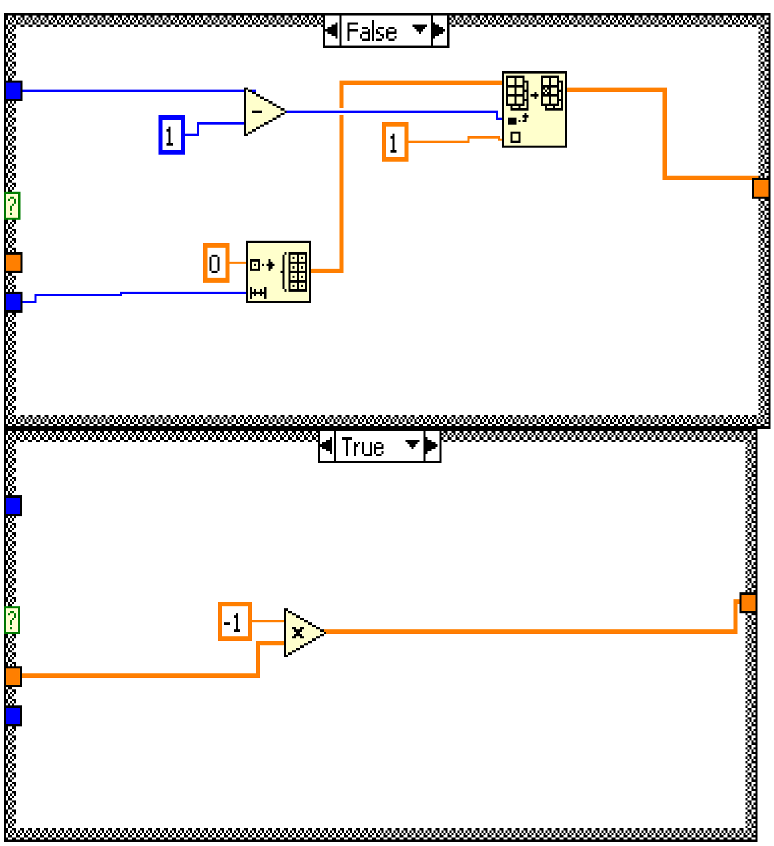
$$\mathbf{y(t)} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x(t)} + \begin{bmatrix} 0 \end{bmatrix} \mathbf{u(t)}$$

Block Diagram:



Case

Structure Details:



68.

a. There are two forwards paths:

$$M_1 = K_h K_{hs} C_s \frac{1}{s^2} \text{ and } M_2 = -K_h T_{es} K_{es} C_s \frac{1}{s}$$

The loops are:

$$L_1 = -K_h K_{hs} C_s \frac{1}{s^2} T_{hs}$$

$$L_2 = +K_h T_{es} K_{es} C_s T_{hs} \frac{1}{s}$$

$$L_3 = -C_j T_{ej} K_{ej}$$

$$L_4 = -K_h C_j T_{hj}$$

There are no non-touching loops. Therefore

$$\Delta = 1 + K_h K_{hs} C_s \frac{1}{s^2} T_{hs} - K_h T_{es} K_{es} C_s T_{hs} \frac{1}{s} + C_j T_{ej} K_{ej} + K_h C_j T_{hj}$$

Also $\Delta_1 = \Delta_2 = 1$

$$\frac{Y_s}{U_h} = \frac{M_1\Delta_1 + M_2\Delta_2}{\Delta} = \frac{K_h K_{hs} C_s \frac{1}{s^2} - K_h T_{es} K_{es} C_s \frac{1}{s}}{1 + K_h K_{hs} C_s \frac{1}{s^2} T_{hs} - K_h T_{es} K_{es} C_s T_{hs} \frac{1}{s} + C_j T_{ej} K_{ej} + K_h C_j T_{hj}}$$

b.

There is only one forward path $M_1 = K_h C_j$

The loops and Δ are the same as in part a. Also $\Delta_1 = 1$. It follows that

$$\frac{Y_j}{U_h} = \frac{M_1\Delta_1}{\Delta} = \frac{K_h C_j}{1 + K_h K_{hs} C_s \frac{1}{s^2} T_{hs} - K_h T_{es} K_{es} C_s T_{hs} \frac{1}{s} + C_j T_{ej} K_{ej} + K_h C_j T_{hj}}$$

69.

a. Assuming $Z_h = 0$ there are two forward paths, $M_1 = Z_m^{-1}$ and $M_2 = C_6 Z_m^{-1}$

The loops are

$$L_1 = -G_s C_s$$

$$L_2 = -Z_m^{-1} C_m$$

$$L_3 = -Z_m^{-1} C_1 G_s C_4$$

$$L_4 = -Z_m^{-1} C_s G_s Z_e C_2$$

There are two non-touching loops L_1 and L_2.

$$\Delta = 1 + G_s C_s + Z_m^{-1} C_m + Z_m^{-1} C_1 G_s C_4 + Z_m^{-1} C_s G_s Z_e C_2 - G_s C_s Z_m^{-1} C_m$$

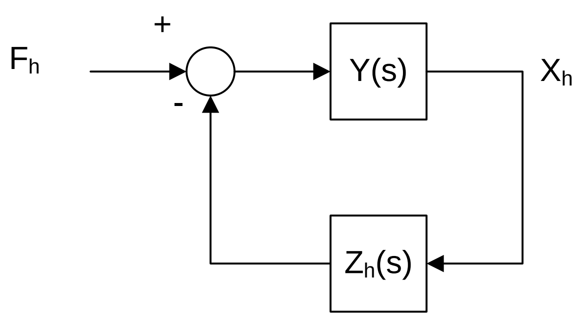
$$= 1 + G_s C_s + Z_m^{-1} C_m + Z_m^{-1} C_s G_s Z_e C_2 = 1 + G_s C_s + Z_m^{-1}(C_m + C_s G_s Z_e C_2)$$

We also have that by eliminating M_1 or M_2

$$\Delta_1 = \Delta_2 = 1 + G_s C_s$$

$$Y(s) = \frac{X_h}{F_h} = \frac{M_1\Delta_1 + M_2\Delta_2}{\Delta} = \frac{(Z_m^{-1} + Z_m^{-1}C_6)(1+G_s C_s)}{1 + G_s C_s + Z_m^{-1}(C_m + C_2 Z_e G_s C_s)} = \frac{Z_m^{-1} C_2 (1 + G_s C_s)}{1 + G_s C_s + Z_m^{-1}(C_m + C_2 Z_e G_s C_s)}$$

b. The system can be described by means of the following diagram:



It follows that

$$\frac{X_h}{F_h} = \frac{Y(s)}{1 + Y(s) Z_h}$$

70.

a. There are three forward paths:

$$M_1 = K_{p2}\left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f C_f \hat{C}_f s^3}$$

$$M_2 = \left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f C_f s^2}$$

$$M_3 = \left(\hat{L}_f s + \hat{R}_f\right)\frac{1}{L_f C_f s^2}$$

The loops are:

$$L_1 = -K_{p2}\left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f C_f s^2}$$

$$L_2 = -\left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f s}$$

$$L_3 = -\frac{R_f}{L_f s}$$

There are no non-touching loops.

$$\Delta = 1 + K_{p2}\left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f C_f s^2} + \left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f s} + \frac{R_f}{L_f s}$$

and $\Delta_1 = \Delta_2 = \Delta_3 = 1$

$$\frac{V_{Load}}{I_{Cf}} = \frac{M_1\Delta_1 + M_2\Delta_2 + M_3\Delta_3}{\Delta} = \frac{K_{p2}\left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f C_f \hat{C}_f s^3} + \left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f C_f s^2} + \left(\hat{L}_f s + \hat{R}_f\right)\frac{1}{L_f C_f s^2}}{1 + K_{p2}\left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f C_f s^2} + (K_{p1} + \frac{K_{i1}}{s})\frac{1}{L_f s} + \frac{R_f}{L_f s}}$$

b. There are three forward paths:

$$M_1 = K_{p2}\left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f C_f \hat{C}_f s^3}$$

$$M_2 = \left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f C_f s^2}$$

$$M_3 = \left(\hat{L}_f s + \hat{R}_f\right)\frac{1}{L_f C_f s^2}$$

The loops are:

$$L_1 = -K_{p2}\left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f C_f s^2}$$

$$L_2 = -\left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f s}$$

$$L_3 = -\frac{R_f}{L_f s}$$

There are no non-touching loops.

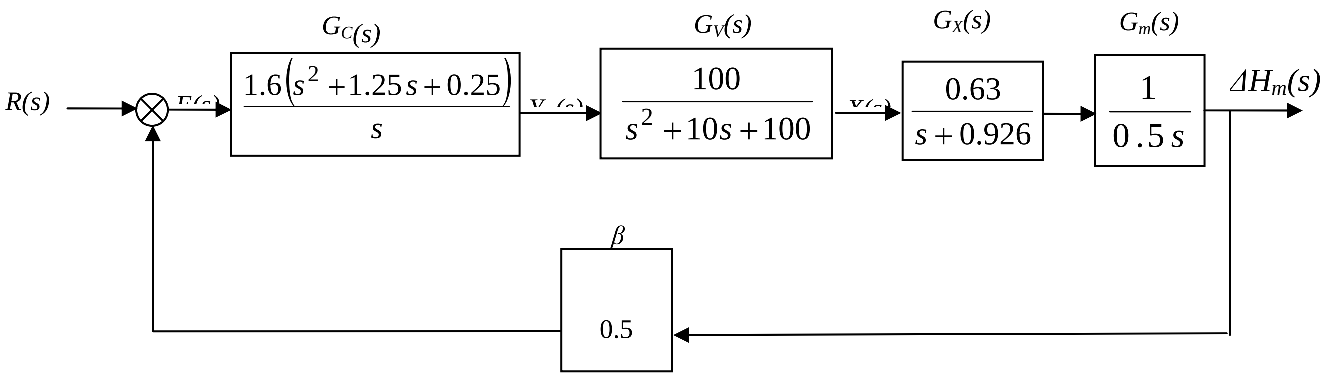
$$\Delta = 1 + K_{p2}\left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f C_f s^2} + \left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f s} + \frac{R_f}{L_f s}$$

and $\Delta_1 = \Delta_2 = \Delta_3 = 1$

$$\frac{V_{Load}}{I_{Cf}} = \frac{M_1\Delta_1 + M_2\Delta_2 + M_3\Delta_3}{\Delta} = \frac{K_{p2}\left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f C_f \hat{C}_f s^3} + \left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f C_f s^2} + \left(\hat{L}_f s + \hat{R}_f\right)\frac{1}{L_f C_f s^2}}{1 + K_{p2}\left(K_{p1} + \frac{K_{i1}}{s}\right)\frac{1}{L_f C_f s^2} + (K_{p1} + \frac{K_{i1}}{s})\frac{1}{L_f s} + \frac{R_f}{L_f s}}$$

71.

a. Substituting the values given above into the block diagram, when $\Delta v_p = 0$, we have:



The Mould Level Block Diagram for $\Delta v_p = 0$

Thus, the closed-loop transfer function is:

$$T(s) = \frac{\Delta H_m(s)}{R(s)} = \frac{G_C(s)\cdot G_V(s)\cdot G_X(s)\cdot G_m(s)}{1+\beta\cdot G_C(s)\cdot G_V(s)\cdot G_X(s)\cdot G_m(s)} =$$

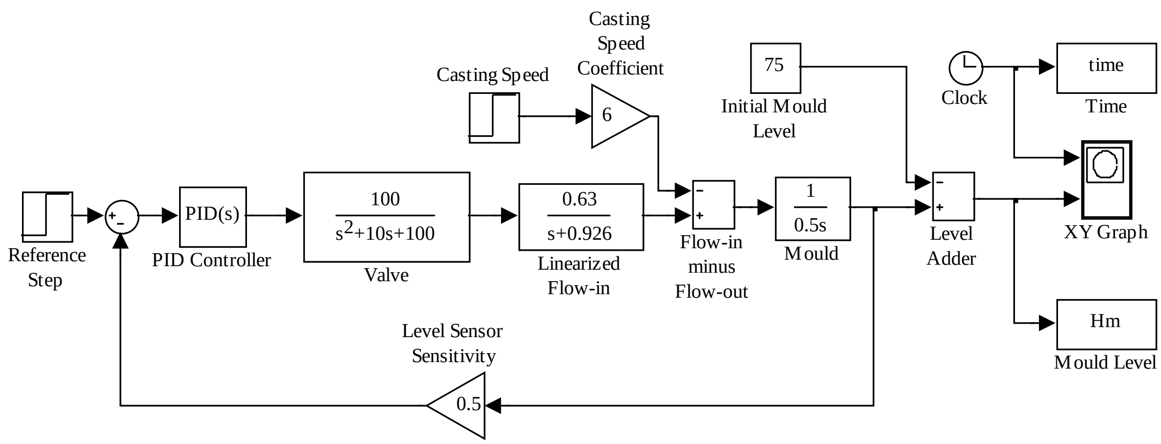
$$= \frac{\frac{1.6\left(s^2+1.25s+0.25\right)}{s}\left(\frac{100}{s^2+10s+100}\right)\left(\frac{0.63}{s+0.926}\right)\frac{1}{0.5s}}{1+0.5\times\frac{1.6\left(s^2+1.25s+0.25\right)}{s}\left(\frac{100}{s^2+10s+100}\right)\left(\frac{0.63}{s+0.926}\right)\frac{1}{0.5s}} =$$

$$= \frac{201.6\left(s^2+1.25s+0.25\right)}{s^2\left(s^2+10s+100\right)\left(s+0.926\right)+0.5\times 201.6\left(s^2+1.25s+0.25\right)} =$$

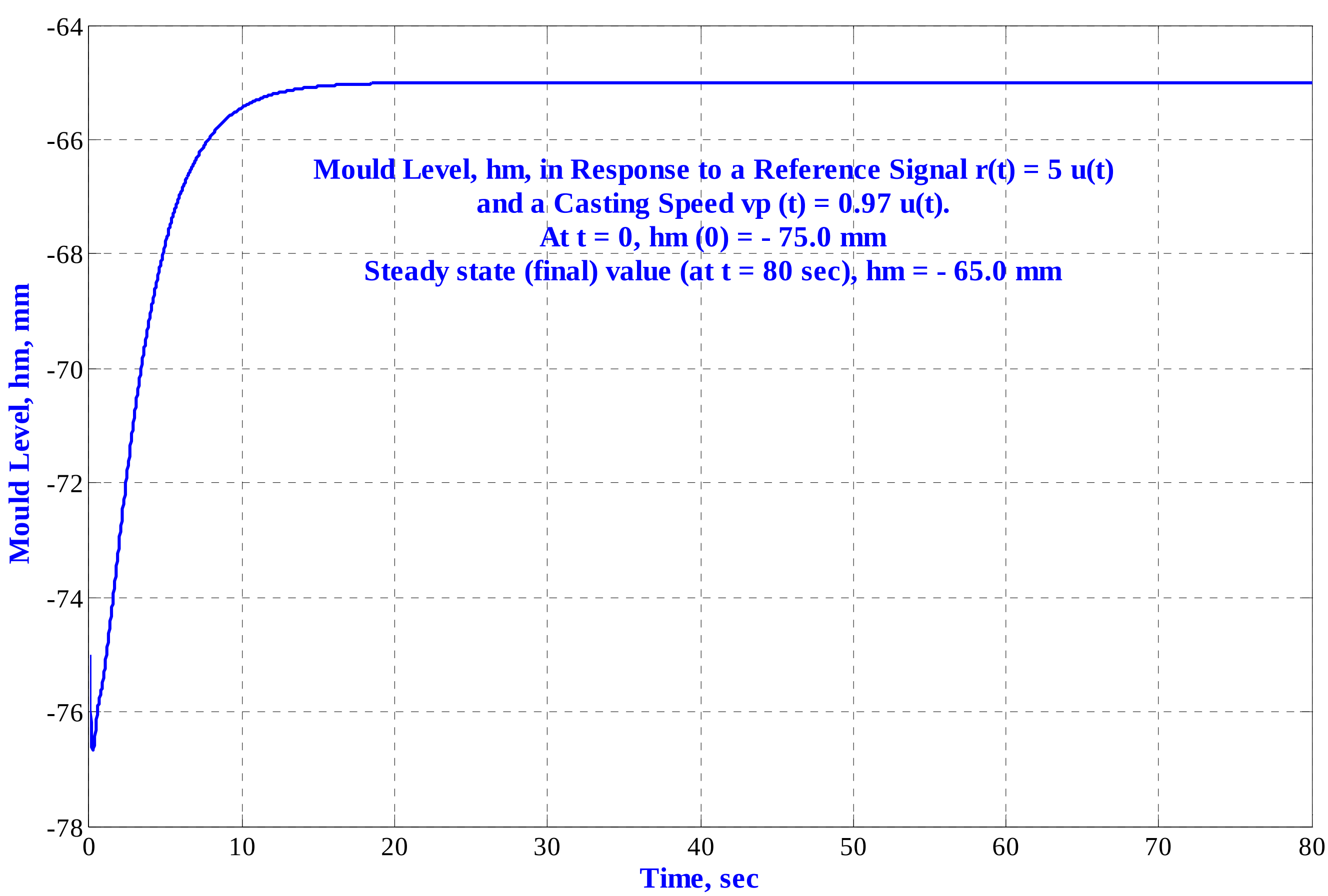
$$= \frac{201.6\left(s^2+1.25s+0.25\right)}{s^5+10.93s^4+109.26s^3+193.4s^2+126s+25.2}$$

b. Simulink was used to simulate the system. The model of that system is shown in Figure P5.x-4*. The parameters of the PID controller were set to: $K_p = 2$, $K_d = 1.6$, and $K_I = 0.4$. The reference step, $r(t) = 5$ u(t), and the casting speed step, v_p (t) = 0.97 u(t) were set to start at t = 0. An adder was used to add the initial value,
$H_m(0^-) = -75$ mm, at the output, to the change in mould level, ΔH_m.

The time and mould level (in array format) were output to "workspace " sinks, each of which was given the respective variable name. After the simulation ended, Matlab plot commands were used to obtain and edit the graph of $h_m(t)$ from 0 to t = 80 seconds.



Simulink Model of the Mould Level Control System



Response of the Mould Level to Simultaneous Step Changes in Reference Input, *r(t)* = 5 u(t), and Casting Speed, v_p (t) = 0.97 u(t) at an Initial Level, H_m (0^-) = – 75

72.

a. Following the procedure described in Chapter 3 we define $\frac{X_1}{\delta} = \frac{aV}{bh}\frac{1}{s^2 - \frac{g}{h}}$ and

$\varphi = (s + \frac{V}{A})X_1(s)$. In time domain $\ddot{x}_1 - \frac{g}{h}x_1 = \frac{aV}{bh}\delta$. $\varphi = \dot{x}_1 + \frac{V}{a}x_1$ and we also define $\dot{x}_1 = x_2$. These equations give

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{g}{h} & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{aV}{bh} \end{bmatrix}\delta$$

$$\varphi = \begin{bmatrix} \frac{V}{a} & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

b. The eigenvalues can be obtained directly from the transfer function poles. Thus $\lambda_{12} = \pm\sqrt{\frac{g}{h}}$

Consider $\lambda_1 = \sqrt{\frac{g}{h}}$, the first eigenvector is found from the solution of $\mathbf{Ax_1} = \lambda_1\mathbf{x_1}$ or

$\begin{bmatrix} 0 & 1 \\ \frac{g}{h} & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \sqrt{\frac{g}{h}}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. This results in $x_2 = \sqrt{\frac{g}{h}x_1}$. Arbitrarily let $x_1 = 1$ so the first eigenvector is $\mathbf{v_1} = \begin{bmatrix} 1 \\ \sqrt{\frac{g}{h}} \end{bmatrix}$.

Similarly for $\lambda_2 = -\sqrt{\frac{g}{h}}$; $\mathbf{Ax}_2 = \lambda_1\mathbf{x}_2$ or $\begin{bmatrix} 0 & 1 \\ \frac{g}{h} & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -\sqrt{\frac{g}{h}}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ resulting in

$x_2 = -\sqrt{\frac{g}{h}x_1}$. Letting arbitrarily $x_1 = 1$ the second eigenvector is $\mathbf{v}_2 = \begin{bmatrix} 1 \\ \sqrt{\frac{g}{h}} \end{bmatrix}$.

c. The similarity transformation matrix is $\mathbf{P} = [\mathbf{v_1} \quad \mathbf{v_2}] = \begin{bmatrix} 1 & 1 \\ \sqrt{\frac{g}{h}} & -\sqrt{\frac{g}{h}} \end{bmatrix}$

$$\mathbf{P}^{-1}=\frac{\begin{bmatrix}-\sqrt{\frac{g}{h}} & -1\\ -\sqrt{\frac{g}{h}} & 1\end{bmatrix}}{-2\sqrt{\frac{g}{h}}}=\begin{bmatrix}\frac{1}{2} & \frac{1}{2}\sqrt{\frac{h}{g}}\\ \frac{1}{2} & -\frac{1}{2}\sqrt{\frac{h}{g}}\end{bmatrix}$$

The matrices for the diagonalized are calculated as follows:

$$\mathbf{A_d}=\mathbf{P}^{-1}\mathbf{AP}=\begin{bmatrix}\frac{1}{2} & \frac{1}{2}\sqrt{\frac{h}{g}}\\ \frac{1}{2} & -\frac{1}{2}\sqrt{\frac{h}{g}}\end{bmatrix}\begin{bmatrix}0 & 1\\ \frac{g}{h} & 0\end{bmatrix}\begin{bmatrix}1 & 1\\ \sqrt{\frac{g}{h}} & -\sqrt{\frac{g}{h}}\end{bmatrix}=\begin{bmatrix}\sqrt{\frac{g}{h}} & 0\\ 0 & -\sqrt{\frac{g}{h}}\end{bmatrix}$$

$$\mathbf{B_d}=\mathbf{P}^{-1}\mathbf{B}=\begin{bmatrix}\frac{1}{2} & \frac{1}{2}\sqrt{\frac{h}{g}}\\ \frac{1}{2} & -\frac{1}{2}\sqrt{\frac{h}{g}}\end{bmatrix}\begin{bmatrix}0\\ \frac{aV}{bh}\end{bmatrix}=\begin{bmatrix}\frac{1}{2}\frac{aV}{b\sqrt{gh}}\\ -\frac{1}{2}\frac{aV}{b\sqrt{gh}}\end{bmatrix}$$

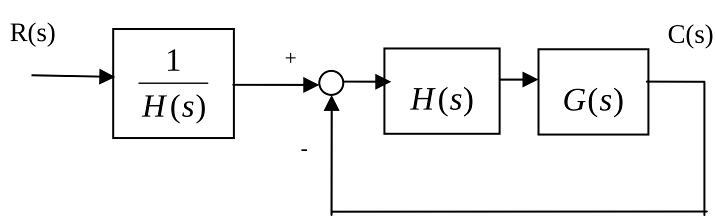
$$\mathbf{C_d}=\mathbf{CP}=\begin{bmatrix}\frac{V}{a} & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ \sqrt{\frac{g}{h}} & -\sqrt{\frac{g}{h}}\end{bmatrix}=\begin{bmatrix}\frac{V}{a}+\sqrt{\frac{g}{h}} & \frac{V}{a}-\sqrt{\frac{g}{h}}\end{bmatrix}$$

The diagonalized representation is:

$$\begin{bmatrix}\dot{z}_1\\ \dot{z}_2\end{bmatrix}=\begin{bmatrix}\sqrt{\frac{g}{h}} & 0\\ 0 & -\sqrt{\frac{g}{h}}\end{bmatrix}\begin{bmatrix}z_1\\ z_2\end{bmatrix}+\begin{bmatrix}\frac{1}{2}\frac{aV}{b\sqrt{gh}}\\ -\frac{1}{2}\frac{aV}{b\sqrt{gh}}\end{bmatrix}\delta$$

$$\varphi=\begin{bmatrix}\frac{V}{a}+\sqrt{\frac{g}{h}} & \frac{V}{a}-\sqrt{\frac{g}{h}}\end{bmatrix}\begin{bmatrix}z_1\\ z_2\end{bmatrix}$$

73.



It can be easily verified that the closed loop transfer function for this system is identical to the original.

SOLUTIONS TO DESIGN PROBLEMS

74.

$J_e = J_a+J_L(\frac{1}{20})^2 = 2+2 = 4$; $D_e = D_a+D_L(\frac{1}{20})^2 = 2+D_L(\frac{1}{20})^2$. Therefore, the forward-path transfer function is,

$$G(s) = (1000)\left(\frac{\frac{1}{4}}{s(s+\frac{1}{4}(D_e+2))}\right)\left(\frac{1}{20}\right).\ \text{Thus, } T(s) = \frac{G}{1+G} = \frac{\frac{25}{2}}{s^2+\frac{1}{4}(D_e+2)s+\frac{25}{2}}.$$

Hence, $\zeta = \frac{-\ln(\frac{\%OS}{100})}{\sqrt{\pi^2+\ln^2(\frac{\%OS}{100})}} = 0.456$; $\omega_n = \sqrt{\frac{25}{2}}$; $2\zeta\omega_n = \frac{D_e+2}{4}$. Therefore $D_e = 10.9$; from which $D_L = 3560$.

75.

a. $T(s) = \frac{25}{s^2+s+25}$; from which, $2\zeta\omega_n = 1$ and $\omega_n = 5$. Hence, $\zeta = 0.1$. Therefore,

$\%OS = e^{-\zeta\pi/\sqrt{1-\zeta^2}}x100 = 72.92\%$; $T_s = \frac{4}{\zeta\omega_n} = 8$.

b. $T(s) = \frac{25K_1}{s^2+(1+25K_2)s+25K_1}$; from which, $2\zeta\omega_n = 1+25K_2$ and $\omega_n = 5\sqrt{K_1}$. Hence,

$\zeta = \frac{-\ln(\frac{\%OS}{100})}{\sqrt{\pi^2+\ln^2(\frac{\%OS}{100})}} = 0.504$. Also, $T_s = \frac{4}{\zeta\omega_n} = 0.2$, Thus, $\zeta\omega_n = 20$; from which $K_2 = \frac{39}{25}$ and $\omega_n = 39.68$. Hence, $K_1 = 62.98$.

76.

The equivalent forward path transfer function is $G_e(s) = \frac{K}{s(1+(1+K_2))}$. Thus, $T(s) = \frac{G_e(s)}{1+G_e(s)}$ = $\frac{K}{s^2+(1+K_2)s+K}$. Prior to tachometer compensation ($K_2 = 0$), $T(s) = \frac{K}{s^2+s+K}$. Therefore $K = \omega_n^2 =$

100. Thus, after tachometer compensation, $T(s) = \dfrac{100}{s^2+(1+K_2)s+100}$. Hence, $\omega_n = 10$; $2\zeta\omega_n = 1+K_2$.

Therefore, $K_2 = 2\zeta\omega_n - 1 = 2(0.69)(10) - 1 = 12.8$.

77.

At the N_2 shaft, with rotation, $\theta_L(s)$

$$(J_{eq}s^2 + D_{eq}s)\theta_L(s) + F(s)r = T_{eq}(s)$$
$$F(s) = (Ms^2 + f_v s)X(s)$$

Thus,

$$(J_{eq}s^2 + D_{eq}s)\theta_L(s) + (Ms^2 + f_v s)X(s)r = T_{eq}(s)$$

But, $X(s) = r\theta_L(s)$. Hence,

$$\left[(J_{eq} + Mr^2)s^2 + (D_{eq} + f_v r^2)s\right]\theta_L(s) = T_{eq}(s)$$

where

$$J_{eq} = J_a(2)^2 + J = 5$$
$$D_{eq} = D_a(2)^2 + D = 4 + D$$
$$r = 2$$

Thus, the total load inertia and load damping is

$$J_L = J_{eq} + Mr^2 = 5 + 4M$$
$$D_L = D_{eq} + f_v r^2 = 4 + D + (1)(2)^2 = 8 + D$$

Reflecting J_L and D_L to the motor yields,

$$J_m = \frac{(5+4M)}{4};\ D_m = \frac{(8+D)}{4}$$

Thus, the motor transfer function is

$$\frac{\theta_m(s)}{E_a(s)} \frac{\dfrac{K_t}{R_a J_m}}{s(s + \dfrac{1}{J_m}(D_m + \dfrac{K_t K_a}{R_a}))} = \frac{\dfrac{1}{J_m}}{s(s + \dfrac{1}{J_m}(D_m + 1))}$$

The gears are (10/20)(1) = 1/2. Thus, the forward-path transfer function is

$$G_e(s) = (500)\left(\frac{\dfrac{1}{J_m}}{s(s + \dfrac{1}{J_m}(D_m + 1))}\right)\frac{1}{2}$$

Finding the closed-loop transfer function yields,

$$T(s)=\frac{G_e(s)}{1+G_e(s)}=\frac{250/J_m}{s^2+\frac{D_m+1}{J_m}s+\frac{250}{J_m}}$$

For T_s = 2, $\frac{D_m+1}{J_m}=4$. For 20% overshoot, ζ = 0.456. Thus,

$$2\zeta\omega_n=2(0.456)\omega_n=\frac{D_m+1}{J_m}=4$$

Or, $\omega_n=4.386=\sqrt{\frac{250}{J_m}}$; from which $J_m=13$ and hence, $D_m=51$. But,

$J_m=\frac{(5+4M)}{4}$; $D_m=\frac{(8+D)}{4}$. Thus, M = 11.75 and D = 196.

78.

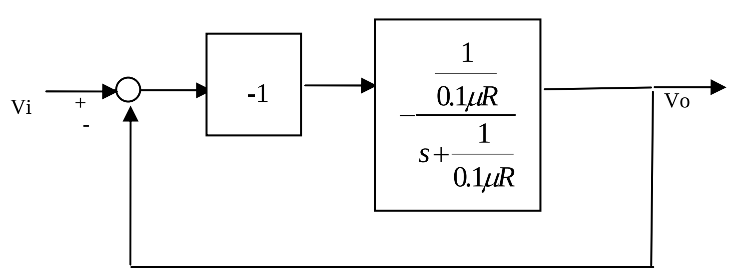
a. The leftmost op-amp equation can be obtained by superposition. Let $v_o=0$, then the circuit is an inverting amplifier thus $v_1=-\frac{10k}{10k}v_{in}=-v_{in}$. Now let $v_{in}=0$, the circuit is a non-inverting amplifier with an equal resistor voltage divider at its input, thus $v_1=\frac{10k}{10k+10k}(1+\frac{10k}{10k})v_o=v_o$. Adding both input components $v_1=v_o-v_{in}$

b. The two equations representing the system are:

$v_1=v_o-v_{in}$ and

$$\frac{v_o}{v_{in}}=-\frac{\frac{1}{0.1\mu s}}{R+\frac{1}{0.1\mu s}}=-\frac{\frac{1}{0.1\mu R}}{s+\frac{1}{0.1\mu R}}$$

The block diagram is:



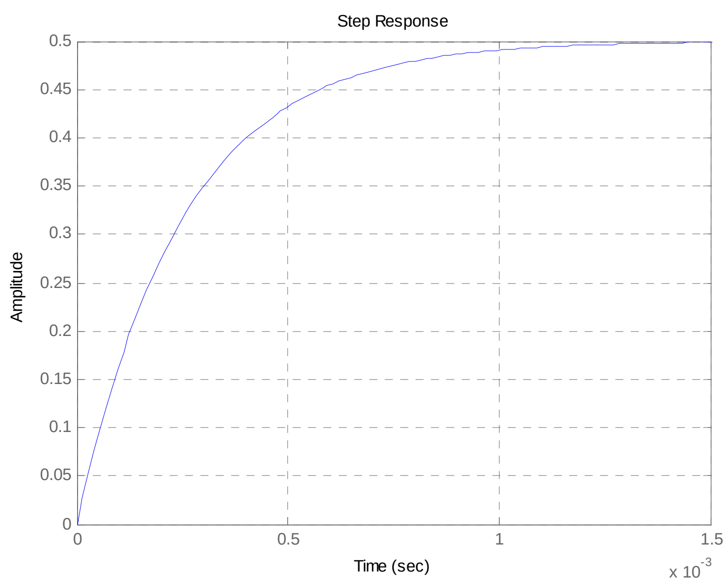
c. From the figure

$$\frac{v_o}{v_{in}} = \frac{\dfrac{\dfrac{1}{0.1\mu R}}{s+\dfrac{1}{0.1\mu R}}}{1+\dfrac{\dfrac{1}{0.1\mu R}}{s+\dfrac{1}{0.1\mu R}}} = \frac{\dfrac{1}{0.1\mu R}}{s+\dfrac{2}{0.1\mu R}}$$

d. The system is first order so $T_s = \dfrac{4}{\dfrac{2}{0.1\mu R}} = 0.2\mu R = 1m\sec$ from which

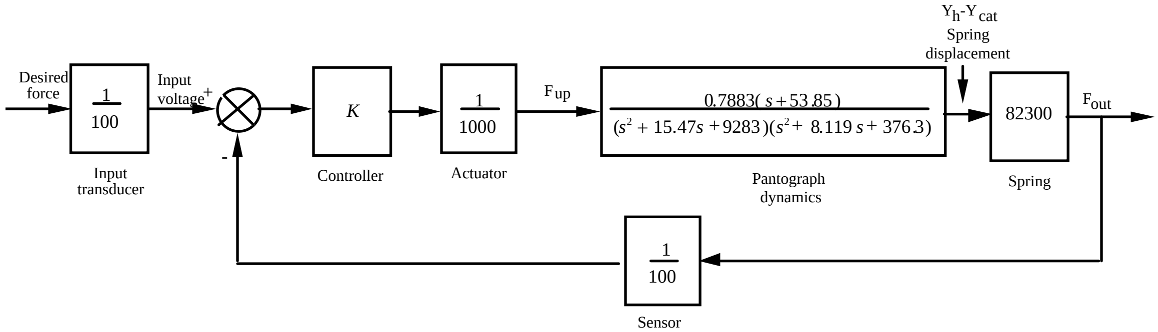
$$R = \frac{1m}{0.2\mu} = 5k\Omega$$

e. $\dfrac{v_o}{v_i} = \dfrac{2000}{s+4000}$ For a unit step input the output will look as follows



79.

a.



b. $G(s) = \dfrac{Y_h(s) - Y_{cat}(s)}{F_{up}(s)} = \dfrac{0.7883(s+53.85)}{(s^2+15.47s+9283)(s^2+8.119s+376.3)}$

$$G_e(s) = (K/100)*(1/1000)*G(s)*82.3e3 = \frac{648.7709\ (s+53.85)}{(s^2 + 8.119s + 376.3)\ (s\text{^}2 + 15.47s + 9283)}$$

$$T(s) = Ge/(1+Ge) = \frac{648.7709\ (s+53.85)}{(s\text{^}2 + 8.189s + 380.2)\ (s^2 + 15.4s + 9279)}$$

$$= \frac{648.8\ s + 3.494e04}{s^4 + 23.59\ s^3 + 9785\ s^2 + 8.184e04\ s + 3.528e06}$$

c.

For G(s) = (yh-ycat)/Fup

Phase-variable form

Ap =

0	1	0	0
0	0	1	0
0	0	0	1
-3.493e6	-81190	- 9785	-23.59

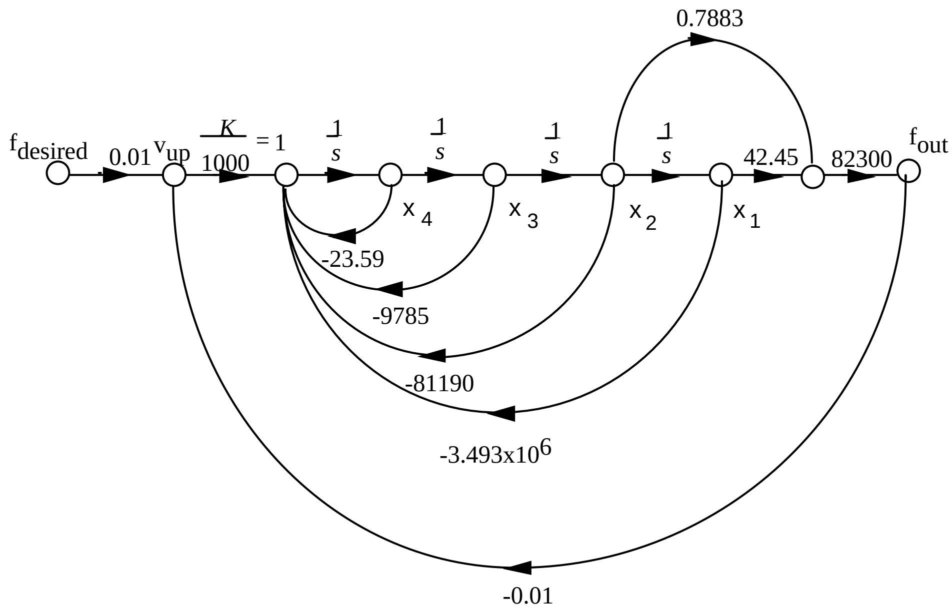
Bp =

0
0
0
1

Cp =

42.45	0.7883	0	0

Using this result to draw the signal-flow diagram,



Writing the state and output equations

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = x_4$$

$$\dot{x}_4 = -23.59x_4 - 9785x_3 - 81190x_2 - 3493000x_1 + 0.01f_{desired} - 0.01 f_{out}$$

But,

$$f_{out} = 42.45*82300x_1 + 0.7883*82300x_2$$

Substituting f_{out} into the state equations yields

$$\dot{x}_4 = -3527936.35x_1 - 81838.7709x_2 - 9785x_3 - 23.59x_4 + 0.01f_{desired}$$

Putting the state and output equations into vector-matrix form.

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -3.528x10^6 & -81840 & -9785 & -23.59 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0.01 \end{bmatrix} f_{desired}$$

$$y = f_{out} = \begin{bmatrix} 3494000 & 64880 & 0 & 0 \end{bmatrix}\mathbf{x}$$

80.

a. The transfer function obtained in Problem 3.30 is

$$\frac{Y}{U_1} = \frac{-520s - 10.3844}{s^3 + 2.6817s^2 + 0.11s + 0.0126}$$ by inspection we write the phase-variable form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -0.0126 & -0.11 & -2.6817 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_1$$

$$y = \begin{bmatrix} -10.3844 & -520 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

b. We renumber the phase-variable form state variables in reverse order

$$\begin{bmatrix} \dot{x}_3 \\ \dot{x}_2 \\ \dot{x}_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -0.0126 & -0.11 & -2.6817 \end{bmatrix} \begin{bmatrix} x_3 \\ x_2 \\ x_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_1$$

$$y = \begin{bmatrix} -10.3844 & -520 & 0 \end{bmatrix} \begin{bmatrix} x_3 \\ x_2 \\ x_1 \end{bmatrix}$$

And we rearrange in ascending numerical order to obtain the controller canonical form:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -2.6817 & -0.11 & -0.0126 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u_1$$

$$y = \begin{bmatrix} 0 & -520 & -10.3844 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

c. To obtain the observer canonical form we rewrite the system's transfer function as:

$$\frac{Y}{U_1} = \frac{-\frac{520}{s^2} - \frac{10.3844}{s^3}}{1 + \frac{2.6817}{s} + \frac{0.11}{s^2} + \frac{0.0126}{s^3}}$$

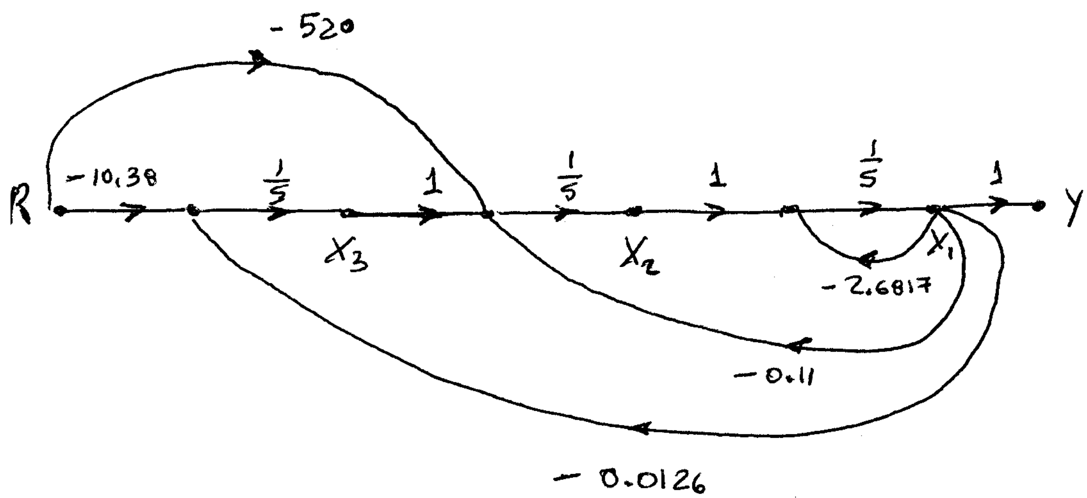
We cross-multiply to obtain

$$\left[-\frac{520}{s^2}-\frac{10.3844}{s^3}\right]U_1 = Y\left[1+\frac{2.6817}{s}+\frac{0.11}{s^2}+\frac{0.0126}{s^3}\right]$$

Combining terms with like powers of integration:

$$Y = \frac{1}{s}[-2.6817Y]+\frac{1}{s^2}[-520R-0.11Y]+\frac{1}{s^3}[-10.3844R-0.0126Y]$$
$$= \frac{1}{s}\left[-2.6817Y+\frac{1}{s}\left([-520R-0.11Y]+\frac{1}{s}[-10.3844R-0.0126Y]\right)\right]$$

We draw the signal flow graph:



The following equations follow:

$$\dot{x}_1 = -2.6817x_1 + x_2$$

$$\dot{x}_2 = -0.11x_1 + x_3 - 520r$$

$$\dot{x}_3 = -0.0126x_1 - 10.38r$$

$$y = x_1$$

Which lead to observer canonical form:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix}=\begin{bmatrix}-2.6817 & 1 & 0\\ -0.11 & 0 & 1\\ -0.0126 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}+\begin{bmatrix}1\\ -520\\ -10.38\end{bmatrix}u_1;\ y=\begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

d.

```
>> A=[-0.04167 0 -0.0058; 0.0217 -0.24 0.0058; 0 100 -2.4];

>> B=[5.2;-5.2;0];

>> C=[0 0 1];

>> [V,D]=eig(A);

>> Bd=inv(V)*B


Bd =


  1.0e+002 *


  -0.9936 + 0.0371i

  -0.9936 - 0.0371i

   1.9797


>> Cd = C*V


Cd =


    0.9963    0.9963    1.0000
```

```
>> D

D =

 -0.0192 + 0.0658i        0                  0
       0           -0.0192 - 0.0658i        0
       0                  0           -2.6433
```

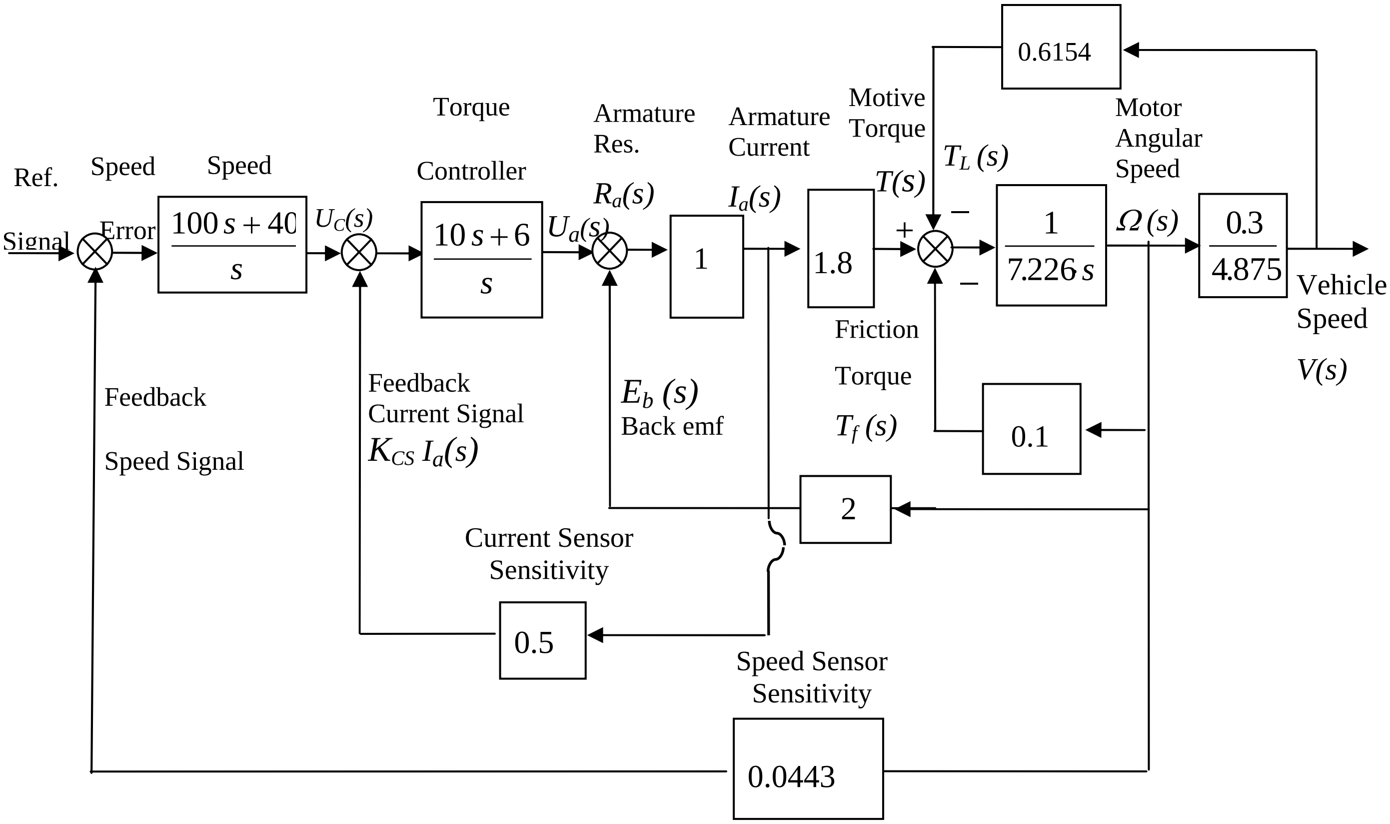
So a diagonalized version of the system is

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -0.0192 + j0.0658 & 0 & 0 \\ 0 & -0.0192 - j0.0658 & 0 \\ 0 & 0 & -2.6433 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} -99.36 + j3.71 \\ -99.36 - j3.71 \\ 197.97 \end{bmatrix} u_1$$

$$y = \begin{bmatrix} 0.9963 & 0.9963 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

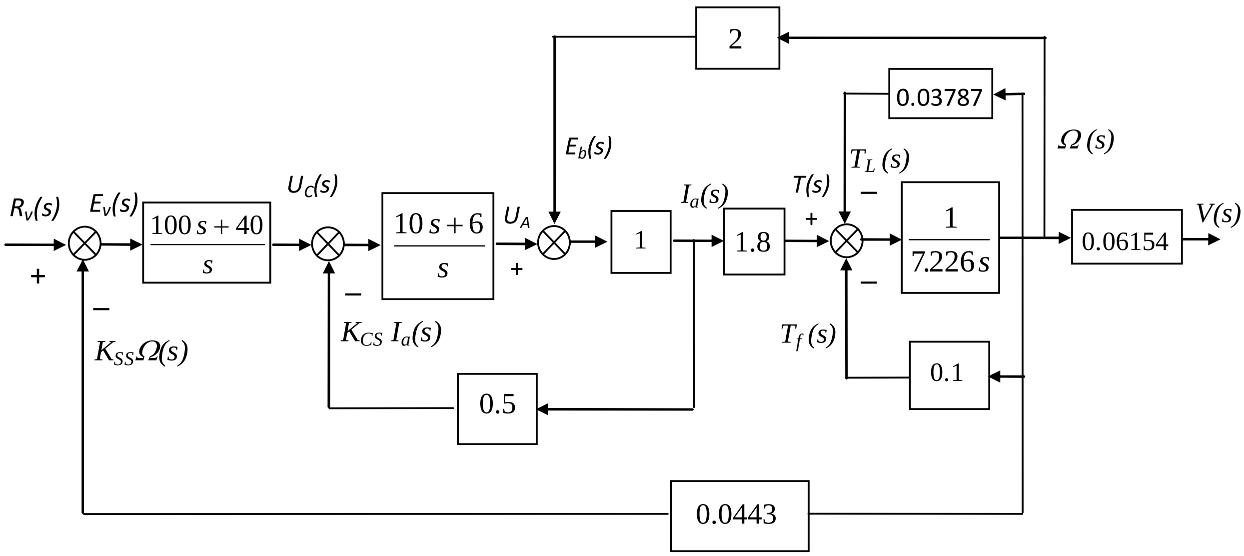
81.

a. Substituting all values and transfer functions into the respective blocks of the system (Figure 4), we get:

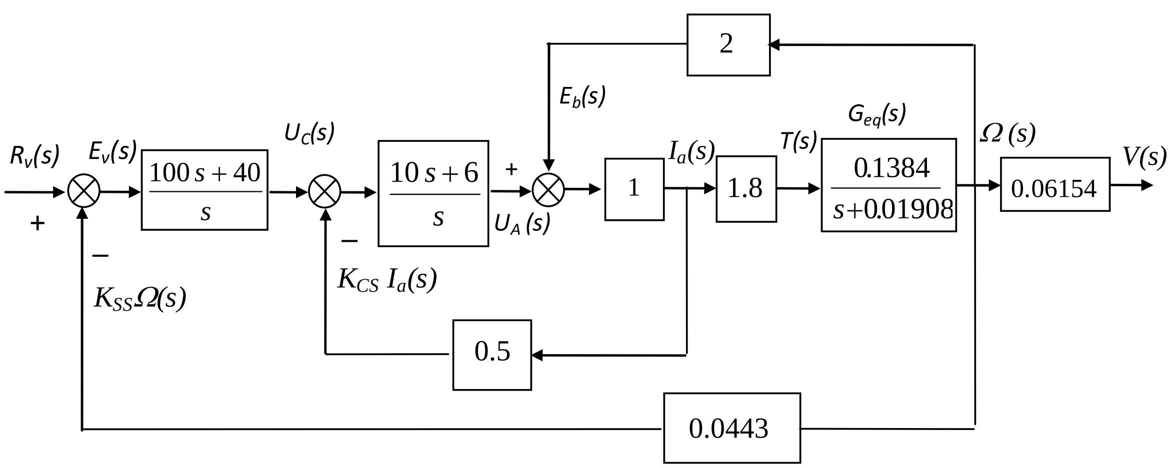


Moving the last pick-off point to the left past the $\dfrac{r}{i_{tot}} = \dfrac{0.3}{4.875} = 0.06154$ block and changing the position of the back-emf feedback pick-off point, so that it becomes an outer loop, we obtain the block-diagram shown below. In that diagram the $\dfrac{1}{7.226\,s}$ block (representing the total inertia) has two parallel feedback blocks. Reducing these two blocks into one, we have the following equivalent feedback transfer function:

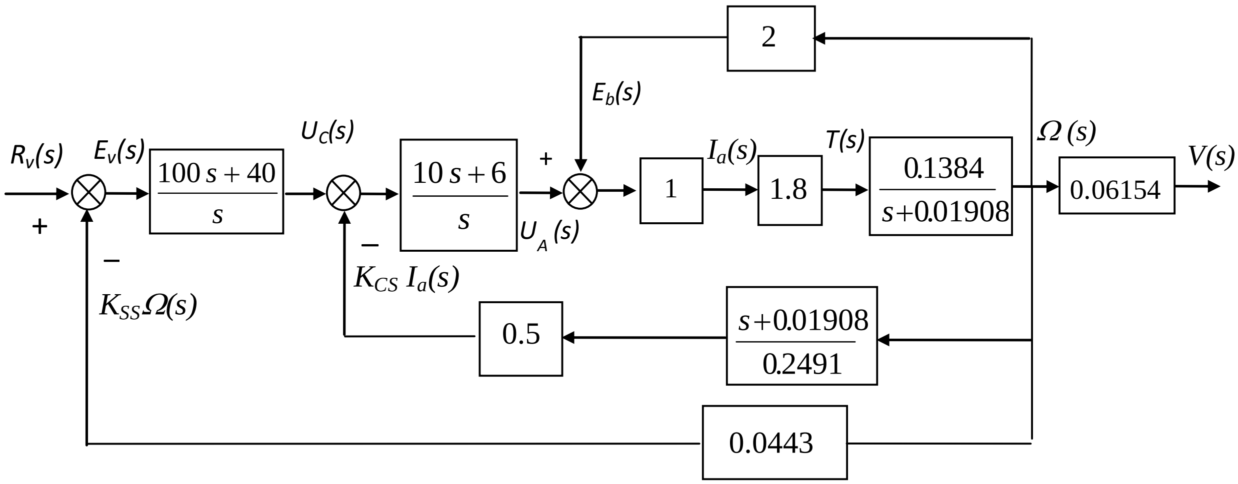
$$G_{eq}(s) = \frac{\Omega(s)}{T(s)} = \frac{\dfrac{1}{7.226\,s}}{1 + \dfrac{0.13787}{7.226\,s}} = \frac{0.1384}{s + 0.01908}$$



Replacing that feedback loop with its equivalent transfer function, $G_{eq}(s)$, we have:

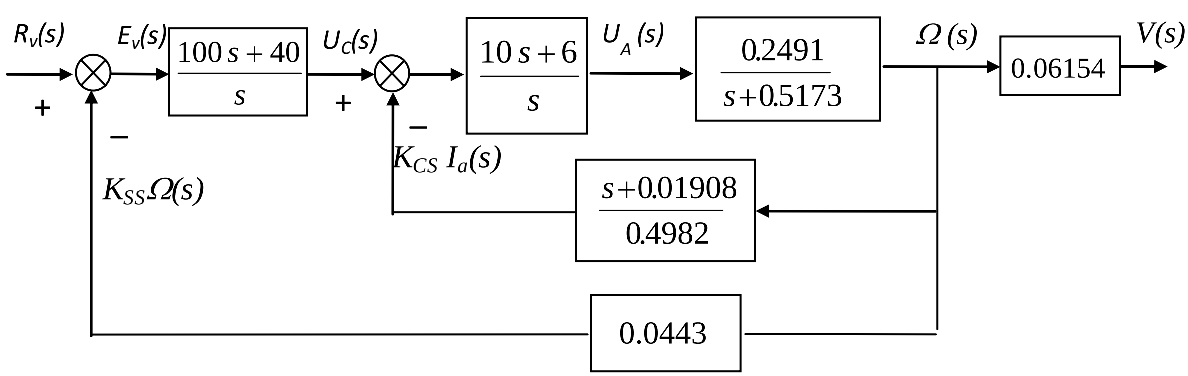


Moving the armature current pick-off point to the right past the $\frac{T(s)}{I_a(s)}$ and $G_{eq}(s)$ blocks, the above block-diagram becomes as shown below.



The latter, in turn, can be reduced to that shown next as the cascaded blocks in the feedback to the torque controller are replaced by the single block: $\dfrac{K_{CS}I_a(s)}{\Omega(s)} = \dfrac{s+0.01908}{0.4982}$ and the inner feedback loop is replaced by its equivalent transfer function:

$$\frac{\Omega(s)}{U_A(s)} = \frac{\dfrac{0.2491}{s+0.01908}}{1+\dfrac{0.2491}{s+0.01908}\times 2} = \frac{0.2491}{s+0.5173}$$



Thus: $$\frac{\Omega(s)}{U_C(s)}=\frac{\left(\frac{10s+6}{s}\right)\left(\frac{0.2491}{s+0.5173}\right)}{1+\left(\frac{10s+6}{s}\right)\left(\frac{0.2491}{s+0.5173}\right)\left(\frac{s+0.01908}{0.4982}\right)}=\frac{0.2491(10s+6)}{s\,(s+0.5173)+\ 0.5\ (10s+6)\ (s+0.01908)}$$

Finally $$\frac{\Omega(s)}{R_v(s)}=\frac{\left(\frac{100\ s+40}{s}\right)\left(\frac{0.2491(10\ s+6)}{s\ (s+0.5173)+0.5\ (10\ s+6)(s+0.01908)}\right)}{1+\ 0.0443\left(\frac{100\,s+40}{s}\right)\left(\frac{0.2491\ (10\ s+6)}{s\ (s+0.5173)+0.5(10\ s+6)(s+0.01908)}\right)}$$ or

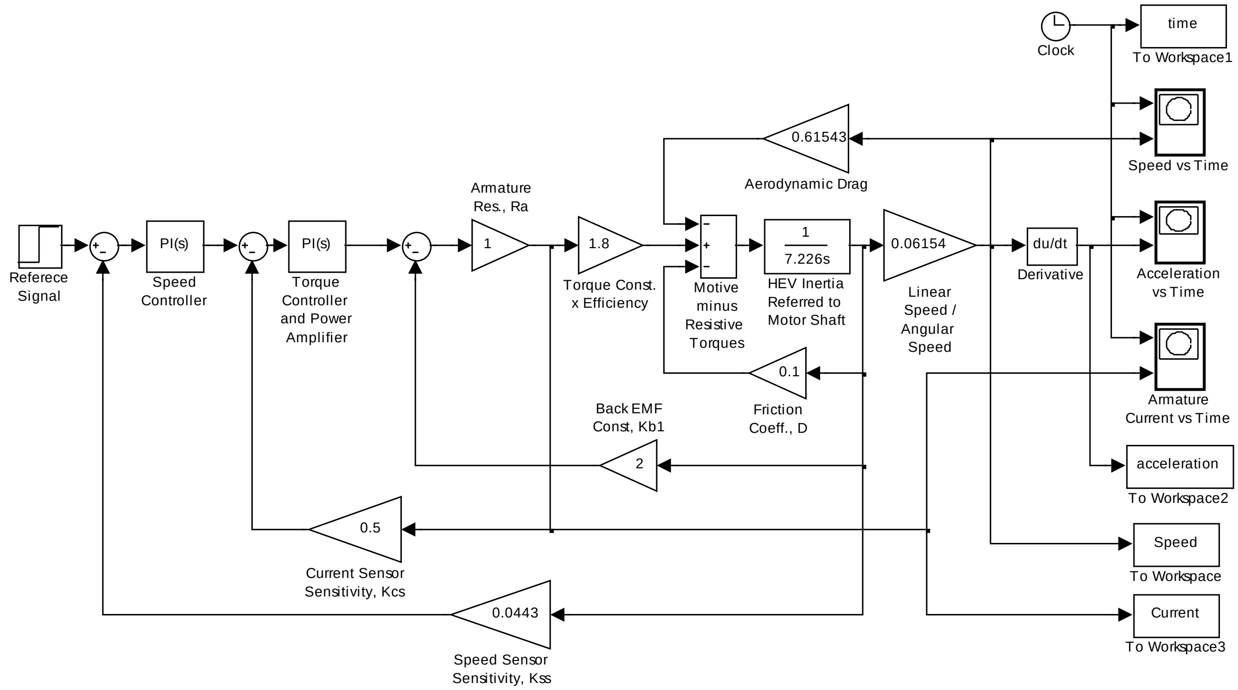
$$\frac{\Omega(s)}{R_V(s)}=\frac{2491\,(s+0.4)\,(s+0.6)}{s\left(6s^2+3.613s+0.0572\right)+11.035\left(s^2+s+0.24\right)}$$

$$=\frac{2491\,(s+0.4)\,(s+0.6)}{6s^3+14.644s^2+11.09s+2.65}$$

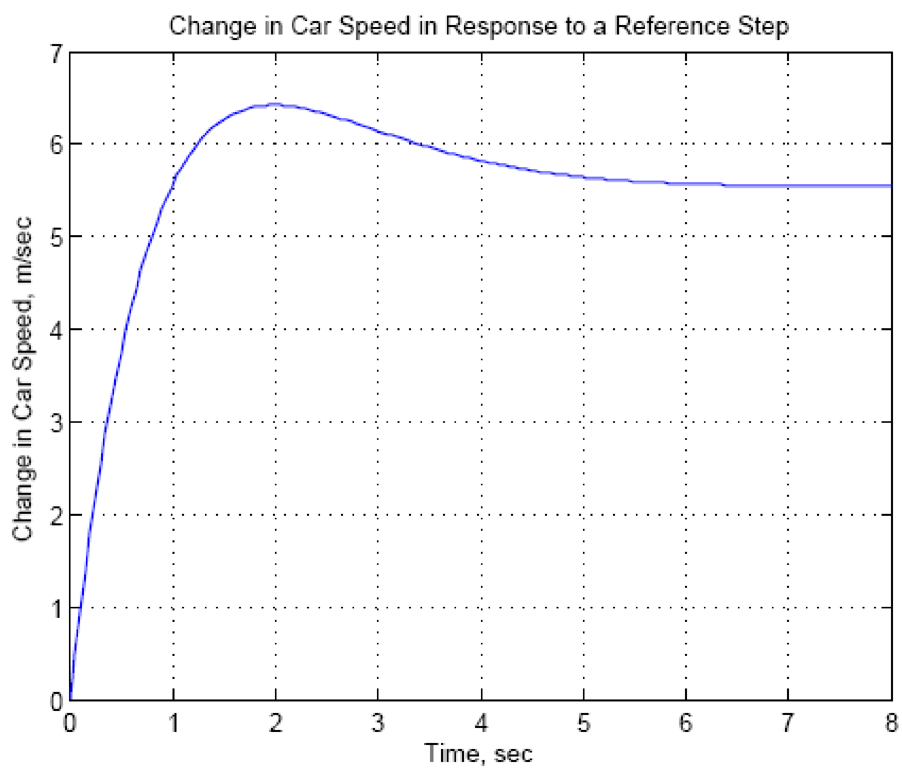
Hence: $$\frac{V(s)}{R_v(s)}=0.06154\frac{\Omega(s)}{R_v(s)}=\frac{15.33\ (s+0.4)\,(s+0.6)}{6\,s^3+14.644\ s^2+11.09\ s+2.65}$$

b. Simulink was used to model the HEV cascade control system. That model is shown below. The reference signal, $r_v\,(t)$, was set as a step input with a zero initial value, a step time = 0 seconds, and a final value equal to 4 volts [corresponding to the desired final car speed, $v\,(\infty)$ = 60 km/h, e.g. a desired final value of the change in car speed, $\Delta v\ (\infty)$ = 5.55 m/s]. The variables of interest [time, change in car speed, acceleration, and motor armature current] were output (in array format) to four "workspace" sinks, each of which was assigned the respective variable name. After the simulation ended, Matlab plot commands were utilized to obtain and edit the required three graphs. These graphs are shown below.

The simulations show that in response to such a speed reference command, car acceleration would go initially to a maximum value of 10.22 m/s^2 and the motor armature current would reach a maximum value of 666.7 A. That would require an electric motor drive rated around 80 kW or using both the electric motor and gas or diesel engine, when fast acceleration is required. Most practical HEV control systems, however, use current-limiting and acceleration-limiting devices or software programs.

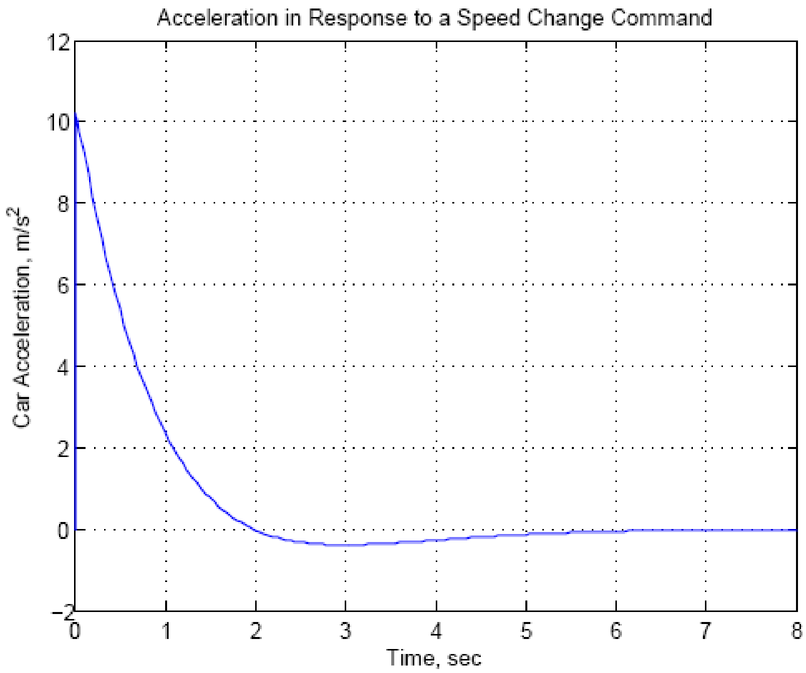


Model of the HEV Cascade Control

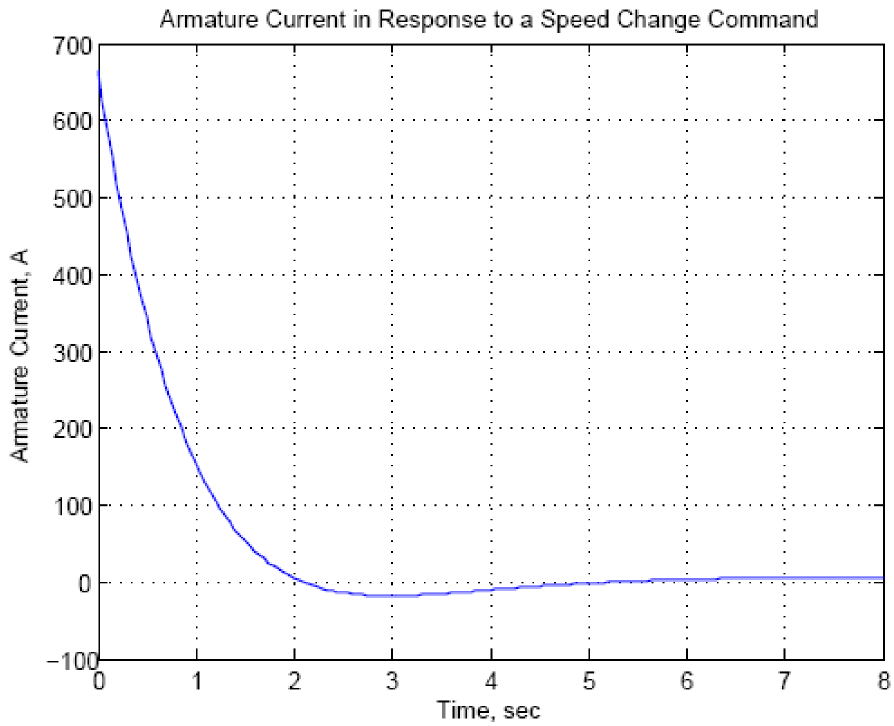


System

Change in car speed in response to a speed reference signal step of 4 volts



Car acceleration reponse to a speed reference signal step of 4 volts



Motor armature current reponse to a speed reference signal step of 4 volts

State Space SS
11. Which two forms of the state-space representation are found using the same method?

State Space SS
12. Which form of the state-space representation leads to a diagonal matrix?

State Space SS
13. When the system matrix is diagonal, what quantities lie along the diagonal?

State Space SS
14. What terms lie along the diagonal for a system represented in Jordan canonical form?

State Space SS
15. What is the advantage of having a system represented in a form that has a diagonal system matrix?

State Space SS
16. Give two reasons for wanting to represent a system by alternative forms.

State Space SS
17. For what kind of system would you use the observer canonical form?

State Space SS
18. Describe state-vector transformations from the perspective of different bases.

State Space SS
19. What is the definition of an eigenvector?

State Space SS
20. Based upon your definition of an eigenvector, what is an eigenvalue?

State Space SS
21. What is the significance of using eigenvectors as basis vectors for a system transformation?

Problems

1. Reduce the block diagram shown in Figure P5.1 to a single transfer function, $T(s) = C(s)/R(s)$ Use the following methods:

a. Block diagram reduction [Section: 5.2]

b. `MATLAB`

MATLAB ML

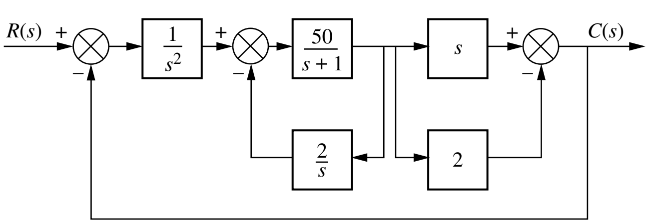


FIGURE P5.1

2. Find the closed-loop transfer function, $T(s) = C(s)/R(s)$ for the system shown in Figure P5.2, using block diagram reduction. [Section: 5.2]

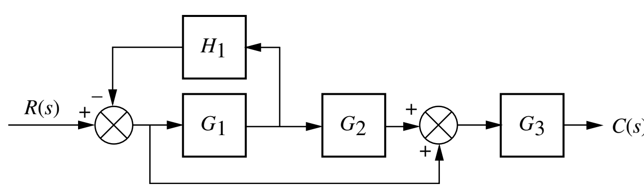


FIGURE P5.2

3. Find the equivalent transfer function, $T(s) = C(s)/R(s)$, for the system shown in Figure P5.3. [Section: 5.2]

WileyPLUS WPCS Control Solutions

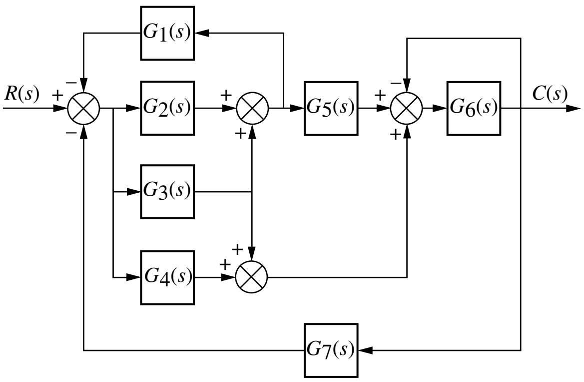


FIGURE P5.3

4. Reduce the system shown in Figure P5.4 to a single transfer function, $T(s) = C(s)/R(s)$. [Section: 5.2]

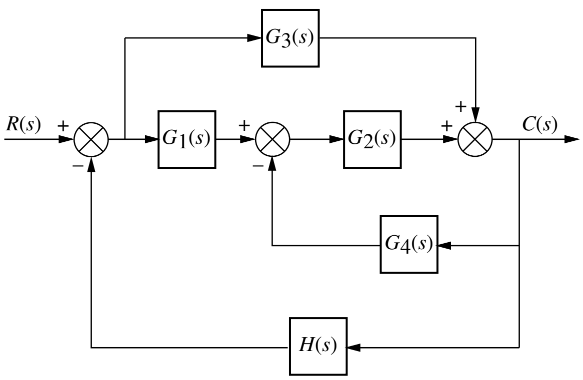


FIGURE P5.4

5. Find the transfer function, $T(s) = C(s)/R(s)$, for the system shown in Figure P5.5. Use the following methods:

   a. Block diagram reduction [Section: 5.2]

   b. MATLAB. Use the following transfer functions: MATLAB ML
   $G_1(s) = 1/(s+7)$, $G_2(s) = 1/(s^2+2s+3)$,
   $G_3(s) = 1/(s+4)$, $G_4(s) = 1/s$,
   $G_5(s) = 5/(s+7)$, $G_6(s) = 1/(s^2+5s+10)$,
   $G_7(s) = 3/(s+2)$, $G_8(s) = 1/(s+6)$.
   Hint: Use the **append** and **connect** commands in MATLAB's Control System Toolbox.

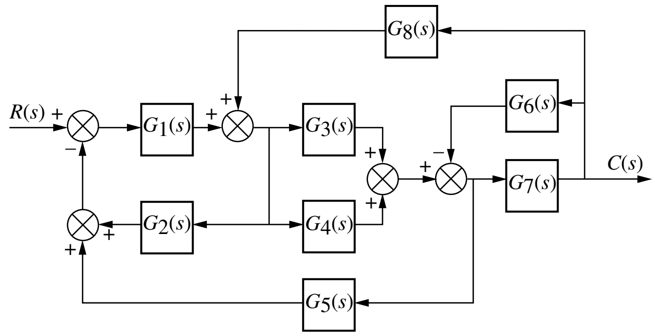


FIGURE P5.5

6. Reduce the block diagram shown in Figure P5.6 to a single block, $T(s) = C(s)/R(s)$. [Section: 5.2]

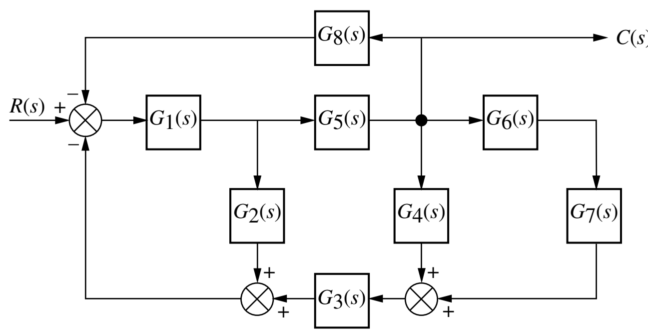


FIGURE P5.6

7. Find the unity feedback system that is equivalent to the system shown in Figure P5.7. [Section: 5.2].

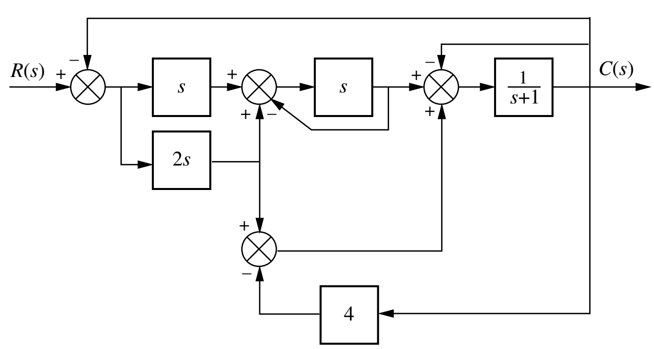


FIGURE P5.7

8. Given the block diagram of a system shown in Figure P5.8, find the transfer function $G(s) = \theta_{22}(s)/\theta_{11}(s)$. [Section: 5.2]

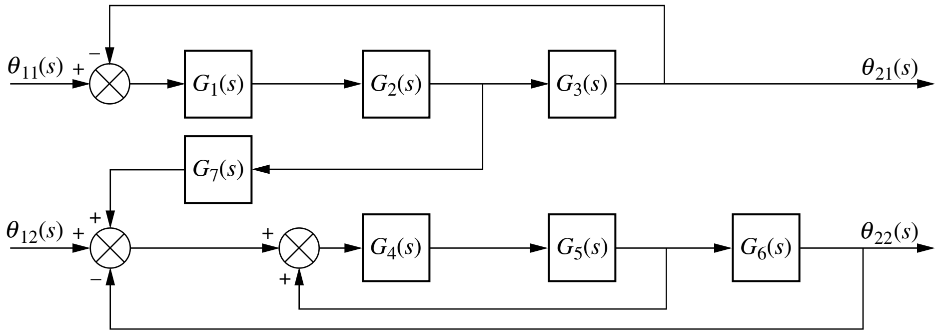


FIGURE P5.8

9. Reduce the block diagram shown in Figure P5.9 to a single transfer function, $T(s) = C(s)/R(s)$. [Section: 5.2]

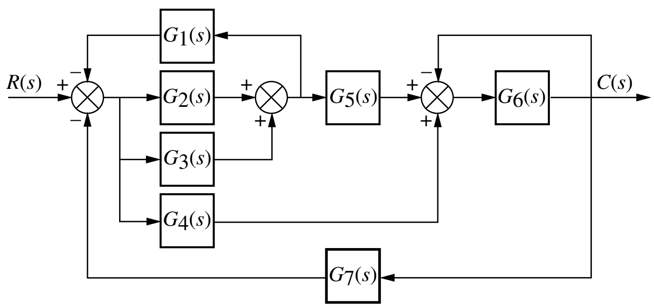


FIGURE P5.9

10. Reduce the block diagram shown in Figure P5.10 to a single block representing the transfer function, $T(s) = C(s)/R(s)$. [Section: 5.2]

WileyPLUS
WPCS
Control Solutions

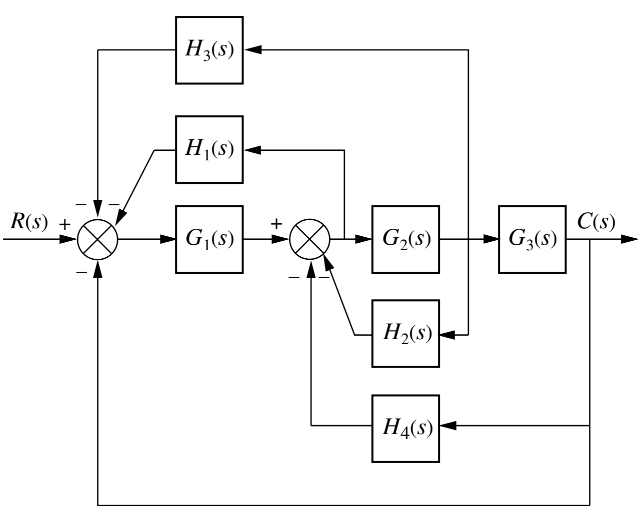


FIGURE P5.10

11. For the system shown in Figure P5.11, find the percent overshoot, settling time, and peak time for a step input if the system's response is underdamped. (Is it? Why?) [Section: 5.3]

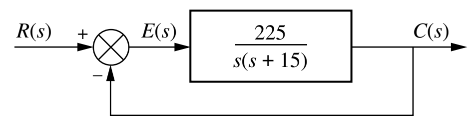


FIGURE P5.11

12. For the system shown in Figure P5.12, find the output, $c(t)$, if the input, $r(t)$, is a unit step. [Section: 5.3]

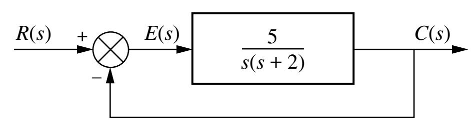


FIGURE P5.12

13. For the system shown in Figure P5.13, find the poles of the closed-loop transfer function, $T(s) = C(s)/R(s)$. [Section: 5.3]

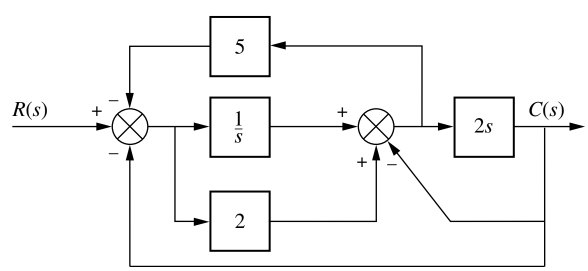


FIGURE P5.13

14. For the system of Figure P5.14, find the value of K that yields 10% overshoot for a step input. [Section: 5.3]

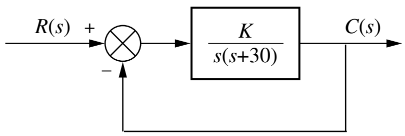


FIGURE P5.14

15. For the system shown in Figure P5.15, find K and α to yield a settling time of 0.15 second and a 30% overshoot. [Section: 5.3]

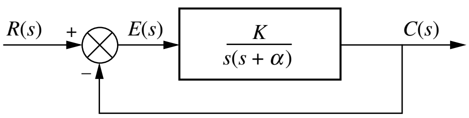


FIGURE P5.15

16. For the system of Figure P5.16, find the values of K_1 and K_2 to yield a peak time of 1.5 second and a settling time of 3.2 seconds for the closed-loop system's step response. [Section: 5.3]

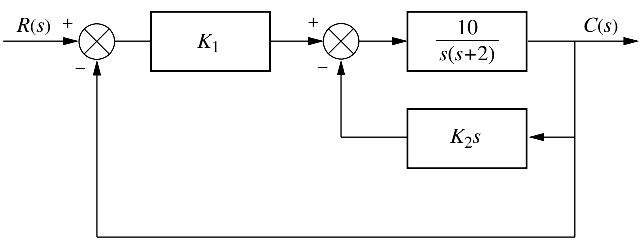


FIGURE P5.16

17. Find the following for the system shown in Figure P5.17: [Section: 5.3]

WileyPLUS WPCS Control Solutions

a. The equivalent single block that represents the transfer function, $T(s) = C(s)/R(s)$.

b. The damping ratio, natural frequency, percent overshoot, settling time, peak time, rise time, and damped frequency of oscillation.

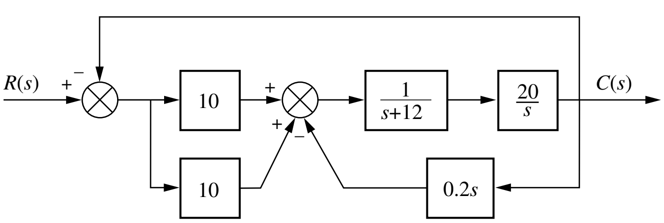


FIGURE P5.17

18. For the system shown in Figure P5.18, find ζ, ω_n, percent overshoot, peak time, rise time, and settling time. [Section: 5.3]

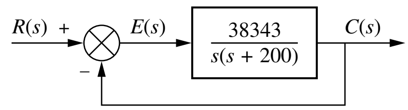


FIGURE P5.18

19. A motor and generator are set up to drive a load as shown in Figure P5.19. If the generator output voltage is $e_g(t) = K_f i_f(t)$, where $i_f(t)$ is the generator's field current, find the transfer function $G(s) = \theta_o(s)/E_i(s)$. For the generator, $K_f = 2\,\Omega$. For the motor, $K_t = 1$ N-m/A, and $K_b = 1$ V-s/rad.

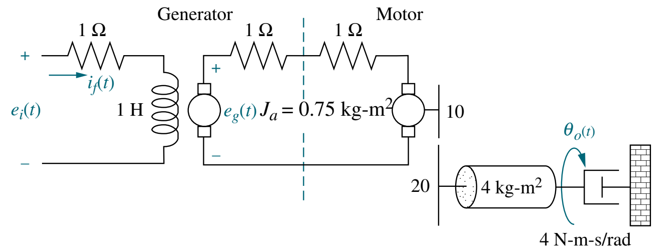


FIGURE P5.19

20. Find $G(s) = E_0(s)/T(s)$ for the system shown in Figure P5.20.

WileyPLUS WPCS Control Solutions

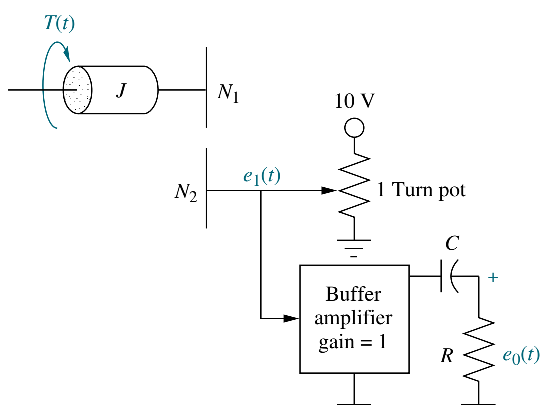


FIGURE P5.20

21. Find the transfer function $G(s) = E_o(s)/T(s)$ for the system shown in Figure P5.21.

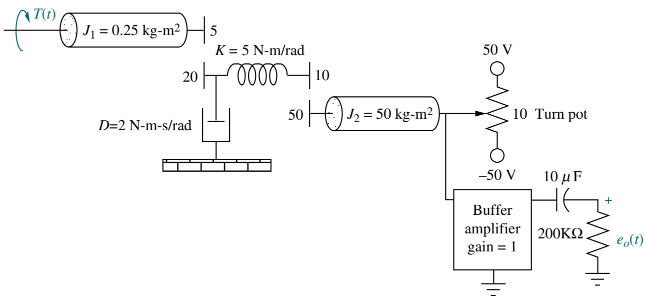


FIGURE P5.21

22. Label signals and draw a signal-flow graph for each of the block diagrams shown in the following problems: [Section: 5.4]

a. Problem 1

b. Problem 3

c. Problem 5

23. Draw a signal-flow graph for each of the following state equations: [Section: 5.6]

State Space **SS**

a. $$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -4 & -6 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}\mathbf{x}$$

b. $$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -3 & 1 \\ -3 & -4 & -5 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}\mathbf{x}$$

c. $$\dot{\mathbf{x}} = \begin{bmatrix} 7 & 1 & 0 \\ -3 & 2 & -1 \\ -1 & 0 & 2 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 3 & 2 \end{bmatrix}\mathbf{x}$$

24. Given the system below, draw a signal-flow graph and represent the system in state space in the following forms: [Section: 5.7]

State Space **SS**

WileyPLUS **WPCS** Control Solutions

a. Phase-variable form

b. Cascade form

$$G(s) = \frac{10}{(s+7)(s+8)(s+9)}$$

25. Repeat Problem 24 for

State Space **SS**

$$G(s) = \frac{20}{s(s-2)(s+5)(s+8)}$$

[Section: 5.7]

26. Using Mason's rule, find the transfer function, $T(s) = C(s)/R(s)$, for the system represented in Figure P5.22. [Section: 5.5]

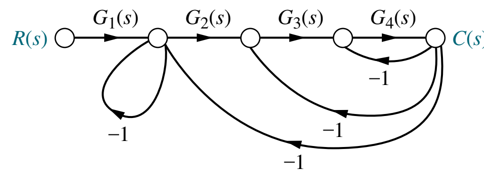


FIGURE P5.22

27. Using Mason's rule, find the transfer function, $T(s) = C(s)/R(s)$, for the system represented by Figure P5.23. [Section: 5.5]

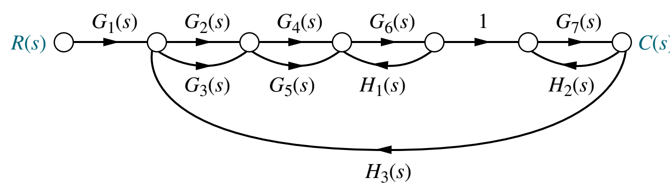


FIGURE P5.23

28. Use Mason's rule to find the transfer function of Figure 5.13 in the text. [Section: 5.5]

29. Use block diagram reduction to find the transfer function of Figure 5.21 in the text, and compare your answer with that obtained by Mason's rule. [Section: 5.5]

30. Represent the following systems in state space in Jordan canonical form. Draw the signal-flow graphs. [Section: 5.7]

State Space **SS**

a. $G(s) = \dfrac{(s+1)(s+2)}{(s+3)^2(s+4)}$

b. $G(s) = \dfrac{(s+2)}{(s+5)^2(s+7)^2}$

c. $G(s) = \dfrac{(s+3)}{(s+2)^2(s+4)(s+5)}$

31. Represent the systems below in state space in phase-variable form. Draw the signal-flow graphs. [Section: 5.7]

State Space **SS**

WileyPLUS **WPCS** Control Solutions

a. $G(s) = \dfrac{s+3}{s^2+2s+7}$

b. $G(s) = \dfrac{s^2 + 2s + 6}{s^3 + 5s^2 + 2s + 1}$

c. $G(s) = \dfrac{s^3 + 2s^2 + 7s + 1}{s^4 + 3s^3 + 5s^2 + 6s + 4}$

32. Repeat Problem 31 and represent each system in controller canonical and observer canonical forms. [Section: 5.7]

State Space **SS**

33. Represent the feedback control systems shown in Figure P5.24 in state space. When possible, represent the open-loop transfer functions separately in cascade and complete the feedback loop with the signal path from output to input. Draw your signal-flow graph to be in one-to-one correspondence to the block diagrams (as close as possible). [Section: 5.7]

State Space **SS**

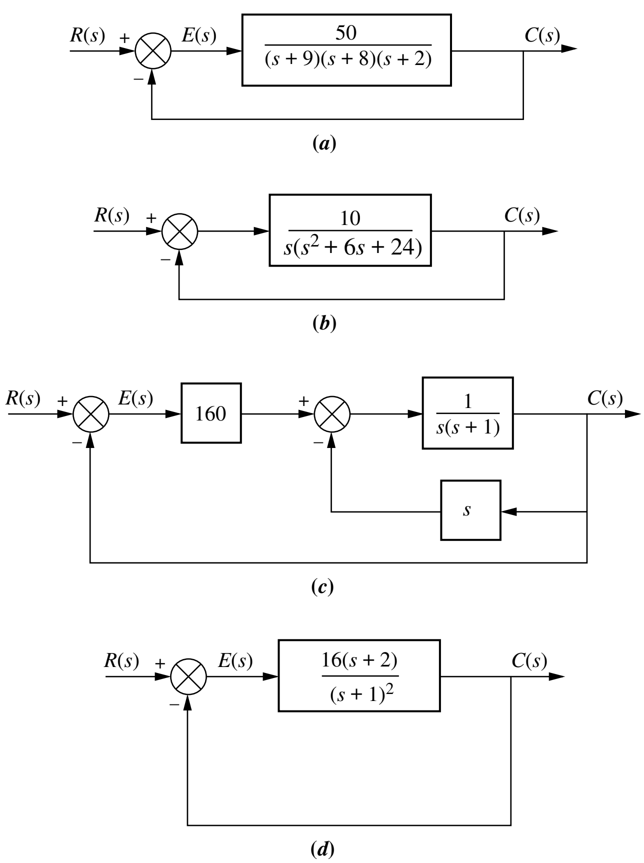


FIGURE P5.24

34. You are given the system shown in Figure P5.25. [Section: 5.7]

State Space **SS**

a. Represent the system in state space in phase-variable form.

b. Represent the system in state space in any other form besides phase-variable.

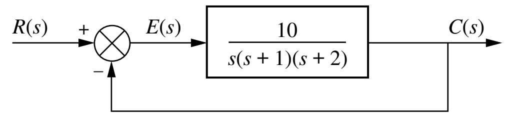


FIGURE P5.25

35. Repeat Problem 34 for the system shown in Figure P5.26. [Section: 5.7]

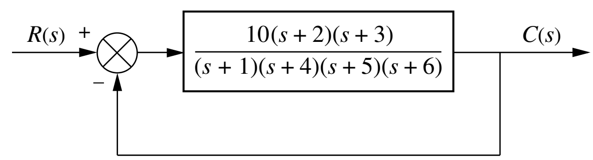


FIGURE P5.26

36. `Use MATLAB to solve Problem 35.`

MATLAB **ML**

37. Represent the system shown in Figure P5.27 in state space where $x_1(t)$, $x_3(t)$, and $x_4(t)$, as shown, are among the state variables, $c(t)$ is the output, and $x_2(t)$ is internal to $X_1(s)/X_3(s)$. [Section: 5.7]

State Space **SS**

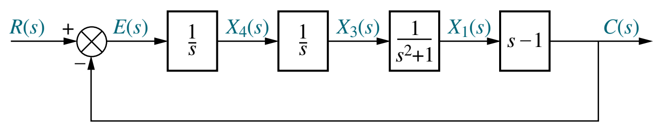


FIGURE P5.27

38. Consider the rotational mechanical system shown in Figure P5.28.

State Space **SS**

a. Represent the system as a signal-flow graph.

b. Represent the system in state space if the output is $\theta_2(t)$.

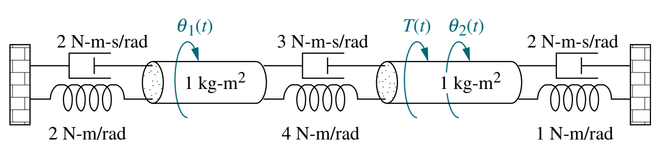


FIGURE P5.28

39. `Given a unity feedback system with the forward-path transfer function`

MATLAB **ML**

State Space **SS**

$$G(s) = \frac{7}{s(s+9)(s+12)}$$

use MATLAB to represent the closed loop system in state space in

a. phase-variable form;

b. parallel form.

40. Consider the cascaded subsystems shown in Figure P5.29. If $G_1(s)$ is represented in state space as

State Space SS

WileyPLUS WPCS Control Solutions

$$\dot{\mathbf{x}}_1 = \mathbf{A}_1\mathbf{x}_1 + \mathbf{B}_1 r$$
$$y_1 = \mathbf{C}_1\mathbf{x}_1$$

and $G_2(s)$ is represented in state space as

$$\dot{\mathbf{x}}_2 = \mathbf{A}_2\mathbf{x}_2 + \mathbf{B}_2 y_1$$
$$y_2 = \mathbf{C}_2\mathbf{x}_2$$

show that the entire system can be represented in state space as

$$\begin{bmatrix} \dot{\mathbf{x}}_1 \\ \cdots \\ \dot{\mathbf{x}}_2 \end{bmatrix} = \left[\begin{array}{c:c} \mathbf{A}_1 & \mathbf{0} \\ \hdashline \mathbf{B}_2\mathbf{C}_1 & \mathbf{A}_2 \end{array}\right] \begin{bmatrix} \mathbf{x}_1 \\ \cdots \\ \mathbf{x}_2 \end{bmatrix} + \begin{bmatrix} \mathbf{B}_1 \\ \cdots \\ \mathbf{0} \end{bmatrix} r$$

$$y_2 = \left[\begin{array}{c:c} \mathbf{0} & \mathbf{C}_2 \end{array}\right] \begin{bmatrix} \mathbf{x}_1 \\ \cdots \\ \mathbf{x}_2 \end{bmatrix}$$

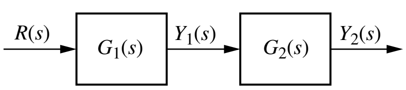


FIGURE P5.29

41. Consider the parallel subsystems shown in Figure P5.30. If $G_1(s)$ is represented in state space as

State Space SS

$$\dot{\mathbf{x}}_1 = \mathbf{A}_1\mathbf{x}_1 + \mathbf{B}_1 r$$
$$y_1 = \mathbf{C}_1\mathbf{x}_1$$

and $G_2(s)$ is represented in state space as

$$\dot{\mathbf{x}}_2 = \mathbf{A}_2\mathbf{x}_2 + \mathbf{B}_2 r$$
$$y_2 = \mathbf{C}_2\mathbf{x}_2$$

show that the entire system can be represented in state space as

$$\begin{bmatrix} \dot{\mathbf{x}}_1 \\ \cdots \\ \dot{\mathbf{x}}_2 \end{bmatrix} = \left[\begin{array}{c:c} \mathbf{A}_1 & \mathbf{0} \\ \hdashline \mathbf{0} & \mathbf{A}_2 \end{array}\right] \begin{bmatrix} \mathbf{x}_1 \\ \cdots \\ \mathbf{x}_2 \end{bmatrix} + \begin{bmatrix} \mathbf{B}_1 \\ \cdots \\ \mathbf{B}_2 \end{bmatrix} r$$

$$y = \left[\begin{array}{c:c} \mathbf{C}_1 & \mathbf{C}_2 \end{array}\right] \begin{bmatrix} \mathbf{x}_1 \\ \cdots \\ \mathbf{x}_2 \end{bmatrix}$$

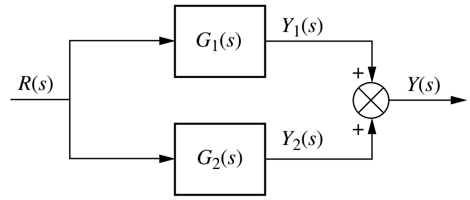


FIGURE P5.30

42. Consider the subsystems shown in Figure P5.31 and connected to form a feedback system. If $G(s)$ is represented in state space as

State Space SS

$$\dot{\mathbf{x}}_1 = \mathbf{A}_1\mathbf{x}_1 + \mathbf{B}_1 e$$
$$y = \mathbf{C}_1\mathbf{x}_1$$

and $H_2(s)$ is represented in state space as

$$\dot{\mathbf{x}}_2 = \mathbf{A}_2\mathbf{x}_2 + \mathbf{B}_2 y$$
$$p = \mathbf{C}_2\mathbf{x}_2$$

show that the closed-loop system can be represented in state space as

$$\begin{bmatrix} \dot{\mathbf{x}}_1 \\ \cdots \\ \dot{\mathbf{x}}_2 \end{bmatrix} = \left[\begin{array}{c:c} \mathbf{A}_1 & -\mathbf{B}_1\mathbf{C}_2 \\ \hdashline \mathbf{B}_2\mathbf{C}_1 & \mathbf{A}_2 \end{array}\right] \begin{bmatrix} \mathbf{x}_1 \\ \cdots \\ \mathbf{x}_2 \end{bmatrix} + \begin{bmatrix} \mathbf{B}_1 \\ \cdots \\ \mathbf{0} \end{bmatrix} r$$

$$y = \left[\begin{array}{c:c} \mathbf{C}_1 & \mathbf{0} \end{array}\right] \begin{bmatrix} \mathbf{x}_1 \\ \cdots \\ \mathbf{x}_2 \end{bmatrix}$$

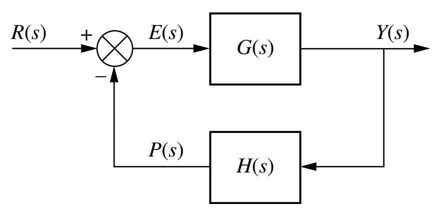


FIGURE P5.31

43. Given the system represented in state space as follows: [Section: 5.8]

State Space SS

$$\dot{\mathbf{x}} = \begin{bmatrix} -1 & -7 & 6 \\ -8 & 4 & 8 \\ 4 & 7 & -8 \end{bmatrix}\mathbf{x} + \begin{bmatrix} -5 \\ -7 \\ 5 \end{bmatrix} r$$

$$y = \begin{bmatrix} -9 & -9 & -8 \end{bmatrix}\mathbf{x}$$

convert the system to one where the new state vector, $\mathbf{z}$, is

$$\mathbf{z} = \begin{bmatrix} -4 & 9 & -3 \\ 0 & -4 & 7 \\ -1 & -4 & -9 \end{bmatrix}\mathbf{x}$$

44. Repeat Problem 43 for the following system: [Section: 5.8]

State Space **SS**

$$\dot{\mathbf{x}} = \begin{bmatrix} -5 & 1 & 1 \\ 9 & -9 & -9 \\ -9 & -1 & 8 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 9 \\ -4 \\ 0 \end{bmatrix} r$$

$$y = [-2 \quad -4 \quad 1]\mathbf{x}$$

and the following state-vector transformation:

$$\mathbf{z} = \begin{bmatrix} 5 & -4 & 9 \\ 6 & -7 & 6 \\ 6 & -5 & -3 \end{bmatrix}\mathbf{x}$$

45. Diagonalize the following system: [Section: 5.8]

State Space **SS** WileyPLUS **WPCS** Control Solutions

$$\dot{\mathbf{x}} = \begin{bmatrix} -5 & -5 & 4 \\ 2 & 0 & -2 \\ 0 & -2 & -1 \end{bmatrix}\mathbf{x} + \begin{bmatrix} -1 \\ 2 \\ -2 \end{bmatrix} r$$

$$y = [-1 \quad 1 \quad 2]\mathbf{x}$$

46. Repeat Problem 45 for the following system: [Section: 5.8]

State Space **SS**

$$\dot{\mathbf{x}} = \begin{bmatrix} -10 & -3 & 7 \\ 18.25 & 6.25 & -11.75 \\ -7.25 & -2.25 & 5.75 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} r$$

$$y = [1 \quad -2 \quad 4]\mathbf{x}$$

47. `Diagonalize the system in Problem 46 using MATLAB.`

MATLAB **ML**

48. During ascent the space shuttle is steered by commands generated by the computer's guidance calculations. These commands are in the form of vehicle attitude, attitude rates, and attitude accelerations obtained through measurements made by the vehicle's inertial measuring unit, rate gyro assembly, and accelerometer assembly, respectively. The ascent digital autopilot uses the errors between the actual and commanded attitude, rates, and accelerations to gimbal the space shuttle main engines (called thrust vectoring) and the solid rocket boosters to effect the desired vehicle attitude. The space shuttle's attitude control system employs the same method in the pitch, roll, and yaw control systems. A simplified model of the pitch control system is shown in Figure P5.32.[4]

a. Find the closed-loop transfer function relating actual pitch to commanded pitch. Assume all other inputs are zero.

b. Find the closed-loop transfer function relating actual pitch rate to commanded pitch rate. Assume all other inputs are zero.

c. Find the closed-loop transfer function relating actual pitch acceleration to commanded pitch acceleration. Assume all other inputs are zero.

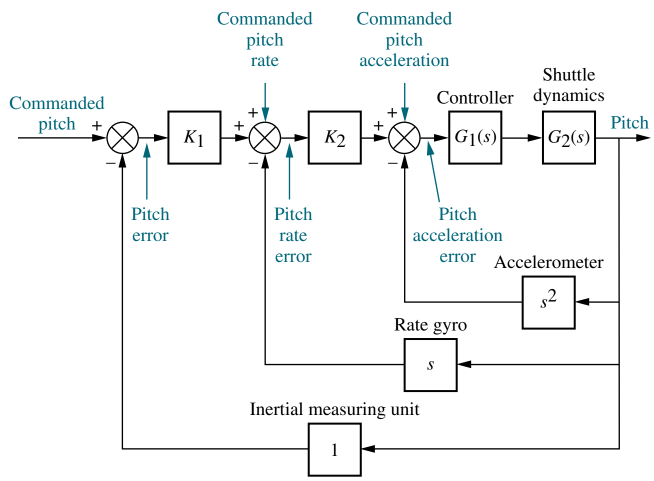


FIGURE P5.32 Space shuttle pitch control system (simplified)

49. An AM radio modulator generates the product of a carrier waveform and a message waveform, as shown in Figure P5.33 (*Kurland, 1971*). Represent the system in state space if the carrier is a sinusoid of frequency $\omega = a$, and the message is a sinusoid of frequency $\omega = b$. Note that this system is nonlinear because of the multiplier.

State Space **SS**

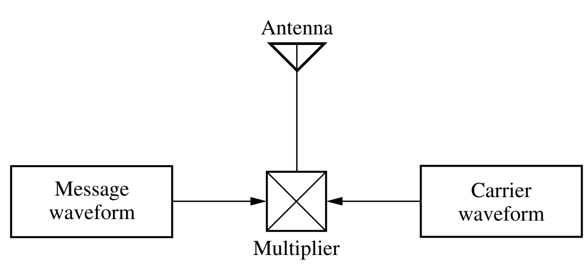


FIGURE P5.33 AM modulator

50. A model for human eye movement consists of the closed-loop system shown in Figure P5.34, where an

[4] Source of background information for this problem: Rockwell International.

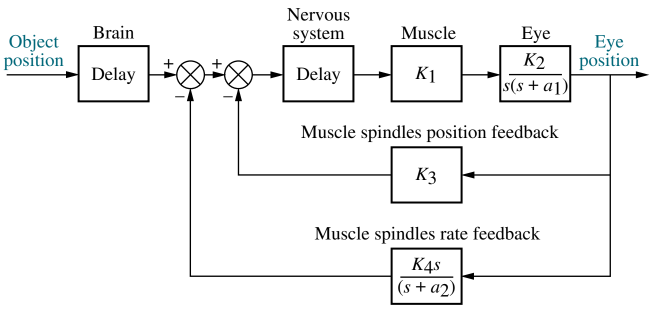


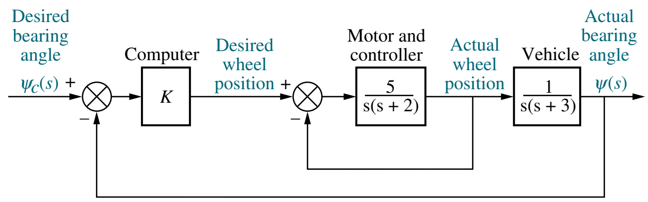
FIGURE P5.34 Feedback control system representing human eye movement

object's position is the input and the eye position is the output. The brain sends signals to the muscles that move the eye. These signals consist of the difference between the object's position and the position and rate information from the eye sent by the muscle spindles. The eye motion is modeled as an inertia and viscous damping and assumes no elasticity (spring) (*Milhorn, 1966*). Assuming that the delays in the brain and nervous system are negligible, find the closed-loop transfer function for the eye position control.

51. A HelpMate transport robot, shown in Figure P5.35(*a*), is used to deliver goods in a hospital setting. The robot can deliver food, drugs, laboratory materials, and patients' records (*Evans, 1992*). Given the simplified block diagram of the robot's bearing angle control system, as shown in Figure P5.35(*b*), do the following:

WileyPLUS WPCS Control Solutions

(*a*)



(*b*)

FIGURE P5.35 **a**. HelpMate robot used for in-hospital deliveries; **b**. simplified block diagram for bearing angle control

a. Find the closed-loop transfer function.

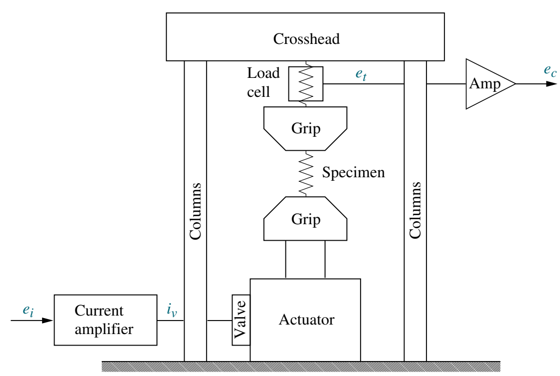
b. Represent the system in state space, where the input is the desired bearing angle, the output is the actual bearing angle, and the actual wheel position and actual bearing angle are among the state variables.

State Space SS

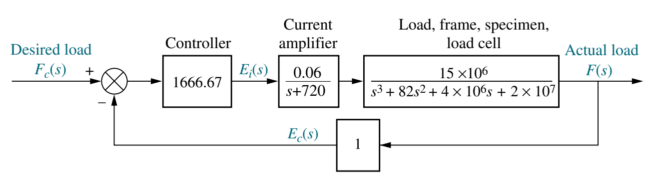
c. `Simulate the closed-loop system using MATLAB. Obtain the unit step response for different values of K that yield responses from overdamped to underdamped to unstable.`

MATLAB ML

52. Automatically controlled load testers can be used to test product reliability under real-life conditions. The tester consists of a load frame and specimen as shown in Figure P5.36(*a*). The desired load is



(*a*)



(*b*)

FIGURE P5.36 **a**. Load tester, (© 1992 IEEE) **b.** approximate block diagram input via a voltage, $e_i(t)$, to a current amplifier. The output load is measured via a voltage, $e_i(t)$, from a load cell measuring the load on the specimen. Figure P5.36(b) shows an approximate model of a load testing system without compensation (*Bailey, 1992*).

a. Model the system in state space. State Space **SS**

b. `Simulate the step response using MATLAB. Is the response predominantly first or second order? Describe the characteristics of the response that need correction.` MATLAB **ML**

53. Consider the F4-E aircraft of Problem 22, Chapter 3. If the open-loop transfer function relating normal acceleration, $A_n(s)$, to the input deflection command, $\delta_c(s)$, is approximated as State Space **SS**

$$\frac{A_n(s)}{\delta_c(s)} = \frac{-272(s^2+1.9s+84)}{(s+14)(s-1.8)(s+4.9)}$$

(*Cavallo, 1992*), find the state-space representation in

a. Phase-variable form

b. Controller canonical form

c. Observer canonical form

d. Cascade form

e. Parallel form

54. Find the closed-loop transfer function of the Unmanned Free-Swimming Submersible vehicle's pitch control system shown on the back endpapers (*Johnson, 1980*).

55. `Repeat Problem 54 using MATLAB.` MATLAB **ML**

56. `Use Simulink to plot the effects of nonlinearities upon the closed-loop step response of the antenna azimuth position control system shown on the front endpapers, Configuration 1. In particular, consider individually each of the following nonlinearities: saturation (±5 volts), backlash (dead-band width 0.15), deadzone (−2 to +2), as well as the linear response. Assume the preamplifier gain is 100 and the step input is 2 radians.` Simulink **SL**

57. Problem 12 in Chapter 1 describes a high-speed proportional solenoid valve. A subsystem of the valve is the solenoid coil shown in Figure P5.37. Current through the coil, L, generates a magnetic field that produces a force to operate the valve. Figure P5.37can be represented as a block diagram (*Vaughan, 1996*)

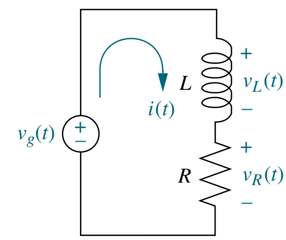


FIGURE P5.37 Solenoid coil circuit

a. Derive a block diagram of a feedback system that represents the coil circuit, where the applied voltage, $v_g(t)$, is the input, the coil voltage, $v_L(t)$, is the error voltage, and the current, $i(t)$, is the output.

b. For the block diagram found in Part **a**, find the Laplace transform of the output current, $I(s)$.

c. Solve the circuit of Figure P5.37 for $I(s)$, and compare to your result in Part **b**.

58. Ktesibios' water clock (see Section 1.2) is probably the first man-made system in which feedback was used in a deliberate manner. Its operations are shown in Figure P5.38(a). The clock indicates time progressively on scale D as water falls from orifice A toward vessel B. Clock accuracy depends mainly on water height h_f in the water reservoir G, which must be maintained at a constant level h_r by means of the conical float F that moves up or down to control the water inflow. Figure P5.38(b) shows a block diagram describing the system (*Lepschy, 1992*).

Let $q_i(t)$ and $q_o(t)$ represent the input and output water flow, respectively, and h_m the height of water in vessel B. Use Mason's rule to find the following transfer functions, assuming α and β are constants:

a. $\dfrac{H_m(s)}{H_r(s)}$

b. $\dfrac{H_f(s)}{H_r(s)}$

c. $\dfrac{Q_i(s)}{H_r(s)}$

d. $\dfrac{Q_o(s)}{H_r(s)}$

e. Using the above transfer functions, show that if $h_r(t) = \text{constant}$, then $q_o(t) = \text{constant}$ and $h_m(t)$ increases at a constant speed.

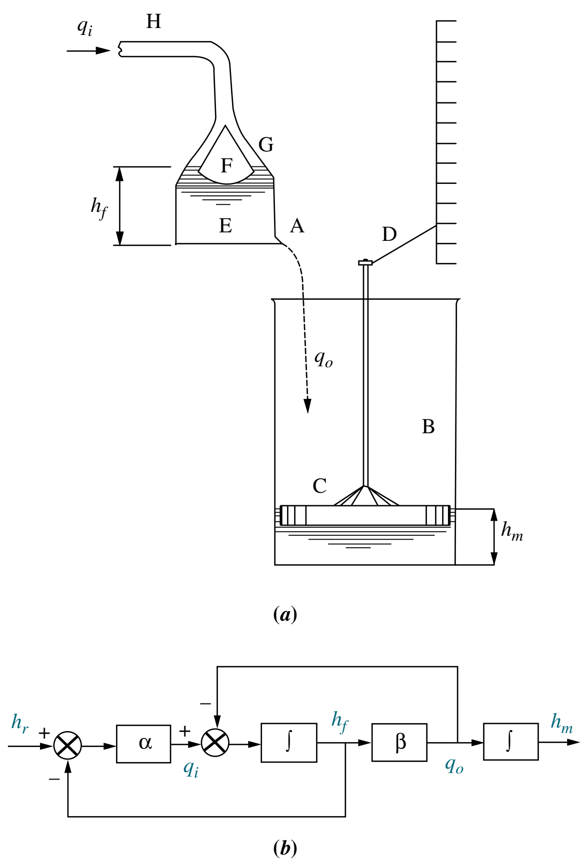


FIGURE P5.38 **a.** Ktesibios' water clock; **b.** water clock block diagram (© 1992 IEEE)

59. Some robotic applications can benefit from actuators in which load position as well as exerted force are controlled. Figure P5.39 shows the block diagram of such an actuator, where u_1 and u_2 are voltage inputs to two coils, each of which controls a pneumatic piston, and y represents the load displacement.

The system's output is u, the differential pressure acting on the load. The system also has a disturbance input f_{ext}, which represents external forces that are not system generated, but are acting on the load. A is a constant (*Ben-Dov, 1995*). Use any method to obtain:

a. An expression for the system's output in terms of the inputs u_1 and u_2 (Assume $f_{ext} = 0$.)

b. An expression for the effect of f_{ext} on the output u (Assume u_1 and $u_2 = 0$.)

c. What condition on the inputs u_1 and u_2 will result in $u = 0$?

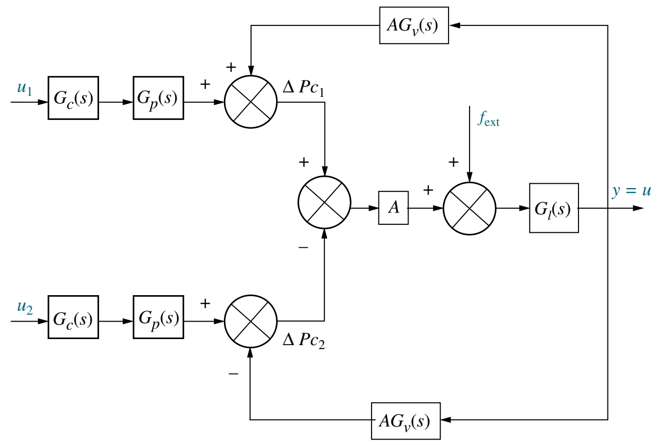


FIGURE P5.39 Actuator block diagram (© 1995 IEEE)

60. Figure P5.40 shows a noninverting operational amplifier.

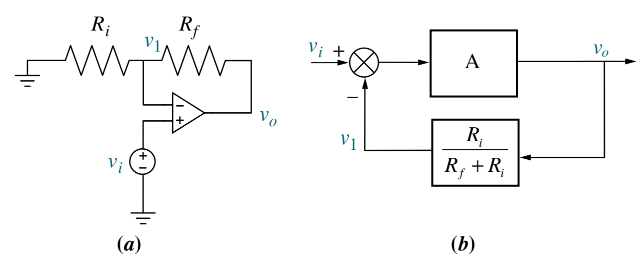


FIGURE P5.40 **a.** Noninverting amplifier; **b.** block diagram

Assuming the operational amplifier is ideal,

a. Verify that the system can be described by the following two equations:

$$v_o = A(v_i - v_o)$$

$$v_1 = \frac{R_i}{R_i + R_f} v_o$$

b. Check that these equations can be described by the block diagram of Figure P5.40(*b*)

c. Use Mason's rule to obtain the closed-loop system transfer function $\dfrac{V_o(s)}{V_i(s)}$

d. Show that when $A \to \infty$, $\dfrac{V_o(s)}{V_i(s)} = 1 + \dfrac{R_f}{R_i}$

61. Figure P5.41 shows the diagram of an inverting operational amplifier.

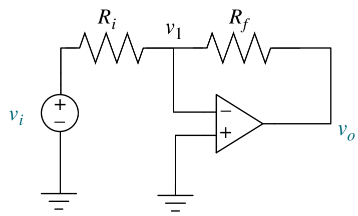


FIGURE P5.41 Inverting operational amplifier

a. Assuming an ideal operational amplifier, use a similar procedure to the one outlined in Problem 60 to find the system equations.

b. Draw a corresponding block diagram and obtain the transfer function $\dfrac{V_o(s)}{V_i(s)}$.

c. Show that when $A \to \infty$, $\dfrac{V_o(s)}{V_i(s)} = -\dfrac{R_f}{R_i}$.

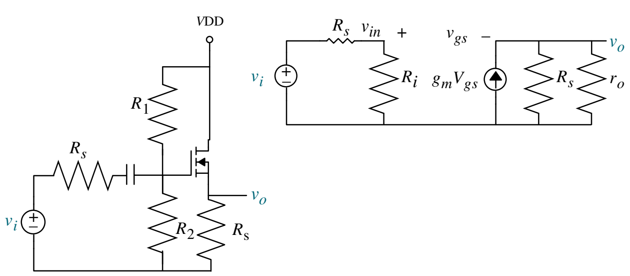
62. Figure P5.42(a) shows an n-channel enhancement-mode MOSFET source follower circuit. Figure P5.42(b) shows its small-signal equivalent (where $R_i = R_1 \| R_2$) (*Neamen, 2001*).

a. Verify that the equations governing this circuit are

$$\frac{v_{in}}{v_i} = \frac{R_i}{R_i + R_s}; \quad v_{gs} = v_{in} - v_o; \quad v_o = g_m(R_s \| r_o)v_{gs}$$

b. Draw a block diagram showing the relations between the equations.

c. Use the block diagram in Part **b** to find $\dfrac{V_o(s)}{V_i(s)}$.



(a) (b)

FIGURE P5.42 **a.** An n-channel enhancement-mode MOSFET source follower circuit; **b.** small-signal equivalent

63. A car active suspension system adds an active hydraulic actuator in parallel with the passive damper and spring to create a dynamic impedance that responds to road variations. The block diagram of Figure P5.43 depicts such an actuator with closed-loop control.

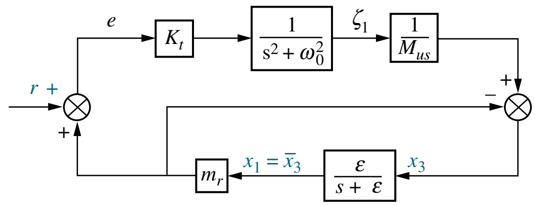


FIGURE P5.43 (© 1997 IEEE)

In the figure, K_t is the spring constant of the tire, M_{US} is the wheel mass, r is the road disturbance, x_1 is the vertical car displacement, x_3 is the wheel vertical displacement, $\omega_0^2 = \frac{K_t}{M_{US}}$ is the natural frequency of the unsprung system and ε is a filtering parameter to be judiciously chosen (*Lin, 1997*). Find the two transfer functions of interest:

a. $\dfrac{X_3(s)}{R(s)}$

b. $\dfrac{X_1(s)}{R(s)}$

64. The basic unit of skeletal and cardiac muscle cells is a *sarcomere*, which is what gives such cells a striated (parallel line) appearance. For example, one bicep cell has about 10^5 sarcomeres. In turn, sarcomeres are composed of protein complexes. Feedback mechanisms play an important role in sarcomeres and thus muscle contraction. Namely, Fenn's law says that the energy liberated during muscle contraction depends on the initial conditions and the load encountered. The following linearized model describing sarcomere contraction has been developed for cardiac muscle:

$$\begin{bmatrix} \dot{A} \\ \dot{T} \\ \dot{U} \\ \dot{SL} \end{bmatrix} = \begin{bmatrix} -100.2 & -20.7 & -30.7 & 200.3 \\ 40 & -20.22 & 49.95 & 526.1 \\ 0 & 10.22 & -59.95 & -526.1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} A \\ T \\ U \\ SL \end{bmatrix} + \begin{bmatrix} 208 \\ -208 \\ -108.8 \\ -1 \end{bmatrix} u(t)$$

$$y = \begin{bmatrix} 0 & 1570 & 1570 & 59400 \end{bmatrix} \begin{bmatrix} A \\ T \\ U \\ SL \end{bmatrix} - 6240u(t)$$

where

A = density of regulatory units with bound calcium and adjacent weak cross bridges (μM)
T = density of regulatory units with bound calcium and adjacent strong cross bridges (M)
U = density of regulatory units without bound calcium and adjacent strong cross bridges (M)
SL = sarcomere length (m)

The system's input is $u(t)$ = the shortening muscle velocity in meters/second and the output is $y_{(t)}$ = muscle force output in Newtons (*Yaniv, 2006*).

Do the following:

a. Use MATLAB to obtain the transfer function $\frac{Y(s)}{U(s)}$. MATLAB **ML**

b. Use MATLAB to obtain a partial-fraction expansion for $\frac{Y(s)}{U(s)}$. MATLAB **ML**

c. Draw a signal-flow diagram of the system in parallel form. State Space **SS**

d. Use the diagram of Part **c** to express the system in state-variable form with decoupled equations. State Space **SS**

65. An electric ventricular assist device (EVAD) has been designed to help patients with diminished but still functional heart pumping action to work in parallel with the natural heart. The device consists of a brushless dc electric motor that actuates on a pusher plate. The plate movements help the ejection of blood in systole and sac filling in diastole. System dynamics during systolic mode have been found to be:

$$\begin{bmatrix} \dot{x} \\ \dot{v} \\ \dot{P}_{ao} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -68.3 & -7.2 \\ 0 & 3.2 & -0.7 \end{bmatrix} \begin{bmatrix} x \\ v \\ P_{ao} \end{bmatrix} + \begin{bmatrix} 0 \\ 425.4 \\ 0 \end{bmatrix} e_m$$

The state variables in this model are x, the pusher plate position, v, the pusher plate velocity, and P_{ao}, the aortic blood pressure. The input to the system is e_m, the motor voltage (*Tasch, 1990*).

a. Use MATLAB to find a similarity transformation to diagonalize the system. MATLAB **ML**

b. Use MATLAB and the obtained similarity transformation of Part a to obtain a diagonalized expression for the system. MATLAB **ML**

66. In an experiment to measure and identify postural arm reflexes, subjects hold with their hands a linear hydraulic manipulator. A load cell is attached to the actuator handle to measure resulting forces. At the application of a force, subjects try to maintain a fixed posture. Figure P5.44 shows a block diagram for the combined arm-environment system.

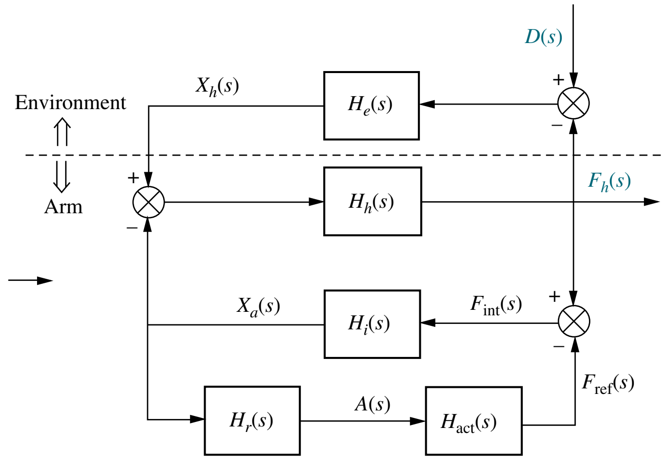


FIGURE P5.44

In the diagram, $H_r(s)$ represents the reflexive length and velocity feedback dynamics; $H_{act}(s)$ the activation dynamics, $H_i(s)$ the intrinsic act dynamics; $H_h(s)$ the hand dynamics; $H_e(s)$ the environmental dynamics; $X_a(s)$ the position of the arm; $X_h(s)$ the measured position of the hand; $F_h(s)$ the measured interaction force applied by the hand; $F_{int}(s)$ the intrinsic force; $F_{ref}(s)$ the reflexive force; $A(s)$ the reflexive activation; and $D(s)$ the external force perturbation (*de Vlugt, 2002*).

a. Obtain a signal-flow diagram from the block diagram.

b. Find $\frac{F_h(s)}{D(s)}$.

67. Use LabVIEW's Control Design and Simulation Module to obtain the controller and the observer canonical forms for: State Space **SS** LabVIEW **LV**

$$G(s) = \frac{s^2 + 7s + 2}{s^3 + 9s^2 + 26s + 24}$$

68. A virtual reality simulator with haptic (sense of touch) feedback was developed to simulate the control of a submarine driven through a joystick input. Operator haptic feedback is provided through joystick position constraints and simulator movement (*Karkoub, 2010*). Figure P5.45 shows the block diagram of the haptic feedback system in which the input u_h is the force exerted by the muscle of the human arm; and the outputs are y_s, the position of the simulator, and y_j, the position of the joystick.

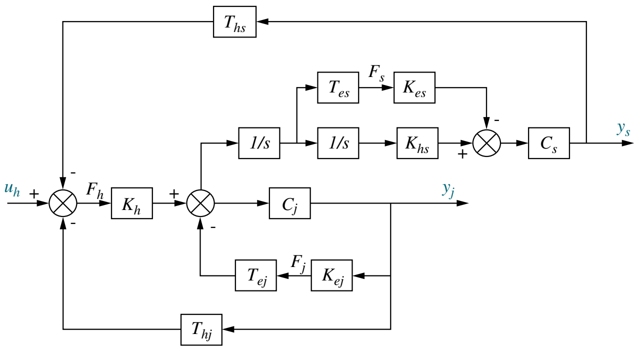


FIGURE P5.45 Copyright © 2010 Cambridge University Press. Reprinted with permission.

a. Find the transfer function $\dfrac{Y_s(s)}{U_h(s)}$.

b. Find the transfer function $\dfrac{Y_j(s)}{U_h(s)}$.

69. Some medical procedures require the insertion of a needle under a patient's skin using CT scan monitoring guidance for precision. CT scans emit radiation, posing some cumulative risks for medical personnel. To avoid this problem, a remote control robot has been developed (*Piccin, 2009*). The robot controls the needle in position and angle in the constraint space of a CT scan machine and also provides the physician with force feedback commensurate with the insertion opposition encountered by the type of tissue in which the needle is inserted. The robot has other features that give the operator the similar sensations and maneuverability as if the needle was inserted directly. Figure P5.46 shows the block diagram of the force insertion mechanism, where F_h is the input force and X_h is the output displacement. Summing junction inputs are positive unless indicated with a negative sign. By way of explanation, Z = impedance; G = transfer function; C_i = communication channel transfer functions; F = force; X = position. Subscripts h and m refer to the master manipulator. Subscripts s and e refer to the slave manipulator.

a. Assuming $Z_h = 0$, $C_1 = C_s$, $C_2 = 1 + C_6$ and $C_4 = -C_m$ use Mason's Rule to show that the transfer function from the operators force input F_h to needle displacement X_h is given by

$$Y(s) = \frac{X_h(s)}{F_h(s)} = \frac{Z_m^{-1}C_2(1 + G_sC_s)}{1 + G_sC_s + Z_m^{-1}(c_m + C_2Z_eG_sC_s)}$$

b. Now with $Z_h \neq 0$ show that $\dfrac{X_h(s)}{F_h(s)} = \dfrac{Y(s)}{1 + Y(s)Z_h}$

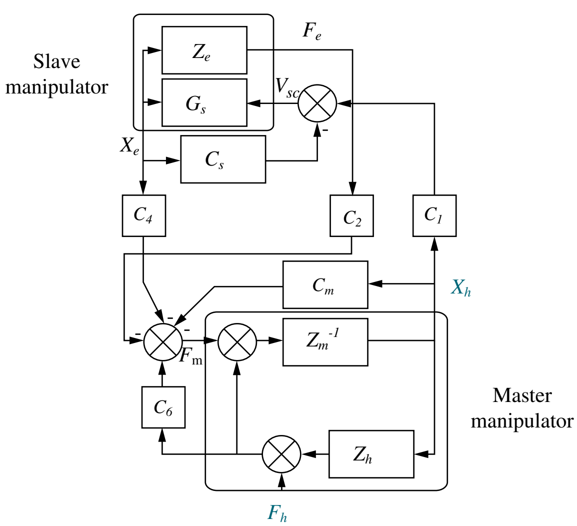


FIGURE P5.46

70. A hybrid solar cell and diesel power distribution system has been proposed and tested (*Lee, 2007*). The system has been shown to have a very good uninterruptible power supply as well as line voltage regulation capabilities. Figure P5.47 shows a signal-flow diagram of the system. The output, V_{Load}, is the voltage across the load. The two inputs are I_{Cf}, the reference current, and I_{Dist}, the disturbance representing current changes in the supply.

a. Refer to Figure P5.47 and find the transfer function $\dfrac{V_{Load}(s)}{I_{Cf}(s)}$.

b. Find the transfer function $\dfrac{V_{Load}(s)}{I_{Dist}(s)}$.

71. Continuous casting in steel production is essentially a solidification process by which molten steel is

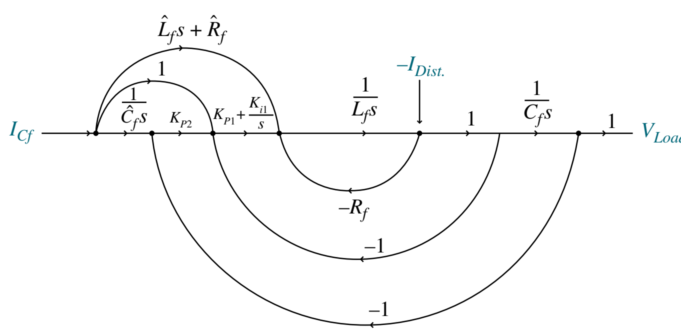


FIGURE P5.47

solidified into a steel slab after passing through a mold, as shown in Figure P5.48(a). Final product dimensions depend mainly on the casting speed V_p (in m/min), and on the stopper position X(in %) that controls the flow of molten material into the mold (*Kong, 1993*). A simplified model of a casting system is shown in Figure P5.48(b) (*Kong, 1993*) and (*Graebe, 1995*). In the model, H_m = mold level (in mm); H_t = assumed constant height of molten steel in the tundish; D_z = mold thickness = depth of nozzle immerged into molten steel; and W_t = weight of molten steel in the tundish.

For a specific setting let $A_m = 0.5$ and

$$G_x(s) = \frac{0.63}{s + 0.926}$$

Also assume that the valve positioning loop may be modeled by the following second-order transfer function:

$$G_V(s) = \frac{X(s)}{Y_C(s)} = \frac{100}{s^2 + 10s + 100}$$

and the controller is modeled by the following transfer function:

$$G_C(s) = \frac{1.6(s^2 + 1.25s + 0.25)}{s}$$

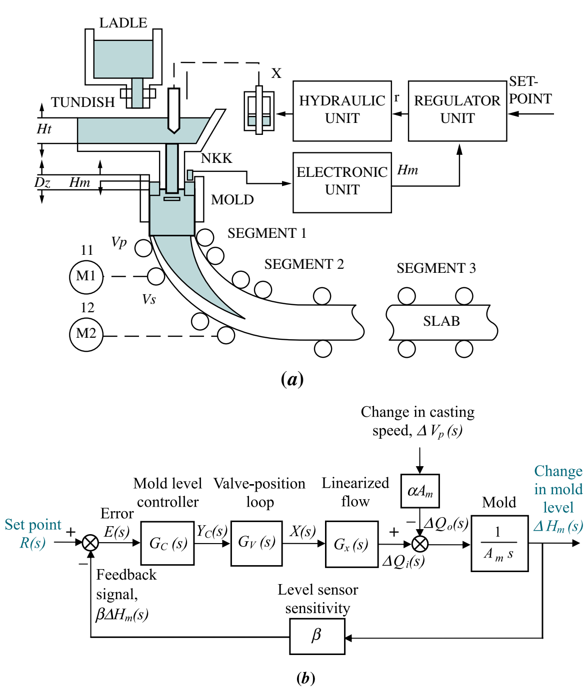


FIGURE P5.48 Steel mold process: **a.** process (© 1993 IEEE); **b.** block diagram

The sensitivity of the mold level sensor is $\beta = 0.5$ and the initial values of the system variables at $t = 0^-$ are: $R(0^-) = 0$; $Y_C(0^-) = X(0^-) = 41.2$; $\Delta H_m(0^-) = 0$; $H_m(0^-) = -75$; $\Delta V_p(0^-) = 0$; and $V_p(0^-) = 0$. Do the following:

a. Assuming $v_p(t)$ is constant $[\Delta v_p = 0]$, find the closed-loop transfer function $T(s) = \Delta H_m(s)/R(s)$.

b. For $r(t) = 5\,u(t)$, $v_p(t) = 0.97\,u(t)$, and $H_m(0^-) = -75$ mm, use Simulink to simulate the system. Record the time and mold level (in array format) by connecting them to **Workspace** sinks, each of which should carry the respective variable name. After the simulation ends, utilize MATLAB plot commands to obtain and edit the graph of $h_m(t)$ from $t = 0$ to 80 seconds.

Simulink **SL**

72. A simplified second-order transfer function model for bicycle dynamics is given by

State Space **SS**

$$\frac{\varphi(s)}{\delta(s)} = \frac{aV}{bh}\frac{\left(s + \frac{V}{a}\right)}{\left(s^2 - \frac{g}{h}\right)}$$

The input is $\delta(s)$, the steering angle, and the output is $\varphi(s)$, the tilt angle (between the floor and the bicycle longitudinal plane). In the model parameter a is the horizontal distance from the center of the back wheel to the bicycle center of mass; b is the horizontal distance between the centers of both wheels; h is the vertical distance from the center of mass to the floor; V is the rear wheel velocity (assumed constant); and g is the gravity constant. It is also assumed that the rider remains at a fixed position with respect to the bicycle so that the steer axis is vertical and that all angle deviations are small (*Åstrom, 2005*).

a. Obtain a state-space representation for the bicycle model in phase-variable form.

b. Find system eigenvalues and eigenvectors.

c. Find an appropriate similarity transformation matrix to diagonalize the system and obtain the state-space system's diagonal representation.

73. It is shown in Figure 5.6(c) that when negative feedback is used, the overall transfer function for the system of Figure 5.6(b) is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

Develop the block diagram of an alternative feedback system that will result in the same closed-loop transfer function, $C(s)/R(s)$, with $G(s)$ unchanged and unmoved. In addition, your new block diagram must have unity gain in the feedback path. You can add input transducers and/or controllers in the main forward path as required.

DESIGN PROBLEMS

74. The motor and load shown in Figure P5.49(a) are used as part of the unity feedback system shown in Figure P5.49(b). Find the value of the coefficient of viscous damping, D_L, that must be used in order to yield a closed-loop transient response having a 20% overshoot.

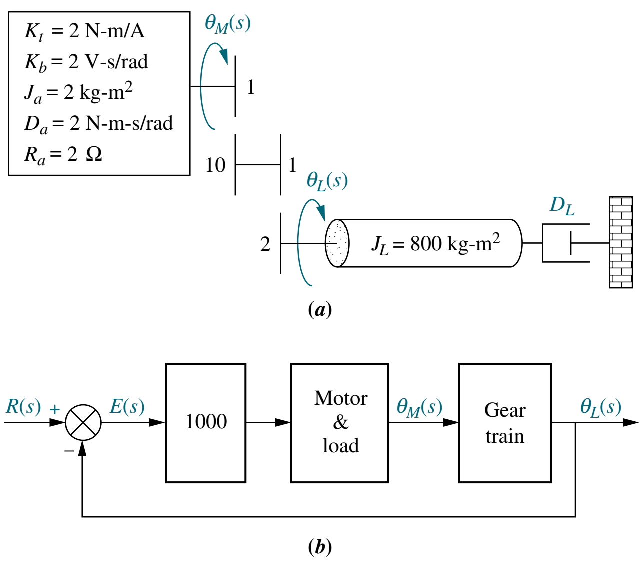


FIGURE P5.49 Position control: **a.** motor and load; **b.** block diagram

75. Assume that the motor whose transfer function is shown in Figure P5.50(a) is used as the forward path of a closed-loop, unity feedback system.

a. Calculate the percent overshoot and settling time that could be expected.

b. You want to improve the response found in Part **a**. Since the motor and the motor constants cannot be changed, an amplifier and a tachometer (voltage generator) are inserted into the loop, as shown in Figure P5.50. Find the values of K_1 and K_2 to yield a 16% overshoot and a settling time of 0.2 second.

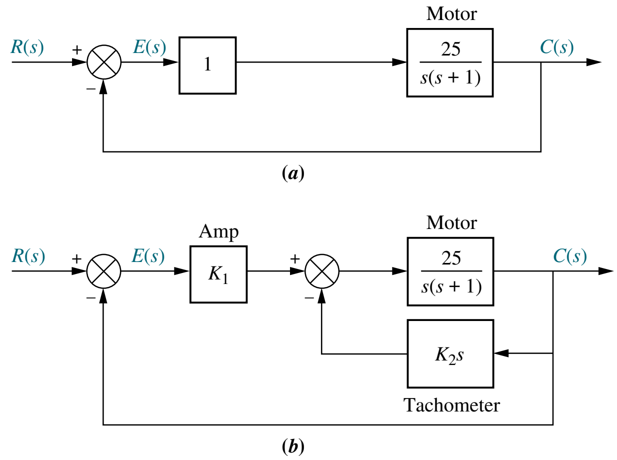


FIGURE P5.50 **a.** Position control; **b.** position control with tachometer

76. The system shown in Figure P5.51 will have its transient response altered by adding a tachometer. Design K and K_2 in the system to yield a damping ratio of 0.69. The natural frequency of the system before the addition of the tachometer is 10 rad/s.

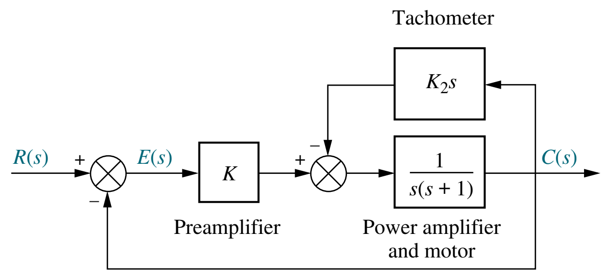


FIGURE P5.51 Position control

77. The mechanical system shown in Figure P5.52(a) is used as part of the unity feedback system shown in Figure P5.52(b). Find the values of M and D to yield 20% overshoot and 2 seconds settling time.

WileyPLUS WPCS Control Solutions

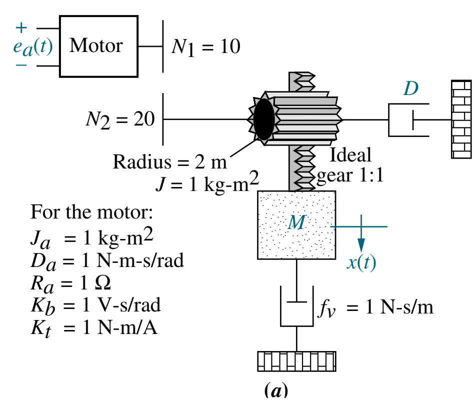
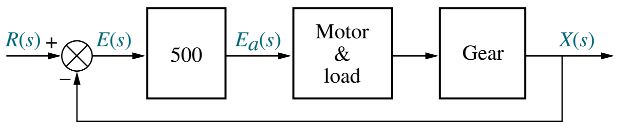


FIGURE P5.52 **a.** Motor and load; (*figure continues*)



(*b*)

FIGURE P5.52 (*Continued*) **b.** motor and load in feedback system

78. Assume ideal operational amplifiers in the circuit of Figure P5.53.

a. Show that the leftmost operational amplifier works as a subtracting amplifier. Namely, $v_1 = v_o - v_{in}$.

b. Draw a block diagram of the system, with the subtracting amplifier represented with a summing junction, and the circuit of the rightmost operational amplifier with a transfer function in the forward path. Keep R as a variable.

c. Obtain the system's closed-loop transfer function.

d. For a unit step input, obtain the value of R that will result in a settling time $T_s = 1$ msec.

e. Using the value of R calculated in Part **d**, make a sketch of the resulting unit step response.

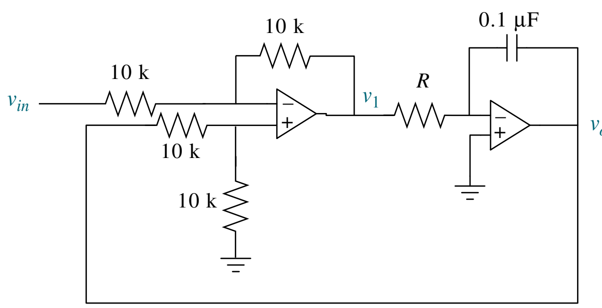


FIGURE P5.53

PROGRESSIVE ANALYSIS AND DESIGN PROBLEMS

79. High-speed rail pantograph. Problem 21 in Chapter 1 discusses the active control of a pantograph mechanism for high-speed rail systems. In this problem you found a functional block diagram relating the output force (*actual*) to the input force (*desired* output). In Problem 67, Chapter 2, you found the transfer function for the pantograph dynamics, that is, the transfer function relating the displacement of the spring that models the head to the applied force, or $G(s) = (Y_h(s) - Y_{cat}(s))/F_{up}(s)$ (*O'Connor, 1997*). We now create a pantograph active-control loop by adding the following components and following your functional block diagram found in Problem 21, Chapter 1: input transducer ($G_i(s) = 1/100$), controller ($G_c(s) = K$), actuator ($G_a(s) = 1/1000$), pantograph spring ($K_s = 82.3 \times 10^3$ N/m), and sensor ($H_o(s) = 1/100$).

a. Using the functional block diagram from your solution of Problem 21 in Chapter 1, and the pantograph dynamics, $G(s)$, found in Problem 67, Chapter 2, assemble a block diagram of the active pantograph control system.

b. Find the closed-loop transfer function for the block diagram found in Part **a** if $K = 1000$.

c. Represent the pantograph dynamics in phase-variable form and find a state-space representation for the closed-loop system if $K = 1000$). State Space SS

80. Control of HIV/AIDS. Given the HIV system of Problem 82 in Chapter 4 and repeated here for convenience (*Craig, 2004*): State Space SS

$$\begin{bmatrix} \dot{T} \\ \dot{T}^* \\ \dot{v} \end{bmatrix} = \begin{bmatrix} -0.04167 & 0 & -0.0058 \\ 0.0217 & -0.24 & 0.0058 \\ 0 & 100 & -2.4 \end{bmatrix} \begin{bmatrix} T \\ T^* \\ v \end{bmatrix} + \begin{bmatrix} 5.2 \\ -5.2 \\ 0 \end{bmatrix} u_1$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} T \\ T^* \\ v \end{bmatrix}$$

Express the system in the following forms:

a. Phase-variable form

b. Controller canonical form

c. Observer canonical form

Finally,

d. `Use MATLAB to obtain the system's diagonalized representation.` MATLAB ML

81. Hybrid vehicle. Figure P5.54 shows the block diagram of a possible cascade control scheme for an HEV driven by a dc motor (*Preitl, 2007*).

Let the speed controller $G_{SC}(s) = 100 + \frac{40}{s}$, the torque controller and power amp $K_A G_{TC}(s) = 10 + \frac{6}{s}$, the current sensor sensitivity $K_{CS} = 0.5$, the speed sensor sensitivity $K_{SS} = 0.0433$. Also following the development in previous chapters $\frac{1}{R_a} = 1$; $\eta_{tot} K_t = 1.8$; $k_b = 2$; $D = k_f = 0.1$; $\frac{1}{J_{tot}} = \frac{1}{7.226}$; $\frac{r}{i_{tot}} = 0.0615$; and $\rho C_w A v_0 \frac{r}{i_{tot}} = 0.6154$.

a. Substitute these values in the block diagram, and find the transfer function, $T(s) = V(s)/R_v(s)$, using block-diagram reduction rules. [Hint: Start

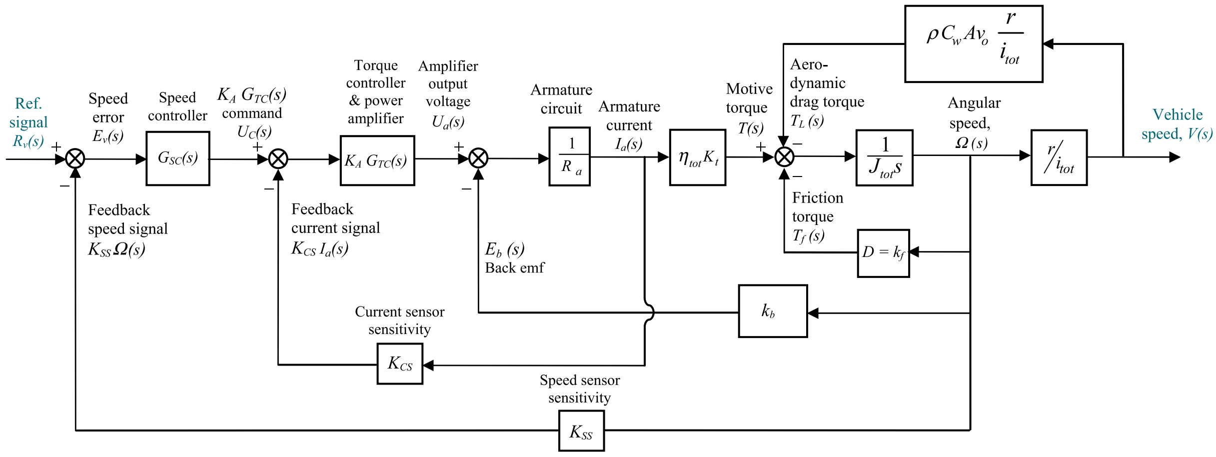


FIGURE P5.54

by moving the last $\frac{r}{i_{tot}}$ block to the right past the pickoff point.]

b. Develop a Simulink model for the original system in Figure P5.54. Set the reference signal input, $r_v(t) = 4\,u(t)$, as a step input with a zero initial value, a step time = 0 seconds, and a final value of 4 volts. Use X-Y graphs to display (over the period from 0 to 8 seconds) the response of the following variables to the step input: (1) change in car speed (m/s), (2) car acceleration (m/s^2), and (3) motor armature current (A).

To record the time and the above three variables (in array format), connect them to four **Workspace** sinks, each of which carry the respective variable name. After the simulation ends, utilize MATLAB plot commands to obtain and edit the three graphs of interest.

Cyber Exploration Laboratory

Experiment 5.1

Objectives To verify the equivalency of the basic forms, including cascade, parallel, and feedback forms. To verify the equivalency of the basic moves, including moving blocks past summing junctions, and moving blocks past pickoff points.

Minimum Required Software Packages MATLAB, Simulink, and the Control System Toolbox

Prelab

1. Find the equivalent transfer function of three cascaded blocks, $G_1(s) = \dfrac{1}{s+1}$, $G_2(s) = \dfrac{1}{s+4}$, and $G_3(s) = \dfrac{s+3}{s+5}$.
2. Find the equivalent transfer function of three parallel blocks, $G_1(s) = \dfrac{1}{s+4}$, $G_2(s) = \dfrac{1}{s+4}$, and $G_3(s) = \dfrac{s+3}{s+5}$.
3. Find the equivalent transfer function of the negative feedback system of Figure P5.55 if $G(s) = \dfrac{s+1}{s(s+2)}$, and $H(s) = \dfrac{s+3}{s+4}$.

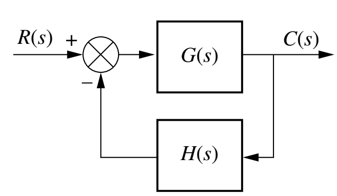


FIGURE P5.55